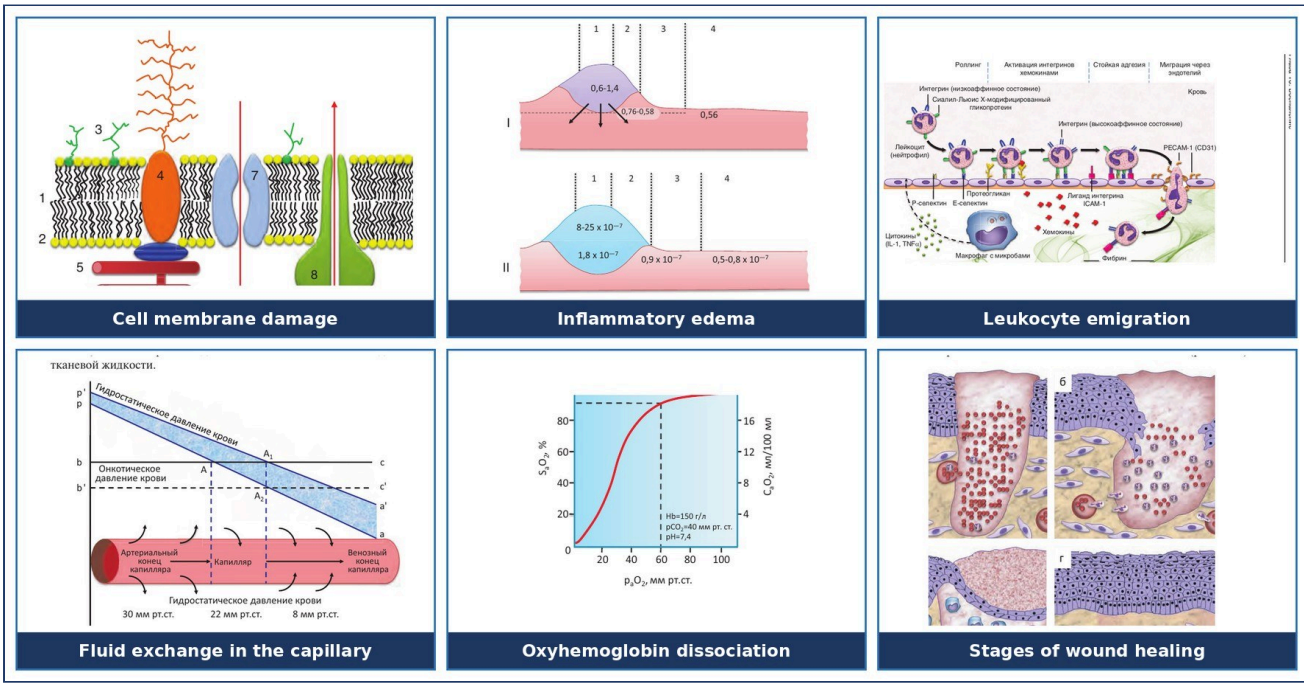
MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Normal and Pathological Physiology

PATHOLOGICAL PHYSIOLOGY

complete study manual

A study manual for practical classes



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Tashkent — 2026

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Pathological Physiology: Complete Study Manual. A study manual for practical classes. — Tashkent, 2026. English edition, translated from the Uzbek original.

This manual is based on the 27 practical class topics of the calendar-thematic plan (semesters V–VI) of the course “Pathological Physiology” approved for the 2026–2027 academic year. It covers general nosology, etiology and pathogenesis, typical pathological processes (local circulatory disorders, inflammation, allergy, tumors, hypoxia), metabolic disorders, and the pathophysiology of the blood, cardiovascular, respiratory, digestive, hepatobiliary, renal and endocrine systems.

Each chapter consists of learning objectives, a theoretical part (with “Remember”, “Clinical significance”, “Memory aid” and “Caution” boxes), an experimental (practical) part, practical skill algorithms, review questions, 15 test questions, 3 clinical cases and a quick review table. The answers to the test questions are given at the end of the book.

The manual is based on the textbooks “Pathological Physiology” by N.H. Abdullayev (2008), “Pathological Physiology” by A.A. Haqberdiyev and “Pathophysiology” by R.Q. Azimov (2010), supplemented by the main Russian-language textbooks of pathophysiology (V.V. Novitskiy, Ye.D. Goldberg; N.N. Zayko, Yu.V. Bits; A.V. Ataman). The figures are taken from these sources or drawn on their basis, and the source is given under each figure. Labels inside some figures remain in the language of the original source.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before clinical information is applied in practice, it is recommended to compare it with the main textbooks.

The manual is intended for 3rd-year students of the faculties of general medicine of higher medical education institutions.

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Preface

Dear student! Pathological physiology (pathophysiology) is the science that studies the vital activity of the diseased organism and the causes, mechanisms of development and outcomes of diseases. It serves as a bridge between the biomedical sciences (physiology, biochemistry, histology) and the clinical disciplines: the clinical thinking that answers the question “what is happening to the patient?” is formed precisely in this subject.

The manual follows the 27 practical class topics of the department's calendar-thematic plan. Every chapter is built in the same order:

- **Learning objectives** — what you should know and be able to do by the end of the class;
- **theoretical part** — the main concepts of the topic, tables, diagrams and figures;
- **experimental part** — the experiments performed in the class and their analysis;
- **practical skills** — step-by-step algorithms;
- **review questions, test questions and clinical cases** — for self-assessment;
- **quick review** — a table for reviewing the topic at a glance before an exam.

The colored boxes in the text mean the following:

REMEMBER
The most important definitions, numbers and classifications — they must be memorized.

CLINICAL SIGNIFICANCE
The significance of the topic in clinical practice, examples related to patients.

MEMORY AID
Techniques and abbreviations that make memorizing easier.

CAUTION
Common mistakes, dangerous situations and discrepancies between textbooks.

The animal experiments in the experimental part are classic models accepted in the department's practice; they are carried out under the teacher's supervision in compliance with the requirements of bioethics. It is recommended to read each topic of the manual together with the main textbook.

The compiler

TOPIC 1

The effect of etiologic factors on the organism. Pathophysiology: subject, methods and tasks. General nosology, etiology and pathogenesis

Learning objectives

To introduce the student to the subject, methods and history of pathological physiology; to master the basic concepts of the general doctrine of disease (nosology), etiology and pathogenesis; to understand the role of causes and conditions in the origin of disease; and to teach the student to analyze any disease according to the scheme “cause — conditions — pathogenesis — outcome”.

By the end of the session, the student should know:

- the subject of pathophysiology, its components, methods and links with other disciplines;
- the main stages in the history of the discipline and its founders;
- the concepts of health, disease, the “predisease” state, pathological reaction, pathological process and pathological state;
- the periods and outcomes of disease and the mechanisms of recovery;
- the doctrines of etiology and the difference between causes and conditions;
- the concept of pathogenesis, its main link and the “vicious circle”;
- the principles of the nomenclature and classification of diseases (ICD-10, ICD-11).

The student should be able to:

- distinguish the etiologic factor and the conditions in any disease;
- identify the cause-and-effect chain in pathogenesis, the main link and the vicious circle;
- distinguish a pathological reaction, process and state with examples;
- justify the principles of etiotropic, pathogenetic and symptomatic treatment.

1. Subject and tasks of pathophysiology

Pathological physiology (pathophysiology) is the science that studies the vital activity of the diseased organism and the general laws of the origin, development, course and outcomes of diseases and pathological processes. The term comes from the Greek words pathos — suffering, disease; physis — nature; logos — study.

The object of study of the discipline is the diseased organism. Pathophysiology studies disease at different levels of integration — molecular, subcellular, cellular, tissue, organ, system and whole-organism level. That is why it is considered an integrative science: on the one hand it relies on biology, physiology, biochemistry, histology and pathological anatomy, and on the other hand on the clinical disciplines.

Table 1.1. *Components of pathophysiology*

Part	What it studies	Examples
General nosology	The general doctrine of disease: etiology, pathogenesis, reactivity	Causes and conditions, links of pathogenesis
Typical pathological processes	Processes that recur in the same way in different diseases	Inflammation, fever, hypoxia, edema, tumor growth

Part	What it studies	Examples
Special pathophysiology	Typical forms of dysfunction of organs and systems	Heart failure, anemia, kidney failure

REMEMBER

The main tasks of pathophysiology:

- to identify the causes and conditions of diseases and pathological processes;
- to reveal the mechanisms of their development (pathogenesis);
- to study the protective, adaptive and compensatory reactions of the organism;
- to study the outcomes of diseases and the mechanisms of recovery;
- to develop the scientific (experimental) basis of treatment and prevention.

Pathophysiology and pathological anatomy are closely interrelated: pathological anatomy studies the changes in **structure** in disease, and pathophysiology studies the changes in **function**. Structure and function form an inseparable unity — there is no functional disorder without a change in structure, and vice versa.

2. Research methods of pathophysiology

The main method of pathophysiology is **experiment**, that is, modeling of diseases and pathological processes in animals. Experiment makes it possible to study the cause of a disease, its mechanism of development and methods of treatment without risk to humans, in dynamics and at any level of integration.

Table 1.2. *Research methods of pathophysiology*

Method	Essence	Advantage and limitation
Experiment (modeling)	Artificial induction and observation of a pathological process in an animal	Studies cause and effect directly; the result cannot be fully transferred to humans
Clinical research	Examination of patients by functional, biochemical and immunological methods	Gives information directly about humans; the possibility of intervention is limited
Mathematical and computer modeling	Study of processes through computational models	Predicts situations in which an experiment cannot be performed
Theoretical analysis	Generalization of accumulated data, creation of theories	Gives a comprehensive picture of the disease

The following methods are used in experiments: **removal of an organ** (extirpation — for example, modeling diabetes mellitus by removing the pancreas), **stimulation**, the **isolated organ** method, **parabiosis** (joining the circulations of two animals), **recording** (ECG, EEG, biochemical parameters), cell and tissue cultures, and genetic modeling (transgenic and “knockout” animals).

CAUTION

Animals and humans differ in structure, metabolism and reactivity (interspecies differences). Therefore the result of an experiment is not transferred to humans **directly** — it must be confirmed by clinical observations. For example, rats synthesize vitamin C, whereas humans do not; for this reason a model of scurvy cannot be created in rats.

Bioethical rules are followed when performing experiments: reducing the animal's suffering (analgesia, anesthesia), minimizing the number of animals and, where possible, replacing them with alternative models (cell culture, computer model) (the “3R” principle: Replacement, Reduction, Refinement).

3. History of the discipline

Views on the nature of disease developed together with medicine. Hippocrates (5th–4th centuries BC) regarded disease as an imbalance of four humors (blood, phlegm, yellow bile and black bile) (the humoral theory). In “The Canon of Medicine”, Avicenna (Ibn Sina) divided the causes of diseases into external and internal factors, suggested that infectious diseases are spread by invisible creatures, and divided medicine into theoretical and practical parts.

Table 1.3. *Important stages in the development of pathophysiology*

Scientist	Contribution
Claude Bernard (1813–1878)	Founder of experimental medicine; the concept of the “constancy of the internal environment”
Rudolf Virchow (1821–1902)	The theory of cellular pathology: “every cell from a cell”
V.V. Pashutin (1845–1901)	Founder of the department of general pathology (pathophysiology) in Russia (Kazan, 1874)
I.I. Mechnikov (1845–1916)	Phagocytosis and the comparative pathology of inflammation; Nobel Prize (1908)
I.P. Pavlov (1849–1936)	The role of the nervous system in pathology (nervism), experimental neuroses
W. Cannon (1871–1945)	The concept of homeostasis
H. Selye (1907–1982)	The doctrine of stress and the general adaptation syndrome
A.A. Bogomolets (1881–1946)	Proposed the name “pathological physiology” for the discipline (with S.S. Khalatov, 1924); physiology of connective tissue

REMEMBER

The three main methodological principles of pathophysiology: **experiment** (modeling), **the systems approach** (disease is the response of the whole organism) and **cause-and-effect analysis** (determinism).

4. General nosology. Health and disease

Nosology (Greek nosos — disease, logos — study) is the doctrine of disease. **General nosology** studies the general laws of the nature, causes, development and classification of disease, whereas **special nosology** studies individual diseases (nosological entities).

Health, according to the definition of the World Health Organization (WHO, 1948), is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.

The norm is the state of the organism that is optimally adapted to the conditions of the external environment. The concept of the norm is relative: an increased heart rate during physical work is normal, but at rest it may be a sign of disease. The reference ranges of laboratory parameters cover 95% of the healthy population, that is, in 5% of healthy people a parameter may fall “outside the norm”.

Disease is a complex systemic response of the organism to the action of a harmful factor, in which injury (alteration) and protective-adaptive reactions appear at the same time; as a result, the adaptation of the organism to the external environment and its working capacity decrease.

REMEMBER

The main features of disease:

- it has a cause (there is no disease without a cause);
- it has two opposite sides: injury and protection-adaptation;
- it is a disease of the whole organism (even with local signs, the response is systemic);
- the adaptive capacity and working capacity of the organism decrease;
- it is a new qualitative state: the diseased organism is not a “reduced” copy of the healthy one.

The predisease (prenosological) state is an intermediate state between health and disease: the adaptive capacity of the organism is reduced, but there are no definite signs of disease yet. Examples are intermittent rises in arterial pressure, impaired glucose tolerance (prediabetes) and chronic fatigue. Preventive measures are most effective during this period.

Pathological reaction, pathological process, pathological state

Table 1.4. *Main forms of pathology*

Concept	Definition	Examples
Pathological reaction	A disproportionate, short-term, unusual response to a stimulus	A brief rise in arterial pressure, pathological reflexes, allergic itching
Pathological process	A regular combination of injury and protective-adaptive reactions	Inflammation, fever, hypoxia, edema, thrombosis
Pathological state	A slowly developing or stable, long-lasting change	A scar, the state after amputation, congenital defects, contracture
Disease	A combination of one or several pathological processes, with reduced working capacity	Pneumonia, diabetes mellitus, myocardial infarction

There is no sharp boundary between a pathological process and a pathological state: under a new influence a pathological state can again turn into a process (for example, a pigmented spot (nevus) can turn into melanoma under mechanical or ultraviolet influence).

Typical pathological processes are processes formed during evolution that develop according to the same laws in different diseases, different organs and different animal species: inflammation, fever, hypoxia, edema, tumor growth, degeneration, local circulatory disorders and allergy.

MEMORY AID

“R-P-S-D” — Reaction → Process → State → Disease: complexity and duration increase. A reaction is “a moment”, a process is “a movement”, a state is “a trace”, a disease is “the whole organism”.

Protective-adaptive reactions in disease

In disease, the organism activates a number of reactions in response to injury:

- **protective reactions** — removal and neutralization of the pathogenic factor: barriers (skin, mucous membranes, the blood-brain barrier), protective reflexes (cough, sneezing, vomiting, lacrimation), phagocytosis, the immune response;

- **adaptive reactions** — adjustment of the organism's activity to changed conditions (for example, an increase in the number of red blood cells at high altitude);
- **compensatory reactions** — replacement of an impaired function by other mechanisms (hypertrophy of one kidney after removal of the other, tachycardia in anemia);
- **restorative (reparative) reactions** — restoration of the injured structure (regeneration).

> **CAUTION**
> A protective-adaptive reaction can also become harmful when excessive. For example, fever is a protective reaction against infection, but above 41 °C the brain is injured; edema in inflammation delimits the focus, but laryngeal edema leads to asphyxia. This is **the dialectical unity of protection and injury**.

5. Periods and outcomes of disease

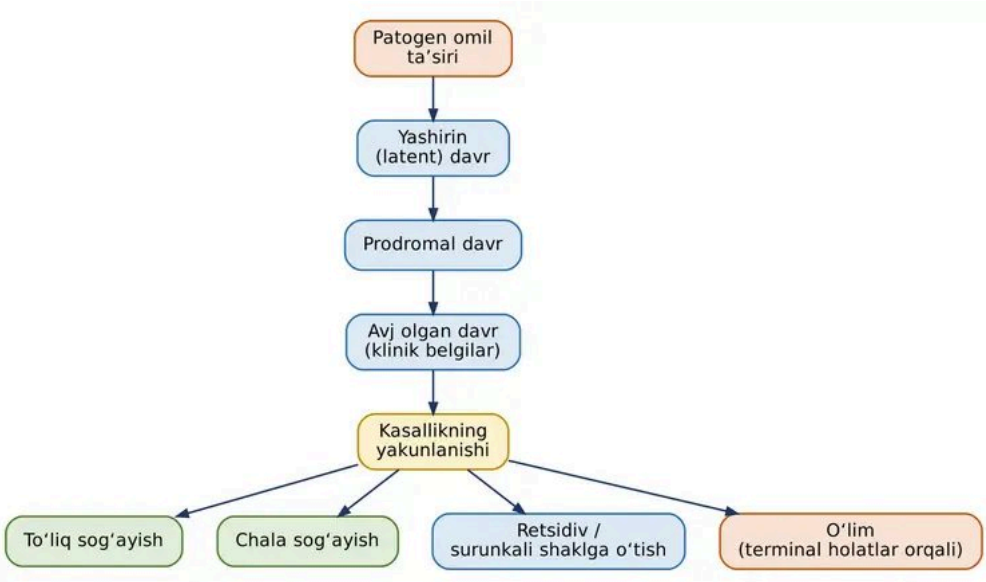


Figure 1.1. *Periods and outcomes of disease (based on data from N.H. Abdullaev, 2008)*

Table 1.5. *Periods of disease*

Period	Essence	Duration
Latent period; in infectious diseases — the incubation period	From the action of the pathogenic factor until the first signs appear	From seconds (trauma, acute poisoning) to years (prion diseases, HIV)
Prodromal period (Greek prodromos — forerunner)	Nonspecific “general” signs: weakness, headache, loss of appetite, low-grade fever	From hours to several days
Full-blown (manifest) period	All signs specific to the disease are fully manifest; a definite diagnosis is made	Depends on the type of disease
Resolution period	The outcome of the disease is formed	Depends on the type of disease

Outcomes of disease:

1. **Complete recovery** — the impaired structures and functions are fully restored. However, the recovered organism does not return exactly to its state before the disease: it acquires new properties (for example, immunity).
2. **Incomplete recovery** — some traces of the disease (residual phenomena) remain: a scar, partial reduction of organ function.

3. **Relapse** (recurrence) — reappearance of the signs of disease after they have disappeared. The period of improvement between relapses is called **remission**.
4. **Transition to a chronic form** — the disease has a long, undulating course.
5. **Death** — transition through terminal states (preagony, agony, clinical death) to biological death.

Mechanisms of recovery

Table 1.6. *Mechanisms of recovery by speed*

Type	Time	Examples
Urgent (“emergency”)	Seconds and minutes	Protective reflexes (cough, vomiting), the stress reaction (epinephrine, glucocorticoids), reflexes maintaining arterial pressure, glucose and osmotic pressure
Relatively stable	Days, weeks	Mobilization of the reserve capacities of organs, reactive leukocytosis, detoxification of poisons (liver), reaction of connective tissue and macrophages
Long-term (persistent)	Months, years	Compensatory hypertrophy, reparative regeneration, immunity, plastic reactions of the central nervous system

CLINICAL SIGNIFICANCE

The reserve capacity of organs is very large: 20–25% of the respiratory surface of the lungs, 20–25% of the renal glomeruli and 12–15% of the liver parenchyma are sufficient for life. That is why chronic diseases run a “silent” course for a long time, and clinical signs appear when the organ reserve is exhausted (for example, in kidney failure when 70–80% of the glomeruli have been lost).

6. Nomenclature and classification of diseases

Nomenclature is the list of names of diseases; **classification** is the grouping of diseases according to certain features. In medical practice the international classification is used — the **International Statistical Classification of Diseases and Related Health Problems (ICD)**. At present **ICD-11** is gradually replacing ICD-10 (it was brought into force by the WHO on 1 January 2022).

Table 1.7. *Principles of classification of diseases*

Principle	Groups
Etiologic	Infectious and noninfectious; hereditary and acquired; occupational diseases, radiation diseases
Topographic-anatomic (by organ)	Diseases of the heart, lungs, liver, kidneys
Pathogenetic	Allergic, neoplastic, inflammatory, degenerative, metabolic diseases
By age and sex	Diseases of children, of the elderly (geriatrics), of women
Ecological	Diseases specific to a certain geographic region (endemic), for example endemic goiter
By course	Acute, subacute, chronic

REMEMBER

Conventional durations by course of disease: **hyperacute** — up to 4 days, **acute** — 5–14 days, **subacute** — 15–40 days, **chronic** — months and years. These durations are defined separately for each disease and are only approximate.

CAUTION

Textbooks give different durations of the course (the textbooks of N.H. Abdullaev and V.V. Novitsky differ slightly). In practice the durations for each disease are defined by the relevant clinical protocol.

7. The doctrine of etiology

Etiology (Greek aitia — cause, logos — study) is the doctrine of the causes and conditions of the origin of diseases.

The cause of disease (etiologic factor) is the factor that produces the disease and gives it its specific features. There is no disease without a cause, and without the action of the cause the disease does not develop.

Conditions are factors that do not themselves produce the disease but promote or hinder the realization of the action of the cause.

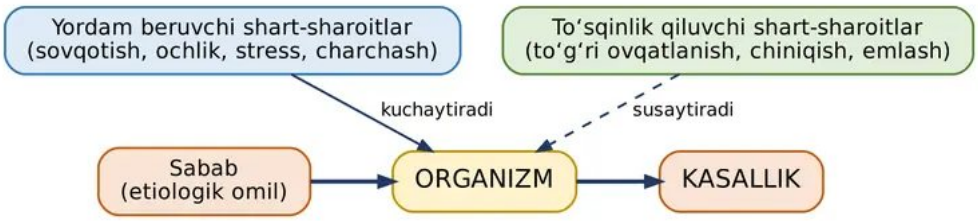


Figure 1.2. *Interrelationship of causes and conditions (based on A.V. Ataman, 2000, Fig. 10)*

Table 1.8. *Difference between cause and condition*

Feature	Cause	Condition
Necessity for the disease	Obligatory (necessary)	Not obligatory
Does it give the disease its specific features?	Yes (it confers specificity)	No
Can it be replaced?	No (one disease — one main cause)	Yes (different conditions may have the same effect)
Example (tuberculosis)	Mycobacterium tuberculosis (Koch's bacillus)	Starvation, poor living conditions, HIV infection, diabetes mellitus, stress

REMEMBER

A large part of the population is infected with Mycobacterium tuberculosis (according to the WHO, about one quarter of the world's population), but only 5–10% of them develop active tuberculosis during their lifetime. Thus, **the cause is necessary, but not always sufficient** — disease arises from the interaction of the cause and the conditions.

Classification of etiologic factors

Table 1.9. *Types of etiologic factors*

Group	Factors
Physical	Mechanical injury, high and low temperature, ionizing radiation, electric current, changes in barometric pressure, acceleration, noise, vibration
Chemical	Acids, alkalis, heavy metals, poisons, drugs (in overdose), alcohol
Biological	Viruses, bacteria, fungi, protozoa, helminths, prions, animal and plant toxins

Group	Factors
Social and psychogenic	Stress, mental trauma, unfavorable living and working conditions, poverty
Nutritional	Starvation, deficiency or excess of protein, vitamins and trace elements
Hereditary (genetic)	Gene and chromosomal mutations

By origin, factors are **exogenous** (from the external environment) or **endogenous** (formed in the organism itself: for example, autoantibodies, tumor cells, mutated genes).

Main schools of thought in etiology

Table 1.10. *Main theories of etiology*

Theory	Main idea	Shortcoming
Monocausalism (R. Koch, late 19th century)	Each disease has a single cause; conditions are of no importance	Does not explain why not every infected person falls ill
Conditionalism (M. Verworn, D. Hansemann; early 20th century)	Disease results from a sum of equivalent conditions; the concept of "cause" is unnecessary	Denies the decisive role of the cause
Constitutionalism (the hereditary-constitutional theory)	Disease depends mainly on the hereditary-constitutional features of the organism	Underestimates the role of external factors
The psychosomatic school (S. Freud and others)	Many diseases result from mental conflicts	Absolutizes the role of the psychological factor
The modern view: multifactoriality and risk factors	Disease results from the interaction of the cause, the conditions and the reactivity of the organism	In many diseases (hypertension, atherosclerosis) a single cause has not been identified

Risk factors are factors that statistically increase the probability of developing a disease. They are mostly used for polyetiologic (multicausal) chronic diseases. For example, the risk factors for ischemic heart disease are smoking, arterial hypertension, hypercholesterolemia, diabetes mellitus, obesity, physical inactivity, hereditary predisposition, male sex and age.

> **CLINICAL SIGNIFICANCE**
> Knowing the etiologic factor makes **etiotropic treatment** (treatment directed against the cause) possible: an antibiotic in bacterial infection, an anthelmintic in helminthiasis, an antidote in poisoning. Knowing the risk factors is the basis of **prevention**: stopping smoking and controlling arterial pressure significantly reduce the risk of cardiovascular diseases.

8. The doctrine of pathogenesis

Pathogenesis (Greek pathos — disease, genesis — origin) is the doctrine of the mechanisms of development, course and resolution of diseases and pathological processes. Etiology answers the question "why?", whereas pathogenesis answers the question "how?".

The role of the etiologic factor in pathogenesis

The etiologic factor takes part in pathogenesis in different ways:

- **it acts only at the onset of the disease** and then disappears (burns, mechanical trauma, ionizing radiation) — the disease continues on its own according to the laws of pathogenesis;
- **it acts throughout the disease** (most infectious diseases, helminthic infestations) — in this case removing the cause leads to recovery.

Basic concepts of pathogenesis

Table 1.11. *Basic concepts of pathogenesis*

Concept	Content	Example
Initial injury	The first change produced by the etiologic factor	Denaturation of tissue proteins in a burn
Cause-and-effect chain	Each effect becomes the cause of the next change	Blood loss → hypovolemia → hypotension → hypoxia
Primary and secondary injury	Primary — the direct action of the factor; secondary — the subsequent links of the chain	Cell injury by a toxin, followed by edema and ischemia
Main (leading) link	The change that determines the development of the disease and on which all other links depend	Insulin deficiency in diabetes mellitus; decreased circulating blood volume in blood loss
Vicious (closed) circle (circulus vitiosus)	The consequence of pathogenesis amplifies its own cause; the process becomes self-amplifying	Myocardial hypoxia → decreased contractility → decreased cardiac output → worsening hypoxia
Local and systemic changes	A local process produces systemic reactions, and vice versa	Furuncle (local) → fever, leukocytosis (systemic)
Specific and nonspecific mechanisms	Specific — characteristic only of this disease; nonspecific — recur in many diseases	Antibody formation (specific); fever, stress, leukocytosis (nonspecific)



Figure 1.3. *The “vicious circle” in blood loss (based on V.V. Novitsky, 2009, Fig. 1.3)*

REMEMBER

The main link is the link of the pathogenetic chain whose elimination breaks the whole chain and leads to recovery. The goal of pathogenetic treatment is precisely to find the main link and act on it. In blood loss the main link is the decrease in circulating blood volume, which is why the main treatment is transfusion of blood and blood substitutes.

MEMORY AID

Vicious circle = "the snake bites its own tail": the consequence comes back and amplifies the cause. To break it, any link of the circle can be targeted, but the most effective is to act on the main link.

The unity of structure and function in pathogenesis

Any functional change is based on a certain structural change — it may be at the molecular or subcellular level and may not be detectable by ordinary methods. The concept of "functional" diseases (for example, neuroses) only means that no structural change has been found with current research methods.

The role of the nervous and endocrine systems in pathogenesis

The nervous system plays a dual role in pathogenesis: it regulates protective reactions (reflexes, stress), and at the same time it can itself be the source of a pathological process (neuroses, pathological reflexes, degenerative processes). The endocrine system regulates the stress reaction (the hypothalamic-pituitary-adrenal system), metabolism and reactivity.

CAUTION

Do not confuse "etiology" and "pathogenesis". "The etiology of pneumonia is the pneumococcus" (cause); "the pathogenesis of pneumonia is inflammation in the alveoli, accumulation of exudate, impaired gas exchange, hypoxemia" (mechanism). A common exam mistake is to answer a question on pathogenesis with a list of causes.

9. General principles of treatment

Table 1.12. *Types of treatment*

Type of treatment	Goal	Examples
Etiotropic (causal)	To remove or weaken the cause of the disease	Antibiotics, antidotes, anthelmintics, removal of a foreign body
Pathogenetic	To break the links of pathogenesis, especially the main link	Insulin in diabetes; infusion in blood loss; diuretics in edema
Symptomatic	To remove the signs that distress the patient	Analgesics, antitussives, antipyretics
Replacement	To supply a lost function from outside	Hemodialysis, hormone replacement therapy, enzyme preparations
Rehabilitative	To restore function and working capacity	Therapeutic exercise, physiotherapy, diet

CAUTION

Symptomatic treatment alone can be dangerous: removing the signs makes diagnosis harder, while the disease continues. For example, if an analgesic is given for "acute abdomen" pain before a diagnosis is made, the signs of acute appendicitis may be masked and peritonitis may be missed.

10. Practical part: pathophysiological analysis of a disease

No experiment is performed in this session; the main practical task is to learn to analyze a clinical situation from the standpoint of etiology and pathogenesis. The analysis is carried out according to the following plan.

Plan for analyzing a disease

1. Identify the etiologic factor (cause) and state its type (physical, chemical, biological, etc.).
2. Distinguish the conditions (external and internal) that promoted and those that hindered the development of the disease.
3. Identify the initial injury (at which level and in which structure).
4. Write down the cause-and-effect chain of pathogenesis in sequence.
5. Mark the main link and, if present, the vicious circle.
6. Distinguish local and systemic, specific and nonspecific changes.
7. State the protective-adaptive and compensatory reactions.
8. Determine the period of the disease and its probable outcome.
9. Justify the directions of etiotropic, pathogenetic and symptomatic treatment.

Example: pathophysiological analysis of lobar pneumonia

Table 1.13. *Analysis of lobar pneumonia*

Stage	Content
Cause	Pneumococcus (Streptococcus pneumoniae) — a biological factor
Conditions	Chilling, fatigue, alcoholism, old age, recent influenza, smoking (impaired bronchial clearance)
Initial injury	Injury of the alveolar epithelium and capillaries by toxins and inflammatory mediators
Chain of pathogenesis	Inflammation → fibrinous exudate in the alveoli → consolidation of the lung lobe → ventilation-perfusion mismatch → hypoxemia → tachypnea, tachycardia
Main link	Inflammation and exudation in the alveoli
Vicious circle	Hypoxemia → myocardial hypoxia → decreased cardiac output → worsening hypoxia
Local and systemic signs	Local: consolidation of the lobe, crepitation, pleuritic pain; systemic: fever, leukocytosis, intoxication
Protective reactions	Cough, phagocytosis, fever, antibody formation
Treatment	Etiotropic — antibiotic; pathogenetic — oxygen therapy, detoxification; symptomatic — antipyretics, expectorants

CLINICAL SIGNIFICANCE

A student who has mastered this plan can independently analyze any disease in the clinical departments. The structure of the clinical diagnosis is based on the same logic: the underlying disease (with its etiology), complications (consequences of pathogenesis) and concomitant diseases.

11. Practical skills

Skill: distinguishing cause and conditions

1. Find the factor that gives the disease its specific features — this is the cause.
2. Ask whether the disease could develop without it: if not, this factor is the cause.
3. Determine whether other factors increase or decrease the probability of the disease; these are conditions.
4. Divide the conditions into external (environment) and internal (organism) groups.
5. Write a conclusion: “Cause — ...; promoting conditions — ...; hindering conditions — ...”.

Skill: drawing a scheme of pathogenesis

1. Write the etiologic factor at the top of the scheme.
2. Indicate the initial injury.
3. Connect each subsequent change with an arrow to build the cause-and-effect chain.
4. Check which link, if removed, makes the others disappear as well — this is the main link.
5. Find the place where the effect amplifies its own cause and mark it with a reverse arrow — this is the vicious circle.
6. Add the protective-compensatory reactions to the scheme in a different color.
7. Mark the points targeted by pathogenetic treatment.

Review questions

1. What are the subject and main tasks of pathological physiology?
2. What are the components of pathophysiology?
3. What are the advantages and limitations of the experimental method?
4. What contributions did C. Bernard, R. Virchow, I.I. Mechnikov and H. Selye make to the development of pathophysiology?
5. Define the concepts of health, norm and disease.
6. What is the “predisease” state and what is its practical significance?
7. Explain the difference between a pathological reaction, a pathological process and a pathological state with examples.
8. What are typical pathological processes? Give examples.
9. What periods and outcomes of disease are there?
10. List the urgent, relatively stable and long-term mechanisms of recovery.
11. According to what principles are diseases classified?
12. How do the cause and the conditions of a disease differ?
13. What are the essence and shortcomings of monocausalism and conditionalism?
14. What are the main link and the vicious circle of pathogenesis? Give examples.
15. What is the difference between etiotropic, pathogenetic and symptomatic treatment?

Test questions

1. What is the main object of study of pathological physiology?
   A) Normal functions of the healthy organism
   B) Structural changes in the dead organism
   C) The vital activity of the diseased organism
   D) The mechanism of action of drugs
   E) The prevalence and statistics of diseases

2. What is the main research method of pathophysiology?
   A) Experiment (modeling of disease in animals)
   B) Autopsy
   C) Epidemiological observation
   D) History taking
   E) X-ray examination

3. Who is the author of the theory of cellular pathology?
   A) I.I. Mechnikov
   B) C. Bernard
   C) I.P. Pavlov
   D) H. Selye
   E) R. Virchow

4. Which of the following is a pathological state?
   A) Inflammation
   B) A scar after a burn
   C) Fever
   D) Thrombosis
   E) A brief rise in arterial pressure

5. How is the prodromal period of disease characterized?
   A) The time from the action of the pathogenic factor until the first signs appear
   B) Full manifestation of all signs specific to the disease
   C) Reappearance of the signs of disease
   D) Appearance of nonspecific general signs
   E) Temporary disappearance of the signs of disease

6. What is the main feature of the cause of a disease (the etiologic factor)?
   A) It is necessary for the disease and gives it its specific features
   B) It only statistically increases the probability of the disease
   C) It can be replaced by another factor
   D) It is not obligatory for the disease
   E) It affects only the course of the disease

7. What is the main idea of the theory of conditionalism?
   A) Each disease has a single cause
   B) Disease depends only on hereditary-constitutional features
   C) Disease results from a sum of equivalent conditions
   D) Diseases arise as a result of mental conflicts
   E) The cause of disease is only microorganisms

8. Which of the following is the cause in the development of tuberculosis?
   A) Starvation
   B) Diabetes mellitus
   C) HIV infection
   D) Poor living conditions
   E) Mycobacterium tuberculosis

9. What is meant by the main link of pathogenesis?
   A) The first action of the etiologic factor on the organism
   B) The link whose elimination breaks the whole chain of pathogenesis
   C) The final outcome of the disease
   D) Local changes only
   E) Nonspecific signs of the disease

10. How is the “vicious circle” (circulus vitiosus) characterized?
    A) Recurrence of the disease
    B) Simultaneous action of several causes
    C) Transition of the disease to a chronic form
    D) A consequence of pathogenesis amplifying its own cause
    E) Activation of protective reactions

11. What is the main link of pathogenesis in blood loss?
    A) Tachycardia
    B) Decrease in circulating blood volume
    C) Spasm of peripheral vessels
    D) Increased erythropoiesis
    E) Rapid breathing

12. Which type of treatment is aimed at removing the cause of the disease?
    A) Symptomatic
    B) Pathogenetic
    C) Replacement
    D) Rehabilitative
    E) Etiotropic

13. Which of the following is a typical pathological process?
    A) Inflammation
    B) Lobar pneumonia
    C) Diabetes mellitus
    D) Myocardial infarction
    E) Congenital heart defect

14. Which of the following belongs to the long-term (persistent) mechanisms of recovery?
    A) The cough reflex
    B) Release of epinephrine
    C) Compensatory hypertrophy
    D) Reflexes maintaining arterial pressure
    E) Vomiting

15. Which example corresponds to the predisease (prenosological) state?
    A) The full-blown period of lobar pneumonia
    B) A scar after myocardial infarction
    C) Acute appendicitis
    D) Impaired glucose tolerance (prediabetes)
    E) Hemorrhagic shock

Clinical cases

Case 1

Four members of one family fell ill with influenza at the same time. The 25-year-old son recovered in 5 days. The 70-year-old grandmother, who has diabetes mellitus, was hospitalized with secondary bacterial pneumonia against the background of influenza. The father (45 years old, smokes 20 cigarettes a day) was treated for a long time for the complication of bronchitis. The mother had previously been vaccinated against influenza and had a mild form of the disease.

Questions: What is the cause of the disease? What factors can explain the different courses of the disease? Which rule of etiology does this example confirm?

Answer: The cause is the influenza virus (a biological etiologic factor); it is the same in all family members. The different courses of the disease depend on the conditions and the reactivity of the organism: in the grandmother, old age and diabetes mellitus (reduced immunity and regeneration) are promoting internal conditions; in the father, smoking has impaired the mucociliary clearance of the bronchi; in the mother, the immunity produced by vaccination is a hindering condition. The example confirms the modern view that “the cause is necessary, but the course and outcome of the disease depend on the interaction of the cause, the conditions and the reactivity of the organism” and shows the limitations of monocausalism.

Case 2

A 40-year-old patient lost 1.5 liters of blood as a result of bleeding from a gastric ulcer. Arterial pressure is 80/50 mm Hg, pulse 120 beats/min, the skin is pale and covered with cold sweat. The ECG shows signs of myocardial ischemia.

Questions: Indicate the main link of pathogenesis. What vicious circle has formed in the patient? What should pathogenetic treatment be aimed at?

Answer: The main link is a sharp decrease in circulating blood volume (hypovolemia). Vicious circle: hypovolemia → decreased venous return to the heart → decreased cardiac output and arterial pressure → decreased coronary blood flow → myocardial hypoxia → decreased contractility → further decrease in cardiac output. Tachycardia and spasm of the skin vessels are compensatory reactions. Pathogenetic treatment is aimed at the main link: restoration of circulating blood volume by transfusion of blood and blood substitutes; this also breaks the vicious circle. Etiotropic treatment is stopping the bleeding (endoscopic or surgical hemostasis).

Case 3

At an exam, a student answered: “The pathogenesis of type 1 diabetes mellitus is hereditary predisposition, viral infections and stress.” The teacher marked the answer as wrong.

Questions: What mistake did the student make? Briefly state the correct pathogenesis of type 1 diabetes mellitus and indicate the main link.

Answer: The student confused etiology (causes and conditions) with pathogenesis (the mechanism of development): hereditary predisposition, viruses and stress are etiologic factors and conditions. Pathogenesis: autoimmune injury of the pancreatic beta cells → insulin deficiency → decreased entry of glucose into tissues, increased gluconeogenesis and lipolysis → hyperglycemia, glucosuria, polyuria, ketoacidosis. The main link is insulin deficiency; therefore the pathogenetic treatment is insulin replacement therapy.

Quick review

Table 1.14. *Key facts on Topic 1*

Question	Answer
Pathophysiology	The science of the vital activity of the diseased organism and the laws of the origin and development of diseases
Main method	Experiment — modeling of disease in animals
Nosology	The doctrine of disease (general and special)
Health (WHO)	A state of complete physical, mental and social well-being
Disease	The unity of injury and protective-adaptive reactions, with reduced adaptation and working capacity
Predisease state	A state with reduced adaptive capacity but no signs of disease yet
Pathological reaction	A disproportionate, short-term response to a stimulus
Pathological process	A regular combination of injury and protective reactions (inflammation, fever)
Pathological state	A stable, slowly changing pathological change (scar)
Periods of disease	Latent, prodromal, full-blown, resolution
Outcomes of disease	Complete and incomplete recovery, relapse, transition to a chronic form, death
Etiology	The doctrine of the causes and conditions of disease
Cause	A factor necessary for the disease that gives it its specific features
Condition	A factor that promotes or hinders the action of the cause
Pathogenesis	The mechanisms of development, course and resolution of disease
Main link	The link whose elimination breaks the whole chain of pathogenesis
Vicious circle	A closed chain in which the consequence amplifies its own cause
Etiotropic treatment	Treatment directed against the cause
Pathogenetic treatment	Action on the links of pathogenesis, above all on the main link

TOPIC 2

The importance of external conditions in the development of pathological processes. The pathogenic action of physical, chemical and biological factors

Learning objectives

To explain to the student the role of external environmental conditions in the origin and development of diseases and pathological processes; to teach the mechanisms of the pathogenic action of physical (temperature, barometric pressure, electric current, ionizing radiation, mechanical action), chemical, biological and social factors; and to teach the student to observe and analyze in an experiment the effect of ambient temperature on the development of hypoxic hypoxia.

By the end of the session, the student should know:

- the classification of external conditions and their role in the development of disease;
- the stages, pathogenesis and outcomes of hyperthermia and hypothermia;
- the mechanisms of the changes caused by decreased and increased atmospheric pressure (mountain sickness, decompression sickness);
- the mechanisms of the pathogenic action of electric current and ionizing radiation, and the forms of acute radiation sickness;
- the features of the action of chemical, biological and socio-ecological factors;
- how ambient temperature changes resistance to hypoxia.

The student should be able to:

- to create a model of hypoxic hypoxia in a pressure chamber and record its stages;
- to compare the resistance to hypoxia of animals kept at different temperatures and explain the result;
- to analyze a pathological state caused by an external factor according to the scheme “cause — conditions — pathogenesis”.

1. External conditions and their classification

As discussed in the previous topic, disease develops from the interaction of the cause and the conditions. Conditions are **external** (related to the environment) or **internal** (related to the organism itself). The same factor may be a cause in one case and a condition in another: for example, low temperature is the **cause** of frostbite but a **condition** of pneumonia (chilling lowers the organism's resistance to infection).

Table 2.1. *Classification of external conditions*

Group	Examples	Effect
Climatic-geographic	Temperature, humidity, atmospheric pressure, solar radiation, seasonality	Change thermoregulation, metabolism and resistance to hypoxia
Nutrition	Starvation, deficiency or excess of protein, vitamins and trace elements	Affects immunity, regeneration and energy reserves
Working and living conditions	Heavy physical work, occupational hazards, housing conditions	Fatigue, chronic intoxication, occupational diseases

Group	Examples	Effect
Ecological	Pollution of air, water and soil, increased background radiation	Risk of allergic, neoplastic and endemic diseases
Social and psychological	Stress, social conflicts, poverty, physical inactivity	Psychosomatic diseases, cardiovascular diseases
Harmful habits	Smoking, alcohol, drug addiction	A risk factor for most chronic diseases

REMEMBER

External conditions act in two ways:

- **promoting** — they amplify the action of the cause (chilling, starvation, fatigue, stress, air pollution);
- **hindering** — they weaken the action of the cause or prevent the development of disease (proper nutrition, hardening, physical activity, vaccination, hygiene).

2. The pathogenic action of temperature

Humans are homeothermic (warm-blooded) organisms: their body temperature is kept at about 36–37 °C by the balance between heat production (thermogenesis) and heat loss (thermolysis). The thermoregulatory center is located in the hypothalamus. When the external temperature is very high or very low, this balance is disturbed and **hyperthermia** or **hypothermia** develops.

The action of high temperature. Hyperthermia

Hyperthermia is a rise in body temperature resulting from insufficient heat loss relative to heat production. It occurs with high external temperature, high air humidity (sweat does not evaporate), heat-insulating clothing and heavy physical work.

Table 2.2. *Stages of hyperthermia*

Stage	Mechanism	Signs
Compensation (adaptation)	Thermolysis increases: skin vessels dilate, sweating increases, breathing becomes rapid; thermogenesis decreases	Body temperature almost normal, flushed skin, profuse sweating, tachycardia
Decompensation	Thermoregulatory mechanisms are insufficient, heat accumulates	Body temperature rises above 40–41 °C, weakness, headache, nausea, dehydration
Heatstroke (severe decompensation)	Injury of the thermoregulatory center, cerebral edema, collapse	Sweating stops, the skin is dry and hot, loss of consciousness, convulsions, disturbed breathing and cardiac activity

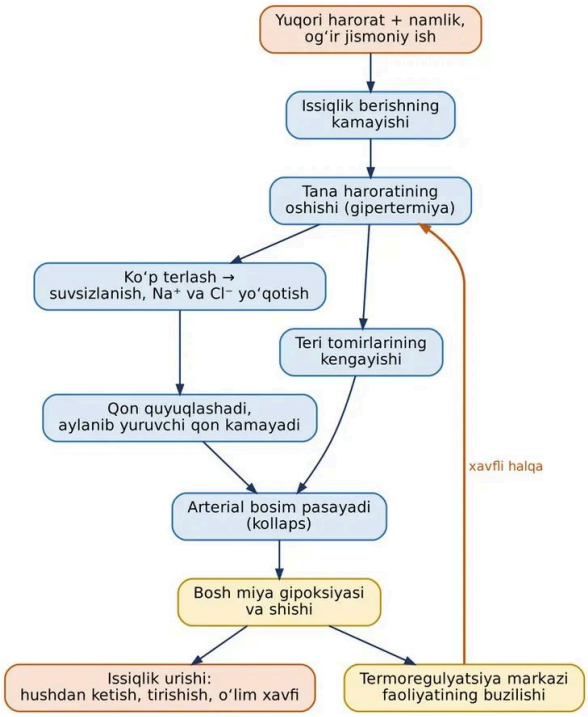


Figure 2.1. *Pathogenesis of heatstroke (based on V.V. Novitsky, 2009, Fig. 2.2)*

CAUTION

Do not confuse **hyperthermia** and **fever**! In fever, the thermoregulatory center is “reset” by pyrogens to maintain the temperature at a higher level (an active process). In hyperthermia, the thermoregulatory center tries to maintain the normal level, but heat loss is insufficient (passive heat accumulation). That is why antipyretics (paracetamol (acetaminophen), ibuprofen) are ineffective in hyperthermia — the body must be cooled physically.

Sunstroke is overheating, hyperemia and edema of the brain and its membranes resulting from the direct action of the sun's rays on the head. Its main signs are severe headache, vomiting and impaired consciousness.

A burn is the local action of high temperature. By depth, degrees I–IV are distinguished. With extensive burns (more than 15–20% of the body surface, and even less in children and the elderly), **burn disease** develops: it runs through the periods of burn shock, toxemia, septicotoxemia and recovery.

MEMORY AID

The **“rule of nines”** for estimating the burned area: head and neck — 9%, each arm — 9%, front of the trunk — 18%, back of the trunk — 18%, each leg — 18%, perineum — 1%. The patient's palm (with the fingers) equals about 1% of the body surface.

The action of low temperature. Hypothermia

Hypothermia is a fall in body temperature below normal (below 35 °C) resulting from heat loss exceeding heat production. In cold water the body cools much faster than in air, because the thermal conductivity of water is about 25 times higher than that of air.

Table 2.3. *Stages of hypothermia*

Stage	Mechanism	Signs
Compensation	Thermogenesis increases (shivering, increased metabolism), thermolysis decreases (skin vessels constrict)	Pale skin, “goose bumps”, shivering, tachycardia, raised arterial pressure
Decompensation	Thermoregulatory mechanisms are exhausted, the central nervous system is inhibited	Body temperature falls, shivering stops, drowsiness, bradycardia, slow breathing, hypoxia
Terminal stage	The activity of the vital centers ceases	Below 24–25 °C — ventricular fibrillation or respiratory arrest

Table 2.4. *Degrees of clinical hypothermia*

Degree	Body temperature	Main signs
Mild	32–35 °C	Shivering, tachycardia, apathy, impaired coordination of movements
Moderate	28–32 °C	Shivering disappears, drowsiness, bradycardia, arrhythmias
Severe	Below 28 °C	Coma, loss of reflexes, risk of ventricular fibrillation

CLINICAL SIGNIFICANCE

Artificial hypothermia is used in medicine: in heart and brain operations, cooling the body to 28–32 °C reduces the oxygen demand of tissues and increases their resistance to hypoxia. In resuscitation after cardiac arrest, targeted temperature management protects the brain. This rule is also confirmed in the experimental part of the session.

Frostbite is the local action of low temperature. Its main mechanism is vascular spasm, slowing of blood flow, stasis and thrombosis, tissue ischemia and necrosis. The fingers, ears, nose and feet are most often affected.

3. Changes in atmospheric (barometric) pressure

Decreased atmospheric pressure. Mountain sickness and altitude sickness

When ascending to altitude, two factors act on the organism: **decreased atmospheric pressure (decompression)** and **decreased partial pressure of oxygen (pO_2) in the air**. As P. Bert and I.M. Sechenov showed, the main pathogenic factor of altitude is the fall in pO_2 and the resulting **hypoxic (exogenous) hypoxia**.

Table 2.5. *Atmospheric pressure and partial pressure of oxygen by altitude (approximate)*

Altitude	Atmospheric pressure	pO_2 in the air
Sea level	760 mm Hg	159 mm Hg
2000 m	596 mm Hg	125 mm Hg
3000 m	526 mm Hg	110 mm Hg
5000 m	405 mm Hg	85 mm Hg
8848 m (Everest)	253 mm Hg	53 mm Hg

Table 2.6. *Zones of the effect of altitude on the human organism (N.H. Abdullaev, 2008)*

Zone	Altitude	State of the organism
Indifferent	Up to 1500–2000 m	No significant changes, working capacity fully preserved
Compensation	2000–4000 m	Pulmonary ventilation and cardiac output increase; heavy physical work becomes difficult
Incomplete compensation	4000–5500 m	General condition deteriorates, headache, euphoria, reduced working capacity
Critical	5500–8000 m	Working capacity is profoundly impaired; an unacclimatized person quickly loses consciousness at 7000 m
Death zone	Above 8000 m	Without oxygen a person remains conscious for 2–3 minutes, followed by profound inhibition and death

Mountain sickness develops gradually, several hours or days after ascent; **altitude sickness**, in contrast, arises rapidly (within minutes and seconds) when the sealing of an aircraft is suddenly lost.

Pathogenesis of mountain sickness: fall in pO_2 → hypoxemia → stimulation of chemoreceptors → hyperventilation → hypocapnia and respiratory (gas) alkalosis → constriction of cerebral vessels → worsening of cerebral hypoxia. In severe cases **pulmonary edema** and **cerebral edema** develop.

Changes related to decompression: intestinal gases expand (at 6000 m the volume of gas roughly doubles) — flatulence, abdominal pain; pain in closed cavities (middle ear, maxillary sinus); above 7000–8000 m, nitrogen dissolved in the blood forms bubbles (gas embolism); at an altitude of about 19 km, water begins to boil at body temperature (“altitude emphysema”).

CLINICAL SIGNIFICANCE

Mechanisms of adaptation (acclimatization): hyperventilation, tachycardia, release of red blood cells from blood depots (urgent); increased production of erythropoietin in the kidneys, increased red blood cell count and hemoglobin, increased 2,3-diphosphoglycerate in red blood cells, and an increased number of capillaries and mitochondria in tissues (long-term). Stepwise ascent into the mountains (N.N. Sirotinin's method) prevents mountain sickness.

Increased atmospheric pressure. Decompression sickness

Under water, pressure increases by about 1 atmosphere for every 10 m of depth. Under high pressure (divers, caisson workers), more gas dissolves in the blood and tissues — **saturation**. Nitrogen accumulates especially in lipid-rich tissues (nervous tissue, adipose tissue) and causes **nitrogen narcosis** (euphoria, then impaired coordination of movements and clouding of consciousness). A high partial pressure of oxygen leads to **oxygen toxicity** (formation of free radicals, injury of the lungs and central nervous system).

Decompression (caisson) sickness is gas embolism, tissue ischemia and compression of nerve endings resulting from the release of nitrogen dissolved in the blood and tissues in the form of bubbles (**desaturation**) during rapid transition from high to normal pressure. Signs: pain in joints and muscles, itching of the skin, shortness of breath, paralysis.

CAUTION

Prevention of decompression sickness: slow, stepwise ascent from high pressure (according to decompression tables) and use of a helium mixture for breathing instead of nitrogen. Treatment: immediate placement of the patient in a pressure chamber, raising the pressure again (recompression) and then lowering it slowly.

4. The action of electric current

Electric current acts on the organism in two ways: **specifically** (electrochemical, electrothermal, electromechanical action) and **nonspecifically** (injury by light, sound and mechanical force). The harmful effect of current depends on the following:

- **current strength** (the main factor) — a current of more than 0.1 A passing through the heart can cause ventricular fibrillation;
- **voltage** — a voltage above 40 V is considered dangerous;
- **type and frequency of current** — alternating current with a frequency of 40–60 Hz is the most dangerous; high-frequency (UHF) current acts mainly as heat and is used in physiotherapy;
- **skin resistance** — dry skin has a high resistance, while the resistance of wet skin falls sharply;
- **the path of the current ("current loop")** — paths through the heart and brain (hand — hand, left hand — legs) are the most dangerous;
- **duration of exposure** and **state of the organism** (resistance changes with fatigue, alcohol and sleep).

Local changes: at the points of entry and exit of the current, **"current marks"** (electrical marks) — gray-yellowish, hard, painless spots; burns, tissue necrosis. Systemic changes: loss of consciousness, convulsions, respiratory arrest (paralysis of the respiratory center or tetanic contraction of the respiratory muscles), disturbance and arrest of the heart rhythm.

> **CLINICAL SIGNIFICANCE**
> First aid for electrical injury: first separate the victim from the source of current without endangering the rescuer (switch off the current, push away with dry wood), then check breathing and circulation and, if necessary, start cardiopulmonary resuscitation. Even with an apparently mild injury, the patient must remain under observation because of the risk of delayed arrhythmias.

5. The action of ionizing radiation

Ionizing radiation (X-rays and gamma rays, alpha and beta particles, neutrons) passes through tissues and ionizes atoms and molecules. The absorbed dose is measured in **grays (Gy)** (1 Gy = 1 J/kg).

Mechanism of action:

- **direct action** — direct injury of DNA, protein and enzyme molecules;
- **indirect action** — radiolysis of water produces active free radicals ($OH^{\bullet}$, $H^{\bullet}$, hydrogen peroxide), which injure biomolecules. Because about 70% of the organism consists of water, the indirect action is the main one.

> **REMEMBER**
> **The law of Bergonié and Tribondeau:** the radiosensitivity of a tissue is directly proportional to the mitotic activity of its cells and inversely proportional to their degree of differentiation. The most sensitive tissues are hematopoietic tissue (bone marrow), lymphoid tissue, intestinal epithelium and germ cells. The most resistant are nervous tissue, muscle and bone.

Table 2.7. *Forms of acute radiation sickness (after a single whole-body exposure)*

Form	Dose	Main injury
Bone marrow form	1–10 Gy	Pancytopenia, infections, bleeding; degrees of severity: mild (1–2 Gy), moderate (2–4 Gy), severe (4–6 Gy), extremely severe (6–10 Gy)

Form	Dose	Main injury
Intestinal form	10–20 Gy	Death of the intestinal epithelium, diarrhea, dehydration, sepsis
Toxemic (vascular) form	20–80 Gy	Severe intoxication, hemodynamic disorders
Cerebral form	More than 80 Gy	Injury of the central nervous system, death within several hours or days

In the bone marrow form of acute radiation sickness, four periods are distinguished: **the primary reaction** (nausea, vomiting, weakness — a few hours after exposure), **the latent period** ("false well-being"), **the full-blown period** (leukopenia, thrombocytopenia, infectious complications, bleeding) and **the recovery period**.

CAUTION

The time of onset of the primary reaction (vomiting) gives an approximate idea of the dose: the earlier vomiting begins, the higher the dose. The patient's feeling of well-being in the latent period is deceptive — it is precisely during this period that the reserve of bone marrow cells is being depleted.

6. The action of mechanical factors

Mechanical factors (impact, compression, stretching, acceleration, vibration) disrupt the integrity of tissues: wounds, contusions, bone fractures. In severe mechanical injury, systemic reactions develop together with local changes — **traumatic shock** (discussed in detail in Topic 27).

Crush syndrome (prolonged compression syndrome) develops as a result of prolonged (several hours) compression of soft tissues (most often of the limbs). Pathogenesis: ischemia and muscle necrosis in the compressed tissues → when the pressure is removed, blood flow is restored and myoglobin, potassium and toxic products pass from the tissues into the blood → hyperkalemia (risk of arrhythmia and cardiac arrest), myoglobinuria and obstruction of the renal tubules → **acute kidney injury**; marked edema of the injured limb → plasma loss and hypovolemia.

CLINICAL SIGNIFICANCE

In crush syndrome, it is recommended to apply a tourniquet to the limb and start intravenous infusion **before** the pressure is removed — this prevents a sudden release of potassium and toxins into the blood. In victims of earthquakes and accidents, this syndrome is one of the main causes of death.

Acceleration (aviation, space flight, transport) leads to redistribution of blood: with acceleration directed toward the head, blood pools in the legs, and cerebral ischemia and blackout are observed. With prolonged exposure, **noise** and **vibration** cause occupational diseases (deafness, vibration disease).

7. Chemical, biological and social factors

Chemical factors

The effect of chemical substances depends on their nature, dose, route of entry into the organism and the state of the organism. The action may be **local** (burns by acids and alkalis) or **systemic** (after absorption — resorptive). Most poisons act selectively on certain organs (organotropism):

carbon monoxide blocks hemoglobin, cyanides block the enzymes of tissue respiration (cytochrome oxidase), organophosphorus compounds inhibit cholinesterase, mercuric chloride injures the kidneys, and carbon tetrachloride injures the liver.

The organism detoxifies poisons mainly in the liver (oxidation, reduction, conjugation with glucuronic and sulfuric acids) and excretes them through the kidneys, intestine and lungs. In liver and kidney diseases the effect of poisons is enhanced.

Biological factors

Biological factors include viruses, bacteria, fungi, protozoa, helminths and prions. Their pathogenicity is determined by **virulence** (the degree of ability to cause disease), **invasiveness** (the ability to penetrate and spread in tissues) and **toxigenicity** (production of exotoxins and endotoxins). The development of disease depends on the number of microorganisms (infectious dose), the route of entry and the resistance of the macroorganism.

Psychogenic and social factors

Prolonged mental strain (distress) activates the hypothalamic-pituitary-adrenal system and the sympathetic nervous system. This promotes the development of psychosomatic diseases such as arterial hypertension, ischemic heart disease and peptic ulcer. Social factors (standard of living, nutrition, education, availability of medical care) largely determine the morbidity and mortality of the population.

> **REMEMBER**
> The pathogenicity of an external factor depends on three things: **the factor itself** (its nature, strength and duration), **the conditions of action** (in what environment and by what route it acts) and **the reactivity of the organism**. The same factor can be harmless under one condition and fatal under another.

8. Experimental part: the effect of ambient temperature on the development of hypoxic hypoxia

Aim: to observe how an external condition — ambient temperature — changes the development and outcome of a pathological process (hypoxic hypoxia).

Equipment: white mice (3, of the same weight), a Komovsky pressure chamber (or a sealed chamber with a pump) and a manometer, a vessel with ice (+5 °C), a thermostat (+40 °C), an electric thermometer, a stopwatch.

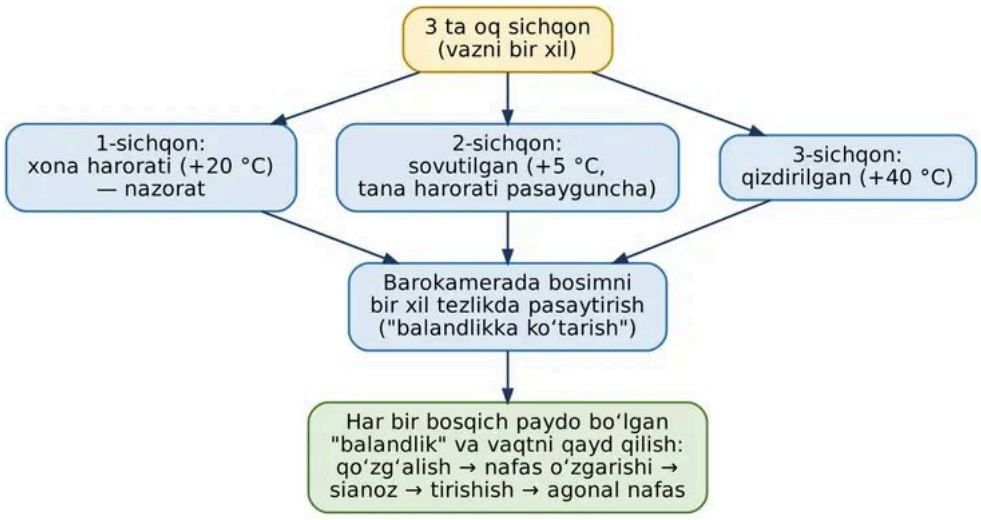


Figure 2.2. *Scheme of the experiment*

Procedure

1. Assess the initial state of the mice: motor activity, respiratory rate (per minute), color of the skin, ears and tail, rectal temperature.
2. Leave mouse 1 at room temperature (+20 °C) — control.
3. Place mouse 2 in the vessel at +5 °C and keep it there until its body temperature falls noticeably (until signs of chilling appear).
4. Place mouse 3 in the thermostat at +40 °C and keep it there until signs of hyperthermia (increased motor activity and rapid breathing) appear.
5. Place each mouse in turn in the pressure chamber and lower the pressure at the same rate (corresponding to an “ascent” of about 1000 m per minute).
6. Using the manometer reading, record the “altitude” and the time at which each stage appears: motor excitation, rapid breathing, cyanosis of the ears and tail, lying on the side, convulsions, agonal (slow, deep) breathing.
7. As soon as agonal breathing appears, immediately return the pressure to normal and observe the animal's recovery time.
8. Enter the results in the table and compare them.

Table 2.8. *Sample table for recording the results of the experiment*

Parameter	Control (+20 °C)	Cooled (+5 °C)	Heated (+40 °C)
Initial respiratory rate			
“Altitude” of excitation			
“Altitude” of cyanosis			
“Altitude” of convulsions			
“Altitude” and time of agonal breathing			
Recovery time			

Expected results and their analysis

- **The heated mouse** is the least resistant to hypoxia: convulsions and agonal breathing appear at a lower “altitude” and sooner. The reason is that in hyperthermia metabolism and the oxygen demand of tissues rise sharply, while oxygen delivery is reduced.
- **The cooled mouse** (with reduced body temperature) is the most resistant to hypoxia: the signs appear at a higher “altitude” and later. The reason is that in hypothermia metabolism slows, the central nervous system is inhibited and oxygen demand decreases.
- **The control mouse** shows an intermediate result.

> **CAUTION**
> The effect of cold has two stages: with brief chilling (while body temperature has not yet fallen), thermogenesis in the mouse — shivering and metabolism — increases, oxygen demand **rises**, and resistance to hypoxia may decrease. Resistance increases only **after body temperature has fallen** (hypothermia). That is why in the experiment it is important to check by body temperature that the mouse has been cooled sufficiently.

Conclusion: the same cause (a decreased partial pressure of oxygen in the air) under different external conditions (ambient temperature) leads to different development and outcomes of hypoxia. This confirms the important role of external conditions in the development of a pathological process.

9. Practical skills

Skill: creating a model of hypoxic hypoxia in a pressure chamber

1. Check the sealing of the pressure chamber and the functioning of the manometer.
2. Record the animal's initial parameters (respiratory rate, movement, color, temperature).
3. Place the animal in the pressure chamber and close the lid tightly.
4. Lower the pressure evenly with the pump; convert the manometer reading into meters using the altitude table.
5. Record the “altitude” and the time whenever a new sign appears.
6. As soon as agonal breathing appears, immediately restore the pressure.
7. Observe the animal's recovery and record the result in the protocol.

Skill: analyzing the results of the experiment

1. Enter in the table the “altitude” and the time at which the signs appeared for each group.
2. Compare with the control group and determine whether resistance increased or decreased.
3. Explain the difference by the level of metabolism and oxygen demand.
4. Distinguish the cause (hypoxia) and the condition (temperature) and write a conclusion.
5. State the clinical significance of the result (artificial hypothermia, the sensitivity of a febrile patient to hypoxia).

Review questions

1. Into what groups are external conditions divided? Give examples.
2. Why can the same factor be a cause in one case and a condition in another?
3. Explain the stages of hyperthermia and the pathogenesis of heatstroke.
4. How does hyperthermia differ from fever?
5. What are the stages and clinical degrees of hypothermia?
6. For what purpose is artificial hypothermia used in medicine?
7. List the zones of the effect of altitude on the organism.
8. Explain the pathogenesis of mountain sickness.
9. How does decompression sickness develop and how is it prevented?
10. On what factors does the harmful effect of electric current depend?
11. What are the direct and indirect actions of ionizing radiation?
12. List the forms and periods of acute radiation sickness.
13. Explain the pathogenesis of crush syndrome.
14. By what properties is the pathogenicity of biological factors determined?
15. Why does the resistance to hypoxia of heated and cooled mice differ in the experiment?

Test questions

1. What is the main difference between hyperthermia and fever?
   A) In hyperthermia, body temperature is always above 42 °C
   B) In hyperthermia, heat loss is insufficient and the thermoregulatory center is not reset
   C) Hyperthermia is caused by pyrogens
   D) There is no sweating in fever
   E) Antipyretics are effective in hyperthermia

2. Which sign is characteristic of the compensation stage of hypothermia?
   A) Bradycardia
   B) Drowsiness
   C) Cessation of shivering
   D) Shivering and constriction of skin vessels
   E) Ventricular fibrillation

3. What is the main pathogenic factor acting on the organism during ascent to altitude?
   A) Decreased partial pressure of oxygen in the air
   B) Decreased air temperature
   C) Increased ultraviolet radiation
   D) Decreased air humidity
   E) Increased wind speed

4. What is the main pathogenetic mechanism of decompression sickness?
   A) Oxygen toxicity
   B) Accumulation of carbon dioxide
   C) Nitrogen narcosis
   D) Expansion of intestinal gases
   E) Gas embolism by nitrogen bubbles during desaturation

5. What is the main factor determining the harmful effect of electric current?
   A) The color of the current
   B) The size of the current source
   C) Current strength
   D) The length of the wire
   E) Air temperature

6. According to the law of Bergonié and Tribondeau, which tissue is the most radiosensitive?
   A) Nervous tissue
   B) Hematopoietic tissue (bone marrow)
   C) Skeletal muscles
   D) Bone tissue
   E) Cartilage

7. What is the indirect action of ionizing radiation based on?
   A) Direct breakage of DNA
   B) Heat denaturation of proteins
   C) Mechanical rupture of the cell membrane
   D) Direct inactivation of enzymes
   E) Formation of free radicals during radiolysis of water

8. Which form of acute radiation sickness develops after a single whole-body exposure to a dose of 2–4 Gy?
   A) Bone marrow form, moderate severity
   B) Intestinal form
   C) Toxemic form
   D) Cerebral form
   E) Radiation sickness does not develop

9. What is the most dangerous cause of death in crush syndrome after the pressure is removed?
   A) Hypoglycemia
   B) Hyperthermia
   C) Hypocalcemia
   D) Hyperkalemia and acute kidney injury
   E) Hypernatremia

10. How does the resistance to hypoxia of a heated (+40 °C) mouse change in the experiment?
    A) It increases because metabolism slows
    B) It does not change
    C) It decreases because the oxygen demand of tissues rises
    D) It increases because breathing becomes faster
    E) It decreases because the number of red blood cells falls

11. What is the reason for the increased resistance to hypoxia of a mouse with reduced body temperature (in hypothermia)?
    A) Decreased metabolism and oxygen demand
    B) Increased erythropoiesis
    C) Decreased affinity of hemoglobin for oxygen
    D) Excitation of the respiratory center
    E) Faster heart rate

12. Why does the body cool faster in cold water than in air?
    A) Water contains little oxygen
    B) Water pressure is high
    C) The thermal conductivity of water is much higher than that of air
    D) Sweating increases in water
    E) Breathing becomes faster in water

13. What change develops as a result of hyperventilation in mountain sickness?
    A) Hypercapnia and respiratory acidosis
    B) Metabolic acidosis
    C) Hypernatremia
    D) Hypoglycemia
    E) Hypocapnia and respiratory alkalosis

14. According to the “rule of nines”, what percentage of the body surface does one leg make up in an adult?
    A) 9 %
    B) 18 %
    C) 1 %
    D) 27 %
    E) 36 %

15. Which of the following is an external condition of pneumonia?
    A) Pneumococcus
    B) Influenza virus
    C) Staphylococcus
    D) Chilling
    E) Klebsiella

Clinical cases

Case 1

In July, at an air temperature of +42 °C, a 35-year-old construction worker did heavy physical work outdoors all day and drank little water. In the evening he complained of severe headache and nausea, and then lost consciousness. On examination: the skin is dry and hot, there is no sweating, body temperature is 41.5 °C, pulse 140 beats/min, arterial pressure 85/50 mm Hg, convulsions.

Questions: What condition has developed in the patient? Explain its pathogenesis. Why are antipyretics ineffective in this condition?

Answer: The patient has developed the decompensation stage of hyperthermia — heatstroke. High external temperature and heavy physical work increased heat production, while heat loss was insufficient. Profuse sweating led to dehydration, loss of sodium and chloride, hemoconcentration and a decrease in circulating blood volume; together with dilatation of the skin vessels, this caused arterial hypotension (collapse). Hypoxia and edema of the brain injured the thermoregulatory center, sweating stopped and the temperature rose even further (a vicious circle). Antipyretics lower the set point of the thermoregulatory center that has been changed by pyrogens; in hyperthermia, however, the set point is unchanged, and the problem is insufficient heat loss. Therefore physical cooling of the body (a cool place, wet wrapping, cold compresses) and restoration of the water-electrolyte balance are necessary.

Case 2

A 28-year-old mountaineer ascended from sea level to an altitude of 4500 m in one day. The next day he complained of headache, weakness, shortness of breath and insomnia. Later a cough with pink frothy sputum appeared.

Questions: What condition has developed? What is the main link of pathogenesis? How can this condition be prevented?

Answer: Mountain sickness has developed, complicated by pulmonary edema. The main link is hypoxic hypoxia and hypoxemia resulting from the decreased partial pressure of oxygen in the air. Hypoxemia causes hyperventilation and respiratory alkalosis, hypoxic spasm of the pulmonary arterioles and increased pressure in the pulmonary artery; as a result, capillary permeability increases and pulmonary edema develops. Prevention: stepwise ascent and time for acclimatization; when severe signs appear, immediate descent and oxygen therapy.

Case 3

In an experiment, students “raised” three mice in a pressure chamber. Convulsions appeared at 9000 m in the control mouse, at 7000 m in the mouse heated at +40 °C, and at 11,000 m in the cooled mouse whose body temperature had fallen to 30 °C.

Questions: Explain the results. What are the cause and the condition in this experiment? What is the clinical significance of the result?

Answer: The cause is the same in all the mice — a decreased partial pressure of oxygen in the air (hypoxic hypoxia). The condition is the ambient and body temperature. In the heated mouse, metabolism and the oxygen demand of tissues increased, so the signs of hypoxia appeared at a lower “altitude” (resistance decreased). In the hypothermic mouse, metabolism slowed, the central nervous system was inhibited and oxygen demand fell, so resistance increased. Clinical significance: febrile patients are more sensitive to hypoxia, and monitoring their oxygen supply is important; in heart and brain operations, artificial hypothermia is used to protect tissues from hypoxia.

Quick review

Table 2.9. *Key facts on Topic 2*

Question	Answer
External condition	An environmental factor that amplifies or weakens the action of the cause
Hyperthermia	A rise in body temperature due to insufficient heat loss; the thermoregulatory set point does not change
Heatstroke	Severe decompensation of hyperthermia: sweating stops, collapse, cerebral edema
Hypothermia	A fall in body temperature below 35 °C; mild 32–35, moderate 28–32, severe below 28 °C
Artificial hypothermia	Controlled cooling to reduce the oxygen demand of tissues
Mountain sickness	Hypoxic hypoxia due to decreased pO_2; in severe cases pulmonary and cerebral edema
Altitude zones	Up to 2000 m indifferent, 2000–4000 m compensation, 4000–5500 m incomplete compensation, 5500–8000 m critical
Decompression sickness	Gas embolism by nitrogen bubbles during desaturation
Electric current	The main factor is current strength; alternating current of 40–60 Hz is the most dangerous
The law of Bergonié and Tribondeau	Tissues with high mitotic activity and low differentiation are the most radiosensitive
Acute radiation sickness	Bone marrow (1–10 Gy), intestinal (10–20 Gy), toxemic (20–80 Gy), cerebral (more than 80 Gy) forms
Crush syndrome	Myoglobinuria, hyperkalemia, acute kidney injury
Conclusion of the experiment	Heating lowers resistance to hypoxia; hypothermia raises it

TOPIC 3

The importance of internal conditions in the development of pathological processes. Reactivity and resistance; the role of constitution and heredity in pathology

Learning objectives

To explain to the student the role of internal conditions — the reactivity and resistance of the organism, the state of the nervous and endocrine systems, constitution, heredity, age and sex — in the origin and course of disease; and to teach the student to observe the development of hypoxic hypoxia in an experiment under different functional states of the central nervous system (narcotic sleep and excitation) and to analyze the result.

By the end of the session, the student should know:

- the concepts of reactivity and resistance, their types and their interrelationship;
- the forms of expression of reactivity (hyperergy, hypoergy, dysergy, anergy);
- the biological barriers of the organism and their significance;
- the role of the nervous and endocrine systems in shaping reactivity, and the types of higher nervous activity;
- the classifications of constitution (Hippocrates, Sigaud, Kretschmer, Chernorutsky) and diatheses;
- the types of hereditary diseases and the difference between the concepts “hereditary” and “congenital”;
- the effect of age and sex on reactivity.

The student should be able to:

- to model hypoxic hypoxia in anesthetized and excited animals and compare the results;
- to assess changes in reactivity and resistance in concrete examples;
- to determine a patient's constitutional type and indicate the predisposition to diseases characteristic of it.

1. Characteristics of internal conditions

Internal conditions are the properties of the organism itself that determine the result of the action of the cause of disease. Internal conditions are the main reason why the same cause (for example, an infection or hypoxia) produces different results in different people.

Table 3.1. *Internal conditions*

Group	Content
Reactivity and resistance	The ability of the organism to respond to a stimulus and to resist it
State of the nervous system	Type of higher nervous activity, the balance of excitation and inhibition, sleep, anesthesia, stress
State of the endocrine system	Levels of hormones of the pituitary, adrenal, thyroid and pancreatic glands
Constitution	Hereditary and acquired, relatively stable morphological and functional features of the organism

Group	Content
Heredity	Hereditary diseases and hereditary predisposition to diseases
Age and sex	Childhood, maturity and old age; features of the female and male organism
Biological barriers	State of the skin, mucous membranes and blood-tissue barriers
Previous state	Previous diseases, immunity, fatigue, nutritional status

2. Reactivity and its types

Reactivity is the property of the organism to respond to influences of the external and internal environment by changing its vital activity. Reactivity is characteristic of all levels of the living organism — molecule, cell, tissue, organ, system and the whole organism.

The concept of “reactivity” was introduced into medicine at the beginning of the 20th century; in this period anaphylaxis and serum sickness were discovered, and C. Pirquet proposed the term “allergy” (altered reactivity). I.I. Mechnikov (immunological reactivity), A.A. Bogomolets and N.N. Sirotinin (constitutional and evolutionary approaches) made a great contribution to the doctrine of reactivity. Avicenna (Ibn Sina) emphasized the need to take into account the patient's “temperament” in treatment.

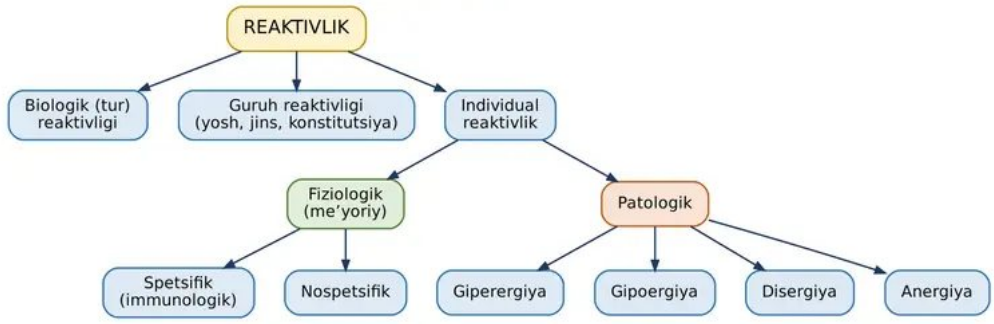


Figure 3.1. *Types of reactivity (based on data from N.H. Abdullaev, 2008)*

Table 3.2. *Types of reactivity*

Type	Definition	Examples
Biological (species) reactivity	Hereditarily determined features of response characteristic of a particular biological species	Resistance of humans to cattle plague and of dogs to measles; hibernation in animals
Group reactivity	Features characteristic of individuals belonging to one group	Differences related to age, sex, constitution and blood group
Individual reactivity	Reactivity of each organism determined by its hereditary and acquired features	Different course of the same infection in different people
Specific (immunological)	Formation of antibodies and sensitized lymphocytes in response to a particular antigen	Immunity, allergy, autoimmune reactions
Nonspecific	Responding in the same way to different factors	Stress, phagocytosis, fever, barrier functions
Physiological	The normal response of a healthy organism	Adaptation to physical work, immunity after vaccination

Type	Definition	Examples
Pathological	An abnormal response with limited adaptive capacity under the influence of a disease factor	Allergy, immunodeficiency, decreased reactivity in traumatic shock

Forms of expression of reactivity

Table 3.3. *Forms of expression of reactivity*

Form	Essence	Clinical picture
Normergy	A normal response appropriate to the stimulus	The usual (typical) course of the disease
Hyperergy	An increased, excessive response	The disease runs a violent, severe course: high fever, intense inflammation, allergic reactions
Hypoergy	A weakened response	The disease is sluggish, with vague signs; “asymptomatic” pneumonia in the elderly and in cachexia
Dysergy	A distorted, abnormal response	Unexpected, unusual reactions (for example, a paradoxical response to drugs)
Anergy	Almost complete absence of a response	Severe sepsis, AIDS, extreme cachexia, a negative tuberculin test

CLINICAL SIGNIFICANCE

A hypoergic course is particularly dangerous: in an elderly patient or a patient with diabetes mellitus, pneumonia may occur without fever or leukocytosis, with only weakness and clouding of consciousness. The “mildness” of the signs reflects not the mildness of the disease but the weakness of the organism's response, and it increases the risk of late diagnosis.

3. Resistance and its relationship to reactivity

Resistance is the ability of the organism to withstand and resist the action of pathogenic factors. Resistance is one of the important manifestations of reactivity.

Table 3.4. *Types of resistance*

Type	Content	Examples
Passive	Resistance related to the structural features of the organism	The barrier of the skin, bones and mucous membranes; the thick skin of some animals
Active	Due to active engagement of protective-adaptive mechanisms	Phagocytosis, antibody formation, detoxification of poisons
Primary (innate)	Hereditarily determined resistance	Species immunity
Secondary (acquired)	Acquired during life	Immunity after disease or vaccination, hardening
Specific	Resistance to a particular factor	Immunity to a certain infection
Nonspecific	General resistance to various factors	Resistance of a hardened person to cold, hypoxia and infections

Reactivity and resistance do not always change in the same direction:

Table 3.5. *Interrelationship of reactivity and resistance*

State	Reactivity	Resistance	Example
Hardening, physical exercise	Increases	Increases	Increased resistance to infections and cold
Hibernation, anesthesia, hypothermia	Decreases	Increases with respect to hypoxia and certain poisons and microbes	Resistance of a hibernating animal to hypoxia
Hyperthyroidism, state of excitation	Increases	Decreases with respect to hypoxia	Rapid death of an excited animal in hypoxia
Cachexia, radiation, severe trauma	Decreases	Decreases	Increased number of infectious complications

REMEMBER

Reactivity is “how does it respond?”, **resistance** is “how much can it withstand?”. A decrease in reactivity does not always mean a decrease in resistance: in anesthesia and hypothermia, reactivity decreases while resistance to hypoxia increases. This is proved in the experimental part of this session.

4. Biological barriers

External barriers: the skin (mechanical protection, acidic environment, bactericidal action of fatty acids, desquamation), mucous membranes (mucus, ciliated epithelium, lysozyme, secretory IgA), the acidic environment of gastric juice, and the normal microflora (competition with pathogenic microbes).

Internal (blood-tissue) barriers: the blood-brain, blood-ocular, blood-testis and blood-placental barriers, lymph nodes, and the liver (detoxifies toxic substances coming from the intestine).

CAUTION

The blood-brain barrier protects the brain, but many drugs (for example, some antibiotics) cross it poorly. In the treatment of meningitis, drugs that cross the barrier well are chosen. In inflamed meninges the permeability of the barrier increases.

5. The role of the nervous system in reactivity

The nervous system is the main system that shapes and regulates reactivity. Its state sharply changes the organism's response to pathogenic influences.

- **In ontogenesis:** in newborns the nervous system is immature, so their fever and inflammatory reactions are weak and infection rapidly becomes generalized.
- **Inhibition of the central nervous system** (sleep, anesthesia, hibernation) reduces metabolism and oxygen demand — resistance to hypoxia and certain poisons increases.
- **Excitation of the central nervous system** (stress, caffeine, amphetamine, hyperthyroidism) increases metabolism — resistance to hypoxia decreases.
- **Injury of the nervous system** (spinal cord transection, denervation) leads to degenerative changes in tissues and impaired inflammation and regeneration.

Table 3.6. *Types of higher nervous activity (I.P. Pavlov) and the temperaments of Hippocrates*

Type according to Pavlov	Properties of the nervous processes	Temperament according to Hippocrates	Predisposition to pathology
Strong, balanced, mobile	Excitation and inhibition are strong, equal and alternate quickly	Sanguine	Relatively resistant
Strong, balanced, slow (inert)	The processes are strong and equal but alternate slowly	Phlegmatic	Relatively resistant
Strong, unbalanced	Excitation predominates over inhibition	Choleric	Arterial hypertension, ischemic heart disease, neurasthenia
Weak	Both processes are weak	Melancholic	Neuroses, psychasthenia, depression

CLINICAL SIGNIFICANCE

Anesthesia and controlled sedation are used in resuscitation practice to protect the brain from hypoxia (by reducing oxygen demand). Conversely, agitation, shivering and seizures sharply increase the patient's oxygen demand and aggravate hypoxia.

6. The role of the endocrine system in reactivity

The endocrine system regulates reactivity together with the nervous system.

- **The hypothalamic-pituitary-adrenocortical system** provides the stress reaction. According to H. Selye's doctrine, the organism responds to strong stimuli with the **general adaptation syndrome**: the alarm stage, the stage of resistance and the stage of exhaustion. Glucocorticoids suppress inflammatory and allergic reactions, but in excess they lower immunity.
- **Thyroid hormones** increase metabolism: in hyperthyroidism resistance to hypoxia decreases, whereas in hypothyroidism reactivity is weakened in general.
- **Insulin deficiency** (diabetes mellitus) weakens phagocytosis and regeneration — susceptibility to purulent infections increases and wounds heal slowly.
- **Sex hormones** determine sex differences in reactivity.

7. Constitution and its role in pathology

Constitution is the complex of hereditary and acquired, relatively stable morphological, functional and psychological features of the organism. Constitution determines the reactivity of the organism and its predisposition to certain diseases.

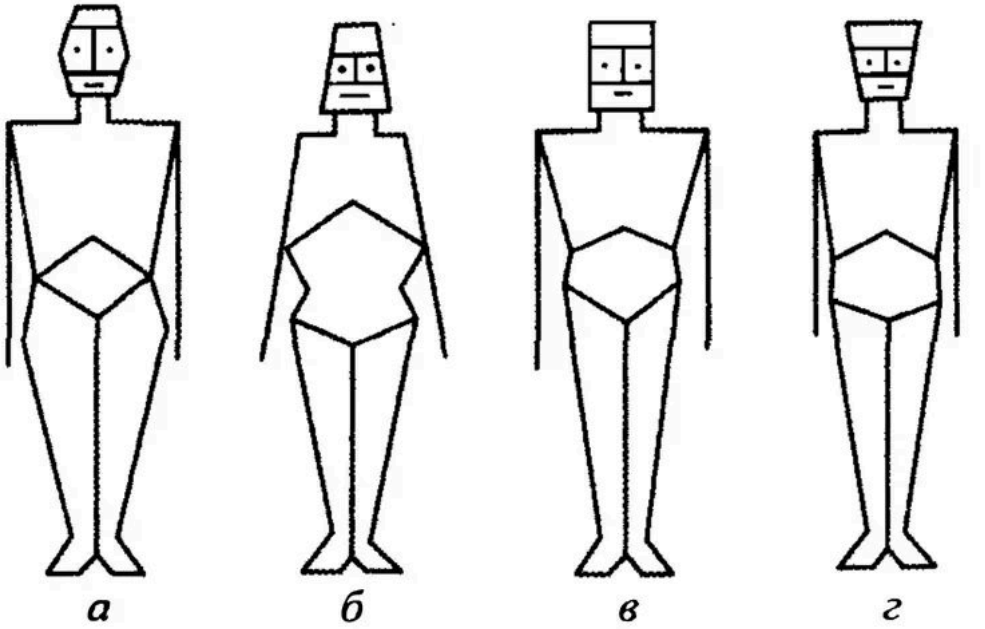


Figure 3.2. *Sigaud's constitutional types (labels а, б, в, г in the figure): respiratory, digestive, muscular and cerebral type (N.N. Zaiko, 2002, Fig. 6)*

Table 3.7. *Classifications of constitution*

Author	Types	Main features
Hippocrates	Sanguine, choleric, phlegmatic, melancholic	Based on temperament (the ratio of the four humors)
C. Sigaud	Respiratory, digestive, muscular, cerebral types	Based on the predominant system in the body (shape of the face and trunk)
E. Kretschmer	Asthenic, athletic, pyknic	Relationship between body build and psychological features
M.V. Chernorutsky	Asthenic (hyposthenic), normosthenic, hypersthenic	Body build, functions and metabolism

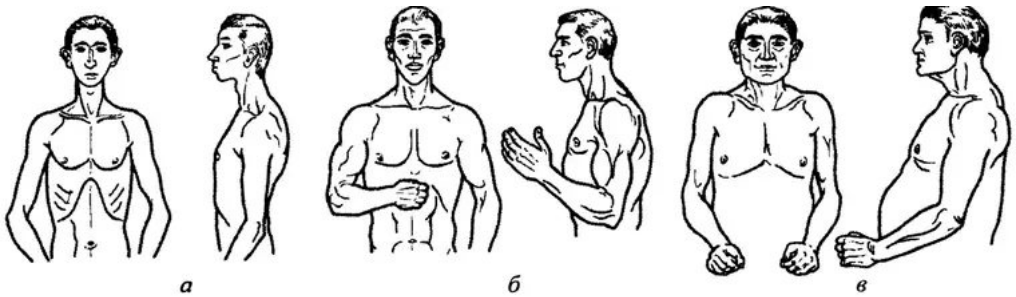


Figure 3.3. *Kretschmer's constitutional types (labels а, б, в in the figure): asthenic, athletic, pyknic (N.N. Zaiko, 2002, Fig. 7)*

Table 3.8. *Constitutional types according to Chernorutsky and predisposition to diseases*

Feature	Asthenic (hyposthenic)	Normosthenic	Hypersthenic
Body build	Tall, thin, long and narrow chest, acute epigastric angle	Medium, proportional; right epigastric angle	Medium or short height, broad-boned, broad and short chest; obtuse epigastric angle
Functional features	Lower arterial pressure, accelerated metabolism, reduced intestinal absorption	Normal	Higher arterial pressure, slowed metabolism, more cholesterol and glucose in the blood

Feature	Asthenic (hyposthenic)	Normosthenic	Hypersthenic
Predisposition to diseases	Gastric and duodenal ulcer, pulmonary tuberculosis, hypotension, neuroses, prolapse of internal organs	Diseases of the respiratory tract and musculoskeletal system	Arterial hypertension, atherosclerosis, ischemic heart disease, obesity, type 2 diabetes mellitus, gallstone disease

MEMORY AID

“Thin — stomach, stout — vessels”: people of the asthenic type are more prone to peptic ulcer and tuberculosis, whereas people of the hypersthenic type are prone to hypertension, atherosclerosis and diabetes.

Diatheses

Diathesis is an anomaly of constitution occurring in childhood, characterized by the organism's predisposition to respond abnormally to certain influences. Diathesis is not a disease but a state of predisposition to disease.

Table 3.9. *Types of diathesis*

Type	Essence	Predisposition to diseases
Exudative-catarrhal (allergic)	Predisposition of the skin and mucous membranes to inflammatory and allergic reactions	Atopic dermatitis, eczema, allergic rhinitis, bronchial asthma
Lymphatic-hypoplastic	Hyperplasia of lymphoid tissue (enlargement of the thymus), reduced adrenal function	Severe course of infections, risk of sudden death under stress
Neuro-arthritic	Impaired purine metabolism, increased excitability of the nervous system	Gout, urolithiasis, obesity, diabetes mellitus (in adulthood)
Hemorrhagic	Tendency to bleed	Hemophilia, thrombocytopathies, vasculitides

8. Heredity and its role in pathology

By the participation of heredity in their origin, diseases are divided into three groups:

Table 3.10. *The ratio of hereditary and external factors in diseases*

Group	Role of heredity and environment	Examples
Hereditary diseases	The mutation is decisive; the external environment affects only the course	Hemophilia, phenylketonuria, Down syndrome
Diseases with hereditary predisposition (multifactorial)	Hereditary predisposition + external factors	Arterial hypertension, diabetes mellitus, bronchial asthma, peptic ulcer disease, atherosclerosis
Diseases caused mainly by external factors	The role of heredity is small	Injuries, burns, most infections

Table 3.11. *Types of hereditary diseases*

Type	Mechanism	Examples
Gene (monogenic) diseases, autosomal dominant	A single mutant allele is sufficient	Marfan syndrome, familial hypercholesterolemia, achondroplasia
Autosomal recessive	Both alleles must be mutant	Phenylketonuria, cystic fibrosis, sickle cell anemia
X-linked recessive	Mainly boys are affected; their mothers are carriers	Hemophilia A and B, Duchenne muscular dystrophy, color blindness
Chromosomal diseases	Change in the number or structure of chromosomes	Down (trisomy 21), Klinefelter (47,XXY), Turner (Shereshevsky–Turner, 45,X0) syndromes
Mitochondrial diseases	Mutation of mitochondrial DNA; transmitted only through the mother	Leber hereditary optic neuropathy

CAUTION

“Congenital” and **“hereditary”** are not the same concept. A congenital disease is present at birth, but it may not be hereditary (for example, defects caused by the rubella virus or alcohol during pregnancy). A hereditary disease, in contrast, may appear long after birth (Huntington chorea — at the age of 30–50). Conditions that develop under the influence of external factors and resemble a hereditary disease but are not passed from generation to generation are called **phenocopies**.

9. The effect of age and sex on reactivity

- **Infancy and early childhood:** the nervous, endocrine and immune systems are immature, and the biological barriers are weak. Inflammatory and febrile reactions are weak, and infections quickly become generalized (predisposition to sepsis). At the same time, newborns are more resistant to hypoxia than older people (high activity of anaerobic glycolysis; fetal hemoglobin binds oxygen well).
- **Maturity:** reactivity is at its highest level.
- **Old age:** the immune response weakens, barriers become weaker, and the reserve capacity of organs decreases; diseases often run a hypoergic, asymptomatic course; neoplastic diseases become more frequent.
- **Sex:** the female organism is more resistant to blood loss, hypoxia and starvation; autoimmune diseases are more common in women, whereas ischemic heart disease occurs more often at an early age in men (the protective effect of estrogens). In women, reactivity changes during the menstrual cycle, pregnancy and menopause.

10. Experimental part: development of hypoxic hypoxia in different states of the central nervous system

Aim: to observe how an internal condition — the functional state of the central nervous system — changes the development and outcome of hypoxia.

Equipment: white mice (3, of the same weight), a Komovsky pressure chamber and a manometer, a solution of a narcotic agent (the drug and dose adopted at the department, for example urethane), 1% caffeine solution, physiological saline, syringes, a stopwatch.

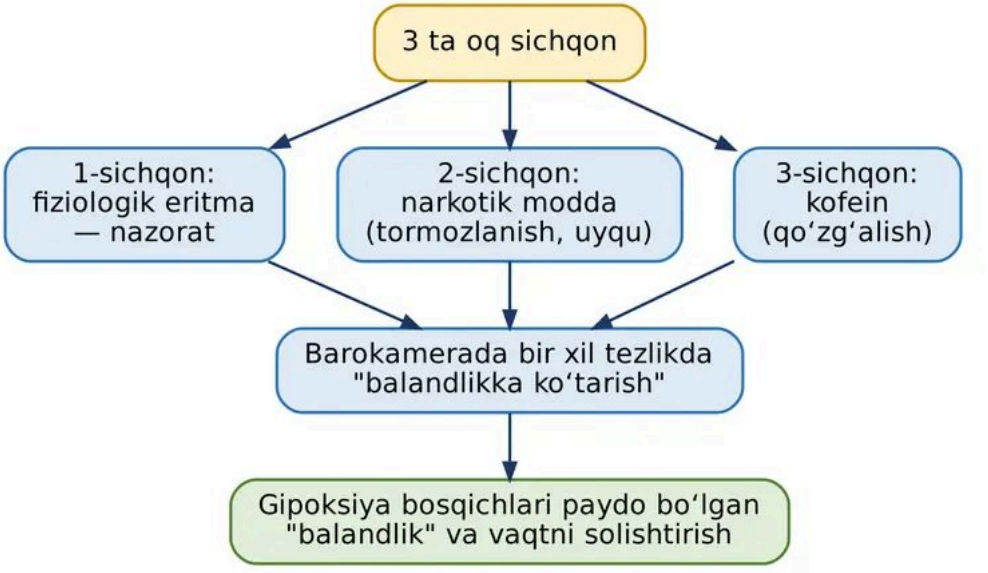


Figure 3.4. *Scheme of the experiment*

Procedure

1. Assess the initial state of all three mice: motor activity, respiratory rate, color of the skin, ears and tail.
2. Inject physiological saline into the peritoneal cavity of mouse 1 (control).
3. Inject the narcotic agent into mouse 2 and wait for signs of narcotic sleep (immobility, weakened reflexes, slower breathing) to appear.
4. Inject the caffeine solution into mouse 3 and wait for signs of excitation (increased motor activity, rapid breathing) to appear.
5. Place each mouse in turn in the pressure chamber and lower the pressure at the same rate.
6. Record the “altitude” and the time at which the stages of hypoxia appear: excitation, change in breathing, cyanosis, lying on the side, convulsions, agonal breathing.
7. As soon as agonal breathing appears, immediately restore the pressure and observe the animal's recovery.
8. Enter the results in the table and compare them.

Table 3.12. *Sample table for recording the results of the experiment*

Parameter	Control	Narcotic sleep	Caffeine (excitation)
Initial respiratory rate			
“Altitude” of cyanosis			
“Altitude” of convulsions			
“Altitude” and time of agonal breathing			
Recovery time			

Expected results and their analysis

- **The mouse in narcotic sleep** is the most resistant to hypoxia: the signs of hypoxia appear at a much higher “altitude”, and convulsions are weak or absent. The reason is that the central nervous system is inhibited, and metabolism and the oxygen demand of neurons are reduced.

- **The mouse given caffeine** is the least resistant to hypoxia: the signs appear at a lower “altitude” and sooner. The reason is that excited neurons and muscles consume much oxygen.
- **The control mouse** shows an intermediate result.

> **CAUTION**
> Although anesthesia increases resistance to hypoxia, narcotic substances in large doses depress the respiratory center and can themselves lead to hypoxia (respiratory hypoxia). That is why the dose is strictly fixed in the experiment, and in the clinic anesthesia is always performed under monitoring of respiration and blood oxygen.

Conclusion: the same cause (a decreased partial pressure of oxygen in the air) gives different results in different states of the central nervous system. Inhibition of the nervous system lowers reactivity but increases resistance to hypoxia, whereas excitation does the opposite. This confirms the role of internal conditions in the development of a pathological process.

11. Practical skills

Skill: modeling hypoxia in narcotic sleep and in excitation

1. Match the animals by weight and label them.
2. Record the initial parameters.
3. Inject the drugs into the peritoneal cavity at the specified dose and wait for the effect to begin.
4. Lower the pressure in the pressure chamber at the same rate.
5. Record the “altitude” and the time at which each stage of hypoxia appears.
6. As soon as agonal breathing appears, restore the pressure.
7. Compare the results with the control group and write a conclusion.

Skill: determining the constitutional type (according to Chernorutsky)

1. Measure height and body weight and calculate the body mass index.
2. Assess the shape, length and width of the chest.
3. Determine the epigastric (costal) angle: less than 90° — asthenic, about 90° — normosthenic, more than 90° — hypersthenic.
4. Calculate the Pignet index: height (cm) − [body weight (kg) + chest circumference (cm)]. Less than 10 — hypersthenic, 10–30 — normosthenic, more than 30 — asthenic.
5. Take into account arterial pressure and pulse as additional signs.
6. Write a conclusion and indicate the predisposition to diseases characteristic of this type.

Review questions

1. What do internal conditions include?
2. What is reactivity? What types of reactivity are there?
3. How do specific and nonspecific reactivity differ?
4. Give examples of hyperergy, hypoergy, dysergy and anergy.
5. What is resistance? Give examples of passive and active resistance.
6. Do reactivity and resistance always change in the same direction? Explain with examples.
7. List the external and internal biological barriers of the organism.

8. How does inhibition of the central nervous system affect resistance to hypoxia?
9. Name the types of higher nervous activity according to I.P. Pavlov and their significance in pathology.
10. What are the stages of the general adaptation syndrome?
11. What classifications of constitution are there?
12. To what diseases are the constitutional types according to Chernorutsky predisposed?
13. What is diathesis? List its types.
14. Into what types are hereditary diseases divided? How do the concepts “congenital” and “hereditary” differ?
15. Why does the resistance to hypoxia of mice in narcotic sleep and mice given caffeine differ in the experiment?

Test questions

1. What is reactivity?
   A) Only the resistance of the organism to pathogenic factors
   B) Only the ability to form antibodies
   C) Hereditary predisposition to disease
   D) The property of the organism to respond to influences by changing its vital activity
   E) Relatively stable morphological features of the organism

2. A violent course of disease with excessively strong reactions is characteristic of which form of reactivity?
   A) Hyperergy
   B) Hypoergy
   C) Anergy
   D) Normergy
   E) Dysergy

3. How does resistance to hypoxia change under anesthesia?
   A) It decreases because the respiratory center is excited
   B) It does not change
   C) It increases because metabolism and oxygen demand decrease
   D) It decreases because metabolism increases
   E) It increases because the number of red blood cells increases

4. In which state does resistance to hypoxia increase while reactivity decreases?
   A) Hyperthyroidism
   B) Cachexia
   C) Ionizing radiation
   D) Severe trauma
   E) Hibernation

5. Which example corresponds to the passive form of resistance?
   A) Phagocytosis
   B) The mechanical barrier function of the skin
   C) Antibody formation
   D) Detoxification of poisons in the liver
   E) The febrile reaction

6. What are the constitutional types according to Kretschmer?
   A) Respiratory, digestive, muscular, cerebral
   B) Sanguine, choleric, phlegmatic, melancholic
   C) Hyposthenic, normosthenic, hypersthenic
   D) Asthenic, athletic, pyknic
   E) Exudative, lymphatic, neuro-arthritic

7. To which diseases are people of hypersthenic constitution more predisposed?
   A) Arterial hypertension, atherosclerosis, diabetes mellitus
   B) Pulmonary tuberculosis and gastric ulcer
   C) Prolapse of internal organs
   D) Arterial hypotension
   E) Neuroses and psychasthenia

8. By what type of inheritance is phenylketonuria transmitted?
   A) Autosomal dominant
   B) Autosomal recessive
   C) X-linked dominant
   D) Mitochondrial
   E) Y-linked

9. What is the cause of Down syndrome?
   A) Monosomy X
   B) Karyotype 47,XXY
   C) Mutation of mitochondrial DNA
   D) Mutation of a single gene
   E) Trisomy 21

10. What is meant by a phenocopy?
    A) The most severe form of a hereditary disease
    B) A type of chromosomal disease
    C) A condition caused by an external factor that resembles a hereditary disease but is not inherited
    D) A disease transmitted only through the mother
    E) The effect of one gene on several traits

11. To which diseases is a child with exudative-catarrhal diathesis predisposed?
    A) Gout and urolithiasis
    B) Bleeding
    C) Enlargement of the thymus and sudden death
    D) Atopic dermatitis and bronchial asthma
    E) Prolapse of internal organs

12. Who is the author of the doctrine of the general adaptation syndrome?
    A) H. Selye
    B) I.P. Pavlov
    C) W. Cannon
    D) C. Pirquet
    E) I.I. Mechnikov

13. Why does the resistance to hypoxia of a mouse given caffeine decrease in the experiment?
    A) Caffeine blocks hemoglobin
    B) Caffeine paralyzes the respiratory center
    C) The oxygen demand of excited neurons and muscles increases
    D) Caffeine sharply lowers blood pressure
    E) Caffeine destroys red blood cells

14. Pneumonia without fever or leukocytosis in an elderly patient is characteristic of which form of reactivity?
    A) Hyperergy
    B) Normergy
    C) Dysergy
    D) Allergy
    E) Hypoergy

15. Which condition is an example of a “congenital but not hereditary” disease?
    A) Hemophilia
    B) A heart defect caused by the rubella virus during pregnancy
    C) Down syndrome
    D) Phenylketonuria
    E) Cystic fibrosis

Clinical cases

Case 1

During an influenza epidemic in a kindergarten, 18 of 30 children fell ill. In 3 of them the disease was severe, with high fever and convulsions; in 2 children (one of whom was often ill and underweight) the disease ran a sluggish course without fever and was complicated by pneumonia. Twelve children did not fall ill at all.

Questions: How do you explain the different courses of the disease under the action of the same cause? Which forms of reactivity were observed? What played the decisive role in the children who did not fall ill?

Answer: The cause of the disease (the influenza virus) is the same, while the different courses depend on internal conditions — individual reactivity and resistance. The severe course with high fever and convulsions reflects hyperergic reactivity (in children, the immaturity and excitability of the nervous system create a predisposition to convulsions). The cases with a sluggish course without fever and with complications reflect hypoergic reactivity (low weight and frequent illness indicate a weakened immune response). The children who did not fall ill had high resistance: specific immunity resulting from vaccination or previous illness, and nonspecific resistance due to good nutrition and hardening.

Case 2

A 52-year-old man, height 168 cm, body weight 98 kg, with a broad and short chest and an obtuse epigastric angle. Arterial pressure is 160/100 mm Hg; blood cholesterol and glucose are elevated. His father died of myocardial infarction at the age of 55.

Questions: Determine the patient's constitutional type. To what diseases is he predisposed? Distinguish internal and external conditions.

Answer: Hypersthenic constitution according to Chernorutsky (the pyknic type according to Kretschmer). This type is characterized by a predisposition to arterial hypertension, atherosclerosis, ischemic heart disease, obesity, type 2 diabetes mellitus and gallstone disease; the patient already has arterial hypertension and metabolic disorders. Internal conditions: constitution, hereditary predisposition (the father's early myocardial infarction), male sex and age. External conditions (to be clarified): overeating, physical inactivity, smoking, stress. Prevention is aimed at eliminating the external factors: weight reduction, diet, physical activity, control of arterial pressure and lipids.

Case 3

In a student experiment in a pressure chamber, convulsions appeared at 9000 m in the control mouse. The mouse in narcotic sleep was “raised” to 11,500 m without convulsions, whereas in the mouse given caffeine convulsions began at 7500 m and agonal breathing developed rapidly.

Questions: Explain the results. How did reactivity and resistance change? What is the clinical significance of the result?

Answer: The cause is the same in all the mice — a decreased partial pressure of oxygen in the air. In the mouse in narcotic sleep, the central nervous system was inhibited, reactivity decreased, and metabolism and oxygen demand fell, so resistance to hypoxia increased. In the mouse given caffeine, the nervous system was excited and reactivity increased, but resistance decreased because oxygen consumption rose. This shows that reactivity and resistance do not always change in the same direction. Clinical significance: in patients at risk of hypoxia (after cardiac arrest, in severe brain injury), sedation and anesthesia are used to protect the brain; agitation, seizures and fever increase oxygen demand and aggravate hypoxia, so they must be eliminated.

Quick review

Table 3.13. *Key facts on Topic 3*

Question	Answer
Internal conditions	Reactivity, resistance, nervous and endocrine state, constitution, heredity, age, sex
Reactivity	The property of the organism to respond to influences by changing its vital activity
Types of reactivity	Biological (species), group, individual; specific and nonspecific; physiological and pathological
Forms of expression	Normergy, hyperergy, hypoergy, dysergy, anergy
Resistance	Resistance to pathogenic factors; passive and active, primary and secondary, specific and nonspecific
Reactivity and resistance	They do not always change in the same direction: under anesthesia reactivity decreases while resistance to hypoxia increases
Biological barriers	External (skin, mucous membranes) and internal (blood-brain, blood-placental and others)
Pavlov's types	Equivalents of the sanguine, phlegmatic, choleric and melancholic temperaments
General adaptation syndrome	Stages of alarm, resistance and exhaustion (H. Selye)
Constitution	The complex of hereditary and acquired, relatively stable morphological and functional features
Chernorutsky's classification	Asthenic — ulcer, tuberculosis; hypersthenic — hypertension, atherosclerosis, diabetes

Question	Answer
Diathesis	An anomaly of constitution in childhood: exudative-catarrhal, lymphatic-hypoplastic, neuro-arthritic, hemorrhagic
Hereditary diseases	Gene (AD, AR, X-linked), chromosomal, mitochondrial
Phenocopy	A condition caused by an external factor that resembles a hereditary disease and is not inherited
Conclusion of the experiment	Anesthesia increases resistance to hypoxia, excitation (caffeine) decreases it

TOPIC 4

Arterial and venous hyperemia. Disorders of the microcirculation and peripheral circulation

Learning objectives

To explain to the student the significance of the microcirculation in health and disease and the mechanisms regulating peripheral circulation; to teach the causes, mechanisms of development, signs and outcomes of arterial and venous hyperemia; and to teach the student to induce arterial and venous hyperemia in the rabbit ear and the frog mesentery in an experiment and to observe the changes in the microcirculation.

By the end of the session, the student should know:

- the structure of the microcirculatory bed and the factors regulating blood flow;
- the classification of disorders of peripheral circulation;
- the types and mechanisms of development of arterial hyperemia (neurotonic, neuroparalytic, neuromyoparalytic, humoral);
- the external signs of arterial hyperemia, the changes in the microcirculation and its outcomes;
- the causes and signs of venous hyperemia, the changes in the microcirculation and its outcomes;
- the distinguishing features of arterial and venous hyperemia.

The student should be able to:

- to induce neuroparalytic arterial hyperemia and venous hyperemia in the rabbit ear;
- to observe the microcirculation in the frog mesentery under a microscope and describe the changes;
- to identify the type of hyperemia by its external signs and explain its mechanism.

1. Microcirculation: structure and regulation

Microcirculation is the movement of blood and lymph through the smallest vessels (less than 100 μm in diameter) and the exchange of substances between blood and tissue. The microcirculatory bed is the functional unit of an organ: it supplies tissues with oxygen and nutrients and removes metabolic products.

Table 4.1. *Components of the microcirculatory bed*

Segment	Function
Arterioles	“Resistance vessels”: regulate blood flow through the tone of smooth muscle
Metarterioles and precapillary sphincters	Regulate the entry of blood into the capillaries (the number of working capillaries)
Capillaries	“Exchange vessels”: the wall consists of a single layer of endothelium; exchange of gases and substances
Postcapillary venules	This is where the exit of fluid and leukocytes from the vessels mainly takes place
Venules	“Capacitance vessels”: collect and carry away the blood
Arteriolovenular anastomoses (shunts)	Carry blood past the capillaries (important in thermoregulation)

Segment	Function
Lymphatic capillaries	Carry away interstitial fluid and proteins

Blood flow through a vessel obeys Poiseuille's law: flow is directly proportional to the pressure difference between the beginning and the end of the vessel and inversely proportional to vascular resistance. Vascular resistance is inversely proportional to the fourth power of the radius — that is, doubling the radius of an arteriole increases blood flow about 16 times.

REMEMBER

Factors regulating the microcirculation:

- **neurogenic** — sympathetic vasoconstrictor nerves (norepinephrine, alpha-adrenergic receptors) and vasodilator nerves in some regions;
- **humoral** — epinephrine, angiotensin II, vasopressin (constrict); histamine, bradykinin, prostacyclin (dilate);
- **local metabolic** — adenosine, CO_2, H^+, K^+, lactate, oxygen deficiency (dilate);
- **endothelial** — nitric oxide (NO) and prostacyclin (dilate), endothelin (constricts);
- **myogenic** — the response of smooth muscle to stretching of the vessel wall by contraction (the Bayliss effect).

2. Classification of disorders of peripheral circulation

Table 4.2. *Disorders of peripheral (local) circulation*

Group	Types
Change in blood filling	Arterial hyperemia, venous hyperemia, ischemia
Cessation of blood flow	Stasis (capillary, venous, ischemic)
Occlusion of the vascular lumen	Thrombosis, embolism
Disorders of the microcirculation	Intravascular (rheological properties of blood, “sludge”), vascular wall (increased permeability, diapedesis), extravascular (edema, impaired lymph flow)
Disruption of the integrity of the vascular wall	Bleeding and hemorrhage

Hyperemia (Greek hyper — excess, haima — blood) is increased blood filling of an organ or tissue. When the inflow of blood increases, **arterial (active)** hyperemia develops; when the outflow of blood is impeded, **venous (passive, congestive)** hyperemia develops.

3. Arterial hyperemia

Arterial hyperemia is an increased inflow of blood through the arteries into an organ or tissue and increased blood filling as a result of arteriolar dilatation, while the outflow of blood remains normal.

Types of arterial hyperemia

Physiological arterial hyperemia is a normal phenomenon observed when the activity of an organ increases:

- **working (functional)** — in a working muscle, in the gastrointestinal tract during digestion, in the brain during mental work;

- **reflex** — reddening of the skin under the action of heat;
- **emotional** — reddening of the face from embarrassment or anger.

Pathological arterial hyperemia develops under the action of pathogenic factors, is not related to increased activity of the organ and is disproportionate to the stimulus.

Mechanisms of development of arterial hyperemia

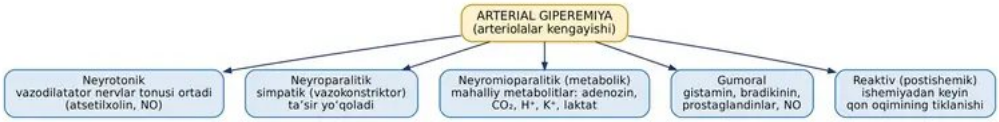


Figure 4.1. *Mechanisms of development of arterial hyperemia (based on V.V. Novitsky, 2009; N.H. Abdullaev, 2008)*

Table 4.3. *Mechanisms of arterial hyperemia*

Mechanism	Essence	Examples
Neurotonic	Increased tone or stimulation of vasodilator nerves	Emotional reddening of the face, some reflex hyperemias, reddening of the skin under the action of heat
Neuroparalytic	Loss of the influence of sympathetic vasoconstrictor nerves; the tone of arteriolar smooth muscle decreases	C. Bernard's experiment: transection of the cervical sympathetic nerve of a rabbit; sympathectomy; injury of the vasoconstrictor centers
Neuromyoparalytic (metabolic)	Decreased tone of smooth muscle in the vessel wall under the action of local metabolites	Working hyperemia, postischemic hyperemia, a hot compress, a mustard plaster
Humoral	Action of vasoactive substances (histamine, bradykinin, prostaglandins, NO)	Hyperemia in an inflammatory focus, allergic reddening

Postischemic (reactive) hyperemia develops after ischemia has been eliminated: metabolites accumulated during ischemia (adenosine, lactate, H^+, K^+) dilate the arterioles, and muscle tone is reduced. An example is reddening of a limb after removal of a tourniquet.

Hyperemia ex vacuo (Latin vacuus — empty) develops when the barometric pressure over a tissue decreases. Examples are the application of medical cups, or the movement of a large amount of blood into the vessels of the abdominal organs when ascitic fluid is rapidly removed from the abdominal cavity.

> **CAUTION**
> Rapid removal of a large volume of ascitic fluid from the abdominal cavity or rapid removal of a large tumor leads to hyperemia ex vacuo: a large part of the blood moves into the abdominal organs, and relative anemia (ischemia) of the brain and fainting develop. That is why paracentesis is performed with slow removal of the fluid.

Signs of arterial hyperemia

Table 4.4. *External signs of arterial hyperemia and their causes*

Feature	Cause
Redness (scarlet-red color)	Increase in arterial blood rich in oxyhemoglobin in the dilated vessels; increased number of working capillaries

Feature	Cause
Increased local temperature	Increased inflow of warm arterial blood and increased metabolism
Increased volume and turgor	Increased blood filling, increased formation of interstitial fluid
Pulsation of small arteries	Dilatation of the arterioles, smaller pressure drop up to the capillaries
Increased lymph formation and flow	Increased filtration pressure in the capillaries

Changes in the microcirculation

- arterioles, precapillaries and capillaries dilate;
- the number of working (open) capillaries increases — previously closed capillaries open;
- the linear and volumetric velocities of blood flow increase;
- hydrostatic pressure in the capillaries rises and filtration increases;
- the arteriovenous oxygen difference decreases (venous blood is also rich in oxygen).

Outcomes of arterial hyperemia

Beneficial outcomes: the supply of the tissue with oxygen and nutrients improves, metabolism, regeneration and phagocytosis increase, and metabolites are rapidly removed. That is why arterial hyperemia is used in physiotherapy (hot compresses, UHF, mustard plasters, cups).

Harmful outcomes: rupture of the wall of altered (atherosclerotic, aneurysmal) vessels and hemorrhage (for example, cerebral hemorrhage in a hypertensive crisis); increased intracranial pressure (headache); relative ischemia in other organs when a large part of the blood collects in one region.

> **CLINICAL SIGNIFICANCE**
> Arterial hyperemia is often observed in the clinic: redness and heat in an inflammatory focus (rubor, calor), warm and moist skin in hyperthyroidism, “hot flashes” in menopause, flushing in carcinoid syndrome. Red, warm, pulsating tissue is always a sign of increased blood inflow.

4. Venous hyperemia

Venous hyperemia (congestion) is increased blood filling of an organ or tissue when the outflow of blood through the veins is impeded; the inflow of blood is normal or reduced.

Causes of venous hyperemia

Table 4.5. *Causes of venous hyperemia*

Group of causes	Examples
External compression of veins	Tumor, scar, pregnant uterus, tight bandage (cast, tourniquet), edema
Occlusion of the venous lumen from within	Thrombosis, embolism, inflammation of the vein wall (phlebitis)
Insufficiency of the vein wall and valves	Varicose dilatation, venous valve insufficiency
Heart failure	Congestion in the pulmonary circulation in left ventricular failure and in the systemic circulation in right ventricular failure
Reduced suction function of the chest	Exudative pleurisy, hydrothorax, pneumothorax

Group of causes	Examples
Gravity	Prolonged standing; in the dependent regions of bedridden severely ill patients (hypostasis)

Signs of venous hyperemia

Table 4.6. *External signs of venous hyperemia and their causes*

Feature	Cause
Bluish color (cyanosis)	Due to slow blood flow, tissues extract more oxygen and the amount of reduced hemoglobin increases (cyanosis is visible when it exceeds 50 g/L)
Decreased local temperature	Decreased inflow of warm arterial blood, weakened metabolism, increased heat loss
Edema	Increased hydrostatic pressure in capillaries and venules, increased permeability of the vessel wall
Increase in volume	Blood filling and edema
Decreased organ function	Hypoxia and impaired metabolism

Changes in the microcirculation

- venules and capillaries dilate and the amount of blood in them increases;
- the velocity of blood flow decreases; in severe cases the blood moves in a **“pendulum-like”** manner (forward in systole, backward in diastole), and then **stasis** develops;
- hydrostatic pressure in the capillaries rises — filtration predominates over reabsorption and edema develops;
- red blood cells pass through the vessel wall (**diapedesis**) — small hemorrhages appear;
- the rheological properties of blood are impaired (aggregation of red blood cells, “sludge”).

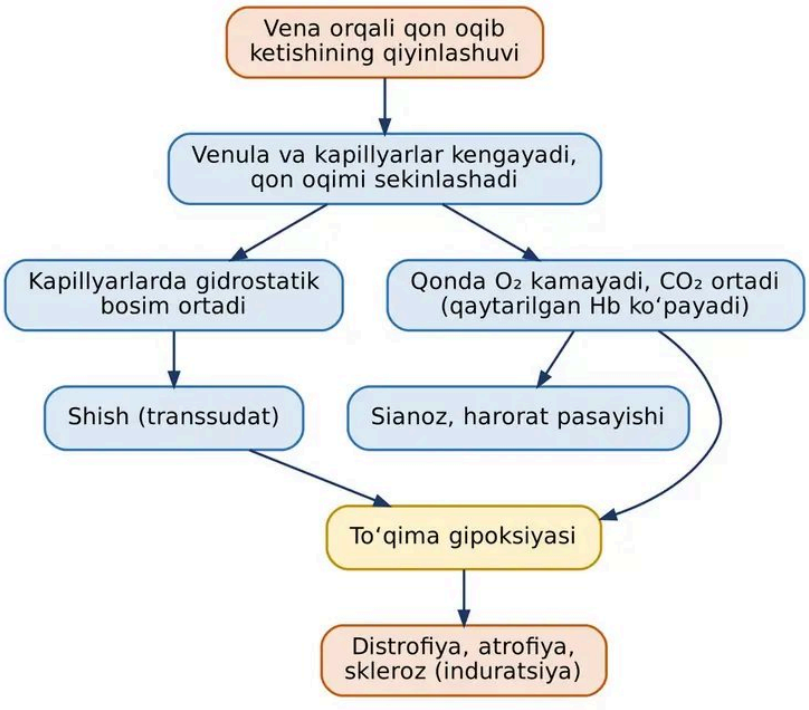


Figure 4.2. *Pathogenesis of venous hyperemia (based on V.V. Novitsky, 2009, Fig. 9.4)*

Outcomes of venous hyperemia

The main harmful factor of venous hyperemia is **hypoxia**. When prolonged, it leads to degeneration and atrophy of parenchymal cells and growth of connective tissue (**sclerosis, induration**).

Table 4.7. *Outcomes of chronic venous hyperemia in various organs*

Organ	Cause	Outcome
Liver	Right ventricular failure	“Nutmeg liver”: congestion in the centers of the lobules and atrophy of hepatocytes, later cardiac sclerosis (cirrhosis)
Lung	Left ventricular failure, mitral stenosis	Brown induration of the lung: diapedesis, accumulation of hemosiderin (“heart failure cells”), sclerosis
Kidney	Generalized venous congestion	“Congested kidney”, proteinuria, oliguria
Legs	Varicose dilatation, venous valve insufficiency	Edema, skin pigmentation, trophic ulcers
Abdominal organs	Portal hypertension (liver cirrhosis)	Ascites, splenomegaly, varicose dilatation of the esophageal and rectal veins

CLINICAL SIGNIFICANCE

Venous hyperemia creates favorable conditions for thrombosis (part of Virchow's triad — slowing of blood flow). That is why measures to prevent deep vein thrombosis and pulmonary embolism are taken in long-term bedridden patients, patients with varicose veins of the legs and postoperative patients: early mobilization, elastic bandaging, anticoagulants.

5. Comparative characteristics of arterial and venous hyperemia

Table 4.8. *Differences between arterial and venous hyperemia*

Feature	Arterial hyperemia	Venous hyperemia
Main mechanism	Increased inflow of blood (dilatation of arterioles)	Impeded outflow of blood
Color	Scarlet-red	Bluish (cyanosis)
Local temperature	Increased	Decreased
Velocity of blood flow	Increased	Slowed, pendulum-like movement, stasis
Working capillaries	Increased in number	Dilated, filled with blood
Arteriovenous oxygen difference	Decreased	Increased
Edema	Slight	Marked
Oxygen supply of the tissue	Improved	Hypoxia
Main outcome	Mostly beneficial; risk of vessel rupture	Hypoxia, degeneration, sclerosis, thrombosis

MEMORY AID

Arterial — “HOT & RED”: red, warm, many open (working) capillaries, pulsation. **Venous — “BLUE & COLD”**: bluish (cyanotic), cold, slow flow, swelling (edema).

6. Experimental part

Experiment 1: neuroparalytic arterial hyperemia in the rabbit ear (C. Bernard's experiment)

Aim: to observe the development of arterial hyperemia when the sympathetic vasoconstrictor influence is lost.

Equipment: a white (albino) rabbit, surgical instruments, a local anesthetic, a thermometer (electric thermometer), a light source.

1. Fix the rabbit on a special table; hold both ears against the light and compare the vascular network, color and temperature.
2. Under local anesthesia, incise the skin and tissues on one side of the neck, find the cervical sympathetic nerve lying next to the carotid artery and transect it.
3. After 5–10 minutes, compare both ears: in the ear on the side of transection, the vessels dilate, the ear reddens, more small vessels become visible and the temperature rises.
4. Record the results in the protocol.

Analysis: when the cervical sympathetic nerve is transected, the constant vasoconstrictor impulses to the vessels of the ear stop, the tone of arteriolar smooth muscle decreases and the arterioles dilate — **neuroparalytic arterial hyperemia** develops.

> **CAUTION**
> This experiment can also be performed with hot water or by rubbing the skin of the ear with xylene — in this case **neuromyoparalytic** (local) hyperemia is observed. The student must be able to name the type of hyperemia correctly according to the method used to induce it.

Experiment 2: venous hyperemia in the rabbit ear

1. Place a rubber ring or ligature at the base of the rabbit ear (where the vessels pass) so that it compresses only the veins and not the artery.
2. Observe for 10–15 minutes: the ear becomes bluish, the veins dilate and fill with blood, the temperature falls and the ear swells.
3. Remove the ring and observe the restoration of circulation.

Experiment 3: changes in the microcirculation in the frog mesentery

1. Under anesthesia (or after destruction of the spinal cord), fix the frog on a cork board, incise the abdominal wall from the side, bring out an intestinal loop, spread the mesentery over the opening and place it under the microscope.
2. In the normal state, observe blood flow in the arterioles, capillaries and venules: in the arterioles blood flows rapidly and pulsates, in the venules slowly and evenly; red blood cells are in the axis of the vessel, and there is a plasma layer next to the wall.
3. **Arterial hyperemia:** apply a drop of xylene (or warm physiological saline) to the mesentery — the arterioles dilate, previously invisible capillaries open and blood flow accelerates.
4. **Venous hyperemia:** tie a mesenteric vein with a ligature — the venules and capillaries dilate, blood flow slows, pendulum-like movement and stasis appear, and diapedesis of red blood cells is observed.
5. Draw your observations in the protocol.

Table 4.9. *Sample table for recording the results of the experiment*

Parameter	Normal	Arterial hyperemia	Venous hyperemia
Diameter of arterioles			
Number of working capillaries			
Velocity and character of blood flow			
State of venules			
Additional phenomena (stasis, diapedesis)			

7. Practical skills

Skill: identifying the type of hyperemia by its external signs

1. Assess the color of the tissue: scarlet-red or bluish.
2. Compare the local temperature with the symmetrical region using the back of the hand or a thermometer.
3. Check for edema (a pit remaining after pressing with a finger).
4. Detect pulsation of small arteries.
5. Determine the cause of hyperemia: whether the inflow of blood has increased or its outflow is impeded.
6. Write a conclusion: the type of hyperemia, its mechanism and probable outcome.

Skill: assessing the microcirculation under a microscope

1. Find the preparation at low magnification and distinguish the arteriole, capillary and venule (by the direction and velocity of flow).
2. Measure the diameter of the vessels with an ocular micrometer.
3. Count the number of working capillaries in the field of view.
4. Describe the character of blood flow: fast, slow, pendulum-like, stasis.
5. Note aggregation of red blood cells, diapedesis and the marginal position of leukocytes.
6. Compare with the normal state and write a conclusion.

Review questions

1. What are the components of the microcirculatory bed?
2. List the neurogenic, humoral and local factors regulating the microcirculation.
3. Why does a small change in the radius of an arteriole sharply change blood flow?
4. How are disorders of peripheral circulation classified?
5. Give examples of physiological and pathological arterial hyperemia.
6. How do the neurotonic and neuroparalytic mechanisms of arterial hyperemia differ?
7. Which substances take part in the neuromyoparalytic (metabolic) mechanism?
8. How do postischemic hyperemia and hyperemia ex vacuo develop?
9. Explain the external signs of arterial hyperemia and their causes.
10. What are the beneficial and harmful outcomes of arterial hyperemia?
11. List the causes of venous hyperemia.
12. Why does the tissue become bluish and its temperature fall in venous hyperemia?

13. What changes in the microcirculation are observed in venous hyperemia?
14. How do “nutmeg liver” and “brown induration of the lung” develop?
15. What type of hyperemia develops in C. Bernard's experiment and why?

Test questions

1. What is meant by arterial hyperemia?
   A) Blood filling of a tissue due to increased inflow of blood through the arteries
   B) Impeded outflow of blood through the veins
   C) Decreased inflow of blood through the arteries
   D) Cessation of blood flow in the capillaries
   E) Occlusion of the vascular lumen by a thrombus

2. What type of hyperemia develops in the ear when the cervical sympathetic nerve of a rabbit is transected?
   A) Neurotonic arterial hyperemia
   B) Venous hyperemia
   C) Neuroparalytic arterial hyperemia
   D) Hyperemia ex vacuo
   E) Humoral arterial hyperemia

3. What is the main mechanism of working (functional) hyperemia?
   A) Stimulation of vasodilator nerves
   B) Decreased vascular tone under the action of local metabolites (adenosine, CO_2, H^+, K^+)
   C) Compression of veins
   D) Transection of the sympathetic nerve
   E) Decreased barometric pressure

4. Which sign is NOT characteristic of arterial hyperemia?
   A) Scarlet-red color
   B) Increased local temperature
   C) Pulsation of small arteries
   D) Increased number of working capillaries
   E) Cyanosis

5. What is the cause of fainting when ascitic fluid is removed rapidly?
   A) Bleeding
   B) Cardiac arrest
   C) Venous hyperemia
   D) Hyperemia ex vacuo in the abdominal organs and relative ischemia of the brain
   E) Allergic reaction

6. What is the decrease in tissue temperature in venous hyperemia related to?
   A) Dilatation of the arterioles
   B) Decreased inflow of warm arterial blood and weakened metabolism
   C) Accelerated blood flow
   D) Increased lymph flow
   E) Increased number of working capillaries

7. Cyanosis in venous hyperemia is related to an increase in which substance?
   A) Reduced (deoxy) hemoglobin
   B) Oxyhemoglobin
   C) Bilirubin
   D) Melanin
   E) Carboxyhemoglobin

8. What change in the microcirculation is observed in venous hyperemia?
   A) Accelerated blood flow
   B) Dilatation of the arterioles
   C) Decreased arteriovenous oxygen difference
   D) Pendulum-like movement of blood and stasis
   E) Decreased lymph formation

9. In which condition does “nutmeg liver” develop?
   A) In left ventricular failure
   B) In arterial hyperemia
   C) In chronic venous congestion due to right ventricular failure
   D) In hepatic artery thrombosis
   E) In viral hepatitis

10. Brown induration of the lung is characteristic of which condition?
    A) Right ventricular failure
    B) Pulmonary embolism
    C) Pneumonia
    D) Bronchial asthma
    E) Mitral stenosis and left ventricular failure

11. What is the main harmful factor of venous hyperemia?
    A) Tissue hypoxia
    B) Overheating of the tissue
    C) Accelerated blood flow
    D) Spasm of the arterioles
    E) Decreased blood pressure

12. How does the arteriovenous oxygen difference change in arterial hyperemia?
    A) Increases
    B) It does not change
    C) It becomes zero
    D) It first increases and then does not change
    E) It decreases

13. How is vascular resistance related to the radius?
    A) Directly proportional to the radius
    B) Inversely proportional to the fourth power of the radius
    C) Directly proportional to the square of the radius
    D) Independent of the radius
    E) Directly proportional to the cube of the radius

14. What is the main mechanism of redness in an inflammatory focus?
    A) Neuroparalytic
    B) Ex vacuo
    C) Venous congestion
    D) Humoral (action of histamine, bradykinin, prostaglandins)
    E) Neurotonic

15. Which condition cannot be a cause of venous hyperemia?
    A) Compression of veins by the pregnant uterus
    B) Venous thrombosis
    C) Transection of the sympathetic nerve
    D) Right ventricular failure
    E) Venous valve insufficiency

Clinical cases

Case 1

In a 35-year-old woman, bleeding was stopped by a tourniquet applied to the right leg for a long time (1.5 hours). After the tourniquet was removed, the skin of the leg became sharply red and warm, and the pulsation of the vessels increased. At the same time, the patient's arterial pressure fell and dizziness appeared.

Questions: What type of hyperemia developed in the leg? What is its mechanism? Why did arterial pressure fall?

Answer: After removal of the tourniquet, postischemic (reactive) arterial hyperemia developed. During ischemia, metabolites (adenosine, lactate, H^+, K^+, CO_2) accumulated in the tissue and the tone of arteriolar smooth muscle decreased — the neuromyoparalytic (metabolic) mechanism. When blood flow was restored, a large volume of blood passed into the dilated vessels. Because a considerable part of the blood collected in the vessels of the leg, the circulating blood volume decreased relatively, arterial pressure fell and the blood supply of the brain deteriorated. That is why a tourniquet is released gradually and the patient's hemodynamics are monitored.

Case 2

A 62-year-old patient has had mitral stenosis for several years. He complains of shortness of breath, cough and sometimes blood-streaked sputum. Examination of the sputum revealed hemosiderin-laden macrophages (“heart failure cells”).

Questions: What disorder has developed in the patient, and in which circulation? How do hemosiderin-laden macrophages appear? What is the outcome of this condition?

Answer: In mitral stenosis, the passage of blood from the left atrium into the left ventricle is impeded, and as a result venous hyperemia (congestion) develops in the pulmonary circulation. Increased pressure in the pulmonary capillaries leads to the exit of fluid and red blood cells into the alveoli (diapedesis) — that is why there is blood in the sputum. The red blood cells in the alveoli are phagocytosed by macrophages, and hemosiderin is formed from hemoglobin. Prolonged congestion and hypoxia lead to growth of connective tissue in the lungs — brown induration of the lung.

Case 3

A 45-year-old woman, a shop assistant, works standing all day. Toward evening, heaviness and swelling appear in her legs. Examination revealed dilated, tortuous veins on the calves, a bluish-

brown color of the skin in the ankle region and a trophic ulcer.

Questions: What circulatory disorder has developed? Explain the pathogenesis of the edema and the trophic ulcer. What dangerous complication can this condition cause?

Answer: As a result of venous valve insufficiency and gravity, chronic venous hyperemia has developed in the legs (varicose disease). Because the outflow of blood is impeded, hydrostatic pressure in the capillaries and venules rises — filtration predominates over reabsorption and edema appears. Prolonged congestion and edema lead to tissue hypoxia, diapedesis of red blood cells and accumulation of hemosiderin (brown pigmentation), impaired trophism of the skin and subcutaneous tissue, and a trophic ulcer. Slowing of blood flow increases the risk of deep vein thrombosis and pulmonary embolism.

Quick review

Table 4.10. *Key facts on Topic 4*

Question	Answer
Microcirculation	Movement of blood and lymph through vessels less than 100 µm in diameter and exchange of substances with the tissue
Arterial hyperemia	Blood filling of a tissue due to increased inflow of blood
Neurotonic mechanism	Increased tone of vasodilator nerves
Neuroparalytic mechanism	Loss of the sympathetic vasoconstrictor influence (Bernard's experiment)
Neuromyoparalytic mechanism	Decreased smooth muscle tone under the action of local metabolites
Postischemic hyperemia	Reactive hyperemia after ischemia has been eliminated
Hyperemia ex vacuo	When barometric pressure decreases (cups, rapid removal of ascites)
Signs of arterial hyperemia	Redness, increased temperature, pulsation, increased volume and turgor
Venous hyperemia	Blood filling of a tissue due to impeded outflow of blood
Signs of venous hyperemia	Cyanosis, decreased temperature, edema, increased volume
Pendulum-like movement	Back-and-forth oscillation of blood in venous congestion, the state preceding stasis
Diapedesis	Exit of red blood cells through the intact vessel wall
Nutmeg liver	Chronic venous congestion in right ventricular failure
Brown induration of the lung	Chronic congestion in the pulmonary circulation, hemosiderin and sclerosis
Main harmful factor	In venous hyperemia — hypoxia

TOPIC 5

Ischemia, stasis, thrombosis and embolism

Learning objectives

To explain to the student the causes, mechanisms of development, manifestations and outcomes of ischemia, stasis, thrombosis and embolism; to teach the significance of collateral circulation; and to teach the student to observe in an experiment local anemia in the frog tongue and the rabbit ear, thrombus formation caused by injury of the vessel wall, and fat embolism.

By the end of the session, the student should know:

- the causes (compression, obturation, angiospasm, redistribution), signs and outcomes of ischemia;
- the types of collateral circulation and the factors determining the outcome of ischemia;
- the mechanisms of reperfusion injury;
- the types and mechanism of development of stasis;
- the causes of thrombus formation (Virchow's triad), its stages, the types of thrombi and their outcomes;
- the types of embolism (by the nature of the embolus and the path of its movement) and their outcomes.

The student should be able to:

- to induce and observe ischemia in the frog tongue and the rabbit ear using epinephrine;
- to injure the vessel wall with a crystal of table salt and observe thrombus formation;
- to model fat embolism and analyze its outcomes.

1. Ischemia

Ischemia (Greek ischo — I hold back, haima — blood) is local anemia that develops as a result of decreased or stopped inflow of blood through the arteries into an organ or tissue.

Causes of ischemia

Table 5.1. *Types of ischemia by cause*

Type	Mechanism	Examples
Compression	External compression of an artery	Tumor, scar, tourniquet, edema, hematoma, tight bandage
Obturation	Occlusion of the arterial lumen from within	Thrombus, embolus, atherosclerotic plaque, endarteritis
Angiospastic	Prolonged spasm of an artery	Cold, stress, epinephrine, vasopressin; Raynaud syndrome; cocaine
Redistributive	Movement of a large amount of blood to another region	Cerebral ischemia during rapid removal of ascites; the “steal” syndrome
Hemodynamic	Sharp fall in arterial pressure	Shock, collapse, massive blood loss

Signs of ischemia

Table 5.2. *External signs of ischemia and their causes*

Feature	Cause
Pallor (loss of color)	Less blood in the capillaries, fewer working capillaries
Decreased local temperature	Decreased inflow of warm blood, weakened metabolism
Decreased volume and turgor	Decreased amount of blood and interstitial fluid
Weakened or absent arterial pulsation	Decreased inflow of blood into the artery
Pain, numbness, pins-and-needles sensation (paresthesia)	Hypoxia and stimulation of nerve endings by metabolites (H^+, K^+, lactate, bradykinin)
Decreased function	Hypoxia, energy deficiency

Changes in the microcirculation

- the arterioles constrict or the inflow of blood into them decreases;
- the number of working capillaries decreases and blood flow slows;
- hydrostatic pressure in the capillaries falls — fluid passes from the tissue into the blood (the tissue “dries”);
- lymph formation decreases;
- the arteriovenous oxygen difference increases (the tissue extracts the maximum oxygen from the incoming blood).

Factors determining the outcome of ischemia

The outcome of ischemia can range from complete recovery to **infarction** (ischemic necrosis of the tissue). The outcome depends on the following:

1. **The development of collateral circulation** — the most important factor.
2. **The rate of development of ischemia** — with slow narrowing the collaterals have time to develop; with sudden occlusion they do not.
3. **The duration of ischemia.**
4. **The sensitivity of the tissue to hypoxia** and the intensity of its metabolism.
5. **The state of the general circulation** (heart failure and hypotension aggravate the outcome).

Table 5.3. *Tolerance time of tissues to complete ischemia (approximate)*

Tissue	Time at which irreversible changes begin
Neurons of the cerebral cortex	4–6 minutes
Myocardium	20–30 minutes
Liver, kidney	1–2 hours
Skeletal muscles	4–6 hours
Skin, connective tissue, bone	More than several hours

REMEMBER

By their collateral circulation, organs are divided into three groups:

- **organs with absolutely sufficient collaterals** — many anastomoses (limbs, mesentery, chest wall): ischemia is slight when one artery is occluded;
- **organs with relatively sufficient (or relatively insufficient) collaterals** — anastomoses exist but open slowly (intestine, stomach, lungs, skin);
- **organs with absolutely insufficient collaterals** — "end" (terminal) arteries (myocardium, brain, kidney, spleen, retina): occlusion of an artery leads to infarction.

Cell injury in ischemia and reperfusion injury

The main mechanisms of cell injury in ischemia: cessation of oxidative phosphorylation → ATP deficiency → failure of the Na^+/K^+ pump → entry of sodium and water into the cell (cellular swelling) → anaerobic glycolysis and lactic acidosis → accumulation of Ca^{2+} inside the cell → activation of phospholipases and proteases → membrane injury → necrosis.

Reperfusion injury is a paradoxical increase in injury when blood flow is restored after ischemia. Its mechanisms: formation of large amounts of reactive oxygen species (free radicals), massive entry of Ca^{2+} into cells, activation of leukocytes and plugging of microvessels (the no-reflow phenomenon), reperfusion arrhythmias.

CLINICAL SIGNIFICANCE

In myocardial infarction, "time is myocardium": the faster blood flow in the coronary artery is restored (thrombolysis or coronary angioplasty), the smaller the volume of necrosis. The same applies to ischemic stroke — thrombolysis is effective only in the first hours after the onset of symptoms (the "therapeutic window"). Along with reperfusion, measures are also taken to reduce its injurious effect.

2. Stasis

Stasis is the slowing and cessation of blood flow in the vessels of the microcirculation (mainly in the capillaries).

Table 5.4. *Types of stasis*

Type	Mechanism	Examples
Ischemic	Sharp decrease in the inflow of blood from the arteries	Arterial occlusion, spasm
Venous (congestive)	Impeded outflow of blood through the veins	The severe stage of venous hyperemia
True (capillary)	Changes in the blood itself and in the capillary wall: aggregation of red blood cells, hemoconcentration	Inflammation, burns, frostbite, poisoning, shock

The main mechanism of true stasis is **aggregation of red blood cells ("sludge")**: red blood cells stick together like "stacks of coins" and plug the capillaries. This is caused by a decrease in the negative surface charge of red blood cells, an increase in high-molecular-weight plasma proteins (fibrinogen, globulins), hemoconcentration (water loss) and increased permeability of the capillary wall.

If stasis is brief, it is a reversible process; if prolonged, it leads to tissue necrosis and thrombus formation.

3. Thrombosis

Thrombosis is the clotting of blood in a vessel or heart cavity in the living organism with the formation of a clot called a thrombus. Unlike a blood clot, a **thrombus** (Greek thrombos — clot) is attached to the vessel wall and forms under conditions of blood flow.

Causes of thrombus formation — Virchow's triad



Figure 5.1. *Virchow's triad (based on N.H. Abdullaev, 2008; V. Kumar et al., 2015)*

Table 5.5. *Virchow's triad*

Factor	Essence	Examples
Endothelial injury	Subendothelial collagen is exposed, tissue factor is released, and the antithrombotic properties of the endothelium (prostacyclin, NO, thrombomodulin) are lost	Atherosclerosis, vasculitis, trauma, intravascular catheter, homocysteinemia, smoking
Abnormal blood flow	Slowing and turbulence: platelets come close to the wall, activated factors are not diluted	Venous congestion, atrial fibrillation, aneurysm, prolonged bed rest, long flights
Hypercoagulability	Increased activity of the coagulation system or weakening of the anticoagulant system	Cancer, pregnancy and the postpartum period, oral contraceptives, hereditary thrombophilias (factor V Leiden mutation, antithrombin III deficiency), antiphospholipid syndrome

Stages of thrombus formation

1. **Platelet adhesion** — at the site of injury, platelets adhere to exposed collagen via von Willebrand factor.
2. **Platelet activation and aggregation** — platelets change shape and release ADP, thromboxane A_2 and serotonin; they stick to one another via fibrinogen bridges (the white head of the thrombus).
3. **Activation of the coagulation cascade** — thrombin is generated via tissue factor (the extrinsic pathway) and factor XII (the intrinsic pathway).
4. **Conversion of fibrinogen to fibrin** — a fibrin network forms, in which red blood cells and leukocytes become trapped.
5. **Clot retraction** — the thrombus becomes denser owing to the contractile proteins of platelets.

Types of thrombi

Table 5.6. *Types of thrombi*

Type	Composition and conditions of formation	Location
White (agglutination)	Mainly platelets and fibrin; forms slowly in fast flow	Arteries, heart valves

Type	Composition and conditions of formation	Location
Red (coagulation)	A fibrin network and many red blood cells; forms rapidly in slow flow	Veins
Mixed (laminated)	White head, mixed body, red tail	Veins, heart cavities, aneurysms
Hyaline	In microvessels; fibrin and fragments of red blood cells and platelets	Capillaries (DIC syndrome, shock)

By whether they completely occlude the vascular lumen, **occlusive** and **mural (parietal)** thrombi are distinguished.

Outcomes of a thrombus

- **Aseptic autolysis** (dissolution) — the thrombus dissolves under the action of the fibrinolytic system (plasmin), and blood flow is restored.
- **Organization** — connective tissue grows into the thrombus.
- **Canalization and vascularization** — new channels form within the organized thrombus, and blood flow is partly restored.
- **Petrification** — calcium salts are deposited in the thrombus ("phleboliths" in veins).
- **Septic (purulent) dissolution** — the thrombus becomes infected; risk of septic embolism and sepsis.
- **Detachment of the thrombus** — thromboembolism.

> **CLINICAL SIGNIFICANCE**
> Thrombus formation is a pathological manifestation of a protective mechanism (hemostasis). Arterial thrombosis leads to ischemia and infarction (myocardial infarction, ischemic stroke), whereas venous thrombosis leads to venous congestion and thromboembolism. Prevention: antiplatelet agents (acetylsalicylic acid (aspirin) — in arterial thrombosis), anticoagulants (heparin, warfarin — in venous thrombosis and atrial fibrillation).

4. Embolism

Embolism (Greek emballo — I throw in) is occlusion of a vascular lumen by particles not normally found in the blood or lymph flow (**emboli**) that are carried along by the flow.

Types of embolism by the nature of the emboli

Table 5.7. *Types of embolism by the nature of the embolus*

Type	Embolus	Conditions of origin
Thromboembolism (the most common)	A detached thrombus or part of it	Deep vein thrombosis of the legs, thrombi in the heart
Fat embolism	Fat droplets	Fractures of long bones, crushing of adipose tissue, accidental intravenous injection of oily solutions
Air embolism	Air bubbles	Injury of the neck veins (negative pressure in the veins), violation of infusion technique, surgery
Gas embolism	Nitrogen bubbles	Decompression sickness, loss of sealing at altitude
Tissue (cellular) embolism	Tumor cells, fragments of tissue	Tumor metastases, trauma

Type	Embolus	Conditions of origin
Amniotic fluid embolism	Elements of the amniotic fluid	During labor; severe shock and DIC syndrome
Microbial (septic) embolism	Clusters of bacteria, an infected thrombus	Sepsis, septic endocarditis — metastatic abscesses
Foreign body embolism	A bullet, a fragment of a catheter	Injuries, medical procedures

Types of embolism by the path of movement of the emboli

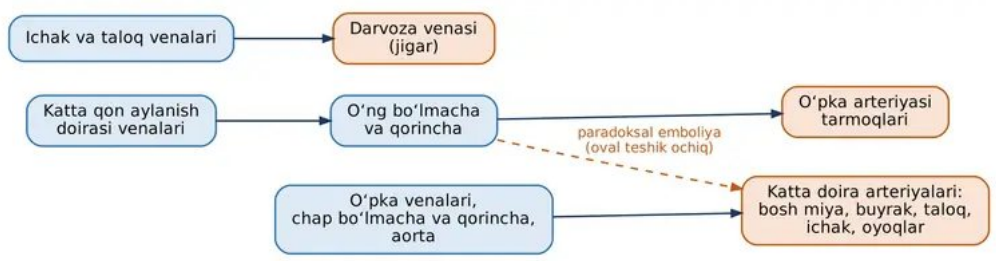


Figure 5.2. *Paths of movement of emboli (based on N.H. Abdullaev, 2008)*

Table 5.8. *Types of embolism by the path of movement*

Type	Path
Embolism of the pulmonary circulation	Systemic veins → right heart → branches of the pulmonary artery
Embolism of the systemic circulation	Pulmonary veins, left heart (thrombi on the valves and in the atrial appendage), aorta → systemic arteries (brain, kidneys, spleen, intestine, legs)
Portal vein embolism	Intestinal and splenic veins → portal vein → liver
Paradoxical embolism	Systemic veins → right heart → left heart through a septal defect between the atria or ventricles → systemic arteries
Retrograde embolism	The embolus moves against the blood flow because of its weight (for example, a heavy embolus in the inferior vena cava descends)

Pulmonary embolism

Pulmonary embolism (PE) is one of the most dangerous types of embolism. The main source of emboli (about 90%) is deep vein thrombosis of the legs and the small pelvis.

Pathogenesis: the embolus occludes the pulmonary artery → pulmonary vascular resistance increases and the right ventricle is overloaded (acute cor pulmonale) → the return of blood to the left heart decreases, and cardiac output and arterial pressure fall → the ventilation-perfusion relationship is disturbed and hypoxemia develops. In addition, serotonin and thromboxane released from platelets cause reflex spasm of the bronchi and pulmonary vessels and spasm of the coronary vessels (the pulmonocoronary reflex).

CAUTION

Massive PE often develops suddenly and unexpectedly: in the days after surgery, when the patient gets up for the first time or strains in the toilet, sudden shortness of breath, chest pain, cyanosis and fainting are observed. Do not forget thromboprophylaxis in patients with risk factors (surgery, prolonged bed rest, cancer, pregnancy).

Fat embolism usually develops 24–72 hours after a fracture of a long bone (the femur): shortness of breath and hypoxemia (occlusion of the pulmonary capillaries), neurological disorders (brain), and a petechial rash on the skin of the chest and axillae.

In **air embolism**, entry of 10–20 mL of air into the blood is usually harmless, but a large volume entering rapidly (more than 100 mL) may form an “air lock” in the right ventricle and lead to cardiac arrest.

5. Experimental part

Experiment 1: ischemia in the frog tongue and the rabbit ear

1. Fix the anesthetized frog on a cork board with its belly down, pull out the tongue, spread it over the opening and observe the vessels under the microscope.
2. Apply 1–2 drops of epinephrine solution (1:1000) to the tongue.
3. Observe the constriction of the arterioles, the decrease in the number of working capillaries, the slowing of blood flow and the blanching of the tongue.
4. In the rabbit ear: press the ear artery with a finger or inject a small amount of epinephrine under the skin — the ear blanches, the vascular network becomes invisible and the temperature falls.
5. After the pressure is removed or the action of epinephrine ends, observe the development of postischemic hyperemia.

Analysis: epinephrine constricts the arterioles via alpha-adrenergic receptors — **angiospastic ischemia** develops. The reddening after ischemia has been eliminated is postischemic arterial hyperemia caused by metabolites.

Experiment 2: observing thrombus formation

1. Place the frog mesentery under the microscope and find a larger vessel (a venule or arteriole).
2. Place a crystal of table salt (sodium chloride) on the vessel wall.
3. Observe the injury of the vessel wall at the site of the crystal, the adhesion and accumulation of platelets (in the frog — cells that perform the function of platelets) on the wall, and then the growth of the thrombus and the narrowing of the blood flow.
4. Record the time of thrombus formation and the degree to which it has occluded the vascular lumen.

Analysis: the hypertonic solution injures the endothelium (the first factor of Virchow's triad), as a result of which platelet adhesion and aggregation begin.

Experiment 3: fat embolism

1. Fix the frog on the board and place the mesentery under the microscope.
2. Inject 0.2–0.5 mL of warm vegetable oil into the abdominal vein of the frog (v. abdominalis), taking care not to inject into the lymph sac.
3. After a few minutes, observe the movement of fat droplets in the mesenteric vessels, their becoming lodged in small vessels and the cessation of blood flow along that vessel.
4. Observe the general state of the frog (breathing, movement).

CAUTION

In the frog, fat injected into the abdominal vein also enters the portal system and the liver; in humans, fat entering through a vein is first trapped in the pulmonary capillaries. This difference shows once again that interspecies differences must be taken into account when transferring the results of animal experiments to humans.

6. Practical skills

Skill: assessing the signs of ischemia

1. Compare the color of the limb on the affected and healthy sides (pallor, mottling).
2. Compare skin temperature at symmetrical points with the back of the hand.
3. Check the pulse in the peripheral arteries (femoral, popliteal, foot arteries).
4. Assess capillary refill: press the nail plate for 5 seconds and release it — if the color returns more slowly than in 2 seconds, perfusion is reduced.
5. Detect pain and impaired sensation and movement.
6. Determine the type of ischemia (compression, obturation, angiospasm) and its probable cause.

Skill: assessing the risk of thrombosis and embolism

1. Examine all three factors of Virchow's triad separately in the patient.
2. Identify factors of endothelial injury: trauma, surgery, catheter, atherosclerosis.
3. Identify factors of slowed blood flow: prolonged bed rest, varicose veins, heart failure, arrhythmia.
4. Identify factors of hypercoagulability: cancer, pregnancy, hormonal contraceptives, a family history of thrombosis.
5. Determine the source of the embolus and its probable path (pulmonary or systemic circulation).
6. Write a conclusion and indicate the direction of prevention.

Review questions

1. What is ischemia? What types of ischemia are there?
2. Explain the external signs of ischemia and the reasons for their appearance.
3. What changes in the microcirculation are observed in ischemia?
4. On what factors does the outcome of ischemia depend?
5. Into what groups are organs divided according to their collateral circulation?
6. What are the main mechanisms of cell injury in ischemia?
7. What is reperfusion injury and how does it develop?
8. Explain the types of stasis and the mechanism of true stasis.
9. How does a thrombus differ from a blood clot?
10. Explain Virchow's triad and give an example of each factor.
11. What stages does thrombus formation consist of?
12. Where and under what conditions do white, red and mixed thrombi form?
13. List the outcomes of a thrombus.
14. List the types of embolism by the nature of the embolus and by the path of movement.
15. Explain the pathogenesis of pulmonary embolism.

Test questions

1. Ischemia caused by an atherosclerotic plaque belongs to which type?
   A) Compression
   B) Angiospastic
   C) Redistributive
   D) Obturation
   E) Hemodynamic

2. Which sign is characteristic of ischemia?
   A) Cyanosis and increased local temperature
   B) Pallor and decreased local temperature
   C) Redness and increased pulsation
   D) Edema and increased tissue volume
   E) Increased lymph formation

3. The arteries of which organ are considered “end” arteries (with absolutely insufficient collaterals)?
   A) Limbs
   B) Mesentery
   C) Chest wall
   D) Skin
   E) Myocardium

4. Which tissue is the most sensitive to complete ischemia?
   A) Neurons of the cerebral cortex
   B) Skeletal muscles
   C) Skin
   D) Bone
   E) Connective tissue

5. What is the main mechanism of reperfusion injury?
   A) A further decrease in blood flow
   B) Closure of collaterals
   C) Formation of large amounts of reactive oxygen species and massive entry of Ca^{2+} into cells
   D) Decreased hemoglobin
   E) Excessive accumulation of glucose

6. What is the main mechanism of true (capillary) stasis?
   A) Occlusion of an artery by a thrombus
   B) Compression of a vein
   C) Spasm of the arterioles
   D) Increased arterial pressure
   E) Aggregation of red blood cells (“sludge”)

7. Which factor is NOT part of Virchow's triad?
   A) Endothelial injury
   B) Slowing of blood flow
   C) Thrombocytopenia
   D) Hypercoagulability
   E) Turbulent blood flow

8. What kind of thrombus mainly forms in arteries?
   A) White (agglutination)
   B) Red (coagulation)
   C) Hyaline
   D) Only mixed
   E) No thrombus forms

9. Which factor plays the main role in the adhesion of platelets to an injured vessel wall?
   A) Heparin
   B) Von Willebrand factor
   C) Prostacyclin
   D) Plasmin
   E) Antithrombin III

10. What is the dissolution of a thrombus under the action of plasmin called?
    A) Organization
    B) Petrification
    C) Septic dissolution
    D) Aseptic autolysis (fibrinolysis)
    E) Canalization

11. What is the main source of the embolus in pulmonary embolism?
    A) The left ventricle
    B) The deep veins of the legs and small pelvis
    C) The aorta
    D) The portal vein
    E) The pulmonary veins

12. If a thrombus in the left atrium detaches, where can the embolus lodge?
    A) In the pulmonary artery
    B) In the portal vein
    C) In the inferior vena cava
    D) In the cerebral arteries
    E) In the right ventricle

13. In which situation does paradoxical embolism develop?
    A) When there is a septal defect in the heart and an embolus passes from the systemic veins into the systemic arteries
    B) When an embolus moves against the blood flow
    C) When an embolus lodges in the portal vein
    D) When an embolus moves only in the lymphatic vessels
    E) When air bubbles enter the pulmonary artery

14. Two days after a fracture of the femur, a patient developed shortness of breath, confusion and petechiae on the skin of the chest. What kind of embolism has developed?
    A) Air embolism
    B) Thromboembolism
    C) Fat embolism
    D) Gas embolism
    E) Microbial embolism

15. What is the most important factor determining the outcome of ischemia?
    A) The patient's sex
    B) Ambient humidity
    C) Blood glucose level
    D) The patient's height
    E) The development of collateral circulation

Clinical cases

Case 1

A 58-year-old man has smoked for 20 years and has diabetes mellitus. He complains of pain in the left calf that appears when walking and disappears when he stops (“intermittent claudication”). On examination, the left foot is pale and cold, and no pulse is detected in the foot arteries. Ultrasound revealed atherosclerotic narrowing of the femoral artery.

Questions: What circulatory disorder has developed? Why does the pain appear only when walking? What factors are important in the development of the disease?

Answer: Obturative ischemia has developed because of an atherosclerotic plaque. At rest, the blood arriving through the narrowed artery and the collaterals covers the needs of the muscle, but during walking the oxygen demand of the muscle increases while the inflow of blood does not — relative ischemia; products of anaerobic metabolism (lactate, H^+, K^+) accumulate, stimulate the nerve endings and cause pain. When the patient stops, demand decreases and the pain passes. Smoking (endothelial injury, vascular spasm) and diabetes mellitus (endothelial dysfunction, accelerated atherosclerosis) are important as etiologic factors and conditions. Because the disease developed slowly, the collaterals have had time to develop partially — that is why there is no necrosis at rest.

Case 2

A 67-year-old woman underwent surgery for a fracture of the femoral neck. On day 6, when she got up for the first time, she suddenly developed severe shortness of breath and chest pain, her lips turned blue, and arterial pressure fell to 80/50 mm Hg. The left calf is swollen and painful on palpation.

Questions: What complication has developed? What is the source of the embolus? Which factors of Virchow's triad took part in thrombus formation?

Answer: Pulmonary embolism has developed. The source of the embolus is a thrombus in the deep veins of the left leg (swelling and pain of the calf). All three factors of Virchow's triad took part: endothelial injury (trauma and surgery), slowing of blood flow (prolonged bed rest, old age) and hypercoagulability (the state after surgery and trauma). When the patient got up and her muscles contracted, the thrombus detached and was carried with venous blood through the right heart into the pulmonary artery. Occlusion of the pulmonary artery led to right ventricular strain, decreased cardiac output and arterial hypotension, and the ventilation-perfusion mismatch led to hypoxemia and cyanosis.

Case 3

A 70-year-old patient has atrial fibrillation (permanent form) and does not take anticoagulants. Paralysis of the right arm and leg and a speech disorder suddenly appeared.

Questions: What condition has developed? Where did the embolus come from and by what path? Could this condition have been prevented?

Answer: Ischemic stroke has developed as a result of thromboembolism of a cerebral artery (the territory of the left middle cerebral artery). In atrial fibrillation the left atrium does not contract effectively; blood flow slows and becomes turbulent, especially in the left atrial appendage, and a thrombus forms (the abnormal blood flow factor of Virchow's triad). The detached thrombus passed through the left ventricle and aorta into the arteries of the systemic circulation — into a cerebral artery. Because the cerebral arteries are “end” arteries with insufficient collaterals, ischemic necrosis developed. In atrial fibrillation, anticoagulant therapy significantly reduces the risk of such embolism.

Quick review

Table 5.9. *Key facts on Topic 5*

Question	Answer
Ischemia	Decreased or stopped inflow of blood through an artery — local anemia
Types of ischemia	Compression, obturation, angiospastic, redistributive, hemodynamic
Signs of ischemia	Pallor, decreased temperature, decreased volume, weakened pulse, pain, paresthesia
Collateral circulation	The most important factor determining the outcome of ischemia
“End” arteries	Myocardium, brain, kidney, spleen, retina
Infarction	Ischemic necrosis of the tissue
Reperfusion injury	Increased injury due to free radicals and Ca^{2+} when blood flow is restored
Stasis	Cessation of blood flow in the capillaries: ischemic, venous, true
“Sludge”	Aggregation of red blood cells — the basis of true stasis
Thrombosis	Formation in the living organism of a clot attached to the wall inside a vessel
Virchow's triad	Endothelial injury, abnormal blood flow, hypercoagulability
White thrombus	Platelets and fibrin; in arteries
Red thrombus	Fibrin and red blood cells; in veins
Embolism	Occlusion of a vascular lumen by foreign particles carried in the bloodstream
Paradoxical embolism	Passage of a venous embolus into an artery through a septal defect in the heart
Source of PE	Deep vein thrombosis of the legs and small pelvis

TOPIC 6

Pathophysiology of inflammation. Alteration and vascular reactions

Learning objectives

To explain to the student the essence and biological significance of inflammation, its etiology, components and stages; to teach alteration, inflammatory mediators, metabolism in the inflammatory focus and the changes in vascular reactions; and to teach the student to observe the stages of local circulation in inflammation in the frog mesentery using Cohnheim's experiment.

By the end of the session, the student should know:

- the definition, biological significance and classical signs of inflammation;
- the etiology and classification of inflammation;
- the components of inflammation: alteration, exudation (with vascular reactions), proliferation;
- the mechanisms of primary and secondary alteration;
- the origin and action of inflammatory mediators;
- the metabolic and physicochemical changes in the inflammatory focus;
- the stages of vascular reactions in inflammation.

The student should be able to:

- to perform Cohnheim's experiment and identify the stages of vascular reactions in the inflammatory focus under the microscope;
- to explain the external signs of inflammation in relation to their mechanisms;
- to group inflammatory mediators by origin and action.

1. The concept of inflammation and its biological significance

Inflammation (Latin inflammatio) is a typical pathological process formed during evolution that manifests in response to injury by changes in local circulation and microcirculation in the tissue, exudation, emigration of leukocytes and proliferation of cells. Its goal is to delimit and eliminate the injurious factor and to clean and restore the injured tissue.

The protective significance of inflammation: delimiting the pathogenic factor and the injured tissue from the rest of the organism (forming a barrier), neutralizing and eliminating it, and creating conditions for regeneration. I.I. Mechnikov regarded inflammation as a protective reaction based on phagocytosis.

The harmful side of inflammation: it may itself intensify injury (secondary alteration), impair the function of vital organs (laryngeal edema, cerebral edema, injury of the heart valves), produce scars, and underlie chronic diseases (rheumatoid arthritis, atherosclerosis).

> **REMEMBER**
> The classical signs of inflammation: the four signs of **Celsus** — **rubor** (redness), **tumor** (swelling), **calor** (heat), **dolor** (pain); the fifth sign added by **Galen** (or Virchow) — **functio laesa** (loss of function).

Table 6.1. *External signs of inflammation and their mechanisms*

Feature	Mechanism
Redness (rubor)	Arterial hyperemia, dilatation of arterioles and capillaries
Swelling (tumor)	Exudation, cellular infiltration and proliferation
Heat (calor)	Inflow of warm arterial blood, increased metabolism and heat production
Pain (dolor)	Stimulation of nerve endings by mediators (bradykinin, prostaglandin E_2, histamine) and by H^+ and K^+ ions; compression of tissues by edema
Loss of function (functio laesa)	Tissue injury, pain, edema and disturbed microcirculation

2. Etiology and classification of inflammation

Phlogogenic factors (Greek phlogosis — inflammation) are factors that cause inflammation:

- **exogenous**: biological (bacteria, viruses, fungi, parasites — the most common), physical (trauma, burns, frostbite, radiation), chemical (acids, alkalis, turpentine, croton oil);
- **endogenous**: foci of necrosis (infarction), hemorrhage, salt crystals (urate crystals in gout), immune complexes, tumor breakdown.

Table 6.2. *Classification of inflammation*

Basis of classification	Types
By course	Acute (from several days to 2–3 weeks), subacute, chronic (months, years)
By the predominant component	Alterative, exudative, proliferative (productive)
By the type of exudate	Serous, fibrinous, purulent, putrefactive, hemorrhagic, mixed
By reactivity	Normergic, hyperergic (allergic), hypoergic
By cause	Infectious (specific and nonspecific), noninfectious (aseptic)

Specific inflammation runs with morphological features characteristic of a particular causative agent (granulomas): tuberculosis, syphilis, leprosy, scleroma.

3. Components and pathogenesis of inflammation

Inflammation consists of three interrelated components: **alteration** (injury), **exudation** (disturbed microcirculation, exit of fluid and cells from the vessels) and **proliferation** (multiplication of cells and repair). These components begin in sequence but then proceed simultaneously.

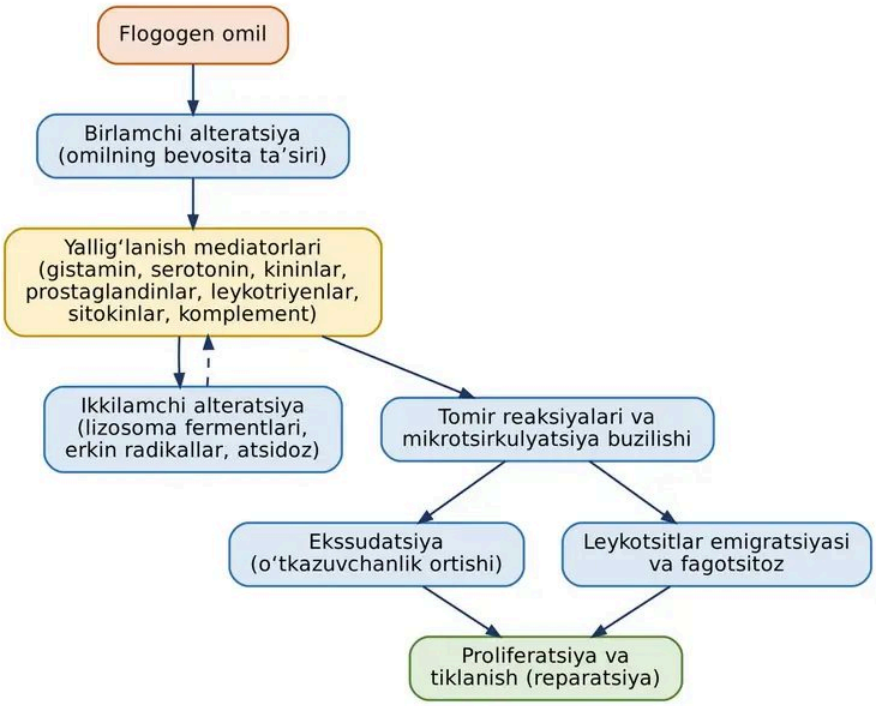


Figure 6.1. *General scheme of the pathogenesis of inflammation (based on V.V. Novitsky, 2009, Fig. 10.4)*

4. Alteration

Alteration (Latin alteratio — change) is injury of the structure, metabolism and function of a tissue. Alteration is the initial stage of inflammation.

Primary alteration is the direct injurious action of the phlogogenic factor on the tissue. Its extent depends on the strength, nature and duration of action of the factor.

Secondary alteration is the spread and deepening of injury under the action of factors formed as a result of primary alteration:

- release of hydrolytic enzymes (proteases, lipases, glycosidases) from the lysosomes of injured cells;
- release of reactive oxygen species and enzymes from activated leukocytes;
- hypoxia due to disturbed microcirculation;
- acidosis and hyperosmia;
- excessive formation of mediators.

> **CAUTION**
> Secondary alteration is often stronger than primary alteration and continues even after the phlogogenic factor has disappeared. For example, in acute pancreatitis the activated pancreatic enzymes themselves "digest" the gland tissue and its surroundings (autolysis). One of the goals of anti-inflammatory treatment is precisely to limit secondary alteration.

Disturbances of metabolism in the inflammatory focus

In the inflammatory focus, metabolism initially increases and oxygen consumption rises; then, as a result of disturbed microcirculation and mitochondrial injury, **anaerobic glycolysis** predominates.

Table 6.3. *Metabolic and physicochemical changes in the inflammatory focus*

Change	Essence and consequence
Carbohydrate metabolism	Glycolysis increases, lactate and pyruvate accumulate
Fat metabolism	Breakdown of fats, accumulation of fatty acids and ketone bodies
Protein metabolism	Proteolysis increases, polypeptides and amino acids accumulate
Acidosis	First compensated, then decompensated; pH falls to 6.5–5.5 (in a purulent focus); it intensifies pain and permeability
Hyperosmia	The number of particles increases because of the breakdown of molecules, osmotic pressure rises — water is drawn into the focus
Hyperonkia	Oncotic pressure rises owing to proteins leaving the vessels and to protein breakdown
Hyperionia	K^+ leaves injured cells, and the ratio of K^+ to H^+ and Ca^{2+} in the tissue increases

5. Inflammatory mediators

Inflammatory mediators are biologically active substances that regulate the development of inflammation and determine its manifestations. They are divided into **cell-derived** and **plasma-derived** mediators.

Table 6.4. *Cell-derived mediators*

Mediator	Source	Main action
Histamine	Mast cells, basophils, platelets	Dilates arterioles, rapidly increases venular permeability (early phase), itching
Serotonin	Platelets, enterochromaffin cells	Increases permeability, changes vascular tone
Prostaglandins (PGE_2, PGI_2)	Many cells (from arachidonic acid by the cyclooxygenase pathway)	Dilate vessels, intensify pain and fever
Leukotrienes (LTB_4, LTC_4, LTD_4)	Leukocytes, mast cells (lipoxygenase pathway)	LTB_4 — strong chemotaxis; LTC_4 and LTD_4 — increased permeability, bronchospasm
Platelet-activating factor	Leukocytes, endothelium, mast cells	Platelet aggregation, increased permeability, bronchospasm
Cytokines (IL-1, TNF-alpha, IL-6)	Macrophages, lymphocytes, endothelium	Activate the endothelium (adhesion molecules), fever, acute-phase proteins
Chemokines (IL-8)	Macrophages, endothelium	Chemotaxis of neutrophils
Nitric oxide (NO)	Endothelium, macrophages	Dilates vessels, kills microbes
Lysosomal enzymes, reactive oxygen species	Neutrophils, macrophages	Kill microbes but also injure the tissue

Table 6.5. *Plasma-derived mediators*

System	Mediators	Main action
Kinin system	Bradykinin, kallidin	Dilate vessels, increase permeability, cause severe pain

System	Mediators	Main action
Complement system	C3a, C5a (anaphylatoxins), C3b, the membrane attack complex	C5a — strong chemotaxis; C3a and C5a — release histamine from mast cells; C3b — opsonization
Coagulation and fibrinolytic system	Fibrinopeptides, fibrin degradation products, factor XIIa	Increase permeability, chemotaxis, activate the kinin system

MEMORY AID

To remember the main actions of mediators: **vasodilators** — “HPN” (Histamine, Prostaglandins, NO); **increase permeability** — “HBL” (Histamine, Bradykinin, Leukotrienes C_4/D_4); **chemotaxis** — “C5a, LTB_4, IL-8”; **pain** — “BP” (Bradykinin, Prostaglandin E_2); **fever** — “IL-1, TNF, IL-6, PGE_2”.

CLINICAL SIGNIFICANCE

Nonsteroidal anti-inflammatory drugs (ibuprofen, diclofenac, acetylsalicylic acid (aspirin)) block cyclooxygenase and reduce the synthesis of prostaglandins — that is why they weaken pain, fever and inflammation. Glucocorticoids inhibit phospholipase A_2 (via lipocortin) and the synthesis of cytokines, that is, they reduce the formation of both prostaglandins and leukotrienes.

6. Vascular reactions in inflammation

In the inflammatory focus, local circulation changes in a certain sequence. These changes were first described by J. Cohnheim (1867) in the frog mesentery.

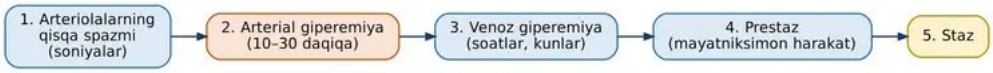


Figure 6.2. *Stages of vascular reactions in inflammation (according to Cohnheim)*

Table 6.6. *Stages of vascular reactions in inflammation*

Stage	Mechanism	Significance
Brief spasm of arterioles	Reflex stimulation of vasoconstrictor nerves, action of catecholamines; lasts a few seconds	Of little significance; sometimes absent
Arterial hyperemia	Dilatation of arterioles by neuroparalytic and metabolic mechanisms and under the action of mediators (histamine, bradykinin, prostaglandins, NO)	Redness and heat; the inflow of blood, leukocytes and oxygen into the focus increases
Venous hyperemia	Hemoconcentration due to exudation, aggregation of red blood cells, compression of venules by edema fluid, endothelial swelling, margination of leukocytes and microthrombi	Blood flow slows, conditions are created for the emigration of leukocytes
Prestasis	“Pendulum-like” movement of blood	The state preceding stasis
Stasis	Complete cessation of blood flow	The inflammatory focus is delimited; but if prolonged, tissue necrosis

REMEMBER

Arterial hyperemia in inflammation is short (from 10–30 minutes to several hours), whereas venous hyperemia lasts long and is the main vascular manifestation of inflammation. The main reasons for the transition to venous hyperemia are extravascular (pressure of the exudate), vascular wall (endothelial swelling, adhesion of leukocytes) and intravascular (hemoconcentration, aggregation, microthrombi) factors.

Exudation

Exudation is the exit of the liquid part of blood (plasma), proteins and blood cells from the vessels into the inflammatory focus. The fluid that leaves the vessels is called **exudate**.

Mechanisms of exudation:

1. **Increased permeability of the vessel wall** — the main mechanism. Under the action of mediators (histamine, bradykinin, leukotrienes), endothelial cells contract and gaps form between them (mainly in postcapillary venules); the endothelium is injured directly; activated leukocytes injure the endothelium.
2. **Increased hydrostatic pressure in capillaries and venules** — as a result of hyperemia.
3. **Increased osmotic and oncotic pressure in the tissue** — hyperosmia and hyperonkia draw fluid into the focus.

Table 6.7. *Differences between exudate and transudate*

Parameter	Exudate (in inflammation)	Transudate (in edema, without inflammation)
Protein content	More than 30 g/L	Less than 20 g/L
Specific gravity	Above 1.018	Below 1.015
Cells	Many (leukocytes, red blood cells)	Few
Rivalta test	Positive	Negative
pH	Decreased	Close to normal
Clarity	Turbid	Clear

Table 6.8. *Types of exudate*

Type	Composition	Examples
Serous	Protein 3–5%, few cells	Blisters in burns, serous pleurisy
Fibrinous	Much fibrinogen; fibrin threads form	Diphtheria, lobar pneumonia, fibrinous pericarditis
Purulent	Many neutrophils (pus corpuscles), disintegrated tissue	Abscess, phlegmon, empyema
Putrefactive	In anaerobic infection; foul-smelling	Gangrene, anaerobic infection
Hemorrhagic	Many red blood cells	Influenza pneumonia, anthrax, plague
Catarrhal (mucous)	On mucous membranes, much mucus	Rhinitis, bronchitis

7. Experimental part: Cohnheim's experiment

Aim: to observe under the microscope the stages of the changes in local circulation in the inflammatory focus in the frog mesentery.

Equipment: a frog, ether or urethane for anesthesia (or a needle for destroying the spinal cord), a cork board with an opening, pins, surgical instruments, a microscope, physiological saline.

Procedure

1. Under anesthesia (or after destruction of the spinal cord), fix the frog on its back on the board with an opening.
2. Make a 1–1.5 cm incision on the side of the abdominal wall and carefully pull out an intestinal loop.
3. Spread the mesentery over the opening in the board and fix it with pins (do not compress the vessels).
4. At low magnification, find the arterioles, capillaries and venules; describe normal blood flow (fast, axial flow, a plasma layer next to the wall).
5. Because the mesentery is in contact with air, inflammation gradually develops. Observe the changes every 5–10 minutes and record them in the protocol (1–1.5 hours in total).
6. Apply drops of physiological saline from time to time so that the mesentery does not dry.

Sequence of the changes observed

1. **Arterial hyperemia** (the first 10–20 minutes): arterioles and capillaries dilate, previously invisible capillaries open and blood flow accelerates.
2. **Venous hyperemia**: venules dilate, blood flow slows and the plasma layer disappears.
3. **Margination of leukocytes**: leukocytes roll along the venular wall and adhere to it.
4. **Emigration of leukocytes**: leukocytes pass through the venular wall into the surrounding tissue (in detail in the next topic).
5. **Prestasis and stasis**: in some capillaries blood moves in a pendulum-like manner and stops; diapedesis of red blood cells.

Table 6.9. *Recording the results of Cohnheim's experiment*

Time	Arterioles	Capillaries	Venules	Blood flow	Leukocytes
Beginning					
10 minutes					
30 minutes					
60 minutes					

CAUTION
For the experiment to succeed, the mesentery must not be stretched so as to compress the vessels, and it must not be allowed to dry. Otherwise venous hyperemia and stasis will appear not because of inflammation but as a result of mechanical compression or drying, and the experiment will be misinterpreted.

8. Practical skills

Skill: preparing the frog mesentery

1. Under anesthesia, fix the frog on its back on the board.
2. Cut the skin and muscles on the side of the abdominal wall without injuring large vessels.
3. Carefully bring out an intestinal loop with forceps.

4. Spread the mesentery over the opening of the board without stretching it and fix it with pins.
5. Keep it moist with physiological saline.
6. Place the preparation on the microscope stage and find the vessels at low magnification.

Skill: explaining the signs of inflammation in relation to their mechanisms

1. Identify the five classical signs of inflammation in the patient.
2. Write the main mechanism for each sign (redness — arterial hyperemia; swelling — exudation, etc.).
3. Determine the type of inflammation by the character of the exudate.
4. Distinguish local and systemic signs (fever, leukocytosis, increased ESR).
5. Write the conclusion in the form of a scheme of pathogenesis.

Review questions

1. What is inflammation? What is its biological significance?
2. Explain the classical signs of inflammation and their mechanisms.
3. Give examples of phlogogenic factors.
4. How is inflammation classified?
5. What are the components of inflammation?
6. What is the difference between primary and secondary alteration?
7. How does metabolism change in the inflammatory focus?
8. What is the significance of acidosis, hyperosmia and hyperonkia in inflammation?
9. List the cell-derived mediators and state their main actions.
10. Which systems belong to the plasma-derived mediators?
11. What is the mechanism of action of nonsteroidal anti-inflammatory drugs and glucocorticoids?
12. List the stages of vascular reactions in inflammation.
13. What are the reasons for the transition from arterial to venous hyperemia in inflammation?
14. Explain the mechanisms of exudation. How does exudate differ from transudate?
15. What sequence of changes is observed in Cohnheim's experiment?

Test questions

1. Which of the following is NOT a classical sign of inflammation?
   A) Redness
   B) Edema
   C) Cyanosis
   D) Pain
   E) Loss of function

2. What is the main cause of secondary alteration?
   A) The direct action of the phlogogenic factor
   B) Arterial hyperemia
   C) Proliferation
   D) Phagocytosis
   E) The action of lysosomal enzymes, reactive oxygen species and acidosis

3. What physicochemical change is observed in the inflammatory focus?
   A) Acidosis and hyperosmia
   B) Alkalosis and hypoosmia
   C) Hypoonkia
   D) Decrease in potassium in the tissue
   E) Rise in pH to 8

4. What is the main mediator that rapidly increases venular permeability and takes part in the early phase?
   A) Interleukin-1
   B) Leukotriene B_4
   C) Nitric oxide
   D) Histamine
   E) Fibrinopeptides

5. Which complement component is a strong chemoattractant for neutrophils?
   A) C3b
   B) C5a
   C) C1q
   D) C4
   E) C2

6. Which plasma-derived mediator causes severe pain?
   A) Bradykinin
   B) Serotonin
   C) Leukotriene B_4
   D) Interleukin-8
   E) Heparin

7. Which enzyme do nonsteroidal anti-inflammatory drugs block?
   A) Lipoxygenase
   B) Phospholipase A_2
   C) Monoamine oxidase
   D) Cyclooxygenase
   E) Kallikrein

8. What is the correct sequence of vascular reactions in inflammation in Cohnheim's experiment?
   A) Venous hyperemia → arterial hyperemia → stasis
   B) Brief spasm → arterial hyperemia → venous hyperemia → prestasis → stasis
   C) Stasis → arterial hyperemia → venous hyperemia
   D) Arterial hyperemia → stasis → venous hyperemia
   E) Ischemia → venous hyperemia → arterial hyperemia

9. Which factor does NOT cause the transition from arterial to venous hyperemia in inflammation?
   A) Compression of venules by the exudate
   B) Hemoconcentration and aggregation of red blood cells
   C) Adhesion of leukocytes to the venular wall
   D) Endothelial swelling
   E) Increased tone of arterioles

10. What is the main difference between exudate and transudate?
    A) Protein content less than 20 g/L
    B) Specific gravity below 1.010
    C) Protein content more than 30 g/L and many cells
    D) Negative Rivalta test
    E) Being clear

11. What kind of exudate forms in diphtheria and lobar pneumonia?
    A) Serous
    B) Purulent
    C) Hemorrhagic
    D) Putrefactive
    E) Fibrinous

12. Which are the main cytokines causing fever in inflammation?
    A) Histamine and serotonin
    B) IL-1, TNF-alpha, IL-6
    C) Bradykinin and kallidin
    D) C3a and C5a
    E) Heparin and plasmin

13. What is the main mechanism of exudation?
    A) Decreased hydrostatic pressure in capillaries
    B) Decreased oncotic pressure in the tissue
    C) Increased permeability of the vessel wall
    D) Increased lymph flow
    E) Constriction of arterioles

14. What is the main mechanism of the anti-inflammatory action of glucocorticoids?
    A) Inhibition of phospholipase A_2 and of cytokine synthesis
    B) Blockade of cyclooxygenase only
    C) Blockade of histamine receptors
    D) Killing bacteria
    E) Dilating vessels

15. With which process is redness (rubor) in the inflammatory focus associated?
    A) Ischemia
    B) Stasis
    C) Venous hyperemia
    D) Arterial hyperemia
    E) Thrombosis

Clinical cases

Case 1

A 25-year-old man got a thorn in his finger. Two days later the fingertip became red, swollen and warm, severe “throbbing” pain appeared, and bending the finger is difficult. Body temperature is 37.8 °C.

Questions: What pathological process has developed? Explain the mechanism of each local sign. What does the systemic sign indicate?

Answer: Acute inflammation of the finger (paronychia/felon) has developed; the phlogogenic factor is the microbes introduced with the thorn and the foreign body. Redness and heat — arterial hyperemia (dilatation of arterioles under the action of mediators, inflow of warm blood, increased metabolism). Swelling — exudation: increased permeability of the vessel wall, hyperosmia and hyperonkia. Pain — stimulation of nerve endings by bradykinin, prostaglandin E_2, H^+ and K^+ ions, and compression of the nerve endings by exudate in the dense tissue of the fingertip (hence “throbbing”). Loss of function — due to pain and swelling. Fever — a systemic reaction under the action of cytokines (IL-1, IL-6, TNF-alpha) released from the inflammatory focus; it indicates that the local process is affecting the whole organism.

Case 2

Students performed Cohnheim's experiment. After 40 minutes, one student saw that blood flow in the venules had slowed and leukocytes had accumulated along the wall. In the second student's preparation, stasis was observed in all vessels after only 10 minutes, and the stage of arterial hyperemia was not seen at all.

Questions: Which stage of inflammation did the first student observe? What does the second student's result indicate?

Answer: The first student observed the stage of venous hyperemia and the margination of leukocytes — the regular development of inflammation. In the second student's preparation, rapidly developing stasis and the absence of arterial hyperemia indicate a fault in the experimental technique: the mesentery was stretched and the vessels were compressed by the pins, or the mesentery dried. Such stasis developed not because of inflammation but as a result of a mechanical factor, so the preparation must be made again.

Case 3

Fluid was obtained by puncture of the pleural cavity in a 40-year-old patient. The fluid is turbid, protein content 42 g/L, specific gravity 1.024, many neutrophils, positive Rivalta test.

Questions: Is this an exudate or a transudate? Explain the mechanism of its formation. What type of exudate could it be?

Answer: This is an exudate: protein more than 30 g/L, specific gravity above 1.018, many cells, positive Rivalta test, turbid. It formed as a result of inflammation of the pleura (pleurisy). Mechanism: under the action of mediators (histamine, bradykinin, leukotrienes), venular permeability increases, hydrostatic pressure rises because of hyperemia, and hyperosmia and hyperonkia develop in the inflammatory focus — as a result, protein-rich fluid and leukocytes pass into the pleural cavity. Because there are many neutrophils, it may be a serous-purulent or purulent exudate (if it progresses — pleural empyema); this is characteristic of bacterial infection.

Quick review

Table 6.10. *Key facts on Topic 6*

Question	Answer
Inflammation	A typical process developing in response to injury with changes in the microcirculation, exudation, emigration and proliferation
Classical signs	Rubor, tumor, calor, dolor, functio laesa
Phlogogenic factors	Exogenous and endogenous factors causing inflammation

Question	Answer
Components	Alteration, exudation, proliferation
Primary alteration	The direct injurious action of the phlogogenic factor
Secondary alteration	Increased injury under the action of lysosomal enzymes, free radicals, acidosis and hypoxia
Physicochemical changes	Acidosis, hyperosmia, hyperonkia, hyperkalia (excess K^+ in the tissue)
Cell-derived mediators	Histamine, serotonin, prostaglandins, leukotrienes, cytokines, NO
Plasma-derived mediators	Kinins, complement, products of coagulation and fibrinolysis
Chemoattractants	C5a, LTB_4, IL-8, bacterial peptides
Vascular reactions	Spasm → arterial hyperemia → venous hyperemia → prestasis → stasis
Exudate	Protein more than 30 g/L, specific gravity above 1.018, cellular, Rivalta positive
Types of exudate	Serous, fibrinous, purulent, putrefactive, hemorrhagic, catarrhal
Cohnheim's experiment	Observation of the vascular changes in inflammation in the frog mesentery

TOPIC 7

Emigration phenomena in inflammation. Phagocytosis and chemotaxis. Proliferation and outcomes of inflammation

Learning objectives

To explain to the student the stages and molecular mechanisms of leukocyte emigration, the theories of emigration, and the significance of phagocytosis and chemotaxis in inflammation; to teach the role of inflammatory mediators in these processes, proliferation, and the systemic signs and outcomes of inflammation; and to teach the student to observe leukocyte emigration and the effect of physicochemical factors on it in an experiment and to assess phagocytic activity.

By the end of the session, the student should know:

- the stages of leukocyte emigration and the role of adhesion molecules;
- the sequence of emigration (Mechnikov's law) and its causes;
- the theories of emigration;
- chemotaxis and chemoattractants;
- the stages of phagocytosis and the mechanisms of killing microbes;
- the causes of insufficient phagocytosis;
- the mechanisms of proliferation and repair, and the systemic signs and outcomes of inflammation.

The student should be able to:

- to observe leukocyte emigration in the frog mesentery and describe its stages;
- to determine experimentally the effect of a change in ionic composition on emigration;
- to calculate and evaluate the phagocytic number and the phagocytic index.

1. Emigration of leukocytes

Emigration is the exit of leukocytes from a blood vessel into the inflammatory focus. Emigration takes place mainly in **postcapillary venules**, because here blood flow is slow, the endothelium has many adhesion molecules, and the junctions between endothelial cells are looser.



Figure 7.1. *Stages of leukocyte emigration (based on V.V. Novitsky, 2009, Fig. 10.11)*

Table 7.1. *Stages of leukocyte emigration*

Stage	Mechanism	Key molecules
Margination	When blood flow slows, red blood cells collect along the axis, and leukocytes are pushed to the periphery, next to the wall	Hemodynamic factor (venous hyperemia, stasis)
Rolling	The leukocyte forms loose bonds with the endothelial surface and rolls slowly	Selectins: P-selectin (rapidly, under the action of histamine), E-selectin (under the action of IL-1 and TNF), L-selectin (on the leukocyte)
Firm adhesion	Leukocyte integrins are activated and bind firmly to endothelial ligands	Integrins (LFA-1, Mac-1, VLA-4) and ICAM-1, VCAM-1

Stage	Mechanism	Key molecules
Diapedesis (transmigration)	The leukocyte extends pseudopodia, passes between endothelial cells and breaks down the basement membrane with enzymes	PECAM-1 (CD31), collagenases
Chemotaxis	Directed movement in the tissue toward the focus along the concentration gradient of chemoattractants	C5a, LTB_4, IL-8, bacterial N-formyl peptides

CLINICAL SIGNIFICANCE

Leukocyte adhesion deficiency is a rare disease associated with a hereditary defect of integrins (CD18). In such children leukocytes cannot leave the vessels: there are many neutrophils in the blood, but no pus is formed, the umbilical stump separates late, and recurrent severe bacterial infections are observed. This example clearly shows the protective significance of emigration.

Sequence of emigration

As I.I. Mechnikov established, in acute inflammation leukocytes leave the vessels in a certain order:

1. **Neutrophils** predominate in the first 6–24 hours (the most mobile, numerous and fast-responding).
2. **Monocytes (which turn into macrophages)** predominate after 24–48 hours.
3. **Lymphocytes** come later, especially in chronic and immune inflammation.

This sequence is not always maintained: in viral infections lymphocytes, and in allergic inflammation and helminthic infestations eosinophils, predominate early.

Theories of emigration

Table 7.2. *Theories of emigration*

Theory	Content
Mechanical (J. Cohnheim)	Leukocytes are passively “squeezed out” through the vessel wall by the pressure difference
Theory of chemotaxis (I.I. Mechnikov)	Leukocytes actively move toward chemical substances in the focus
Physicochemical (surface tension) theory	Changes in pH, ionic composition and surface tension in the focus cause the leukocyte membrane to extend pseudopodia on one side
Modern (molecular) view	Emigration is a regulated process with the active participation of adhesion molecules, chemokines and the leukocyte cytoskeleton

REMEMBER

The role of physicochemical factors in emigration: **acidosis** and **an increase in K^+ ions** in the inflammatory focus change the surface properties of the endothelium and leukocytes and intensify emigration; **Ca^{2+} ions**, in contrast, make the vessel wall denser and weaken permeability and emigration. This rule is tested in the experimental part.

2. Chemotaxis

Chemotaxis is the directed movement of cells along a concentration gradient of chemical substances. If leukocytes move toward the focus, it is **positive** chemotaxis; if they move away from it, **negative** chemotaxis.

Table 7.3. *Main chemoattractants*

Group	Substances
Exogenous	Bacterial N-formyl peptides, components of the bacterial cell wall (lipopolysaccharide)
Complement fragments	C5a (the strongest), C3a
Lipid mediators	Leukotriene B_4, platelet-activating factor
Chemokines	IL-8 (CXCL8) — for neutrophils; MCP-1 — for monocytes; eotaxin — for eosinophils
Other	Breakdown products of fibrin and collagen

The chemoattractant binds to a receptor on the leukocyte membrane → a signal is transmitted via a G protein → intracellular Ca^{2+} increases and actin polymerizes → the leukocyte extends pseudopodia and moves toward the focus.

3. Phagocytosis

Phagocytosis (Greek phagein — to devour, kytos — cell) is the active engulfment and digestion of foreign particles, microorganisms and dead cells by specialized cells (phagocytes). Phagocytosis was discovered by I.I. Mechnikov in 1883, who regarded it as the main protective mechanism of inflammation.

Phagocytes:

- **microphages** — neutrophil granulocytes (mainly phagocytose bacteria);
- **macrophages** — monocytes and tissue macrophages (histiocytes, Kupffer cells, alveolar macrophages, microglia, osteoclasts); they engulf large particles and dead cells and present antigen.

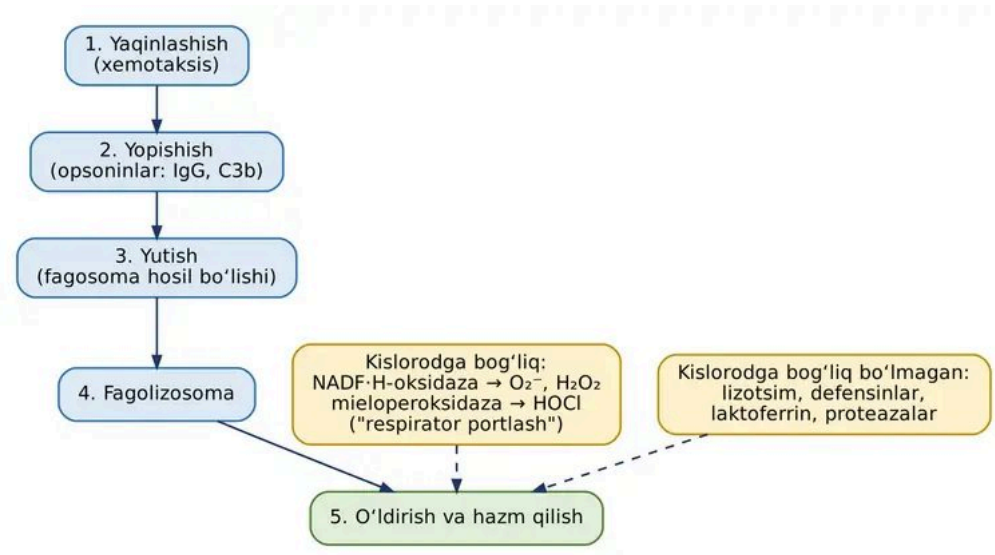


Figure 7.2. *Stages of phagocytosis and mechanisms of killing microbes*

Table 7.4. *Stages of phagocytosis*

Stage	Essence
Approach	The phagocyte moves toward the object by chemotaxis
Attachment (recognition)	Phagocyte receptors recognize the object; opsonins (IgG, the C3b fragment of complement, lectins) sharply accelerate this process
Engulfment	The phagocyte surrounds the object with pseudopodia, and a phagosome forms

Stage	Essence
Formation of the phagolysosome	The phagosome fuses with lysosomes
Killing and digestion	The microbe is killed by oxygen-dependent and oxygen-independent mechanisms and broken down by hydrolytic enzymes

Mechanisms of killing microbes

Oxygen-dependent mechanism ("respiratory burst"): oxygen consumption rises sharply during phagocytosis; NADPH oxidase forms the superoxide anion (O_2^-), from which hydrogen peroxide (H_2O_2) and the hydroxyl radical arise; myeloperoxidase forms a very powerful bactericidal substance, hypochlorous acid (HOCl), from H_2O_2 and chloride ion.

Oxygen-independent mechanism: lysozyme (breaks down the bacterial wall), defensins (form pores in the membrane), lactoferrin (binds iron and stops bacterial growth), cationic proteins, proteases, the acidic environment in the phagolysosome.

MEMORY AID

The myeloperoxidase system = "household bleach": $H_2O_2 + Cl^- \rightarrow HOCl$. The greenish-yellow color of pus is due precisely to myeloperoxidase (an enzyme with a green pigment).

Completeness of phagocytosis

- **Complete phagocytosis** — the engulfed microbe is killed and fully digested.
- **Incomplete phagocytosis** — the engulfed microbe does not die; it survives and even multiplies inside the phagocyte (Mycobacterium tuberculosis, gonococcus, Brucella, Leishmania). In this case the phagocyte may spread the microbe throughout the organism.

Insufficiency of phagocytosis

Table 7.5. *Causes of insufficient phagocytosis*

Impaired stage	Causes and examples
Decreased number of phagocytes	Neutropenia (cytostatics, radiation, agranulocytosis)
Impaired chemotaxis	Diabetes mellitus, Chediak–Higashi syndrome, glucocorticoids
Impaired opsonization	Hypogammaglobulinemia, complement deficiency (C3)
Impaired adhesion	Leukocyte adhesion deficiency (integrin defect)
Impaired killing	Chronic granulomatous disease (NADPH oxidase defect), myeloperoxidase deficiency

CLINICAL SIGNIFICANCE

In chronic granulomatous disease, neutrophils engulf the microbe but cannot kill it because there is no "respiratory burst". Patients develop recurrent abscesses and granulomas caused by catalase-positive microbes (staphylococcus, Aspergillus). The nitroblue tetrazolium (NBT) test is used for diagnosis.

4. Proliferation and repair

Proliferation is the multiplication of cells in the inflammatory focus; it is the final stage of inflammation and ensures the restoration of the injured tissue.

Proliferation mainly involves **local stem (cambial) cells** (fibroblasts, endothelium, epithelium) and monocytes arriving from the blood. Macrophages release growth factors: platelet-derived growth factor (PDGF), fibroblast growth factor (FGF), transforming growth factor beta (TGF-beta) and vascular endothelial growth factor (VEGF).

Granulation tissue is young connective tissue consisting of newly formed capillaries, fibroblasts and inflammatory cells. It fills the bottom of the wound, and later, as collagen fibers increase, it turns into a scar.

Table 7.6. *Types of repair*

Type	Essence	Condition
Complete regeneration (restitution)	The injured tissue is restored by the same kind of tissue	The injury is small and the tissue is capable of regeneration (epithelium, liver)
Incomplete regeneration (substitution)	The defect is filled with connective tissue — a scar	Extensive injury, tissue with low regenerative capacity (myocardium, neurons)

5. Systemic signs of inflammation

Although inflammation is a local process, it is accompanied by a response of the whole organism.

Table 7.7. *Systemic signs of inflammation*

Feature	Mechanism
Fever	IL-1, IL-6, TNF-alpha → PGE_2 in the hypothalamus → raised set point of the thermoregulatory center
Leukocytosis (neutrophilia, left shift)	Increased release and production of leukocytes in the bone marrow under the action of cytokines and colony-stimulating factors
Increased ESR (erythrocyte sedimentation rate)	An increase in blood fibrinogen and globulins enhances aggregation of red blood cells
Increase in acute-phase proteins	Under the action of IL-6, the liver increases the synthesis of C-reactive protein, fibrinogen, ceruloplasmin and haptoglobin
Change in the protein composition of blood	Alpha- and gamma-globulins increase, albumin decreases
General intoxication	Weakness, loss of appetite, drowsiness (the action of cytokines on the central nervous system)

6. Outcomes of inflammation and chronic inflammation

Outcomes of acute inflammation:

- **complete resolution** — the phlogogenic factor disappears, the exudate is absorbed and the tissue is restored;
- **scar formation (fibrosis)** — in extensive injury;
- **abscess formation** — a delimited collection of pus in purulent infection;
- **transition to chronic inflammation** — if the phlogogenic factor persists.

Chronic inflammation is long-lasting inflammation in which alteration, inflammation and repair proceed simultaneously. Macrophages, lymphocytes and plasma cells predominate in its cellular composition; fibrosis and angiogenesis are marked. Causes: microbes that are difficult to kill (Mycobacterium tuberculosis, fungi), long-acting exogenous substances (silica dust — silicosis), autoimmune reactions.

Granulomatous inflammation is a special type of chronic inflammation in which **granulomas** consisting of activated macrophages (epithelioid cells) and multinucleated giant cells are formed (tuberculosis, sarcoidosis, around foreign bodies).

7. Experimental part

Experiment 1: observing leukocyte emigration in the frog mesentery

1. Prepare the frog mesentery by Cohnheim's method (see Topic 6).
2. Find a venule under the microscope and observe it for 40–60 minutes.
3. Record the following stages in sequence: movement of leukocytes to the wall (margination), slow rolling along the wall, adhesion to the wall, passage through the wall by extending pseudopodia, and accumulation outside the vessel.
4. Count the number of leukocytes that have left the vessel in the field of view every 10 minutes.

Experiment 2: the effect of a change in ionic composition on emigration (physicochemical factors)

This experiment appears in the department plan as “Danilevsky's experiment” and is aimed at studying the effect of physicochemical shifts in the inflammatory focus on emigration. The preparation and concentrations are specified according to the department's methodological guidelines.

1. Prepare mesentery preparations from two frogs.
2. Keep applying drops of potassium chloride solution (or a weakly acidic buffer solution) to the first preparation.
3. Keep applying drops of calcium chloride solution to the second preparation.
4. Compare the intensity of emigration (the number of leukocytes that have left the vessels) in both preparations over the same period of time.

Expected results and analysis: K^+ ions and acidosis loosen the junctions between endothelial cells and increase the mobility of leukocytes — emigration increases. Ca^{2+} ions make the vessel wall denser and stabilize membranes — emigration weakens. This confirms that hyperkalia and acidosis in the inflammatory focus are factors that intensify emigration.

> **CAUTION**
> Old ideas about the “anti-inflammatory” and “antiallergic” action of calcium preparations were based on experiments like this one. In modern clinical practice calcium preparations are not considered effective for this purpose; they are used only in hypocalcemia.

Experiment 3: determining phagocytic activity

1. Place 0.2 mL of heparinized blood (or peritoneal exudate of an animal) and 0.1 mL of a microbial suspension (for example, killed staphylococci or yeast cells) in a test tube.
2. Keep it in a thermostat at 37 °C for 30 minutes, mixing from time to time.
3. Prepare a smear from the mixture, dry it, fix it and stain it by the Romanowsky–Giemsa method.
4. Under the oil immersion microscope, examine 100 neutrophils: count how many of them have engulfed microbes and how many microbes each contains.
5. Calculate the parameters.

Table 7.8. *Parameters of phagocytosis*

Parameter	Calculation	Norm (approximate)
Phagocytic index (phagocytic activity)	Percentage of neutrophils that have engulfed microbes	40–80 %
Phagocytic number	Average number of microbes per phagocyte	4–9
Index of completeness of phagocytosis	Ratio of killed microbes to engulfed microbes	Greater than 1 — complete

CAUTION

The reference values of phagocytosis parameters depend on the method used in the laboratory and on the test object (type of microbe, exposure time). Always compare the result with the control values of the same laboratory.

8. Practical skills

Skill: identifying the stages of emigration under the microscope

1. Distinguish a venule from an arteriole and a capillary by the direction of flow and the diameter.
2. Note the disappearance of the plasma layer and the movement of leukocytes to the wall.
3. Observe the slow rolling of a leukocyte along the wall (slower movement than red blood cells).
4. Count the leukocytes firmly attached to the wall and immobile.
5. Find and draw a leukocyte passing through the wall by extending pseudopodia.
6. Describe the movement of leukocytes outside the vessel toward the focus.

Skill: calculating the phagocytic index and phagocytic number

1. Examine 100 consecutive neutrophils in the stained smear.
2. Count the neutrophils that have engulfed at least one microbe — this number equals the phagocytic index (in percent).
3. Calculate the total number of microbes inside all neutrophils.
4. Divide the total number of microbes by the number of neutrophils that have engulfed microbes — the phagocytic number.
5. Compare the result with the norm and write whether phagocytic activity has decreased or increased.

Review questions

1. What is leukocyte emigration and in which vessels does it take place?
2. List the stages of emigration and the molecules involved at each stage.
3. What is the role of selectins and integrins in emigration?
4. What is the sequence of leukocyte emigration according to Mechnikov's law?
5. Explain the theories of emigration.
6. How do physicochemical changes in the inflammatory focus affect emigration?
7. What is chemotaxis? List the main chemoattractants.
8. Into what groups are phagocytes divided?
9. Explain the stages of phagocytosis. What are opsonins?

10. What is the “respiratory burst” and what is its significance?
11. List the oxygen-independent bactericidal mechanisms.
12. In which microbes is incomplete phagocytosis observed?
13. For what reasons does insufficient phagocytosis develop?
14. What is the significance of proliferation and granulation tissue?
15. List the systemic signs and outcomes of inflammation.

Test questions

1. In which vessels does leukocyte emigration mainly take place?
   A) Arterioles
   B) Postcapillary venules
   C) Arteriolovenular shunts
   D) Large arteries
   E) Lymphatic vessels

2. Which molecules provide the rolling of leukocytes along the endothelium?
   A) Selectins
   B) Integrins
   C) Immunoglobulins
   D) Collagenases
   E) Defensins

3. Which leukocytes predominate in the focus in the first hours of acute inflammation?
   A) Lymphocytes
   B) Monocytes
   C) Eosinophils
   D) Neutrophils
   E) Basophils

4. Which molecules play the main role in the firm adhesion of leukocytes to the endothelium?
   A) Selectins only
   B) Histamine and serotonin
   C) Integrins and ICAM-1
   D) Lysozyme and lactoferrin
   E) Prostaglandins

5. Which is one of the strongest chemoattractants for neutrophils?
   A) Histamine
   B) Heparin
   C) Prostacyclin
   D) Serotonin
   E) C5a

6. What function do opsonins perform?
   A) They kill microbes directly
   B) They stop the adhesion of leukocytes to the vessel wall
   C) They reduce the release of histamine
   D) They accelerate the attachment of the microbe to the phagocyte and its engulfment
   E) They stop the formation of the phagolysosome

7. Which enzyme forms the superoxide anion in the “respiratory burst”?
   A) Lysozyme
   B) NADPH oxidase
   C) Collagenase
   D) Cyclooxygenase
   E) Lactoferrin

8. What powerful bactericidal substance does myeloperoxidase form?
   A) Lactate
   B) Ammonia
   C) Nitric oxide
   D) Bradykinin
   E) Hypochlorous acid

9. Incomplete phagocytosis is characteristic of which microbe?
   A) Mycobacterium tuberculosis
   B) Pneumococcus
   C) Escherichia coli
   D) Streptococcus
   E) Klebsiella

10. Which stage of phagocytosis is impaired in chronic granulomatous disease?
    A) Chemotaxis
    B) Attachment
    C) Killing of the microbe (“respiratory burst”)
    D) Engulfment
    E) Margination

11. Which ion weakens emigration in the inflammatory focus?
    A) K^+
    B) H^+
    C) Ca^{2+}
    D) Cl^-
    E) Na^+

12. What does the phagocytic index indicate?
    A) The percentage of neutrophils that have engulfed microbes
    B) The number of microbes inside one neutrophil
    C) The total number of leukocytes in the blood
    D) The number of killed microbes
    E) The erythrocyte sedimentation rate

13. Which cytokine mainly increases the synthesis of acute-phase proteins in the liver?
    A) IL-2
    B) IL-4
    C) Interferon gamma
    D) IL-10
    E) IL-6

14. What does granulation tissue consist of?
    A) Only neutrophils
    B) New capillaries, fibroblasts and inflammatory cells
    C) Only collagen fibers
    D) Mature bone tissue
    E) Fibrin and red blood cells

15. Which cells are characteristic of granulomatous inflammation?
    A) Neutrophils and eosinophils
    B) Red blood cells and platelets
    C) Basophils and mast cells
    D) Epithelioid and multinucleated giant cells
    E) Only plasma cells

Clinical cases

Case 1

A 3-year-old child has had recurrent severe bacterial infections (pneumonia, otitis, skin abscesses) since birth. Staphylococcus and Aspergillus were isolated from the abscesses. The neutrophil count in the blood is normal, and the neutrophils have preserved their ability for chemotaxis and engulfment, but the NBT test is negative.

Questions: What disease does the child have? Which stage of phagocytosis is impaired? Why are catalase-positive microbes particularly dangerous?

Answer: Chronic granulomatous disease — a hereditary defect of NADPH oxidase. Neutrophils engulf microbes, but because there is no “respiratory burst”, they cannot form the superoxide anion and hydrogen peroxide; that is, the killing stage of phagocytosis is impaired, and the negative NBT test shows exactly this. Catalase-negative microbes “deliver” the hydrogen peroxide they produce into the phagolysosome, and myeloperoxidase uses it to kill the microbe. Catalase-positive microbes (staphylococcus, Aspergillus) break down their own peroxide, so in such patients they are not killed and cause recurrent abscesses and granulomas.

Case 2

In a 60-year-old patient with diabetes mellitus, a small wound on a toe has not healed for a long time, and purulent inflammation is spreading to the surroundings. Glycated hemoglobin is 10%.

Questions: Why do inflammation and infections run a severe course in diabetes mellitus? Which stages of leukocyte function are impaired?

Answer: Prolonged hyperglycemia impairs several stages of leukocyte function: chemotaxis and adhesion of neutrophils weaken, phagocytosis and the “respiratory burst” decrease (NADPH deficiency — glucose is consumed by the polyol pathway), and opsonization is impaired. In addition, diabetic microangiopathy and macroangiopathy worsen the blood supply of the tissue, neuropathy causes the wound to go unnoticed, and regeneration and the formation of granulation tissue slow down. As a result, the infection is not delimited and the wound does not heal for a long time; control of glycemia is an important part of treatment.

Case 3

A 30-year-old patient has acute appendicitis. Blood test: leukocytes 15×10^9/L, band neutrophils 12%, ESR 28 mm/h, C-reactive protein elevated. Body temperature 38.1 °C.

Questions: What signs of inflammation are these changes? Explain their mechanism.

Answer: These are systemic signs of inflammation. Leukocytosis and an increase in band neutrophils (a left shift) result from the release of immature neutrophils from the bone marrow and increased granulopoiesis under the action of cytokines (IL-1, TNF-alpha) and colony-stimulating factors released from the inflammatory focus. Increased ESR — an increase in fibrinogen and globulins in the blood enhances aggregation of red blood cells. C-reactive protein is an acute-phase protein synthesized in the liver under the action of IL-6. Fever — cytokines increase the formation of PGE_2 in the hypothalamus and raise the set point of the thermoregulatory center.

Quick review

Table 7.9. *Key facts on Topic 7*

Question	Answer
Emigration	Exit of leukocytes from the vessel into the inflammatory focus (mainly in postcapillary venules)
Stages of emigration	Margination, rolling, firm adhesion, diapedesis, chemotaxis
Selectins	Provide rolling (P-, E-, L-selectin)
Integrins	Provide firm adhesion (with ICAM-1, VCAM-1)
Mechnikov's law	First neutrophils, then monocytes, then lymphocytes
Ions that intensify emigration	K^+ and H^+ (acidosis); weakening — Ca^{2+}
Chemoattractants	C5a, LTB_4, IL-8, bacterial formyl peptides
Stages of phagocytosis	Approach, attachment, engulfment, phagolysosome, killing and digestion
Opsonins	IgG and C3b — accelerate phagocytosis
Respiratory burst	Formation of O_2^-, H_2O_2 and HOCl via NADPH oxidase and myeloperoxidase
Incomplete phagocytosis	The microbe survives inside the phagocyte (tuberculosis, gonococcus, Brucella)
Chronic granulomatous disease	NADPH oxidase defect — the killing stage is impaired
Phagocytic index	Percentage of neutrophils that have engulfed microbes
Phagocytic number	Average number of microbes per phagocyte
Granulation tissue	New capillaries, fibroblasts and inflammatory cells
Systemic signs	Fever, leukocytosis, increased ESR, acute-phase proteins

TOPIC 8

Allergy: etiology, pathogenesis and types. Immunodeficiency states

Learning objectives

To explain to the student the concepts of allergy and sensitization, the types of allergens, the stages and types of allergic reactions (the Gell–Coombs classification), the pathogenesis of immediate and delayed allergic reactions, anaphylaxis and its local manifestations, the mechanisms of desensitization, and primary and secondary immunodeficiency states; and to teach the student to observe anaphylactic shock and desensitization in a guinea pig in an experiment.

By the end of the session, the student should know:

- the concepts of allergy, sensitization and allergen, and the classification of allergens;
- the three stages of allergic reactions;
- the mechanisms and examples of types I–IV (V) of allergic reactions according to Gell and Coombs;
- the differences between immediate and delayed allergic reactions;
- the pathogenesis of anaphylactic shock and the principles of emergency care;
- the mechanisms of specific and nonspecific desensitization;
- primary and secondary immunodeficiencies and the pathogenesis of AIDS.

The student should be able to:

- to model anaphylactic shock in a guinea pig and describe its stages;
- to perform desensitization by Besredka's method and explain its mechanism;
- to identify the type of allergic reaction in a clinical situation.

1. The concept of allergy

Allergy (Greek allos — other, ergon — action) is an altered (enhanced) immune response of the organism on repeated exposure to an allergen, accompanied by injury of its own tissues. The term was proposed in 1906 by the Austrian pediatrician C. Pirquet.

Immunity and allergy are based on the same immune mechanisms; the difference lies in the result: in immunity the antigen is neutralized and the tissues are not injured, whereas in allergy the immune response injures the organism's own tissues.

Sensitization is an increase in the organism's sensitivity to an allergen after its first entry (formation of specific antibodies or sensitized T lymphocytes). Sensitization itself gives no clinical signs; it forms in 7–14 days and can persist for years.

- **Active sensitization** — formation of the organism's own antibodies when an allergen is introduced.
- **Passive sensitization** — when the serum (antibodies) or lymphocytes of a sensitized organism are transferred to another organism.

Allergens and their classification

An allergen is an antigen that causes an allergic reaction. Complete antigens (proteins, glycoproteins) and haptens (small molecules that are not antigens themselves but become allergens when bound to a protein: drugs, nickel, chromium) are distinguished.

Table 8.1. *Classification of allergens*

Group	Examples
Household	House dust, dust mites (Dermatophagoides), cockroaches
Epidermal	Hair and dander of cats, dogs and horses; bird feathers
Plant pollen (pollinosis)	Ragweed, wormwood, grasses, tree pollen
Food	Cow's milk, eggs, fish, nuts, peanuts, citrus fruits
Drugs	Penicillins, sulfonamides, local anesthetics, vaccines and sera
Insects	Bee and wasp venom
Infectious (bacterial, fungal)	Staphylococcus, streptococcus, molds, Mycobacterium tuberculosis
Industrial and occupational	Formaldehyde, latex, chromium and nickel salts, dyes
Endogenous (autoallergens)	Altered own proteins; “sequestered” tissues (lens of the eye, thyroid colloid, sperm)

2. Stages of allergic reactions

According to A.D. Ado, every allergic reaction passes through three stages.

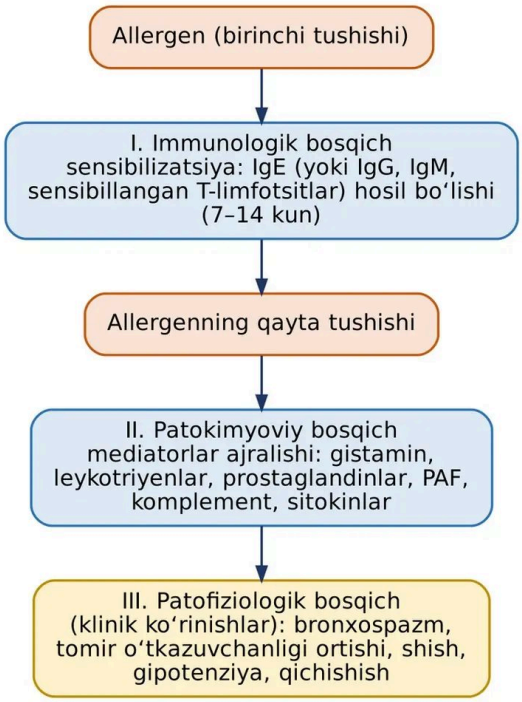


Figure 8.1. *Stages of allergic reactions (according to A.D. Ado; based on V.V. Novitsky, 2009)*

Table 8.2. *Stages of an allergic reaction*

Stage	Essence
I. Immunological	On first entry of the allergen, antibodies (IgE, IgG, IgM) or sensitized T lymphocytes are formed; on repeated entry the allergen binds to them
II. Pathochemical (biochemical)	As a result of antigen-antibody or antigen-lymphocyte binding, biologically active substances (mediators) are formed and released

Stage	Essence
III. Pathophysiological (clinical manifestations)	Under the action of mediators, the function of tissues and organs is impaired: contraction of smooth muscle, increased permeability, edema, inflammation, itching

3. Types of allergic reactions (the Gell and Coombs classification)

P. Gell and R. Coombs (1963) divided allergic reactions into four types according to the mechanism of immune injury; type V (stimulatory) was added later.

Table 8.3. *Types of allergic reactions*

Type	Mechanism	Time of development	Examples
I — anaphylactic (reaginic, IgE-dependent)	The allergen binds to IgE on the surface of mast cells and basophils → degranulation, release of histamine, leukotrienes and other mediators	15–30 minutes (a few minutes)	Anaphylactic shock, atopic bronchial asthma, allergic rhinitis (hay fever), urticaria, Quincke edema (angioedema)
II — cytotoxic (cytolytic)	IgG or IgM binds to an antigen on the cell surface → cell lysis via complement, phagocytosis, antibody-dependent cellular cytotoxicity	Minutes — hours	Hemolytic anemias (incompatible blood transfusion, hemolytic disease of the newborn), drug-induced thrombocytopenia, Goodpasture syndrome
III — immune complex (Arthus type)	Soluble antigen-antibody complexes are deposited in vessel walls and tissues → complement is activated → neutrophils accumulate and injure the tissue	4–8 hours	Serum sickness, the Arthus phenomenon, glomerulonephritis, systemic lupus erythematosus, “farmer's lung”
IV — cell-mediated (delayed type)	Sensitized T lymphocytes (Th1, cytotoxic T cells), the lymphokines they release, and activation of macrophages	24–72 hours	The tuberculin reaction (Mantoux), contact dermatitis, transplant rejection, tuberculous granuloma
V — stimulatory	Antibodies bind to a cell receptor and stimulate it	—	Diffuse toxic goiter (autoantibodies to the TSH receptor)

MEMORY AID
To remember the types: **“ACID”** — **A**naphylactic (I), **C**ytotoxic (II), **I**mmune complex (III), **D**elayed (IV). Only type IV proceeds without antibodies, via T lymphocytes.

Immediate and delayed allergic reactions

Table 8.4. *Differences between immediate and delayed allergic reactions*

Feature	Immediate (I–III)	Delayed (IV)
Time of onset	Minutes — hours	24–72 hours
Main factor	Antibodies (IgE, IgG, IgM)	Sensitized T lymphocytes
Passive transfer	Possible via serum	Only with lymphocytes
Main mediators	Histamine, leukotrienes, complement	Lymphokines (interferon gamma, IL-2, TNF)

Feature	Immediate (I–III)	Delayed (IV)
Effect of antihistamines	Marked in type I	No
Histological picture	Edema, eosinophils, neutrophils	Mononuclear (lymphocyte and macrophage) infiltrate, granuloma

4. Anaphylaxis and anaphylactic shock

Anaphylaxis (Greek ana — reverse, phylaxis — protection) — the systemic (generalized) form of a type I allergic reaction. It was discovered in 1902 by Ch. Richet and P. Portier: dogs that had received a small dose of sea anemone toxin died instead of being protected when it was injected a second time.

Anaphylactic shock — the most severe, life-threatening form of a type I allergic reaction.

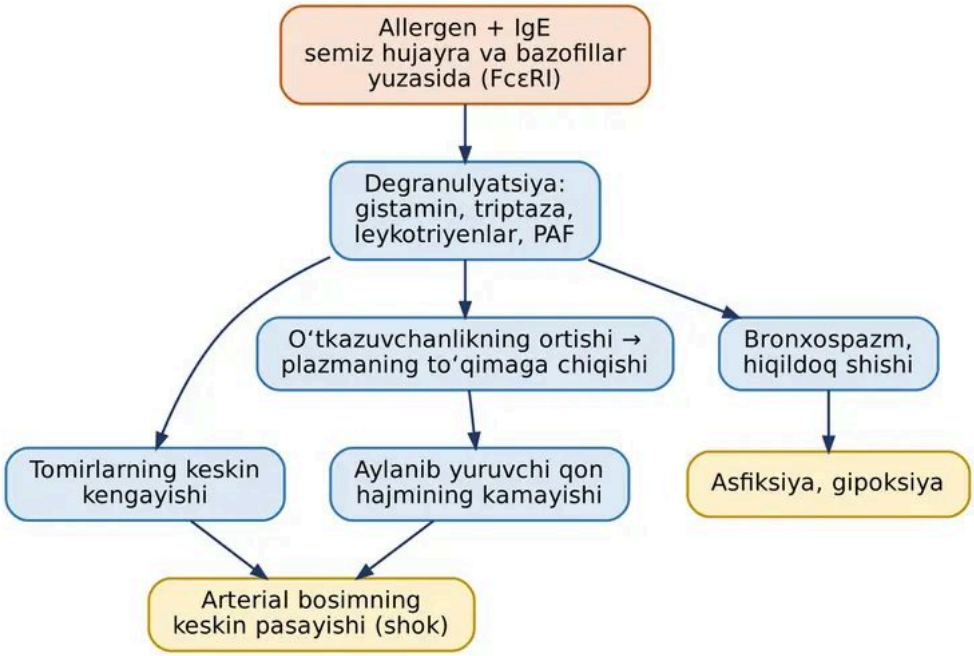


Figure 8.2. *Pathogenesis of anaphylactic shock (based on Ataman A.V., 2000, figure 27)*

Main mediators of anaphylactic shock and their effects:

- **histamine** — dilates arterioles, increases venular permeability, constricts the bronchi, causes itching;
- **leukotrienes C_4, D_4, E_4** (formerly called the “slow-reacting substance of anaphylaxis”) — strong and prolonged bronchospasm, increased permeability;
- **platelet-activating factor** — hypotension, bronchospasm, platelet aggregation;
- **prostaglandin D_2** — bronchospasm, vasodilation;
- **tryptase** — a laboratory marker of mast cell activation.

In humans the main “target organs” of anaphylactic shock are the cardiovascular system (hypotension, collapse) and the airways (laryngeal edema, bronchospasm). In the guinea pig, by contrast, bronchospasm and asphyxia predominate; in the dog — spasm of the hepatic veins and pooling of blood in the portal system; in the rabbit — spasm of the pulmonary arteries. This interspecies difference is explained by the concept of the “shock organ”.

CLINICAL SIGNIFICANCE

The first and main drug in anaphylactic shock is **epinephrine (adrenaline)** (0.1 % solution, 0.3–0.5 ml into the thigh muscle in adults, repeated after 5–15 minutes if needed). Via alpha-adrenoceptors it constricts vessels and raises arterial pressure; via beta2-receptors it dilates the bronchi and stops mediator release from mast cells. Additional measures: stop the allergen, lay the patient down with legs raised, oxygen, intravenous crystalloid solutions; antihistamines and glucocorticoids are second-line agents.

Local manifestations of allergy

- **Hives (urticaria)** — itchy wheals on the skin; increased vascular permeability in the dermis.
- **Quincke's edema (angioedema)** — circumscribed deep edema of the subcutaneous and submucosal tissue; risk of asphyxia if it develops in the larynx.
- **Allergic rhinitis and conjunctivitis (pollinosis)** — seasonal, occurring when plants are in bloom.
- **Atopic bronchial asthma** — bronchial hyperreactivity, bronchospasm, mucus hypersecretion and eosinophilic inflammation.
- **Arthus phenomenon** — when an allergen is re-injected subcutaneously into a sensitized animal, immune complexes deposit at that site and hyperemia, edema and necrosis develop (type III).

CAUTION

Atopy is a hereditary predisposition to IgE production. In atopic patients (those with a family history of asthma, hay fever, atopic dermatitis), allergic diseases are common. **Pseudoallergic reactions** resemble allergy in appearance but proceed without an immunological stage: mediators are released directly (radiocontrast agents, opiates, some foods — strawberries, chocolate). They develop even on first exposure and do not require sensitization.

5. Desensitization

Desensitization (hyposensitization) — reduction or elimination of the body's sensitivity to an allergen.

Specific desensitization — directed at a particular allergen:

- **Besredka method (rapid desensitization)** — the allergen (e.g. a therapeutic serum) is first given in very small doses, then in gradually increasing doses at 20–30-minute intervals. The small doses gradually “bind” the IgE on mast cells and release mediators in small, safe portions; the full dose then does not cause shock. The effect is temporary.
- **Allergen-specific immunotherapy (slow desensitization)** — the allergen is given in increasing doses over months and years. Mechanism: formation of “blocking” IgG antibodies (they capture the allergen before IgE does), a shift of the Th2 response towards Th1 and regulatory T cells, and reduced IgE synthesis.

Nonspecific desensitization — acting on the common mechanisms of allergic reactions: antihistamines, mast cell membrane stabilizers (cromoglicate), glucocorticoids, leukotriene receptor antagonists, adrenomimetics, monoclonal anti-IgE antibodies (omalizumab).

6. Immunodeficiency states

Immunodeficiency states — reduced resistance of the body to infections and tumors as a result of a defect in one or more components of the immune system.

Table 8.5. *Primary (congenital) immunodeficiencies*

Affected component	Examples	Clinical features
B lymphocytes (humoral)	X-linked agammaglobulinemia (Bruton disease), selective IgA deficiency	Recurrent bacterial infections after 6 months of life (once maternal antibodies are exhausted)
T lymphocytes (cellular)	DiGeorge syndrome (thymic aplasia)	Viral and fungal infections, hypocalcemia (the parathyroid glands are also absent)
Combined	Severe combined immunodeficiency, Wiskott–Aldrich syndrome	Severe infections with early onset
Phagocytosis	Chronic granulomatous disease, leukocyte adhesion deficiency	Bacterial and fungal abscesses
Complement	C1-inhibitor deficiency (hereditary angioedema), C5–C9 deficiency	Recurrent swelling; neisserial infections

Secondary (acquired) immunodeficiencies — far more common. Causes: HIV infection, starvation and protein deficiency, diabetes mellitus, tumors (especially lymphoproliferative), ionizing radiation, treatment with cytostatics and glucocorticoids, splenectomy, old age, severe burns and injuries, chronic kidney failure.

Acquired immunodeficiency syndrome (AIDS)

The causative agent of AIDS is the **human immunodeficiency virus (HIV)**, a member of the retrovirus family. The virus binds to the CD4 molecule (and the CCR5 or CXCR4 coreceptors) and mainly damages **$CD4^+$ T helper cells**, as well as macrophages and dendritic cells.

Pathogenesis: the virus enters the cell → a DNA copy is made from RNA by reverse transcriptase → it is integrated into the cell genome by integrase → the virus replicates and $CD4^+$ cells die → the number of $CD4^+$ lymphocytes gradually falls → cellular and humoral immunity are profoundly impaired → opportunistic infections and tumors develop.

> **REMEMBER**
> Stages of HIV infection: acute infection (mononucleosis-like syndrome), a long clinically latent period (years), and the AIDS stage. In the AIDS stage, the $CD4^+$ lymphocyte count is less than 200 cells/μL, and Pneumocystis pneumonia, tuberculosis, candidiasis, toxoplasmosis, cytomegalovirus infection, Kaposi sarcoma and lymphomas develop. Modern antiretroviral therapy stops viral replication and restores immunity.

7. Experimental part

Experiment 1: anaphylactic shock in the guinea pig

Aim: to model and observe the systemic form of a type I allergic reaction — anaphylactic shock.

Equipment: guinea pigs (at least 2 sensitized and 1 control), horse serum (allergen), syringes, stopwatch.

1. **Sensitization** (performed in advance by the teacher): 14–21 days earlier, 0.1 ml of horse serum (sensitizing dose) was injected subcutaneously or intraperitoneally into the guinea pig.
2. Assess the baseline state of the sensitized and control (non-sensitized) animals: movement, respiratory rate, condition of the fur.

3. Inject both animals intravenously (or intracardially) with 0.5–1 ml of horse serum (challenge dose).
4. In the sensitized animal, record in sequence the signs that develop within a few minutes: restlessness, rubbing the nose, sneezing, ruffled fur, passage of urine and feces, difficulty breathing (expiratory dyspnea), trembling, convulsions, falling onto its side.
5. Record that no such changes are observed in the control animal.
6. On autopsy of the animal that died, observe the markedly distended (emphysematous), non-collapsing lungs — air is trapped in the alveoli because of bronchospasm.

Analysis: in the sensitized guinea pig, IgE (and IgG1) antibodies are bound to mast cells.

When the challenge dose is given, mast cell degranulation and release of mediators (mainly histamine) cause severe bronchiolar spasm and asphyxia — the guinea pig's “shock organ” is the lung.

Experiment 2: desensitization by the Besredka method

1. Before the challenge dose, inject a second sensitized guinea pig subcutaneously with a very small dose (0.01–0.05 ml) of horse serum.
2. After 20–30 minutes give a larger dose, then after another 20–30 minutes inject the full challenge dose intravenously.
3. Observe the animal — anaphylactic shock does not develop or is mild.

Analysis: the small doses gradually bind the antibodies on mast cells, and mediators are released in small, harmless amounts; when the full dose is given, not enough active antibodies remain to cause shock.

This is the basis of the Besredka method, which was used clinically when administering therapeutic sera (e.g. antitetanus or antivenom serum).

> **CAUTION**
> Before administering preparations based on horse serum, an intradermal test and stepwise administration by Besredka's method are mandatory. At the same time, a resuscitation kit (epinephrine, glucocorticoids, infusion agents) must always be ready.

8. Practical skills

Skill: determining the type of allergic reaction

1. Determine the time of onset of the reaction: minutes (type I), hours (types II–III) or 1–3 days (type IV).
2. Assess the main clinical picture: bronchospasm and hypotension (I), hemolysis or cytopenia (II), vasculitis, glomerulonephritis, arthralgia (III), contact dermatitis or infiltrate (IV).
3. Determine the nature of the allergen: soluble protein, cell-surface antigen, hapten.
4. Take laboratory findings into account: blood IgE and eosinophils (I), Coombs test (II), immune complexes and decreased complement (III).
5. Write a conclusion and indicate the main direction of treatment.

Skill: emergency care algorithm for anaphylactic shock

1. Stop exposure to the allergen (stop the drug infusion, remove the bee sting).

2. Call for help; lay the patient on their back and raise the legs (semi-sitting position if breathing is difficult).
3. Inject epinephrine into the middle outer surface of the thigh muscle; if there is no effect, repeat after 5–15 minutes.
4. Ensure airway patency and give oxygen.
5. Establish venous access and infuse crystalloid solutions.
6. Give an antihistamine and a glucocorticoid (second line).
7. Monitor arterial pressure, pulse and breathing continuously; the patient must be observed for at least 24 hours (risk of a delayed wave).

Review questions

1. What is allergy? How does it differ from immunity?
2. What is sensitization? What is the difference between active and passive sensitization?
3. Into what groups are allergens divided? What is a hapten?
4. Explain the three stages of an allergic reaction.
5. List the types of allergic reactions according to Gell and Coombs and give an example of each.
6. Explain the mechanism of a type I allergic reaction.
7. How do type II and type III allergic reactions differ?
8. What is distinctive about a type IV allergic reaction?
9. List the differences between immediate and delayed allergic reactions.
10. What are the pathogenesis and main mediators of anaphylactic shock?
11. Why is epinephrine the main drug in anaphylactic shock?
12. How do pseudoallergic reactions differ from true allergy?
13. Explain the mechanisms of specific and nonspecific desensitization.
14. List the main types of primary immunodeficiency.
15. Explain the pathogenesis of AIDS.

Test questions

1. Who proposed the term “allergy”?
   A) Ch. Richet
   B) R. Koch
   C) I.I. Mechnikov
   D) A.M. Besredka
   E) C. Pirquet

2. Which antibody plays the main role in a type I (anaphylactic) allergic reaction?
   A) IgM
   B) IgA
   C) IgG
   D) IgE
   E) IgD

3. Hemolytic disease of the newborn belongs to which type of allergic reaction?
   A) Type I
   B) Type IV
   C) Type II (cytotoxic)
   D) Type III
   E) Type V

4. The tuberculin reaction (Mantoux test) belongs to which type?
   A) Type I
   B) Type IV (cell-mediated)
   C) Type II
   D) Type III
   E) Pseudoallergic

5. Serum sickness is an example of which type of allergic reaction?
   A) Type III (immune complex)
   B) Type I
   C) Type II
   D) Type IV
   E) Type V

6. What happens in the pathochemical stage of an allergic reaction?
   A) Antibodies are formed
   B) Clinical signs appear
   C) Mediators are formed and released
   D) The allergen enters the body for the first time
   E) Sensitized lymphocytes proliferate

7. Which drug is used first in anaphylactic shock?
   A) An antihistamine
   B) A glucocorticoid
   C) Calcium chloride
   D) A diuretic
   E) Epinephrine (adrenaline)

8. What is the main “shock organ” in anaphylactic shock in the guinea pig?
   A) The lung (bronchiolar spasm)
   B) Liver
   C) Kidney
   D) Skin
   E) The intestine

9. What is the essence of the Besredka method?
   A) Giving the allergen in a single large dose
   B) Using only antihistamines
   C) Giving the allergen over several months
   D) Giving the allergen step by step, starting with a small dose, at short intervals
   E) Giving the serum only by mouth

10. What is needed for passive transfer of a delayed-type allergic reaction?
    A) Serum of a sensitized animal
    B) Sensitized lymphocytes
    C) Erythrocytes
    D) Platelets
    E) Histamine

11. In diffuse toxic goiter, autoantibodies to the TSH receptor belong to which type of reaction?
    A) Type I
    B) Type II
    C) Type III
    D) Type V (stimulatory)
    E) Type IV

12. What is the main distinguishing feature of a pseudoallergic reaction?
    A) It occurs only by type IV
    B) It occurs only in children
    C) It occurs without an immunological stage and develops even on first exposure
    D) It occurs only via IgE
    E) It is never dangerous

13. Which cells does HIV mainly damage?
    A) B lymphocytes
    B) $CD4^+$ T helper cells
    C) Erythrocytes
    D) Neutrophils
    E) Platelets

14. Which component of the immune system is affected in DiGeorge syndrome?
    A) B lymphocytes
    B) Complement system
    C) Neutrophils
    D) Macrophages only
    E) T lymphocytes (thymic aplasia)

15. Quincke's edema is a local manifestation of which type of allergic reaction?
    A) Type I
    B) Type II
    C) Type III
    D) Type IV
    E) Type V

Clinical cases

Case 1

A 28-year-old woman was given an anesthetic for local anesthesia at the dentist. After 5 minutes she complained of shortness of breath, tightness in the throat and itching all over the body. Red wheals appeared on the skin, her lips swelled, arterial pressure was 70/40 mm Hg, pulse 130 beats/min, breathing wheezy.

Questions: What condition has developed? Explain its pathogenesis. In what order is emergency care provided?

Answer: Anaphylactic shock — the systemic form of a type I (IgE-mediated) allergic reaction. The patient had previously been sensitized to this anesthetic: IgE specific to it is present on the surface of mast cells and basophils. On re-exposure the allergen binds to IgE and causes mast cell degranulation; histamine, leukotrienes and platelet-activating factor are released. They sharply dilate the vessels and increase their permeability (hypotension, reduced circulating blood volume, compensatory tachycardia), constrict the bronchi (wheezing, dyspnea), and cause edema (lips, larynx) and urticaria. Care: stop the anesthetic, immediately inject epinephrine into the thigh muscle, lay the patient down with legs raised, oxygen, intravenous crystalloid solutions; then an antihistamine and a glucocorticoid; observation for at least 24 hours.

Case 2

9 days after receiving antitetanus horse serum, a patient developed fever, skin rashes, joint pain and enlarged lymph nodes. Protein was found in the urine, and the blood complement level was decreased.

Questions: What condition has developed? Which type of allergic reaction does it belong to? Why did the signs appear after 9 days?

Answer: Serum sickness — a type III (immune complex) allergic reaction. Foreign (horse) proteins persist in the body for a long time; within about a week antibodies (mainly IgG) are formed against them and create soluble immune complexes with the antigen still in the blood. These complexes deposit in vessel walls, renal glomeruli and joint synovium, activate complement (which is why blood complement falls), attract neutrophils and cause vasculitis, glomerulonephritis (proteinuria) and arthritis. The signs appeared after the time needed to form antibodies (7–14 days); because there was no prior sensitization, the reaction developed on the very first injection.

Case 3

A 32-year-old patient has had weight loss, prolonged fever, oral candidiasis and recurrent pneumonias over a year. Pneumocystis was found in the latest pneumonia. Blood $CD4^+$ lymphocytes are 120 cells/µl.

Questions: What condition can be suspected? Explain the mechanism of the immunodeficiency.

Answer: The AIDS stage of HIV infection (secondary immunodeficiency). HIV enters T helper cells via the CD4 molecule and coreceptors, inserts its genome into the cell DNA by means of reverse transcriptase, replicates and kills the cells. Loss of $CD4^+$ T helper cells impairs cellular immunity (macrophage activation, cytotoxic T cells) and humoral immunity (help to B lymphocytes). When $CD4^+$ cells fall below 200/µl, opportunistic infections develop — pneumocystis pneumonia, candidiasis, tuberculosis. The diagnosis should be confirmed by detecting antibodies to HIV and viral RNA, and antiretroviral therapy should be started.

Quick review

Table 8.6. *Key facts on Topic 8*

Question	Answer
Allergy	An altered immune response on repeated exposure to an allergen, accompanied by damage to the body's own tissues

Question	Answer
Sensitization	Increased sensitivity after the first exposure to an allergen (7–14 days)
Hapten	A small molecule that becomes an allergen by binding to a protein
Stages	Immunological, pathochemical, pathophysiological
Type I	IgE, mast cells, histamine: anaphylaxis, asthma, pollinosis, urticaria
Type II	IgG/IgM, cell lysis: hemolytic anemias, Goodpasture syndrome
Type III	Immune complexes: serum sickness, Arthus phenomenon, glomerulonephritis
Type IV	T lymphocytes, 24–72 hours: Mantoux, contact dermatitis, transplant rejection
Type V	Stimulatory antibodies: diffuse toxic goiter
Anaphylactic shock	Systemic form of type I; main drug — epinephrine
Shock organ	In the guinea pig — the lung (bronchospasm)
Pseudoallergy	No immunological stage; develops even on first exposure
Besredka method	Rapid desensitization by giving the allergen stepwise, starting with a small dose
Primary immunodeficiencies	Bruton, DiGeorge, severe combined, chronic granulomatous
AIDS	HIV damages $CD4^+$ T helper cells; $CD4^+$ below 200/µl

TOPIC 9

Disorders of water and electrolyte metabolism. Pathogenetic factors of edema

Learning objectives

To explain to the student the balance of water and electrolytes in the body, the mechanisms that regulate it, the types of water balance disorders (dehydration and hyperhydration), their causes, pathogenesis and consequences, disorders of the main electrolytes (sodium, potassium, calcium), and the pathogenetic factors and types of edema; to teach the student to observe experimentally the role of the osmotic and colloid-osmotic factors in edema formation in the frog lymph sac.

By the end of the session, the student should know:

- the distribution of water in the body, water balance and its regulation;
- the types of dehydration and hyperhydration (iso-, hypo-, hyperosmolar), their causes and consequences;
- the main manifestations of disorders of sodium, potassium and calcium metabolism;
- the Starling forces and the pathogenetic factors of edema;
- the pathogenesis of cardiac, renal, hepatic, starvation, inflammatory and allergic edema.

The student should be able to:

- observe the role of the osmotic factor in edema formation by introducing a hypertonic solution into the frog lymph sac;
- determine the type of water balance disorder from clinical and laboratory signs;
- determine the type of edema and its leading pathogenetic factor.

1. Distribution of water in the body and water balance

In the adult body, water makes up about 60 percent of body weight (more in men, less in women and the elderly, up to 75–80 percent in newborns).

Table 9.1. *Distribution of water in the body (approximate, for a 70 kg adult)*

Compartment	Share of body weight	Volume
Total water	60 %	42 l
Intracellular fluid	40 %	28 l
Extracellular fluid	20 %	14 l
— interstitial fluid	15 %	10.5 l
— blood plasma	5 %	3.5 l
— transcellular fluid (CSF, pleural, synovial, digestive juices)	1–2 %	about 1 l

Table 9.2. *Daily water balance in an adult (approximate)*

Intake	Volume	Output	Volume
Drink	1.2–1.5 l	Urine	1.2–1.5 l
In food	0.8–1 l	Through the skin (evaporation and sweat)	0.5–0.6 l
Endogenous (metabolic) water	0.3 l	Through the lungs	0.4 l

Intake	Volume	Output	Volume
		With feces	0.1–0.2 l
Total	2.3–2.8 l	Total	2.3–2.8 l

Normal plasma osmolarity is 280–295 mOsm/kg; most of it is provided by sodium (135–145 mmol/l) and its accompanying anions. Sodium is the main cation of the extracellular fluid, and potassium (140–150 mmol/l inside cells, 3.5–5.0 mmol/l in plasma) is the main cation of the intracellular fluid.

Regulation of water and electrolyte balance

REMEMBER

The main systems regulating water–salt homeostasis:

- **antidiuretic hormone (ADH, vasopressin)** — formed in the hypothalamus when osmoreceptors sense a rise in plasma osmolarity and released from the posterior pituitary; it increases water reabsorption in the renal collecting ducts; thirst appears at the same time;
- **renin–angiotensin–aldosterone system (RAAS)** — renin is released when renal blood flow falls, angiotensin II constricts vessels and increases aldosterone secretion; aldosterone retains sodium and excretes potassium;
- **natriuretic peptides** — released when the atria and ventricles are stretched, they increase sodium and water excretion;
- **thirst mechanism** — regulates water intake.

2. Disorders of water balance

Water balance disorders are divided into **dehydration (hypohydration, water loss)** and **hyperhydration (excess water accumulation)**. Each is subdivided into three types according to the osmolarity of the extracellular fluid.

Dehydration (water loss)

Table 9.3. *Types of dehydration*

Type	Causes	Pathogenesis	Main features
Isoosmolar	Blood loss, early stage of diarrhea and vomiting, burns, polyuria (diuretics)	Water and sodium are lost in equal amounts; extracellular fluid decreases, intracellular fluid hardly changes	Hypovolemia, tachycardia, hypotension, oliguria
Hyperosmolar (water deficit)	Restricted water intake (unconscious patient, impaired swallowing), profuse sweating, hyperventilation, diabetes insipidus, osmotic diuresis (hyperglycemia)	Water is lost in excess of salt; plasma osmolarity rises, water leaves the cells — cellular dehydration	Intense thirst, dry mucous membranes, fever, impaired consciousness (dehydration of neurons), hypernatremia
Hypoosmolar (salt deficit)	Profuse vomiting and diarrhea when losses are replaced with plain water only; adrenal insufficiency, salt-wasting nephropathy	Salt is lost in excess of water; plasma osmolarity falls, water moves into the cells — cellular swelling, extracellular fluid decreases sharply	No thirst, nausea, muscle cramps, hypotension, collapse, risk of cerebral edema, hyponatremia

CLINICAL SIGNIFICANCE

In diarrhea (cholera, rotavirus infection), dehydration develops rapidly in children. Oral rehydration solution (a mixture of glucose and salts) is based on the co-absorption of sodium and glucose in the intestine and sharply reduces the risk of death. “Rehydrating” a dehydrated patient with plain water alone can lead to hypoosmolar dehydration.

Hyperhydration (excess water accumulation)

Table 9.4. *Types of hyperhydration*

Type	Causes	Pathogenesis	Main features
Isoosmolar	Excessive infusion of isotonic solutions, heart failure, nephrotic syndrome, liver cirrhosis	Extracellular fluid increases, osmolarity is normal	Edema, raised venous pressure, hypervolemia
Hypoosmolar (“water intoxication”)	Drinking large amounts of water or infusing salt-free solutions when renal water excretion is impaired; syndrome of inappropriate ADH secretion	Plasma osmolarity falls, water moves into the cells — swelling of cells (especially neurons)	Headache, vomiting, convulsions, coma, cerebral edema; hyponatremia
Hyperosmolar	Drinking sea water, excessive infusion of hypertonic solutions, hyperaldosteronism	Plasma osmolarity rises, water leaves the cells — cellular dehydration with excess water in the extracellular compartment	Thirst, edema, impaired consciousness, arterial hypertension, hypernatremia

MEMORY AID

Water always moves towards the side of **higher osmolarity**: “**water follows salt**”. If plasma salt increases, the cell “dries out” (hyperosmolar state); if plasma salt decreases, the cell “swells” (hypoosmolar state).

3. Disorders of the main electrolytes

Table 9.5. *Disorders of sodium and potassium metabolism*

Disorder	Normal value and threshold	Main causes	Main features
Hyponatremia	Na^+ below 135 mmol/l	Hypoosmolar states, diuretics, excess ADH secretion, adrenal insufficiency	Headache, nausea, convulsions, cerebral edema
Hypernatremia	Na^+ above 145 mmol/l	Water deficit, diabetes insipidus, hypertonic solutions	Thirst, impaired consciousness, convulsions
Hypokalemia	K^+ below 3.5 mmol/l	Vomiting, diarrhea, diuretics, hyperaldosteronism, alkalosis, insulin therapy	Muscle weakness, intestinal atony, arrhythmias; on the ECG — flattening of the T wave and appearance of a U wave
Hyperkalemia	K^+ above 5.0–5.5 mmol/l	Kidney failure, tissue breakdown (crush syndrome, hemolysis), acidosis, potassium-sparing diuretics, hypoaldosteronism	Muscle weakness, bradycardia, cardiac arrest; on the ECG — tall peaked T wave

Disorders of calcium metabolism: total plasma calcium is 2.2–2.6 mmol/l. **Hypocalcemia** (hypoparathyroidism, vitamin D deficiency, acute pancreatitis) increases neuromuscular excitability:

tetany, Chvostek and Trousseau signs, laryngospasm. **Hypercalcemia** (hyperparathyroidism, bone metastases) causes weakness, constipation, polyuria, kidney stones and arrhythmias.

4. Edema and its pathogenetic factors

Edema — excess accumulation of fluid in the interstitial space. Accumulation of fluid in serous cavities is called **dropsy**: in the abdominal cavity — ascites, in the pleura — hydrothorax, in the pericardium — hydropericardium, in the testicular membranes — hydrocele. Widespread edema of the whole body is called **anasarca**.

Starling forces

Fluid exchange between capillary and tissue obeys E. Starling's law. The forces that **move** fluid out of the capillary into the tissue are capillary hydrostatic pressure and the oncotic pressure of tissue fluid; the forces that **return** it to the capillary are plasma oncotic pressure and tissue hydrostatic pressure.

Table 9.6. *Filtration and reabsorption in the capillary (approximate values)*

Parameter	Arterial end of the capillary	Venous end of the capillary
Capillary hydrostatic pressure	30–35 mm Hg	15 mm Hg
Plasma oncotic pressure	25 mm Hg	25 mm Hg
Result	Filtration (fluid moves out into the tissue)	Reabsorption (fluid returns to the blood)

Normally filtration slightly exceeds reabsorption; excess fluid and proteins are removed by the lymphatic vessels (2–4 l of lymph per day).

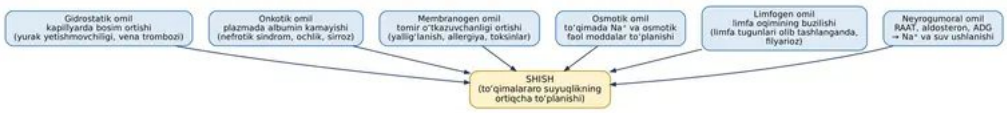


Figure 9.1. *Pathogenetic factors of edema (based on Abdullayev N.H., 2008; Novitskiy V.V., 2009)*

Table 9.7. *Pathogenetic factors of edema*

Factor	Mechanism	Examples
Hydrostatic	Pressure in capillaries and venules rises and filtration increases	Heart failure, venous thrombosis, venous congestion
Oncotic	Plasma protein (albumin) falls and the force holding fluid in the vessels decreases; edema appears when albumin is below 25–30 g/l	Nephrotic syndrome (proteinuria), liver cirrhosis (reduced synthesis), starvation (cachectic edema), enteropathies
Membranogenic	Vessel wall permeability increases and protein leaks into the tissue	Inflammation, allergy, burns, toxins (snake venom), hypoxia
Osmotic	Sodium and osmotically active substances accumulate in the tissue and draw water	Inflammatory focus, renal edema (sodium retention)
Lymphogenic	Impaired lymph flow; protein-rich fluid accumulates	After removal of lymph nodes (mastectomy), filariasis (“elephantiasis”), tumor metastases
Neurohumoral	Sodium and water are retained under the action of RAAS, aldosterone and ADH	As a secondary factor in cardiac, renal and hepatic edema

REMEMBER

Most edemas develop with the participation of **several factors**, one of which is leading (initiating). For example, in cardiac edema the leading factor is hydrostatic, in nephrotic edema — oncotic, in inflammatory edema — membranogenic, in lymphedema — lymphogenic.

5. Main types of edema

Cardiac edema

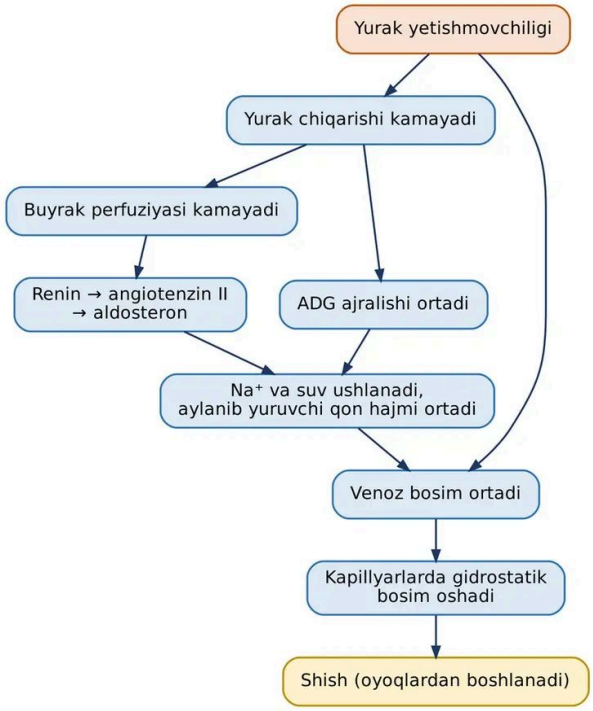


Figure 9.2. *Pathogenesis of cardiac edema (based on Novitskiy V.V., 2009, figure 12.50)*

In right ventricular failure, edema begins in the legs (ankles and shins), increases towards evening and decreases overnight; in bedridden patients it is located in the sacral and lumbar region (effect of gravity). The skin is cold and cyanotic. In severe cases ascites, hydrothorax and anasarca develop.

Renal edema

Table 9.8. *Differences between nephrotic and nephritic edema*

Feature	Nephrotic edema	Nephritic (glomerulonephritis) edema
Leading factor	Oncotic: heavy proteinuria (more than 3.5 g per day) → hypoalbuminemia	Sodium and water retention (reduced glomerular filtration) + increased vascular permeability
Location	Begins on the face and eyelids, spreads quickly, anasarca	Face, eyelids; more in the morning
Blood proteins	Sharply reduced (albumin down to 20 g/l)	Little changed
Arterial pressure	Usually normal	Often raised
In the urine	Heavy proteinuria, with hyperlipidemia	Hematuria, moderate proteinuria

Renal edema is soft, the skin is pale and warm, and pressure easily leaves a pit; eyelid edema is especially marked in the morning (loose subcutaneous tissue).

Other types of edema

- **Hepatic edema and ascites** (cirrhosis): portal hypertension (hydrostatic factor), reduced albumin synthesis in the liver (oncotic factor), secondary hyperaldosteronism (aldosterone is poorly broken down in the liver), increased lymph formation.
- **Cachectic (starvation) edema**: protein deficiency — hypoalbuminemia; kwashiorkor in children.
- **Inflammatory edema**: membranogenic, osmotic and oncotic factors (see topic 6).
- **Allergic edema** (Quincke's edema, urticaria): a sharp increase in permeability under the action of histamine and other mediators.
- **Toxic edema** (toxic pulmonary edema, edema after a snakebite): direct damage to the capillary endothelium.

> **CAUTION**
> Edema is sometimes life-threatening: **laryngeal edema** leads to asphyxia, **cerebral edema** to herniation of the brainstem into the foramina of the skull, **pulmonary edema** to severe hypoxemia. Such edemas require emergency treatment.

Consequences of edema

Edema mechanically compresses tissues and impairs microcirculation and oxygen diffusion (the diffusion distance increases) — hypoxia, dystrophy and susceptibility to infection result; if prolonged, connective tissue overgrowth (sclerosis) develops. At the same time, inflammatory edema also has a protective role: it dilutes toxins and walls off the focus.

6. Experimental part: role of the osmotic and colloid-osmotic factors in edema formation

Aim: to observe how a rise in tissue osmotic pressure affects edema formation.

Equipment: frogs (of similar weight, 2–3), 10 % sodium chloride (hypertonic) solution, 0.6 % saline (isotonic for the frog), scales, syringes.

Procedure

1. Weigh the frogs and record their weights.
2. Inject 1–2 ml of hypertonic sodium chloride solution into the dorsal lymph sac of frog 1.
3. Inject the same volume of isotonic solution into the lymph sac of frog 2 (control).
4. Place both frogs in a container of water (the frog absorbs water through its skin).
5. For 1–2 hours, weigh the frogs every 30 minutes and record the change in weight and appearance (swelling of the lymph sac and legs).

Table 9.9. *Recording the experimental results*

Time	Frog 1 (hypertonic solution), weight	Frog 2 (isotonic solution), weight
Beginning		
30 minutes		
60 minutes		

Time	Frog 1 (hypertonic solution), weight	Frog 2 (isotonic solution), weight
90 minutes		

Expected results and analysis: the weight of the frog given the hypertonic solution increases noticeably and its lymph sac and legs swell; the weight of the control frog hardly changes. Accumulation of an osmotically active substance — sodium — in the tissue (lymph sac) raises osmotic pressure, and water moves along the osmotic gradient from the vessels and the external medium into the tissue. This confirms the role of the **osmotic factor** in edema formation. By performing the same experiment with a protein solution (e.g. gelatin), the **colloid-osmotic (oncotic)** factor can also be demonstrated.

> **CLINICAL SIGNIFICANCE**
> The osmotic factor is used clinically in two ways: sodium retention aggravates edema (which is why table salt is restricted in cardiac and renal edema), while osmotically active agents (mannitol, hypertonic sodium chloride) are used in cerebral edema to draw water from the tissue into the vessels and reduce the edema.

7. Practical skills

Skill: determining the type of water balance disorder

1. From the history, identify the source of water and salt loss or excess intake (vomiting, diarrhea, polyuria, sweating, infusions).
2. Assess the clinical signs: thirst, mucous membranes, skin turgor, pulse, arterial pressure, diuresis, edema, neurological signs.
3. Determine plasma sodium and osmolarity (osmolarity ≈ 2 × Na + glucose + urea, mmol/l).
4. Assess hematocrit and total protein (increased in dehydration, decreased in hyperhydration).
5. Determine the type of disorder (dehydration or hyperhydration; iso-, hypo- or hyperosmolar).
6. Explain the movement of water between the intracellular and extracellular compartments and write a conclusion.

Skill: determining the type of edema and the leading factor

1. Determine the location and extent of the edema (legs, face, generalized, local).
2. Ask how the edema changes over time (worse in the morning or in the evening).
3. Assess skin color, temperature and whether pressure leaves a pit.
4. Identify heart, kidney and liver diseases and nutritional status.
5. Analyze laboratory values: blood albumin, urine protein, creatinine, liver tests.
6. Indicate the leading pathogenetic factor and write a conclusion.

Review questions

1. Into what compartments is water distributed in the body?
2. What makes up the daily water balance?
3. Which hormones and mechanisms regulate water and electrolyte balance?
4. Describe the types of dehydration by their causes and signs.
5. Why is impaired consciousness seen in hyperosmolar dehydration?
6. How does hypoosmolar hyperhydration (“water intoxication”) develop?

7. State the causes and ECG signs of hypo- and hyperkalemia.
8. What are the main clinical manifestations of hypocalcemia?
9. Explain the Starling forces.
10. List the pathogenetic factors of edema and give an example of each.
11. Explain the pathogenesis of cardiac edema.
12. How do nephrotic and nephritic edema differ?
13. How does ascites develop in liver cirrhosis?
14. What are the consequences of edema?
15. Why does edema develop when a hypertonic solution is introduced into the frog lymph sac?

Test questions

1. Approximately what percentage of body weight does total body water make up in an adult?
   A) 60 %
   B) 30 %
   C) 45 %
   D) 75 %
   E) 90 %

2. Which hormone increases water reabsorption when plasma osmolarity rises?
   A) Aldosterone
   B) Insulin
   C) Natriuretic peptide
   D) Renin
   E) Antidiuretic hormone

3. What happens to cells in hyperosmolar dehydration?
   A) The cells swell
   B) It does not change
   C) The cells break down
   D) Water leaves the cells and cellular dehydration develops
   E) Sodium enters the cells

4. What disorder develops after profuse vomiting when losses are replaced with plain water only?
   A) Hyperosmolar dehydration
   B) Isoosmolar hyperhydration
   C) Hypoosmolar dehydration
   D) Hyperosmolar hyperhydration
   E) Water balance is not disturbed

5. What is the main danger of “water intoxication”?
   A) Arterial hypotension
   B) Swelling of cells (neurons) and cerebral edema
   C) Hypernatremia
   D) Cellular dehydration
   E) Hypokalemia

6. Which ECG sign is typical of hypokalemia?
   A) Tall peaked T wave
   B) Flattening of the T wave and appearance of a U wave
   C) Widening of the QRS complex
   D) ST segment elevation
   E) Loss of the P wave

7. Which condition can cause hyperkalemia?
   A) Profuse vomiting
   B) Hyperaldosteronism
   C) Acute kidney failure and crush syndrome
   D) Insulin therapy
   E) Alkalosis

8. What is the leading pathogenetic factor of nephrotic edema?
   A) Hydrostatic
   B) Membranogenic
   C) Lymphogenic
   D) Osmotic
   E) Oncotic (hypoalbuminemia)

9. What is the leading factor of cardiac edema?
   A) Increased hydrostatic pressure in the capillaries
   B) Decreased plasma albumin
   C) Arrest of lymph flow
   D) Increased vascular permeability
   E) Increased protein synthesis in the tissue

10. What is the leading factor of arm edema after mastectomy and removal of the axillary lymph nodes?
    A) Oncotic
    B) Hydrostatic
    C) Osmotic
    D) Lymphogenic
    E) Membranogenic

11. What is the leading factor of allergic (Quincke's) edema?
    A) Membranogenic (increased vascular permeability)
    B) Oncotic
    C) Lymphogenic
    D) Hydrostatic
    E) Neurohumoral

12. What is the main cause of filtration at the arterial end of the capillary?
    A) High tissue hydrostatic pressure
    B) Capillary hydrostatic pressure exceeding plasma oncotic pressure
    C) High plasma oncotic pressure
    D) Increased lymph flow
    E) Absence of protein in the tissue

13. Which is a clinical manifestation of hypocalcemia?
    A) Polyuria and kidney stones
    B) Constipation
    C) Bradycardia
    D) Tetany and laryngospasm
    E) Muscle atony

14. When hypertonic sodium chloride solution is introduced into the frog lymph sac, which factor causes the edema?
    A) Hydrostatic
    B) Membranogenic
    C) Osmotic
    D) Lymphogenic
    E) Oncotic

15. Which factor does NOT contribute to the development of ascites in liver cirrhosis?
    A) Portal hypertension
    B) Hypoalbuminemia
    C) Secondary hyperaldosteronism
    D) Increased lymph formation
    E) Increased cardiac output

Clinical cases

Case 1

A 1-year-old child has had liquid stools 10–12 times a day and vomiting for 2 days. The mother gave the child only boiled water to drink. On examination: the child is lethargic, not thirsty, the skin fold returns slowly, the fontanelle is sunken, muscle cramps are observed. Plasma sodium is 124 mmol/l.

Questions: What water–electrolyte disorder has developed? Explain its pathogenesis. Why is there no thirst?

Answer: Hypoosmolar (salt-deficit) dehydration. Water and sodium were lost with diarrhea and vomiting; because the losses were replaced only with plain water, sodium became more deficient than water — hyponatremia and plasma hypoosmolarity developed. Water moves along the osmotic gradient into the cells: extracellular fluid (and circulating blood volume) decreases further, and the cells, especially neurons, swell — lethargy and convulsions. Thirst is mainly triggered by a rise in plasma osmolarity; here osmolarity has fallen, so there is no thirst. Treatment — rehydration with saline solutions (oral rehydration solution or intravenous isotonic solutions).

Case 2

3 weeks after tonsillitis, an 8-year-old child developed edema of the face and eyelids (especially in the morning), reduced urine output and “meat-washings”-colored urine. Arterial pressure 140/90 mm Hg. Urine: protein 1.2 g/l, many erythrocytes. Blood albumin is normal.

Questions: What type of edema is this? What are its leading factors? How does it differ from nephrotic edema?

Answer: Nephritic edema in acute glomerulonephritis. Leading factors: sodium and water retention due to reduced glomerular filtration (osmotic and neurohumoral factor — RAAS activation), as well as increased capillary permeability (membranogenic factor). Fluid retention has also led to arterial

hypertension. Difference from nephrotic edema: proteinuria is moderate and blood albumin is normal, i.e. the oncotic factor is not leading; hematuria and hypertension are present. In nephrotic edema, by contrast, daily proteinuria exceeds 3.5 g, hypoalbuminemia is marked and the edema quickly progresses to anasarca.

Case 3

A 55-year-old patient has chronic heart failure. Edema in the legs increases towards evening and decreases somewhat in the morning. The neck veins are distended and the liver is enlarged. Furosemide was prescribed; a week later the patient developed muscle weakness and a heart rhythm disturbance.

Questions: Explain the pathogenesis of the edema. What are the changes after taking furosemide related to?

Answer: Cardiac edema: in right ventricular failure venous pressure rises and capillary hydrostatic pressure increases (leading factor). Reduced cardiac output lowers renal perfusion, RAAS is activated, and aldosterone and ADH secretion increase — sodium and water are retained, circulating blood volume increases and venous pressure rises further (a vicious circle). Under gravity the edema is located in the dependent parts and increases towards evening. Furosemide excretes large amounts of potassium along with sodium and water — hypokalemia has developed; it causes muscle weakness and arrhythmias. Therefore, during diuretic treatment potassium must be monitored and, if needed, replaced, or potassium-sparing diuretics added.

Quick review

Table 9.10. *Key facts on Topic 9*

Question	Answer
Total water	60 % of body weight: 40 % intracellular, 20 % extracellular
Plasma osmolarity	280–295 mOsm/kg; main determinant — sodium
ADH	Increases water reabsorption; released when osmolarity rises
Aldosterone	Retains sodium, excretes potassium
Isoosmolar dehydration	Water and salt lost equally; hypovolemia
Hyperosmolar dehydration	Water deficit; cellular dehydration, thirst
Hypoosmolar dehydration	Salt deficit; cellular swelling, no thirst
Water intoxication	Hypoosmolar hyperhydration; cerebral edema
Hypokalemia	K^+ below 3.5 mmol/l; weakness, U wave
Hyperkalemia	K^+ above 5.5 mmol/l; tall T, risk of cardiac arrest
Starling forces	Balance of hydrostatic and oncotic pressures
Edema factors	Hydrostatic, oncotic, membranogenic, osmotic, lymphogenic, neurohumoral
Cardiac edema	Leading — hydrostatic; starts in the legs, worse in the evening
Nephrotic edema	Leading — oncotic (proteinuria, hypoalbuminemia)
Lymphedema	Leading — lymphogenic
Conclusion of the experiment	Accumulation of sodium in the tissue causes osmotic edema

TOPIC 10

Pathophysiology of carbohydrate metabolism. Hyper- and hypoglycemia, glucosuria

Learning objectives

To explain to the student the regulation of carbohydrate metabolism, the biological effects of insulin and counter-insulin hormones, the main manifestations of carbohydrate metabolism disorders (hyperglycemia, hypoglycemia, glucosuria) and their causes, mechanisms and consequences; to teach the student to model and analyze hyper- and hypoglycemia and glucosuria experimentally.

By the end of the session, the student should know:

- the normal values of blood glucose and the mechanisms that regulate it;
- the effect of insulin on metabolism and the counter-insulin hormones;
- the classification of carbohydrate metabolism disorders by stage (digestion, absorption, storage, intermediate metabolism);
- the types and mechanisms of hyperglycemia;
- the causes and mechanism of hypoglycemia and the signs of hypoglycemic coma;
- the types of glucosuria and the concept of the renal threshold;
- how to interpret the glucose tolerance test.

The student should be able to:

- induce hyperglycemia in a rabbit with epinephrine and determine blood glucose;
- induce hypoglycemia and hypoglycemic convulsions in a mouse with insulin and reverse them with glucose;
- detect glucose in urine qualitatively;
- interpret the results of a glucose tolerance test.

1. Carbohydrate metabolism and its regulation

Glucose is the main energy source for the body's cells; the brain obtains almost all its energy from glucose (partly from ketone bodies during prolonged fasting). Therefore blood glucose is kept within narrow limits.

Table 10.1. *Normal and diagnostic blood glucose values (capillary blood, mmol/l)*

State	Fasting	2 hours after a glucose load
Normal	3,3–5,5	below 7.8
Impaired fasting glycemia	5,6–6,0	below 7.8
Impaired glucose tolerance	below 6.1	7,8–11,0
Diabetes mellitus	6.1 and above	11.1 and above

CAUTION

In fasting venous plasma the thresholds are different: normal up to 6.1 mmol/l, diabetes mellitus — 7.0 mmol/l and above. When making a diagnosis, note which blood sample (capillary blood or venous plasma) was used. Glycated hemoglobin (HbA_{1c}) of 6.5 % or higher is also a criterion for diabetes mellitus.

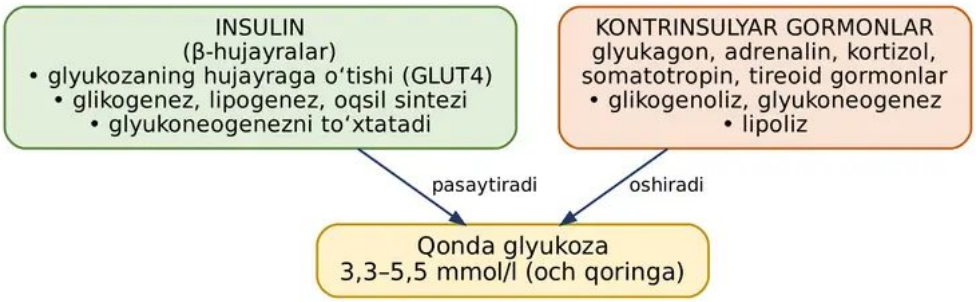


Figure 10.1. *Hormonal regulation of blood glucose*

Biological effects of insulin

Insulin is the only hypoglycemic hormone; it is produced in the beta cells of the pancreatic islets of Langerhans. Its secretion is stimulated mainly by glucose: glucose enters the beta cell via GLUT2 and is metabolized to form ATP, ATP-sensitive potassium channels close, the cell depolarizes, calcium enters and insulin granules are released.

Table 10.2. *Effects of insulin on metabolism*

Type of metabolism	Effect of insulin
Carbohydrates	Increases glucose entry into muscle and adipose tissue (moves GLUT4 transporters to the membrane); stimulates glycogen synthesis; inhibits glycogenolysis and gluconeogenesis; activates glycolysis
Fats	Stimulates lipogenesis, inhibits lipolysis; reduces ketone body formation
Proteins	Increases amino acid entry into cells and protein synthesis, reduces proteolysis (an anabolic hormone)
Electrolytes	Increases potassium entry into cells

Counter-insulin hormones raise blood glucose: **glucagon** (glycogenolysis and gluconeogenesis), **epinephrine** (glycogenolysis, reduced insulin secretion), **glucocorticoids** (gluconeogenesis, reduced glucose use by tissues), **growth hormone** and **thyroid hormones**.

2. Classification of carbohydrate metabolism disorders

Table 10.3. *Carbohydrate metabolism disorders by stage*

Stage	Disorder	Examples
Digestion and absorption	Deficiency of amylase or disaccharidases, impaired absorption	Lactase deficiency (milk intolerance), pancreatitis, malabsorption syndrome
Glycogen synthesis and breakdown	Impaired glycogen storage or mobilization	Glycogenoses (von Gierke disease — glucose-6-phosphatase deficiency; Pompe disease), liver cirrhosis
Intermediate metabolism	Impaired glycolysis, Krebs cycle and gluconeogenesis	Lactic acidosis in hypoxia, vitamin B_1 deficiency (pyruvate accumulation), galactosemia, fructosemia
Hormonal regulation	Imbalance between insulin and counter-insulin hormones	Diabetes mellitus, hypercortisolism, pheochromocytoma, insulinoma

CLINICAL SIGNIFICANCE

Lactase deficiency is a common condition in adults. Undigested lactose passes into the colon, where bacteria ferment it, producing gas and organic acids; because of the osmotic effect, water is drawn into the intestine. As a result, bloating, pain and diarrhea occur after drinking milk.

3. Hyperglycemia

Hyperglycemia — a rise in blood glucose above normal.

Table 10.4. *Types of hyperglycemia*

Type	Mechanism	Examples
Alimentary	Intake of a large amount of rapidly absorbed carbohydrate	After eating sweets; short-lived, physiological
Emotional (stress)	Activation of the sympathoadrenal system, release of epinephrine and increased glycogenolysis	Before an exam, pain, fear; “exam glucosuria”
Hormonal (endocrine)	Insulin deficiency or excess of counter-insulin hormones	Diabetes mellitus, Cushing syndrome, acromegaly, pheochromocytoma, thyrotoxicosis, glucagonoma
Neurogenic	Damage to the central nervous system, irritation of the hypothalamus	Head injury, stroke; C. Bernard's “sugar puncture” (needle puncture of the floor of the fourth ventricle)
Drug-induced	Glucocorticoids, thiazide diuretics, some antipsychotics	Long-term hormonal therapy

Consequences of hyperglycemia

- **Glucosuria and osmotic diuresis** — when glucose exceeds the renal threshold it passes into the urine and takes water with it: polyuria, dehydration, thirst (polydipsia).
- **Increased blood osmolarity** — dehydration of cells; in severe cases hyperosmolar coma.
- **Glycation of proteins** — glucose binds non-enzymatically to the amino groups of proteins (glycated hemoglobin, collagen, crystallin); the basis of the long-term complications.
- **Activation of the polyol (sorbitol) pathway** — in tissues that take up glucose independently of insulin (lens, nerves, kidney), sorbitol accumulates and damages the cells.

4. Hypoglycemia

Hypoglycemia — a fall in blood glucose below 3.3 mmol/l (clinically often below 3.9 mmol/l).

Table 10.5. *Causes of hypoglycemia*

Group	Examples
Excess insulin	Overdose of insulin or sulfonylurea drugs, insulinoma (beta-cell tumor)
Insufficient glucose intake	Prolonged fasting, malabsorption, a diabetic patient not eating after taking insulin
Sharp increase in glucose consumption	Heavy physical work, fever, some large tumors
Reduced glucose production	Liver failure, glycogenoses, alcohol (inhibits gluconeogenesis)

Group	Examples
Deficiency of counter-insulin hormones	Adrenocortical insufficiency (Addison disease), hypopituitarism

Signs of hypoglycemia

Hypoglycemia presents with two groups of signs:

1. **Adrenergic (autonomic) signs** — epinephrine release as a counter-insulin response: hunger, trembling, profuse sweating (cold sweat), tachycardia, palpitations, anxiety, pallor.
2. **Neuroglycopenic signs** — insufficient glucose supply to the brain: headache, impaired concentration, strange behavior (resembling drunkenness), visual disturbances, convulsions, loss of consciousness — **hypoglycemic coma**.

REMEMBER

Hypoglycemia is especially dangerous for the brain, because neurons do not store glucose and obtain their energy mainly from it. Prolonged severe hypoglycemia leads to irreversible changes in the brain. Hypoglycemic coma develops quickly (within minutes), the skin is moist, muscle tone is increased and there are convulsions — this distinguishes it from ketoacidotic coma, which develops slowly and in which the skin is dry.

CLINICAL SIGNIFICANCE

Care in hypoglycemia: if the patient is conscious — 15–20 g of rapidly absorbed carbohydrate (sugar, sweet juice); if unconscious — 40 % glucose solution intravenously or glucagon intramuscularly. In doubtful cases (when the type of coma is unclear) giving glucose is safer: it saves life in hypoglycemia and does not significantly worsen the condition in hyperglycemia.

5. Glucosuria

Glucosuria — the appearance of glucose in the urine. Normally, glucose filtered in the glomeruli is completely reabsorbed in the proximal tubules together with sodium (via the SGLT2 and SGLT1 transporters).

Renal threshold — the blood glucose concentration above which the tubular transporters become saturated and glucose begins to appear in the urine. In adults it is about **8.9–10 mmol/l**.

Table 10.6. *Types of glucosuria*

Type	Mechanism	Examples
Hyperglycemic (extrarenal)	Blood glucose exceeds the renal threshold	Diabetes mellitus, alimentary and emotional hyperglycemia, after epinephrine injection
Renal	Blood glucose is normal, but tubular reabsorption is impaired (the renal threshold is lowered)	Hereditary renal glucosuria, Fanconi syndrome, pregnancy, phlorizin poisoning, treatment with SGLT2 inhibitors

CLINICAL SIGNIFICANCE

SGLT2 inhibitors (empagliflozin, dapagliflozin) are modern drugs used to treat type 2 diabetes mellitus. They reduce glucose reabsorption in the proximal tubules, causing a “drug-induced renal glucosuria”, and thereby lower glycemia. These agents also have cardio- and renoprotective properties.

6. Glucose tolerance test

The glucose tolerance test assesses the body's ability to handle a glucose load.

Procedure: after 3 days of usual diet and an 8–14-hour overnight fast, fasting blood glucose is measured in the morning; the patient drinks 75 g of glucose dissolved in 250–300 ml of water within 5 minutes; after 2 hours (every 30 minutes in the extended version) glucose is measured again.

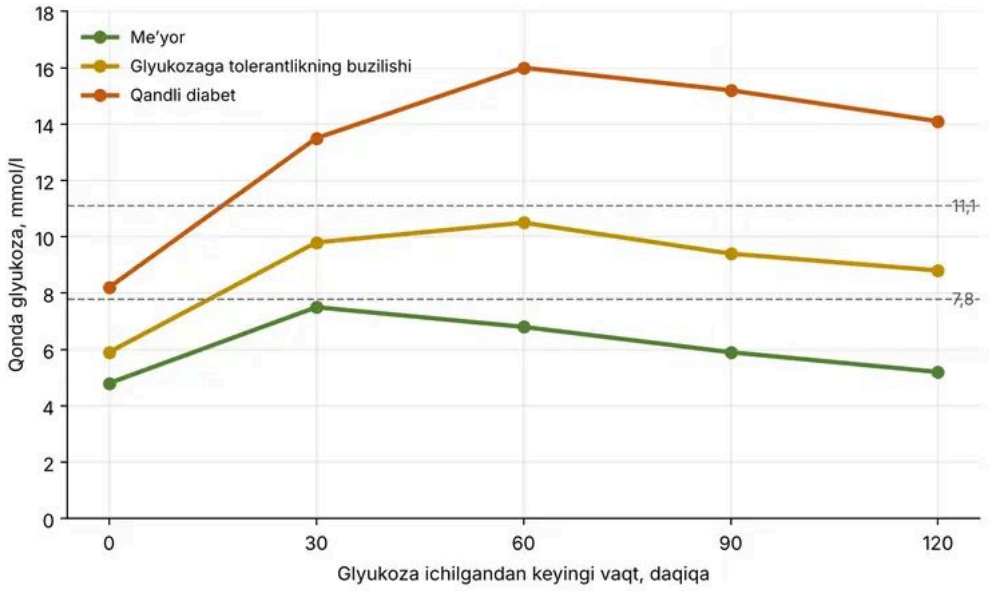


Figure 10.2. *Glucose tolerance test curves (approximate, drawn for explanation)*

Normally glucose peaks at 30–60 minutes (not exceeding 8–9 mmol/l) and returns to the baseline level by 2 hours (below 7.8 mmol/l). In impaired glucose tolerance the 2-hour value is 7.8–11.0 mmol/l, and in diabetes mellitus it is 11.1 mmol/l or higher.

7. Experimental part

Experiment 1: epinephrine hyperglycemia and glucosuria in the rabbit

1. Take blood from the rabbit's ear vein and determine the baseline glucose level (with a glucometer or the glucose oxidase method).
2. Place the rabbit in a metabolic cage or catheterize the bladder, and test the baseline urine for glucose (test strip or Nylander/Fehling test).
3. Inject 0.1 % epinephrine solution (0.1–0.2 ml/kg) subcutaneously.
4. Measure blood glucose again after 30, 60 and 90 minutes.
5. Once glycemia exceeds the renal threshold, test the urine for glucose.

Expected results and analysis: via beta2- and alpha-adrenoceptors in the liver, epinephrine stimulates glycogenolysis and gluconeogenesis and reduces insulin secretion — hyperglycemia develops.

When glucose exceeds the renal threshold, **hyperglycemic glucosuria** appears. This experiment is a model of emotional (stress) hyperglycemia.

Experiment 2: insulin hypoglycemia in the mouse

1. Fast two mice for 12–18 hours beforehand (their glycogen stores will be reduced).

2. Inject each of them intraperitoneally with insulin (at the dose given in the department's guidelines), and a third mouse with saline (control).
3. Observe the mice in a warm place and record the time at which signs appear: immobility, then restlessness, ruffled fur, loss of balance, convulsions, lying on the side.
4. As soon as convulsions begin, inject one mouse intraperitoneally with 0.5–1 ml of 40 % glucose solution; continue to observe the second one (with the teacher's permission if rescue is required).
5. Record the recovery of the glucose-treated mouse within a few minutes.

Expected results and analysis: insulin increases glucose uptake by tissues and stops glucose production in the liver; in the fasted mouse glycogen stores are low, so hypoglycemia develops quickly. The energy supply of neurons is impaired, and first excitation, then convulsions and coma develop. Giving glucose restores the energy supply of neurons and quickly rescues the animal — this proves that the cause of hypoglycemic convulsions is precisely the lack of glucose.

> **CAUTION**
> In the experiment, the fasting period of the animals and the insulin dose are chosen in strict accordance with the department's methodological guidelines. The glucose solution must be ready before the experiment begins.

Experiment 3: qualitative detection of glucose in urine

1. Pour 2–3 ml of urine into a test tube.
2. Dip a test strip (glucose oxidase method) into the urine and, after the specified time, compare its color with the scale.
3. Alternative method — the Nylander test: when the reagent is added to urine and boiled, a black precipitate forms if glucose is present.
4. Compare normal urine, urine with added glucose and the urine of the experimental animal.

8. Practical skills

Skill: interpreting the glucose tolerance test

1. Compare the fasting value with the normal range (3.3–5.5 mmol/l in capillary blood).
2. Determine the value 2 hours after drinking the glucose.
3. A 2-hour value below 7.8 mmol/l is normal; 7.8–11.0 — impaired glucose tolerance; 11.1 and above — diabetes mellitus.
4. Assess the shape of the curve: when and how high the peak is, and the time to return to the baseline level.
5. Write the result taking into account the type of blood sample (capillary blood or venous plasma).

Skill: distinguishing hypoglycemic and hyperglycemic coma

1. Determine the speed of onset: minutes (hypoglycemic) or hours to days (hyperglycemic).
2. Assess the skin: moist with cold sweat (hypoglycemic) or dry (ketoacidotic).
3. Assess breathing: normal, or deep noisy Kussmaul breathing with an acetone odor (ketoacidosis).
4. Check muscle tone and convulsions (increased tone and convulsions in hypoglycemia).
5. Measure glucose immediately with a glucometer.
6. Start first aid according to the diagnosis.

Review questions

1. What are the normal values of blood glucose?
2. How is insulin secretion regulated?
3. Explain the effects of insulin on carbohydrate, fat and protein metabolism.
4. Which hormones have counter-insulin effects?
5. At what stages can carbohydrate metabolism disorders occur?
6. How does diarrhea develop in lactase deficiency?
7. List the types of hyperglycemia and explain their mechanisms.
8. What are the consequences of hyperglycemia?
9. What are the causes of hypoglycemia?
10. List the adrenergic and neuroglycopenic signs of hypoglycemia.
11. Why is hypoglycemia especially dangerous for the brain?
12. Explain the types of glucosuria and the concept of the renal threshold.
13. How is the glucose tolerance test performed and interpreted?
14. What is the mechanism of epinephrine hyperglycemia?
15. What causes the convulsions in a mouse given insulin, and how are they reversed?

Test questions

1. What is the normal fasting glucose level in capillary blood?
   A) 1.5–2.5 mmol/l
   B) 6.1–7.0 mmol/l
   C) 3.3–5.5 mmol/l
   D) 7.8–11.0 mmol/l
   E) above 11.1 mmol/l

2. Which is the only hormone that lowers blood glucose?
   A) Glucagon
   B) Insulin
   C) Epinephrine (adrenaline)
   D) Cortisol
   E) Somatotropin

3. Through which transporter does insulin increase glucose entry into muscle and fat cells?
   A) GLUT4
   B) GLUT2
   C) SGLT2
   D) GLUT1
   E) Na^+/K^+ pump

4. Which hormone does NOT have a counter-insulin effect?
   A) Glucagon
   B) Epinephrine (adrenaline)
   C) Cortisol
   D) Somatotropin
   E) Insulin

5. To which type does hyperglycemia detected in a student before an exam belong?
   A) Alimentary
   B) Hormonal (diabetic)
   C) Neurogenic (head injury)
   D) Emotional (stress)
   E) Drug-induced

6. What is the approximate renal threshold for glucose in adults?
   A) 8.9–10 mmol/l
   B) 3.3–5.5 mmol/l
   C) 5.5–6.1 mmol/l
   D) 15–20 mmol/l
   E) above 25 mmol/l

7. What is the main sign of renal glucosuria?
   A) Blood glucose above 15 mmol/l
   B) There is no polyuria
   C) It occurs only in diabetes mellitus
   D) It is accompanied by insulin deficiency
   E) Glucose in the urine with normal blood glucose

8. Which is a neuroglycopenic sign of hypoglycemia?
   A) Profuse sweating
   B) Tachycardia
   C) Convulsions and impaired consciousness
   D) Trembling
   E) Hunger

9. Which sign distinguishes hypoglycemic coma from ketoacidotic coma?
   A) Dry skin
   B) Acetone odor
   C) Kussmaul breathing
   D) Rapid onset, moist skin and convulsions
   E) Polyuria

10. In a glucose tolerance test, capillary blood glucose 2 hours after drinking 75 g of glucose was 9.5 mmol/l. What does this mean?
    A) Normal
    B) Impaired glucose tolerance
    C) Diabetes mellitus
    D) Hypoglycemia
    E) Renal glucosuria

11. What is the main mechanism of epinephrine hyperglycemia?
    A) Increased intestinal absorption of glucose
    B) Increased hepatic glycogenolysis and gluconeogenesis, decreased insulin secretion
    C) Decreased renal reabsorption of glucose
    D) Increased glucose entry into muscles
    E) Increased number of insulin receptors

12. What is the mechanism of diarrhea after drinking milk in lactase deficiency?
    A) Secretory
    B) Reduced intestinal motility
    C) Inflammation
    D) Osmotic effect and fermentation of undigested lactose
    E) Allergic reaction

13. What is the non-enzymatic binding of proteins to glucose in hyperglycemia called?
    A) Phosphorylation
    B) Hydroxylation
    C) Methylation
    D) Acetylation
    E) Glycation

14. What is given first to a patient lying unconscious with hypoglycemia?
    A) Intravenous glucose solution or intramuscular glucagon
    B) Insulin
    C) Furosemide
    D) Sodium bicarbonate
    E) Epinephrine (adrenaline)

15. What does the cessation of convulsions after glucose injection in an insulin-treated mouse prove?
    A) The convulsions are due to potassium deficiency
    B) The convulsions are due to oxygen deficiency
    C) The cause of the convulsions is the lack of glucose supply to neurons
    D) The convulsions are due to acidosis
    E) The convulsions are due to an allergic reaction

Clinical cases

Case 1

A 22-year-old man has type 1 diabetes mellitus. In the morning he took his usual dose of insulin but went to the gym without having breakfast. During the workout he developed intense hunger, trembling, cold sweat and palpitations, then began to behave inappropriately and lost consciousness; brief convulsions were observed.

Questions: What condition has developed? Explain the mechanism of the signs. What is the first aid?

Answer: Hypoglycemic coma. Insulin was taken, but no glucose was taken in (he did not eat), and physical exertion sharply increased glucose consumption by the muscles — hypoglycemia developed. Hunger, trembling, cold sweat and tachycardia are adrenergic signs: in response to hypoglycemia the sympathoadrenal system is activated and epinephrine is released. Changed behavior, loss of consciousness and convulsions are neuroglycopenic signs: brain neurons do not store glucose, and their function is impaired by lack of energy. First aid: because he is unconscious, give nothing by mouth; give 40 % glucose solution intravenously or glucagon intramuscularly; once he regains consciousness, give carbohydrate food.

Case 2

Glucose was found in the urine of a 30-year-old pregnant woman. Fasting blood glucose is 4.6 mmol/l and the glucose tolerance test is normal.

Questions: Which type of glucosuria can be considered? Explain its mechanism.

Answer: Renal glucosuria. Blood glucose is normal and the tolerance test is not impaired, so there is no hyperglycemia. In pregnancy glomerular filtration increases and more glucose reaches the proximal tubules, while the tubular reabsorptive capacity is relatively reduced — the renal threshold falls and glucose appears in the urine even when blood levels are normal. Nevertheless, glycemia is still monitored to rule out diabetes of pregnancy (gestational diabetes).

Case 3

A 40-year-old patient with a head injury has blood glucose of 11.5 mmol/l and glucose in the urine. He had no previous diabetes mellitus. After two weeks glycemia returned to normal.

Questions: What type of hyperglycemia is this? Explain the mechanism. Which classic experiment models this condition?

Answer: Neurogenic (and stress) hyperglycemia. Brain injury irritates the autonomic centers in the hypothalamus and medulla oblongata and activates the sympathoadrenal system and the hypothalamic–pituitary–adrenal axis: epinephrine and cortisol release increases, and glycogenolysis and gluconeogenesis are enhanced. Because glycemia exceeded the renal threshold, glucosuria appeared. After the acute period of the injury, the hyperglycemia resolved. This condition is modeled by C. Bernard's “sugar puncture” experiment: when a needle is inserted into the floor of the fourth ventricle, the animal develops hyperglycemia and glucosuria.

Quick review

Table 10.7. *Key facts on Topic 10*

Question	Answer
Normal glycemia	Fasting capillary blood 3.3–5.5 mmol/l
Insulin	The only hypoglycemic hormone: GLUT4, glycogenesis, lipogenesis, protein synthesis
Counter-insulin hormones	Glucagon, epinephrine, cortisol, somatotropin, thyroid hormones
Types of hyperglycemia	Alimentary, emotional, hormonal, neurogenic, drug-induced
Hypoglycemia	Glucose below 3.3 mmol/l; adrenergic and neuroglycopenic signs
Hypoglycemic coma	Rapid onset, moist skin, convulsions; treatment — glucose
Renal threshold	8.9–10 mmol/l for glucose
Hyperglycemic glucosuria	Blood glucose above the renal threshold
Renal glucosuria	Blood glucose normal, reabsorption impaired
Glucose tolerance test	75 g glucose; at 2 hours below 7.8 — normal, 11.1 and above — diabetes
Glycation	Non-enzymatic binding of glucose to proteins (HbA_{1c})
Lactase deficiency	Osmotic diarrhea and bloating
Epinephrine hyperglycemia	Increased glycogenolysis and gluconeogenesis
Sugar puncture	C. Bernard: needle into the floor of the fourth ventricle — neurogenic hyperglycemia

TOPIC 11

Pathophysiology of acid–base balance. Acidosis and alkalosis

Learning objectives

To explain to the student the concept of acid–base balance (ABB), its main parameters, the buffer systems and physiological (lung, kidney) regulatory mechanisms, the types of ABB disorders (respiratory and nonrespiratory acidosis and alkalosis), their causes, compensatory mechanisms and consequences; to teach the student in practice to determine titratable acidity and ammonia in urine and the alkaline reserve in blood, and to interpret ABB parameters.

By the end of the session, the student should know:

- the main parameters of ABB and their normal values (pH, pCO_2, HCO_3^-, BE, anion gap);
- the blood buffer systems and their significance;
- the role of the lungs and kidneys in regulating ABB;
- the classification, causes and compensatory mechanisms of acidosis and alkalosis;
- the effects of ABB disorders on the body (heart, nervous system, electrolytes).

The student should be able to:

- determine and interpret titratable acidity and ammonia in urine;
- assess the alkaline reserve of blood;
- determine the type of disorder and the degree of compensation from blood gases and ABB parameters.

1. Acid–base balance and its parameters

Acid–base balance — the relative constancy of the hydrogen ion concentration (pH) in the internal environment of the body. Arterial blood pH is kept within very narrow limits — 7.35–7.45. A pH below 6.8 or above 7.8 is incompatible with life.

Acids are constantly produced in the body: the **volatile acid** — carbon dioxide (about 15,000 mmol per day, excreted by the lungs) — and **nonvolatile (fixed) acids** — sulfuric, phosphoric, lactic and keto acids (50–100 mmol per day, excreted by the kidneys).

Table 11.1. *Main ABB parameters (arterial blood)*

Parameter	Normal	Meaning
pH	7,35–7,45	Active reaction; the final result of the balance
pCO_2	35–45 mm Hg	Partial pressure of carbon dioxide — the respiratory component
HCO_3^- (standard bicarbonate)	22–26 mmol/l	Metabolic component
Base excess (BE)	from −2.3 to +2.3 mmol/l	Deviation of buffer bases from normal; negative — base deficit
Buffer bases (BB)	44–53 mmol/l	Sum of all buffer anions
Anion gap	8–16 mmol/l	$Na^+ - (Cl^- + HCO_3^-)$; unmeasured anions

pH depends on the ratio of carbon dioxide (respiratory) to bicarbonate (metabolic) (Henderson–Hasselbalch equation):

$pH = 6.1 + \lg ([HCO_3^-] / (0.03 \times pCO_2))$

Normally the ratio of HCO_3^- to dissolved CO_2 is **20 : 1**. As long as this ratio is maintained, pH stays normal even when the absolute values change — this is the basis of compensation.

2. Mechanisms maintaining ABB

Buffer systems

Buffer systems act within seconds and convert a strong acid or base into a weak acid or base.

Table 11.2. *Blood buffer systems*

Buffer system	Share of blood buffer capacity	Feature
Hemoglobin buffer system	about 75 % (in erythrocytes)	The strongest; linked to CO_2 transport
Bicarbonate buffer system (H_2CO_3 / HCO_3^-)	10–15 %	An “open” system: CO_2 is regulated by the lungs and HCO_3^- by the kidneys; in practice the most important and quickly adaptable
Protein buffer system	5–10 %	Plasma proteins are amphoteric
Phosphate buffer system	1–2 % (in blood)	Of great importance inside cells and in urine

REMEMBER

The largest share of blood buffer capacity is provided by the **hemoglobin** buffer system, but clinically the most important system, and the one that is assessed, is the **bicarbonate buffer system**, because both of its components are physiologically regulated (CO_2 — by the lungs, HCO_3^- — by the kidneys), and it is exactly this system that is measured in blood gas analysis.

Role of the lungs

The respiratory center is very sensitive to pH and pCO_2. In acidosis (increased H^+) pulmonary ventilation increases, more CO_2 is exhaled and pH rises; in alkalosis ventilation decreases. Pulmonary compensation starts within minutes to hours, but it is incomplete.

Role of the kidneys

The kidneys regulate ABB most powerfully, but slowly (over hours and days):

1. **Bicarbonate reabsorption** — almost all filtered HCO_3^- is reabsorbed (85 percent of it in the proximal tubules).
2. **Acidogenesis** — secretion of H^+ ions and their excretion bound to the phosphate buffer (**titratable acidity**).
3. **Ammoniogenesis** — ammonia (NH_3) is formed from glutamine in the tubular cells, binds H^+ and is excreted as ammonium (NH_4^+).
4. In these processes a new HCO_3^- is formed for each H^+ excreted and returned to the blood.

CLINICAL SIGNIFICANCE

In acidosis the kidneys can excrete up to 500 mmol of acid per day, with **ammoniogenesis** playing the main role, because it can increase many times over within a few days. Therefore, in kidney failure ammonia formation decreases and metabolic acidosis inevitably develops.

Other organs

Bone tissue (in prolonged acidosis, through calcium carbonate and phosphates — risk of osteoporosis), the liver (converts lactic acid to glucose, urea synthesis) and the stomach (HCl secretion) also take part in regulating ABB.

3. Classification of ABB disorders

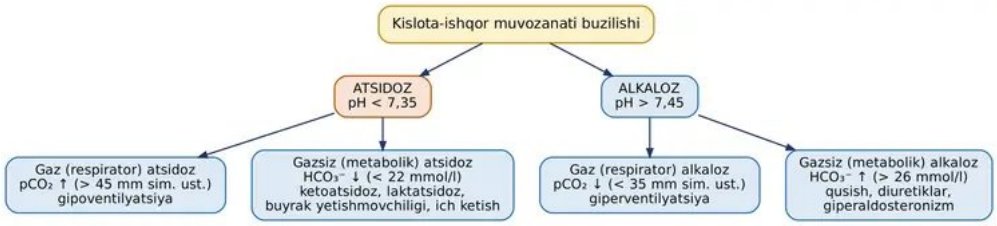


Figure 11.1. *Classification of ABB disorders*

By compensation:

- **compensated** — pH within the normal range (7.35–7.45), but pCO_2 and HCO_3^- are altered;
- **subcompensated and decompensated** — pH outside the normal range.

4. Acidosis

Respiratory (gas) acidosis

Cause — reduced elimination of CO_2 through the lungs (alveolar hypoventilation):

- depression of the respiratory center (opiates, barbiturates, anesthesia, head injury);
- damage to the respiratory muscles and neuromuscular apparatus (myasthenia, poliomyelitis, paralysis of the respiratory muscles);
- diseases of the lungs and airways (chronic obstructive pulmonary disease, severe asthma attack, pulmonary edema, extensive pneumonia);
- chest deformity, pneumothorax;
- breathing CO_2-rich air in a closed room.

Compensation: buffer systems (mainly intracellular and hemoglobin) and the kidneys — H^+ secretion and bicarbonate reabsorption increase, and blood HCO_3^- rises (in the chronic state this is fully developed in 3–5 days).

Nonrespiratory (metabolic) acidosis

Causes are divided into two groups according to the anion gap.

Table 11.3. *Causes of metabolic acidosis*

Group	Mechanism	Examples
High anion gap	Excess production or intake of acids	Ketoacidosis (diabetes mellitus, starvation, alcohol), lactic acidosis (hypoxia, shock, metformin), kidney failure (retention of sulfates and phosphates), poisoning with methanol, ethylene glycol, salicylates
Normal anion gap (hyperchloremic)	Loss of bicarbonate or impaired renal acid excretion	Diarrhea, intestinal fistulas, proximal and distal renal tubular acidosis, carbonic anhydrase inhibitors, infusion of large amounts of sodium chloride

MEMORY AID

Causes of high anion gap metabolic acidosis: **"MUDLES"** (a short form of the classic MUDPILES) — **M**ethanol, **U**remia (kidney failure), **D**iabetic and other ketoacidosis, **L**actic acidosis, **E**thylene glycol (antifreeze), **S**alicylates.

Compensation: buffer systems (HCO_3^- is consumed), **lungs** — hyperventilation, CO_2 elimination increases (deep, noisy **Kussmaul breathing** in severe ketoacidosis), **kidneys** — acidogenesis and ammoniogenesis increase (unless the kidneys themselves are the cause).

Effects of acidosis on the body

- **Cardiovascular system:** myocardial contractility decreases, vessels lose their sensitivity to catecholamines (hypotension), and the risk of arrhythmias increases.
- **Nervous system:** depression, weakness, drowsiness, coma in severe cases (especially in respiratory acidosis — CO_2 crosses the blood–brain barrier rapidly; "CO_2 narcosis").
- **Electrolytes: hyperkalemia** — H^+ enters the cells and K^+ leaves them; ionized calcium increases.
- **Oxyhemoglobin dissociation:** the curve shifts to the right (Bohr effect) — oxygen delivery to tissues is facilitated.
- **Bone:** in prolonged acidosis calcium leaves bone and osteoporosis develops.

5. Alkalosis

Respiratory (gas) alkalosis

Cause — alveolar hyperventilation and excessive elimination of CO_2:

- psychogenic hyperventilation (hysteria, fear, pain);
- hypoxemia (altitude, lung diseases, anemia) — reflex hyperventilation;
- brain injury and inflammation (stimulation of the respiratory center);
- fever, early sepsis, early stage of salicylate poisoning;
- liver failure, pregnancy (progesterone stimulates the respiratory center);
- incorrect settings of mechanical ventilation.

Compensation: the kidneys reduce bicarbonate reabsorption and H^+ secretion, and more HCO_3^- is excreted in the urine.

Nonrespiratory (metabolic) alkalosis

Causes:

- **loss of H^+** — profuse vomiting (HCl is lost with gastric juice), prolonged gastric drainage through a tube;
- **renal loss of H^+ and K^+** — thiazide and loop diuretics, hyperaldosteronism (Conn syndrome), Cushing syndrome;
- **hypokalemia** — K^+ leaves the cells, H^+ enters them;
- **excessive intake of alkali** — excessive administration of sodium bicarbonate, "milk-alkali syndrome".

Compensation: the lungs — hypoventilation (CO_2 retention), but limited by the risk of hypoxemia; the kidneys — increase bicarbonate excretion (provided there is no hypovolemia, hypokalemia or

hypochloremia).

Effects of alkalosis on the body

- **Increased neuromuscular excitability:** ionized calcium decreases (more of it binds to albumin) — paresthesias, muscle cramps, tetany (“obstetrician's hand”), laryngospasm.
- **Brain:** in respiratory alkalosis, hypocapnia constricts the cerebral vessels — dizziness, fainting.
- **Electrolytes: hypokalemia** (K^+ enters the cells and is lost through the kidneys), hypochloremia.
- **Oxyhemoglobin dissociation:** the curve shifts to the left — hemoglobin gives up oxygen to the tissues less readily.
- **Heart:** arrhythmias (especially together with hypokalemia).

Table 11.4. *Changes in the main parameters in ABB disorders*

Type of disorder	pH	pCO_2	HCO_3^-	BE	Compensation
Respiratory acidosis	Decreases	Primary increase	Compensatory increase	Towards positive	Kidneys: HCO_3^- retained
Metabolic acidosis	Decreases	Compensatory decrease	Primary decrease	Negative	Lungs: hyperventilation
Respiratory alkalosis	Increases	Primary decrease	Compensatory decrease	Towards negative	Kidneys: HCO_3^- excreted
Metabolic alkalosis	Increases	Compensatory increase	Primary increase	Positive	Lungs: hypoventilation

CAUTION

A rule of ABB analysis: **the primary change and the compensatory change go in the same direction** (pCO_2 and HCO_3^- both increase or both decrease). If they change in opposite directions (e.g. pCO_2 raised, HCO_3^- decreased), this indicates a **mixed disorder** (respiratory and metabolic acidosis together) — for example, in cardiac arrest.

6. Experimental part

Experiment 1: determining titratable acidity and ammonia in urine

Aim: to assess the role of the kidneys in regulating ABB.

Determining titratable acidity:

1. Put 10 ml of urine and 1–2 drops of phenolphthalein indicator into a flask.
2. Titrate with 0.1 N sodium hydroxide solution until a pale pink color appears.
3. Record the volume of alkali used and recalculate it for the daily urine volume.
4. Normal titratable acidity is about 10–30 mmol per day (part of the H^+ excreted in the urine).

Determining ammonia (formol titration method):

1. Add neutralized formalin solution to the titrated urine — ammonium ions react with formalin, an equivalent amount of acid is released and the pink color disappears.
2. Titrate again with 0.1 N sodium hydroxide until the pink color returns.
3. The volume of alkali used in the second titration corresponds to the amount of ammonium.
4. Normally about 30–50 mmol of ammonium is excreted in the urine per day.

Table 11.5. *Interpreting the experimental results*

State	Titratable acidity	Ammonium excretion
Normal	Normal	Normal
Metabolic acidosis (healthy kidneys)	Increased	Markedly increased
Kidney failure	Decreased	Decreased (despite acidosis)
Alkalosis	Decreased	Decreased

CLINICAL SIGNIFICANCE
If ammonium excretion in the urine is low in a patient with acidosis, the cause of the acidosis is related to the kidneys (renal tubular acidosis or kidney failure). If ammonium excretion is high, the kidneys are doing their job and the acidosis has developed for extrarenal reasons (e.g. diarrhea or ketoacidosis).

Experiment 2: determining the alkaline reserve of blood (Van Slyke method)

Alkaline reserve — the amount of CO_2 that blood plasma can bind as bicarbonate when saturated with carbon dioxide (pCO_2 = 40 mm Hg). Normally 50–65 vol % (22–29 mmol/l).

1. Saturate the plasma at a CO_2 pressure close to that of alveolar air.
2. In the Van Slyke apparatus, add acid to the plasma and measure the volume of CO_2 released from bicarbonate.
3. Correct the result for temperature and pressure and express it in volume percent.

Interpretation: in metabolic acidosis the alkaline reserve decreases (bicarbonate is used up neutralizing acids), and in metabolic alkalosis it increases. In modern practice, standard bicarbonate and BE measured on a blood gas analyzer are used instead of this parameter.

7. Practical skills

Skill: determining an ABB disorder from blood gases

1. **Assess pH:** below 7.35 — acidosis, above 7.45 — alkalosis; if normal, there may be a compensated or mixed disorder.
2. **Assess pCO_2:** if it has changed in the opposite direction to pH (pH decreased, pCO_2 increased), the primary disorder is respiratory.
3. **Assess HCO_3^- and BE:** if they have changed in the same direction as pH (pH decreased, HCO_3^- decreased), the primary disorder is metabolic.
4. **Assess compensation:** has the second component changed in the same direction as the first?
5. **In metabolic acidosis, calculate the anion gap:** $Na^+ - (Cl^- + HCO_3^-)$.
6. Compare with the clinical data and indicate the cause.

Skill: determining titratable acidity and ammonium in urine

1. Measure the volume of the urine sample accurately.
2. Perform the first titration with sodium hydroxide solution and phenolphthalein.
3. Add neutralized formalin and perform the second titration.
4. Recalculate both values for the daily urine.
5. Compare the result with the patient's ABB status and assess renal compensation.

Review questions

1. What is acid–base balance? What is the normal pH of arterial blood?
2. List the main ABB parameters and their normal values.
3. Explain the meaning of the Henderson–Hasselbalch equation.
4. List the blood buffer systems and state their significance.
5. How do the lungs regulate ABB?
6. Explain the mechanisms by which the kidneys regulate ABB.
7. How are ABB disorders classified?
8. What are the causes and compensatory mechanisms of respiratory acidosis?
9. List the causes of metabolic acidosis with high and normal anion gap.
10. In what condition and why does Kussmaul breathing develop?
11. What are the effects of acidosis on the cardiovascular system and electrolytes?
12. List the causes of respiratory alkalosis.
13. What ABB disorder develops with profuse vomiting, and why?
14. Why can tetany develop in alkalosis?
15. What is the clinical significance of determining titratable acidity and ammonia in urine?

Test questions

1. What is the normal pH of arterial blood?
   A) 7,0–7,2
   B) 6,8–7,0
   C) 7,5–7,6
   D) 7,35–7,45
   E) 7,8–8,0

2. Which buffer system provides the largest share of blood buffer capacity?
   A) Bicarbonate
   B) Phosphate
   C) Hemoglobin
   D) Protein (plasma)
   E) Ammonia

3. What is the normal ratio of HCO_3^- to dissolved CO_2 in the bicarbonate buffer system?
   A) 1 : 1
   B) 10 : 1
   C) 1 : 20
   D) 5 : 1
   E) 20 : 1

4. Which renal acid excretion mechanism increases most in acidosis?
   A) Glucose reabsorption
   B) Ammoniogenesis
   C) Potassium secretion
   D) Decreased sodium reabsorption
   E) Water reabsorption

5. What ABB disorder develops in a patient poisoned with morphine?
   A) Respiratory acidosis
   B) Respiratory alkalosis
   C) Metabolic alkalosis
   D) Metabolic acidosis
   E) ABB is not disturbed

6. What is the compensatory mechanism in diabetic ketoacidosis?
   A) Hypoventilation
   B) Increased renal excretion of bicarbonate
   C) Vomiting
   D) Decreased ammoniogenesis
   E) Hyperventilation (Kussmaul breathing)

7. Which is a cause of normal anion gap metabolic acidosis?
   A) Diarrhea (bicarbonate loss)
   B) Diabetic ketoacidosis
   C) Lactic acidosis
   D) Methanol poisoning
   E) Uremia

8. What ABB disorder develops with profuse vomiting?
   A) Metabolic acidosis
   B) Respiratory acidosis
   C) Respiratory alkalosis
   D) Metabolic alkalosis
   E) Mixed acidosis

9. How does potassium metabolism change in acidosis?
   A) Hypokalemia develops
   B) Hyperkalemia develops (K^+ leaves the cells)
   C) Potassium does not change
   D) Potassium is excreted in large amounts in the urine and falls in the blood
   E) Potassium accumulates in bone

10. What disorder develops as a result of hyperventilation during a hysterical attack?
    A) Respiratory acidosis
    B) Metabolic acidosis
    C) Respiratory alkalosis
    D) Metabolic alkalosis
    E) Mixed acidosis

11. What is the cause of tetany in alkalosis?
    A) Decreased blood glucose
    B) Increased sodium
    C) Decreased ionized calcium
    D) Increased potassium
    E) Decreased bicarbonate

12. Arterial blood: pH 7.25, pCO_2 30 mm Hg, HCO_3^- 13 mmol/l. What is the disorder?
    A) Uncompensated respiratory acidosis
    B) Respiratory alkalosis
    C) Metabolic alkalosis
    D) Mixed acidosis
    E) Partially compensated metabolic acidosis

13. What is urinary ammonium excretion like in a patient with acidosis due to kidney failure?
    A) Decreased
    B) Markedly increased
    C) Normal
    D) First increases, then does not change
    E) Increases only at night

14. How does the oxyhemoglobin dissociation curve shift in alkalosis?
    A) To the right
    B) It does not shift
    C) Upwards
    D) To the left (oxygen is given up to the tissues less readily)
    E) Downwards

15. In which condition does the alkaline reserve decrease?
    A) Profuse vomiting
    B) Metabolic acidosis
    C) Hyperaldosteronism
    D) Excessive administration of sodium bicarbonate
    E) Hypokalemia

Clinical cases

Case 1

A 19-year-old girl with type 1 diabetes mellitus has not taken insulin for 2 days. Nausea, vomiting, abdominal pain, deep noisy breathing, acetone odor on the breath. Blood: glucose 26 mmol/l, pH 7.12, pCO_2 22 mm Hg, HCO_3^- 7 mmol/l, BE −20 mmol/l, Na^+ 136, Cl^- 100 mmol/l, K^+ 5.6 mmol/l.

Questions: Determine the type of ABB disorder. Calculate and interpret the anion gap. Explain the changes in breathing and potassium.

Answer: Decompensated metabolic acidosis (pH 7.12; HCO_3^- and BE markedly decreased) with partial respiratory compensation (pCO_2 22 mm Hg — hyperventilation). Anion gap = 136 − (100 + 7) = 29 mmol/l — increased: in insulin deficiency, lipolysis and the formation of ketone bodies (acetoacetic and beta-hydroxybutyric acids) are sharply increased, and these are unmeasured anions. Deep noisy Kussmaul breathing is compensatory hyperventilation resulting from stimulation of the respiratory center by H^+; CO_2 elimination increases. Hyperkalemia — in acidosis H^+ enters the cells and K^+ leaves them; in addition, in insulin deficiency potassium does not move into the cells. During treatment with insulin, potassium returns to the cells and there is a risk of hypokalemia, so potassium is monitored.

Case 2

A 45-year-old patient has had profuse vomiting for 1 week due to narrowing of the gastric outlet (pyloric stenosis). The patient is weak and has cramps in the hands (“obstetrician's hand”). Blood: pH 7.52, pCO_2 48 mm Hg, HCO_3^- 36 mmol/l, K^+ 2.9 mmol/l, Cl^- 85 mmol/l.

Questions: Determine the type of ABB disorder. Explain its pathogenesis and the cause of the cramps.

Answer: Metabolic alkalosis (pH and HCO_3^- increased) with partial respiratory compensation (pCO_2 48 — hypoventilation). With vomiting, HCl (H^+ and Cl^-) is lost in the gastric juice — hypochloremia. Dehydration and hypovolemia increase aldosterone secretion: the kidneys retain sodium and excrete K^+ and H^+ — hypokalemia and maintenance of the alkalosis. In hypokalemia K^+ leaves the cells and H^+ enters them, which deepens the alkalosis. The cramps (“obstetrician's hand”) are due to increased neuromuscular excitability, because in alkalosis more calcium binds to albumin and ionized calcium decreases. Treatment — restore water and electrolyte balance with sodium chloride and potassium chloride solutions.

Case 3

A 62-year-old patient has had chronic obstructive pulmonary disease for many years. His condition is stable. Blood: pH 7.37, pCO_2 60 mm Hg, HCO_3^- 34 mmol/l.

Questions: Determine the type of ABB disorder. Why is the pH normal? Why is it dangerous to give such a patient high-concentration oxygen?

Answer: Compensated chronic respiratory acidosis. The primary disorder is a rise in pCO_2 (60 mm Hg) due to hypoventilation. Over several days the kidneys have increased H^+ secretion and bicarbonate reabsorption, raising HCO_3^- to 34 mmol/l; the HCO_3^-/CO_2 ratio has approached 20 : 1, so the pH is within the normal range. In such patients the respiratory center has adapted to prolonged hypercapnia and lost its sensitivity to CO_2, and breathing is driven mainly by hypoxemia (via the carotid bodies). High-concentration oxygen removes the hypoxic stimulus, reduces ventilation further and creates the risk of a sharp rise in pCO_2, decompensation of the respiratory acidosis and hypercapnic coma. Therefore oxygen is given to them under monitoring, at a low flow rate.

Quick review

Table 11.6. *Key facts on Topic 11*

Question	Answer
Arterial blood pH	7,35–7,45
pCO_2	35–45 mm Hg — respiratory component
HCO_3^-	22–26 mmol/l — metabolic component
BE	from −2.3 to +2.3 mmol/l
Anion gap	$Na^+ - (Cl^- + HCO_3^-) = 8–16$ mmol/l
Key ratio	$HCO_3^- : CO_2 = 20 : 1$
Strongest buffer	Hemoglobin; clinically most important — bicarbonate
Renal mechanisms	Bicarbonate reabsorption, acidogenesis (titratable acidity), ammoniogenesis
Respiratory acidosis	pCO_2 increased; hypoventilation (opiates, COPD)
Metabolic acidosis	HCO_3^- decreased; ketoacidosis, lactic acidosis, uremia, diarrhea

Question	Answer
Kussmaul breathing	Compensatory deep breathing in metabolic acidosis
Respiratory alkalosis	pCO_2 decreased; hyperventilation (hysteria, altitude)
Metabolic alkalosis	HCO_3^- increased; vomiting, diuretics, hyperaldosteronism
Potassium in acidosis	Hyperkalemia
In alkalosis	Hypokalemia, decreased ionized Ca^{2+}, tetany
Alkaline reserve	50–65 vol %; decreases in metabolic acidosis

TOPIC 12

Main metabolic diseases: diabetes mellitus, atherosclerosis, obesity, metabolic syndrome (seminar)

Learning objectives

To explain to the student the etiology, pathogenesis, main manifestations and complications of the most common metabolic diseases — diabetes mellitus, atherosclerosis, obesity and metabolic syndrome; to teach the student to distinguish the types of diabetic coma and to assess lipid parameters and body mass index.

By the end of the session, the student should know:

- the classification of diabetes mellitus and the etiology and pathogenesis of types 1 and 2;
- the metabolic disorders in diabetes mellitus and the mechanisms of its clinical signs;
- the pathogenesis of diabetic comas (ketoacidotic, hyperosmolar, lactic acidotic, hypoglycemic);
- the mechanisms of the late complications of diabetes mellitus (micro- and macroangiopathies, neuropathy);
- the definition, risk factors and pathogenesis of atherosclerosis;
- the types, causes and pathogenesis of obesity;
- the concept of metabolic syndrome and its criteria.

The student should be able to:

- distinguish the types of diabetic coma by clinical and laboratory signs;
- assess lipid profile parameters and the atherogenic index;
- determine the degree of obesity from body mass index and waist circumference.

1. Diabetes mellitus: definition and classification

Diabetes mellitus — a group of metabolic diseases characterized by chronic hyperglycemia resulting from impaired insulin secretion, impaired insulin action, or both. It is accompanied by disturbed carbohydrate, fat and protein metabolism and damage to the vessels, nervous system, kidneys and eyes.

Table 12.1. *Classification of diabetes mellitus (WHO)*

Type	Main mechanism	Features
Type 1	Destruction of beta cells (mainly autoimmune) → absolute insulin deficiency	Usually in childhood and youth; acute onset; prone to ketoacidosis; insulin is essential for life
Type 2	Insulin resistance + secretory dysfunction of beta cells (relative insulin deficiency)	About 90 percent of diabetes; usually after 40 years of age, with obesity; gradual onset
Gestational diabetes	Insulin resistance in pregnancy (due to placental hormones)	Usually resolves after delivery; later risk of type 2 is high
Other specific types	Genetic defects of beta-cell function (MODY), diseases of the pancreas, endocrinopathies, drugs	Pancreatitis, Cushing syndrome, acromegaly, glucocorticoids

Pathogenesis of type 1 diabetes mellitus

1. **Genetic predisposition** — certain alleles of the HLA system (DR3, DR4).
2. **Triggering factor** — viral infections (Coxsackie B, rubella, mumps), toxins.
3. **Autoimmune process** — autoantibodies and cytotoxic T lymphocytes against beta-cell antigens (insulin, glutamate decarboxylase, tyrosine phosphatase) are formed; insulitis develops.
4. Clinical signs appear **after 80–90 percent of beta cells have died**.

Pathogenesis of type 2 diabetes mellitus

1. **Insulin resistance** — reduced sensitivity of target tissues (liver, muscle, fat) to insulin; its main cause is abdominal (visceral) obesity: visceral adipose tissue releases large amounts of free fatty acids and inflammatory cytokines (TNF-alpha, IL-6), and adiponectin decreases; these disrupt post-receptor insulin signaling.
2. At first the beta cells overcome the resistance by producing more insulin (compensatory hyperinsulinemia).
3. Later **beta-cell dysfunction** (gluco- and lipotoxicity, amylin deposition) develops, insulin secretion becomes insufficient and hyperglycemia appears.

Table 12.2. *Differences between type 1 and type 2 diabetes mellitus*

Feature	Type 1	Type 2
Age at onset	Usually before 30 years	Usually after 40 years
Body weight	Often reduced	Often excessive (obesity)
Beginning	Acute, rapid	Gradual, asymptomatic for years
Tendency to ketoacidosis	High	Low
Blood insulin and C-peptide	Very low or absent	Normal or increased (at first)
Autoantibodies to beta cells	Present	No
Hereditary predisposition	Moderate	Strong (high concordance in twins)
Treatment	Insulin	Diet, physical activity, oral agents, later insulin

2. Metabolic disorders and clinical signs in diabetes mellitus

Table 12.3. *Main signs of diabetes mellitus and their mechanisms*

Feature	Mechanism
Hyperglycemia	Reduced glucose uptake by tissues, increased glycogenolysis and gluconeogenesis
Glucosuria	Glycemia exceeding the renal threshold (8.9–10 mmol/l)
Polyuria	Osmotic diuresis due to glucose
Polydipsia (thirst)	Dehydration and increased plasma osmolarity
Polyphagia (hunger) and weight loss (in type 1)	“Energy starvation” of cells, increased lipolysis and proteolysis
Ketonemia and ketonuria	Conversion of fatty acids released by lipolysis into ketone bodies in the liver
Hyperlipidemia	Increased lipolysis, decreased lipoprotein lipase activity

Feature	Mechanism
Susceptibility to infections, slow wound healing	Impaired leukocyte function, microangiopathy

MEMORY AID
The “three P's” of diabetes mellitus: **P**olyuria, **P**olydipsia, **P**olyphagia. The fourth is weight loss (mainly in type 1).

3. Diabetic comas

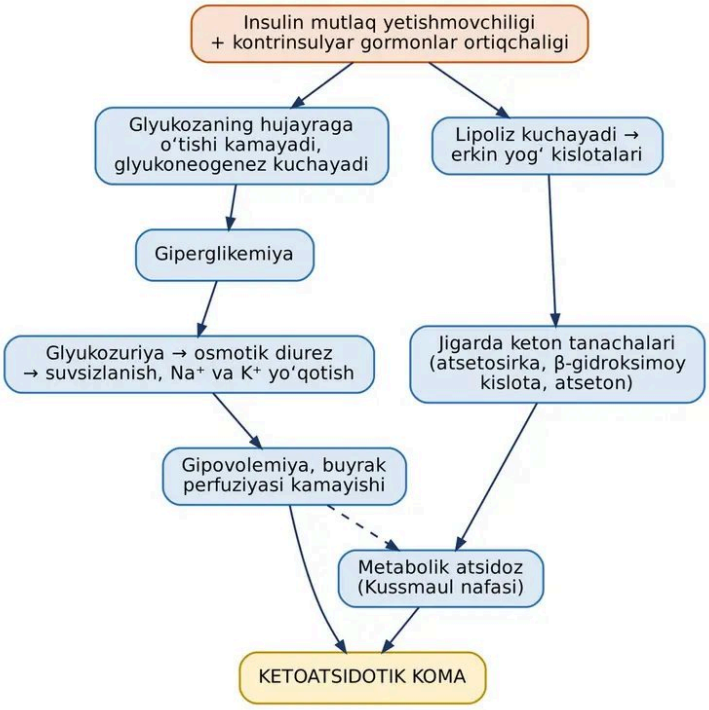


Figure 12.1. *Pathogenesis of diabetic ketoacidosis (based on Novitskiy V.V., 2009, figure 12.29)*

Table 12.4. *Comparative characteristics of diabetic comas*

Feature	Ketoacidotic	Hyperosmolar	Lactic acidotic	Hypoglycemic
Main type of diabetes	Type 1	Type 2, elderly	Type 2 (with metformin, hypoxia)	Those taking insulin or sulfonylureas
Onset	Hours, days	Days, weeks	Hours	Minutes
Glycemia	High (15–30 mmol/l)	Very high (30–50 mmol/l or more)	Normal or slightly raised	below 3.3 mmol/l
Ketone bodies	Many	Absent or few	No	No
Plasma osmolarity	Slightly raised	Very high (more than 330 mOsm/kg)	Normal	Normal
pH	Decreased	Normal	Markedly decreased	Normal
Breathing	Kussmaul, acetone odor	Rapid, shallow	Kussmaul	Normal
Skin	Dry	Very dry	Dry, cold	Moist

Feature	Ketoacidotic	Hyperosmolar	Lactic acidotic	Hypoglycemic
Main pathogenetic factor	Absolute insulin deficiency, ketoacidosis	Severe hyperglycemia and dehydration, no ketosis (a small amount of insulin is enough to suppress lipolysis)	Accumulation of lactic acid	Lack of glucose supply to neurons

CAUTION

There is no ketoacidosis in hyperosmolar coma, because in type 2 diabetes the residual insulin is enough to suppress lipolysis and ketogenesis in the liver, but not enough to ensure glucose uptake by the tissues. The main dangers in this coma are severe dehydration, hypernatremia and thrombosis; mortality is higher than in ketoacidotic coma.

4. Late complications of diabetes mellitus

The main cause of late complications is prolonged hyperglycemia. The main mechanisms are:

1. **Non-enzymatic glycation of proteins** and accumulation of advanced glycation end products (AGE) — basement membranes thicken, the vessel wall stiffens, and inflammation and oxidative stress are enhanced via AGE receptors.
2. **Activation of the polyol (sorbitol) pathway** — in cells that take up glucose independently of insulin (Schwann cells, lens, endothelium), sorbitol accumulates, causing osmotic damage and NADPH deficiency (weakened antioxidant defense).
3. **Activation of protein kinase C** — impaired vascular permeability and contractility.
4. **Oxidative stress** and endothelial dysfunction.

Table 12.5. *Late complications of diabetes mellitus*

Complication	Manifestation	Outcome
Diabetic retinopathy (microangiopathy)	Microaneurysms, hemorrhages, growth of new vessels	One of the main causes of blindness
Diabetic nephropathy (microangiopathy)	Thickening of the glomerular basement membrane, microalbuminuria, then proteinuria	Chronic kidney failure (requiring hemodialysis)
Diabetic neuropathy	Distal symmetric polyneuropathy (“stocking and glove” type), autonomic neuropathy	Loss of sensation, painless ulcers, orthostatic hypotension, gastroparesis
Macroangiopathy	Accelerated atherosclerosis	Ischemic heart disease, stroke, obliterative disease of the leg arteries
Diabetic foot syndrome	Neuropathy + angiopathy + infection	Trophic ulcers, gangrene, amputation

CLINICAL SIGNIFICANCE

Glycated hemoglobin (HbA_{1c}) reflects the average glycemia over the last 2–3 months (the erythrocyte lifespan is about 120 days). Large clinical trials have shown that keeping HbA_{1c} below 7 percent significantly reduces the risk of microangiopathic complications.

5. Atherosclerosis

Atherosclerosis — a chronic disease characterized by accumulation of lipids (mainly cholesterol) in the intima of large and medium-sized arteries of the elastic and muscular-elastic type, connective tissue overgrowth and formation of atherosclerotic plaques. It underlies ischemic heart disease, stroke and peripheral arterial disease — the leading causes of death worldwide.

Lipoproteins and their role

Table 12.6. *Blood lipoproteins*

Lipoprotein	Main function	Atherogenicity
Chylomicrons	Transport dietary triglycerides from the intestine	Few
Very low-density lipoproteins (VLDL)	Transport triglycerides formed in the liver	Moderate (their remnants are atherogenic)
Low-density lipoproteins (LDL)	Transport cholesterol from the liver to the tissues	High (“bad cholesterol”)
High-density lipoproteins (HDL)	Return cholesterol from the tissues to the liver	Antiatherogenic (“good cholesterol”)

Table 12.7. *Risk factors*

Non-modifiable	Modifiable
Age (men over 45, women over 55)	Dyslipidemia (increased LDL, decreased HDL)
Male sex (estrogen protection in women before menopause)	Arterial hypertension
Hereditary predisposition, familial hypercholesterolemia	Smoking
	Diabetes mellitus
	Abdominal obesity, physical inactivity
	Unhealthy diet, chronic stress

Pathogenesis of atherosclerosis

According to the modern view, atherosclerosis is a **chronic inflammatory process** that develops in response to lipid accumulation in the vessel wall (the “response to injury” theory).

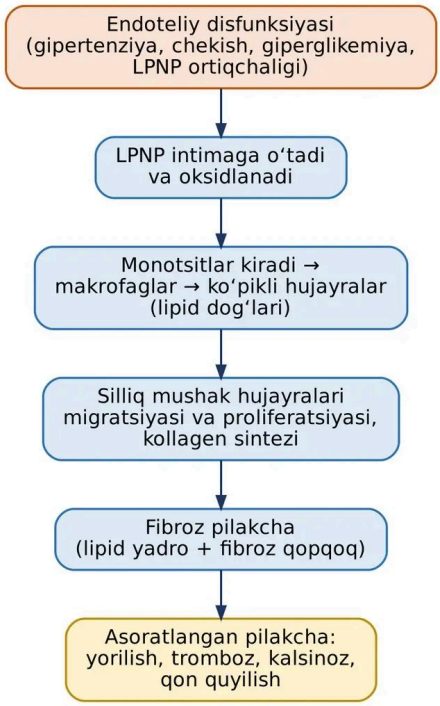


Figure 12.2. *Stages of atherosclerotic plaque formation (based on Kumar V. et al., 2015)*

REMEMBER

The most dangerous type of atherosclerotic plaque is the **unstable plaque**: a large lipid core, a thin fibrous cap and many macrophages. Its rupture leads to platelet adhesion and thrombus formation — this is the main mechanism of acute coronary syndrome (unstable angina, myocardial infarction) and ischemic stroke.

6. Obesity

Obesity — excess accumulation of adipose tissue in the body that poses a risk to health. The main cause is a long-term excess of energy intake over expenditure.

Table 12.8. *Assessment by body mass index (BMI = weight (kg) / height (m)^2) (WHO)*

BMI, kg/m^2	Assessment
below 18.5	Underweight
18,5–24,9	Normal weight
25,0–29,9	Overweight
30,0–34,9	Class I obesity
35,0–39,9	Class II obesity
40 or more	Class III (morbid) obesity

Table 12.9. *Types of obesity*

Basis of classification	Types
Origin	Primary (alimentary-constitutional — more than 90 percent); secondary (hypothalamic, endocrine — hypothyroidism, Cushing syndrome; genetic syndromes; drug-induced)
Fat distribution	Abdominal (android, “apple” type) — high metabolic and cardiovascular risk; gluteofemoral (gynoid, “pear” type)

Basis of classification	Types
Structure of adipose tissue	Hypertrophic (adipocytes enlarge — in adults); hyperplastic (the number of adipocytes increases — when onset is in childhood)

Regulation of appetite and body weight: the arcuate nucleus of the hypothalamus regulates appetite. **Leptin** (secreted by adipocytes) and insulin give the satiety signal and reduce appetite; **ghrelin** (stomach) increases appetite. Most obese people have high leptin levels but reduced sensitivity to it (**leptin resistance**).

CLINICAL SIGNIFICANCE

A simple method of assessing abdominal obesity is **waist circumference**: the risk increases if it exceeds 94 cm in men and 80 cm in women (102 and 88 cm respectively in some guidelines). Visceral adipose tissue is metabolically active: free fatty acids and inflammatory cytokines from it pass directly to the liver via the portal vein and increase insulin resistance.

7. Metabolic syndrome

Metabolic syndrome — the combination of abdominal obesity, insulin resistance, dyslipidemia, arterial hypertension and impaired glucose metabolism. It is based on insulin resistance and compensatory hyperinsulinemia; the syndrome sharply increases the risk of type 2 diabetes mellitus and cardiovascular disease.

Table 12.10. *Criteria for metabolic syndrome (International Diabetes Federation, 2005)*

Criterion	Value
Abdominal obesity (mandatory)	Waist circumference: more than 94 cm in men, more than 80 cm in women (for Europeans)
+ at least two of the following:	
Triglycerides	1.7 mmol/l or higher
HDL cholesterol	Below 1.03 mmol/l in men, below 1.29 mmol/l in women
Arterial pressure	130/85 mm Hg or higher, or antihypertensive treatment
Fasting glucose	5.6 mmol/l or higher, or type 2 diabetes mellitus

CAUTION

The criteria for metabolic syndrome differ slightly between international organizations (WHO, NCEP ATP III, IDF): for example, in NCEP ATP III abdominal obesity is not mandatory and the waist circumference thresholds are higher. When making the diagnosis, the criteria used should be stated.

8. Seminar part: clinical and pathophysiological analysis

Assessing the lipid profile

Table 12.11. *Optimal lipid profile values (for the general population, mmol/l)*

Parameter	Optimal value
Total cholesterol	below 5.0
LDL cholesterol	below 3.0 (a much lower target in high-risk patients)

Parameter	Optimal value
HDL cholesterol	Above 1.0 in men, above 1.2 in women
Triglycerides	below 1.7

Atherogenic index (A.N. Klimov) = (total cholesterol − HDL cholesterol) / HDL cholesterol. Normally below 3; if above 4, the risk of atherosclerosis is high.

Discussion questions for the seminar

1. Why does the patient lose weight in type 1 diabetes, while in type 2 diabetes they are often obese?
2. Why does ketoacidosis develop mainly in type 1 diabetes?
3. Through what mechanisms does abdominal obesity lead to insulin resistance?
4. How does diabetes mellitus accelerate atherosclerosis?
5. Acting on which component of the metabolic syndrome also improves the others?

9. Practical skills

Skill: determining the type of diabetic coma

1. Clarify the history: type of diabetes, treatment, last insulin dose and meal, infection, metformin intake.
2. Assess the speed of onset (minutes or days).
3. Check skin moisture, type of breathing and acetone odor on the breath.
4. Measure blood glucose immediately with a glucometer.
5. Determine ketone bodies (blood or urine), blood gases and pH, and lactate.
6. Calculate plasma osmolarity: $2 \times (Na^+ + K^+)$ + glucose + urea.
7. Determine the type of coma on the basis of the data obtained.

Skill: assessing the degree of obesity and metabolic risk

1. Measure height (m) and weight (kg) and calculate BMI.
2. Measure waist circumference at the level midway between the lowest rib and the iliac crest.
3. Measure arterial pressure.
4. Assess fasting glucose and the lipid profile.
5. Check the criteria for metabolic syndrome.
6. Calculate the atherogenic index and write a conclusion.

Review questions

1. How is diabetes mellitus classified?
2. Explain the etiology and pathogenesis of type 1 diabetes mellitus.
3. How does insulin resistance develop in type 2 diabetes mellitus?
4. Explain the main clinical signs of diabetes mellitus and their mechanisms.
5. How does ketoacidotic coma develop?
6. How does hyperosmolar coma differ from ketoacidotic coma?
7. Under what conditions does lactic acidotic coma develop?

8. Through what mechanisms do the late complications of diabetes mellitus develop?
9. What are the types of lipoproteins and their atherogenicity?
10. List the risk factors for atherosclerosis.
11. How is an atherosclerotic plaque formed?
12. What is an unstable plaque and why is it dangerous?
13. How is obesity classified? How is BMI calculated?
14. What is the role of leptin and ghrelin in regulating appetite?
15. What is metabolic syndrome and what are its criteria?

Test questions

1. What is the main pathogenetic mechanism of type 1 diabetes mellitus?
   A) Insulin resistance
   B) Autoimmune destruction of beta cells
   C) Excess of counter-insulin hormones
   D) Glucagon deficiency
   E) Increased number of insulin receptors

2. What is the main cause of insulin resistance in type 2 diabetes mellitus?
   A) Autoimmune insulitis
   B) Viral infection
   C) Antibodies to insulin
   D) Abdominal (visceral) obesity
   E) Removal of the pancreas

3. What is the mechanism of polyuria in diabetes mellitus?
   A) Excess ADH secretion
   B) Decreased glomerular filtration
   C) Osmotic diuresis due to glucose
   D) Excess aldosterone
   E) Increased sodium reabsorption

4. What is the main cause of diabetic ketoacidosis?
   A) Absolute insulin deficiency
   B) Excessive insulin dose
   C) Taking metformin
   D) Severe dehydration and absence of ketosis
   E) Glucagon deficiency

5. Which sign is NOT typical of hyperosmolar coma?
   A) Very high glycemia
   B) Severe dehydration
   C) Increased plasma osmolarity
   D) Occurrence in elderly patients with type 2 diabetes
   E) Large amounts of ketone bodies and Kussmaul breathing

6. Over what period does glycated hemoglobin reflect average glycemia?
   A) The last 24 hours
   B) The last week
   C) The last 2–3 months
   D) The last year
   E) Only the day of the test

7. Which pathway is responsible for sorbitol accumulation in the lens and nerve tissue in diabetes mellitus?
   A) Glycolysis
   B) The polyol pathway
   C) The pentose phosphate pathway
   D) The Krebs cycle
   E) Gluconeogenesis

8. Which lipoprotein is antiatherogenic?
   A) High-density lipoproteins
   B) Low-density lipoproteins
   C) Very low-density lipoproteins
   D) Chylomicron remnants
   E) Intermediate-density lipoproteins

9. From which cells are the “foam cells” of an atherosclerotic plaque formed?
   A) Neutrophils
   B) Erythrocytes
   C) Platelets
   D) Fibroblasts
   E) Macrophages (that have engulfed oxidized LDL)

10. What is the main mechanism of acute coronary syndrome?
    A) Slow growth of a stable plaque
    B) Spasm of the coronary arteries
    C) Anemia
    D) Rupture of an unstable plaque and thrombus formation
    E) Arterial hypotension

11. How is the BMI of a person 1.70 m tall and weighing 92 kg assessed?
    A) Normal weight
    B) Overweight
    C) Class II obesity
    D) Class I obesity
    E) Class III obesity

12. Which hormone is secreted by adipocytes and reduces appetite?
    A) Leptin
    B) Ghrelin
    C) Insulin
    D) Glucagon
    E) Cortisol

13. What mechanism underlies metabolic syndrome?
    A) Absolute insulin deficiency
    B) Hypothyroidism
    C) Insulin resistance and compensatory hyperinsulinemia
    D) Glucagonoma
    E) Autoimmune insulitis

14. How is the atherogenic index calculated?
    A) Triglycerides / HDL
    B) (Total cholesterol − HDL) / HDL
    C) LDL / triglycerides
    D) Total cholesterol / triglycerides
    E) HDL / LDL

15. Which factors are leading in the development of diabetic foot syndrome?
    A) Infection only
    B) Atherosclerosis only
    C) Obesity only
    D) Hypoglycemia only
    E) Neuropathy, angiopathy and infection together

Clinical cases

Case 1

A 72-year-old woman has type 2 diabetes mellitus and takes oral agents. She has had diarrhea for a week and drank little fluid. She was brought to hospital with clouded consciousness. The skin and mucous membranes are very dry, breathing is rapid and shallow, there is no acetone odor. Blood: glucose 48 mmol/l, Na^+ 152 mmol/l, K^+ 4.8 mmol/l, urea 18 mmol/l, pH 7.36; no ketone bodies in the urine.

Questions: What coma has developed? Calculate the plasma osmolarity. Why is there no ketoacidosis?

Answer: Hyperosmolar (hyperglycemic, non-ketoacidotic) coma. Plasma osmolarity = 2 × (152 + 4.8) + 48 + 18 ≈ 380 mOsm/kg (normal 280–295) — markedly increased. Diarrhea and low fluid intake started the dehydration, hyperglycemia aggravated it through osmotic diuresis, and reduced renal perfusion cut glucose excretion, raising glycemia further. Hyperosmolarity led to dehydration of neurons and impaired consciousness. In type 2 diabetes the residual insulin secretion, although insufficient to ensure glucose uptake by tissues, is enough to suppress lipolysis in adipose tissue and ketogenesis in the liver — which is why there are no ketone bodies and no acidosis.

Case 2

A 48-year-old man, height 175 cm, weight 100 kg, waist circumference 112 cm. Arterial pressure 145/90 mm Hg. Blood: fasting glucose 6.3 mmol/l, triglycerides 2.6 mmol/l, total cholesterol 6.2 mmol/l, HDL 0.9 mmol/l.

Questions: What syndrome does the patient have? Calculate the atherogenic index. Explain the pathogenesis.

Answer: Metabolic syndrome: abdominal obesity (waist circumference 112 cm; BMI ≈ 32.7 kg/m^2 — class I obesity) and four additional criteria — raised triglycerides, low HDL, arterial hypertension

and raised fasting glycemia. Atherogenic index = (6.2 − 0.9) / 0.9 ≈ 5.9 — the risk of atherosclerosis is high. Pathogenesis: visceral adipose tissue releases large amounts of free fatty acids and inflammatory cytokines, which cause insulin resistance; compensatory hyperinsulinemia increases renal sodium retention and sympathetic activity, causing arterial hypertension; in the liver, synthesis of VLDL and triglycerides increases and HDL decreases. Because the beta-cell reserve is beginning to be exhausted, glycemia is rising — the risk of type 2 diabetes is high.

Case 3

A 30-year-old patient has had type 1 diabetes mellitus for 12 years, with poor glycemic control (HbA_{1c} 10.5 %). Recently his vision has deteriorated, he has numbness in his feet and reduced sensation in a “stocking” distribution, and persistent protein is found in the urine.

Questions: What complications have developed in the patient? What is their common pathogenetic mechanism?

Answer: Late microangiopathic and neuropathic complications of diabetes: diabetic retinopathy (deteriorating vision), diabetic nephropathy (proteinuria) and distal symmetric polyneuropathy (sensory loss in a “stocking” distribution). The common mechanism is many years of poorly controlled hyperglycemia (HbA_{1c} 10.5 %): non-enzymatic glycation of proteins and accumulation of advanced glycation end products (thickening of basement membranes), activation of the polyol pathway and sorbitol accumulation in cells that take up glucose independently of insulin, activation of protein kinase C, oxidative stress and endothelial dysfunction. Good glycemic control slows the development of these complications.

Quick review

Table 12.12. *Key facts on Topic 12*

Question	Answer
Diabetes mellitus	Chronic hyperglycemia due to impaired insulin secretion and/or action
Type 1	Autoimmune destruction of beta cells; absolute insulin deficiency
Type 2	Insulin resistance + beta-cell dysfunction
The “three P's”	Polyuria, polydipsia, polyphagia
Ketoacidotic coma	Type 1; ketonemia, acidosis, Kussmaul breathing, acetone odor
Hyperosmolar coma	Type 2, elderly; very high glycemia and osmolarity, no ketosis
Lactic acidotic coma	Lactic acid accumulation; hypoxia, metformin
Late complications	Retinopathy, nephropathy, neuropathy, macroangiopathy, diabetic foot
HbA_{1c}	Average glycemia over the last 2–3 months
Atherosclerosis	Lipid accumulation and fibrous plaques in the arterial intima — chronic inflammation
LDL	Atherogenic (“bad”) cholesterol
HDL	Antiatherogenic (“good”) cholesterol
Unstable plaque	Large lipid core, thin cap; if it ruptures — thrombosis
BMI	Weight (kg) / height $(m)^2$; above 30 — obesity
Leptin	Adipocyte hormone, reduces appetite
Metabolic syndrome	Abdominal obesity + dyslipidemia, hypertension, hyperglycemia

Question	Answer
Atherogenic index	(Total cholesterol − HDL) / HDL; below 3 — normal

TOPIC 13

Cell pathophysiology. General mechanisms of cell injury (seminar)

Learning objectives

To explain to the student the causes, general mechanisms and main manifestations of cell injury, damage to the cell membrane and organelles, disturbances of energy supply and ion balance, damage to the genetic apparatus, the forms of cell death (necrosis and apoptosis), and the protective and adaptive mechanisms of the cell.

By the end of the session, the student should know:

- the classification of factors that injure cells;
- the general mechanisms of cell injury: disturbances of energy supply, membranes, ion and water balance, the genetic apparatus and regulatory mechanisms;
- free-radical (peroxide) oxidation of lipids and the antioxidant defense systems;
- the damaging effect of calcium overload in the cell;
- the signs of reversible and irreversible injury;
- the differences between necrosis and apoptosis;
- the mechanisms by which the cell adapts to injury.

The student should be able to:

- determine the leading mechanism of cell injury in a specific pathological condition;
- distinguish necrosis from apoptosis by morphological and biochemical signs;
- identify the injured organ from blood markers of cell injury (enzymes).

1. Factors that injure cells

Cell injury — a change in the structure, metabolism, physicochemical properties and function of a cell that disrupts its vital activity. Every disease is ultimately linked to cell injury.

Table 13.1. *Factors that injure cells*

Group	Examples
Hypoxia (the most common cause)	Ischemia, anemia, respiratory failure, carbon monoxide poisoning
Physical	Mechanical trauma, high and low temperature, ionizing radiation, electric current
Chemical	Poisons (cyanides, heavy metals), drugs (a large dose of paracetamol), alcohol, excess oxygen
Biological	Viruses, bacteria and their toxins, parasites
Immune	Autoimmune reactions, allergic reactions, cytotoxic T lymphocytes
Genetic	Enzyme defects, mutations of structural proteins, chromosomal abnormalities
Nutritional disorders	Deficiency or excess of protein, vitamins, trace elements

Injury can be **primary** (direct action of the factor on the cell) or **secondary** (the effect of changes resulting from the primary injury, e.g. hypoxia or accumulation of toxins). **Specific** changes

(characteristic of a particular factor) and **nonspecific** changes (repeated in any injury) are also distinguished.

2. General mechanisms of cell injury

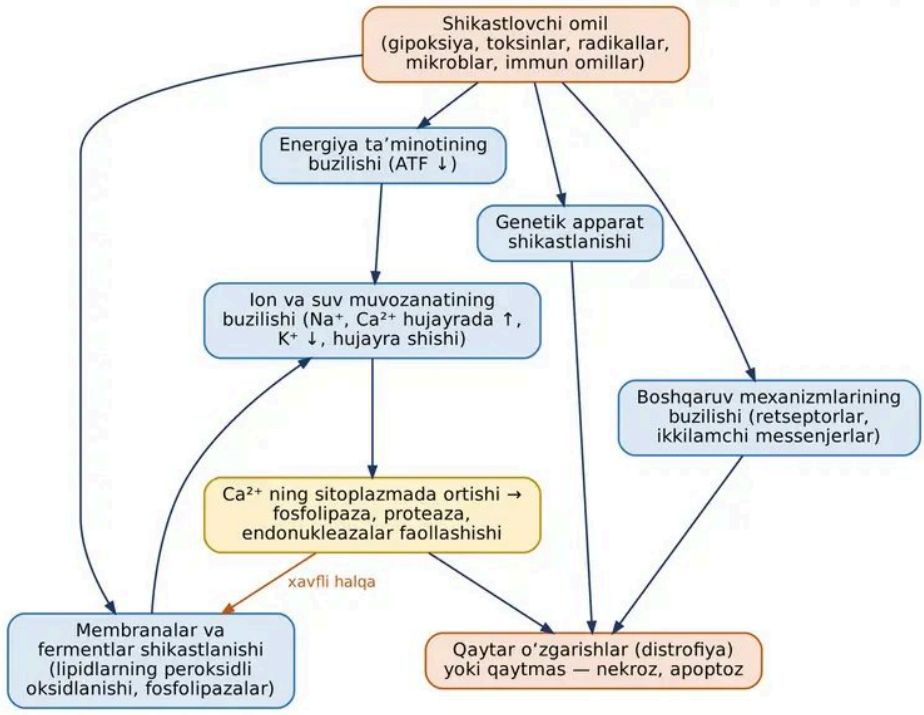


Figure 13.1. *General mechanisms of cell injury (based on Novitskiy V.V., 2009, figure 3.15)*

Disturbance of energy supply

The main source of cell energy is ATP, formed by oxidative phosphorylation in the mitochondria. Energy supply can be disturbed at the following stages:

- **reduced ATP formation** — hypoxia, inhibitors of the respiratory chain (cyanides block cytochrome oxidase), uncoupling of oxidation and phosphorylation (excess thyroid hormones, 2,4-dinitrophenol), lack of substrates (hypoglycemia);
- **impaired ATP transport** — damage to the creatine phosphokinase system;
- **impaired ATP use** — reduced ATPase activity.

Consequences of ATP deficiency: the Na^+/K^+ pump stops → Na^+ and water enter the cell (cell swelling) and K^+ leaves; anaerobic glycolysis increases → lactate accumulates and pH falls; Ca^{2+} pumps stop → cytoplasmic Ca^{2+} rises; protein synthesis decreases (ribosomes detach).

Membrane damage

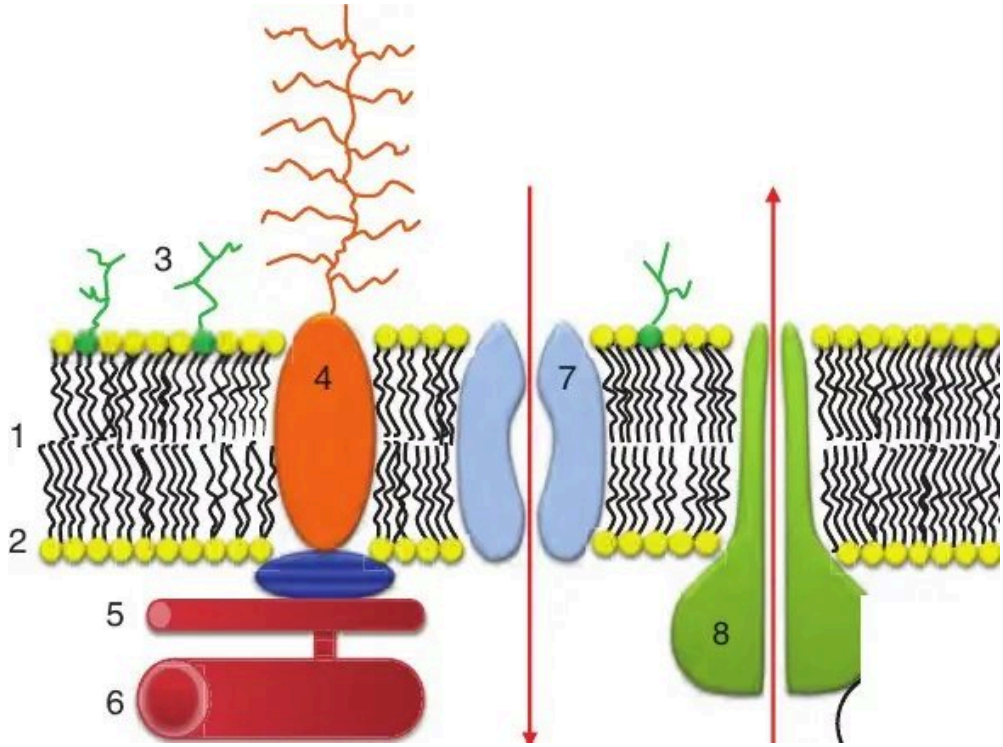


Figure 13.2. *Elements of the biological membrane that are subject to damage: 1 — lipid bilayer, 2 — lipid monolayer, 3 — glycolipids, 4 — glycoproteins, 5 — microfilaments, 6 — microtubules, 7 — ion channel, 8 — ion pump (Novitskiy V.V., 2009, figure 3.1)*

Table 13.2. *Main mechanisms of membrane damage*

Mechanism	Essence
Free-radical (peroxide) oxidation of lipids	Reactive oxygen species oxidize unsaturated fatty acids; a chain reaction creates “holes” in the membrane
Activation of membrane phospholipases	When Ca^{2+} rises, phospholipase A_2 is activated and breaks down phospholipids; lysophospholipids with a detergent action and free fatty acids accumulate
Damage to membrane proteins and enzymes	By proteases, radicals, toxins
Mechanical stretching	Excessive stretching and rupture of the membrane in cell swelling
Immune damage	The membrane attack complex of complement (C5b–C9), perforins of cytotoxic T lymphocytes

Free-radical oxidation of lipids and antioxidant defense

Free radicals — highly reactive molecules with an unpaired electron in the outer orbital. The main oxygen radicals are the superoxide anion (O_2^-) and the hydroxyl radical ($OH^\bullet$); hydrogen peroxide (H_2O_2) and singlet oxygen are also reactive oxygen species.

Sources of radicals: the mitochondrial respiratory chain (electron “leakage”, especially during reperfusion), NADPH oxidase of phagocytes, xanthine oxidase (accumulates in ischemia), ionizing radiation, metabolism of drugs and poisons.

Table 13.3. *Antioxidant defense systems*

Group	Representatives	Effect
Enzymatic	Superoxide dismutase Catalase	Converts O_2^- to H_2O_2 Breaks down H_2O_2 into water and oxygen
	Glutathione peroxidase (contains selenium)	Neutralizes H_2O_2 and lipid peroxides
Non-enzymatic (fat-soluble)	Vitamin E (tocopherol), carotenoids	Breaks the chain reaction in membranes
Non-enzymatic (water-soluble)	Vitamin C (ascorbate), glutathione, uric acid	Bind radicals
Metal-binding proteins	Transferrin, ferritin, ceruloplasmin	Bind free iron and copper, reducing radical formation

REMEMBER

Oxidative stress — a shift in the balance between free radical formation and antioxidant defense in favor of the radicals. It plays an important role in reperfusion injury, atherosclerosis, the complications of diabetes mellitus, aging, radiation injury and many inflammatory diseases.

Disturbance of ion and water balance

- Entry of Na^+ and water into the cell → **cell swelling** (hydropic degeneration) and swelling of organelles;
- Loss of K^+ from the cell → change in membrane potential, impaired excitability;
- **Rise of cytoplasmic Ca^{2+}** — the “final common pathway” of cell injury.

MEMORY AID

The “**4P**” effects of calcium overload: **P**hospholipases (break down the membrane), **P**rotein- and DNA-cleaving enzymes — proteases (disrupt the cytoskeleton) and endonucleases (cleave DNA), **P**hosphorylation uncoupling (mitochondria are damaged, ATP falls), **P**ulling of fibrils (contracture of myofibrils).

Damage to the genetic apparatus

DNA damage (mutations, breaks, chromosomal aberrations) occurs under the action of ionizing radiation, chemical mutagens, viruses and free radicals. Consequences: impaired protein synthesis, arrest of cell division, apoptosis (via the p53 protein), tumor transformation. The cell has DNA repair systems; their hereditary defects (e.g. xeroderma pigmentosum) sharply increase the risk of tumors.

Disturbance of regulatory mechanisms

Disturbance of the systems that regulate cell activity: changes in the number and sensitivity of receptors, impaired G-protein function (e.g. cholera toxin permanently activates the Gs protein, raising cAMP in intestinal cells and causing severe secretory diarrhea), impaired function of second messengers and protein kinases.

3. Damage to organelles

Table 13.4. *Manifestations of organelle damage*

Organelle	Manifestation of damage	Outcome
Mitochondria	Swelling, destruction of cristae, calcium deposition in the matrix; increased membrane permeability	Decreased ATP; release of cytochrome C and initiation of apoptosis
Endoplasmic reticulum	Dilation, detachment of ribosomes	Decreased protein synthesis; accumulation of misfolded proteins (“ER stress”)
Lysosomes	Loss of membrane stability, release of enzymes into the cytoplasm	Autolysis; sometimes they escape from the cell and damage the tissue
Nucleus	Pyknosis (shrinkage), karyorrhexis (fragmentation), karyolysis (dissolution)	Morphological signs of cell death
Cytoskeleton	Disruption of microfilaments and microtubules	Impaired cell shape, movement and division

4. Reversible and irreversible injury. Cell death

Signs of **reversible injury**: swelling of the cell and organelles, blebs on the membrane surface, detachment of ribosomes, accumulation of fat droplets (fatty degeneration). If the injuring factor is removed, the cell recovers.

Signs of **irreversible injury**: marked swelling of mitochondria with dense calcium deposits in the matrix, extensive damage to the plasma membrane, rupture of lysosomes, nuclear changes (pyknosis, karyorrhexis, karyolysis).

CLINICAL SIGNIFICANCE

When the cell membrane is damaged, the enzymes and proteins inside it are released into the blood — they are used diagnostically as **markers of cell injury**: troponins and the MB fraction of creatine phosphokinase in myocardial injury, aminotransferases (ALT, AST) in the liver, creatine phosphokinase in skeletal muscle, amylase and lipase in the pancreas.

Table 13.5. *Differences between necrosis and apoptosis*

Feature	Necrosis	Apoptosis
Definition	“Forced”, disorderly death of the cell under the action of an injuring factor	Genetically programmed, energy-dependent active cell death
Causes	Severe injury: ischemia, toxins, burns	Physiological (embryogenesis, replacement of aged cells) and pathological (DNA damage, viruses, cytotoxic T lymphocytes, hormone deficiency)
Extent	A group of cells, an area of tissue	Individual cells
Cell volume	Increases (swells)	Decreases (shrinks)
Membrane	Ruptures early	Remains intact for a long time; phosphatidylserine appears on the outer surface
Nucleus	Pyknosis, karyorrhexis, karyolysis; DNA is degraded randomly	Chromatin condensation; DNA is cleaved between nucleosomes into regular fragments (a “ladder” on electrophoresis)

Feature	Necrosis	Apoptosis
ATP requirement	None (caused by ATP deficiency)	Required
Cell fragments	Released into the surrounding tissue	Membrane-bound apoptotic bodies, rapidly engulfed by macrophages
Inflammation	Develops	Does not develop
Main enzymes	Lysosomal enzymes, calpains	Caspases

Pathways of apoptosis:

- **intrinsic (mitochondrial) pathway** — DNA damage, oxidative stress, lack of growth factors → imbalance of Bcl-2 family proteins (Bax and Bak are activated) → cytochrome C is released from the mitochondria → apoptosome → caspase-9 → effector caspases (caspase-3);
- **extrinsic (receptor) pathway** — "death receptors" (Fas, TNF receptor) bind their ligand → caspase-8 → effector caspases.

CAUTION

Both **decreased** and **increased** apoptosis cause disease. Insufficient apoptosis — tumors (p53 mutation, Bcl-2 excess), autoimmune diseases (autoreactive lymphocytes are not eliminated). Excessive apoptosis — neurodegenerative diseases (Alzheimer, Parkinson), death of $CD4^+$ lymphocytes in AIDS, part of ischemic injury.

5. Mechanisms of cell adaptation to injury

Table 13.6. *Protective and adaptive mechanisms at the cellular level*

Mechanism	Essence
Compensation of energy supply	Increased glycolysis, increased number and activity of mitochondria
Membrane protection	Activation of antioxidant systems, membrane repair (resynthesis)
Restoration of ion and water balance	Increased activity of ion pumps, buffer systems
Heat shock proteins (chaperones)	Refold damaged proteins or direct them to degradation
DNA repair	Repair of damaged DNA segments; if repair is impossible — apoptosis (p53)
Autophagy	Digestion and recycling of damaged organelles
Cell adaptation	Hypertrophy, hyperplasia, atrophy, metaplasia

Table 13.7. *Forms of cell adaptation*

Form	Definition	Examples
Hypertrophy	Increase in cell size	Myocardium in arterial hypertension, skeletal muscles in athletes
Hyperplasia	Increase in cell number	Uterus in pregnancy, endometrium with excess estrogen
Atrophy	Decrease in cell size and function	Muscles in prolonged immobility, ischemia, starvation
Metaplasia	Replacement of one differentiated cell type by another	Transformation of bronchial epithelium into stratified squamous epithelium in smokers, Barrett esophagus

6. Seminar part: questions and situations for discussion

1. Why is hypoxia the most common cause of cell injury?
2. Why is venous blood also red (rich in oxygen) in cyanide poisoning?
3. How does a large dose of paracetamol damage liver cells, and how does acetylcysteine protect them? (A paracetamol metabolite depletes glutathione; acetylcysteine restores glutathione stores.)
4. Why can reperfusion aggravate injury?
5. Why is calcium called the “final common pathway” of cell injury?
6. How is apoptosis altered in tumors and in neurodegenerative diseases?

7. Practical skills

Skill: determining the leading mechanism of cell injury

1. Determine the type of injuring factor (hypoxia, toxin, radiation, immune, etc.).
2. Determine the primary target of the factor: mitochondria, membrane, DNA or receptor.
3. Assess signs of disturbed energy supply (lactic acidosis, cell swelling).
4. Determine the involvement of oxidative stress and calcium overload.
5. Establish whether the injury is reversible or irreversible.
6. Indicate the form of cell death (necrosis or apoptosis) and write a conclusion.

Skill: interpreting markers of cell injury

1. Determine which enzyme or protein is raised.
2. Assess its organ specificity: troponin — myocardium, ALT — liver, lipase — pancreas.
3. Take into account the degree of rise and its dynamics (when it rises, when it returns to normal).
4. Compare with the clinical signs.
5. Write a conclusion about the injured organ and the extent of injury.

Review questions

1. List the factors that injure cells.
2. What are the general mechanisms of cell injury?
3. What changes in the cell does ATP deficiency lead to?
4. Explain the main mechanisms of membrane damage.
5. What are free radicals and where are they formed?
6. List the antioxidant defense systems.
7. What is oxidative stress and in which diseases is it important?
8. What is the damaging effect of calcium overload in the cell?
9. What are the consequences of damage to the genetic apparatus?
10. List the manifestations of organelle damage.
11. Explain the signs of reversible and irreversible injury.
12. What are the main differences between necrosis and apoptosis?
13. Explain the intrinsic and extrinsic pathways of apoptosis.
14. What diseases does disturbed apoptosis lead to?

15. List the mechanisms by which the cell adapts to injury.

Test questions

1. What is the most common cause of cell injury?
   A) Ionizing radiation
   B) Electric current
   C) Genetic defects
   D) Immune reactions
   E) Hypoxia

2. What is the result of the Na^+/K^+ pump stopping in ATP deficiency?
   A) Cell swelling (entry of Na^+ and water into the cell)
   B) Cell shrinkage
   C) Increased intracellular potassium
   D) Increased membrane potential
   E) Increased protein synthesis

3. What is the mechanism by which cyanides injure the cell?
   A) They block hemoglobin
   B) They block cytochrome oxidase and stop tissue respiration
   C) They dissolve membrane lipids
   D) They break DNA
   E) They rupture lysosomes

4. Which enzyme converts the superoxide anion to hydrogen peroxide?
   A) Catalase
   B) Glutathione peroxidase
   C) Myeloperoxidase
   D) Superoxide dismutase
   E) Xanthine oxidase

5. Which vitamin is a fat-soluble antioxidant that breaks the chain reaction in membranes?
   A) Vitamin C
   B) Vitamin B_1
   C) Vitamin E
   D) Vitamin B_{12}
   E) Vitamin PP

6. Which of the following is NOT a damaging effect of increased cytoplasmic Ca^{2+}?
   A) Activation of phospholipases
   B) Activation of proteases
   C) Activation of endonucleases
   D) Increased ATP synthesis
   E) Damage to mitochondria

7. By what mechanism does cholera toxin cause diarrhea in intestinal cells?
   A) It destroys the membrane
   B) It damages DNA
   C) It permanently activates the Gs protein and raises cAMP
   D) It damages mitochondria
   E) It induces apoptosis

8. Which is a sign of irreversible cell injury?
   A) Cell swelling
   B) Dense calcium deposits in the mitochondria and karyorrhexis of the nucleus
   C) Detachment of ribosomes
   D) Blebs on the membrane
   E) Accumulation of fat droplets

9. Which sign is characteristic of apoptosis?
   A) Cell shrinkage and formation of apoptotic bodies, without inflammation
   B) Cell swelling and early rupture of the membrane
   C) Death of a group of cells and inflammation
   D) No ATP requirement
   E) Random degradation of DNA

10. Which protein is released from the mitochondria in the intrinsic (mitochondrial) pathway of apoptosis?
    A) Troponin
    B) Myoglobin
    C) Collagen
    D) Albumin
    E) Cytochrome C

11. Which are the effector enzymes of apoptosis?
    A) Lipases
    B) Phosphatases
    C) Peroxidases
    D) Collagenases
    E) Caspases

12. Which pathology does insufficient apoptosis lead to?
    A) Alzheimer disease
    B) Parkinson disease
    C) Tumor growth
    D) Death of $CD4^+$ cells in AIDS
    E) Myocardial ischemia

13. What is the transformation of bronchial epithelium into stratified squamous epithelium in smokers called?
    A) Hypertrophy
    B) Hyperplasia
    C) Atrophy
    D) Metaplasia
    E) Dysplasia

14. Which is the most specific marker of myocardial cell injury?
    A) Troponins
    B) ALT
    C) Amylase
    D) Alkaline phosphatase
    E) Lipase

15. What is the function of heat shock proteins (chaperones) in cell protection?
    A) They cleave DNA
    B) Refold damaged proteins or direct them to degradation
    C) They bring calcium into the cell
    D) They generate free radicals
    E) They rupture lysosomes

Clinical cases

Case 1

A 25-year-old man took a total of 15 g of paracetamol over one day because of toothache. In the first 24 hours he had only nausea. On day 3 jaundice and pain under the right costal margin appeared; blood ALT and AST were sharply raised (50 times the normal value) and the prothrombin time was prolonged.

Questions: Explain the mechanism of hepatocyte injury. What does the rise in ALT and AST indicate? Why is acetylcysteine an antidote?

Answer: A large dose of paracetamol forms a toxic metabolite in the liver via cytochrome P450 — N-acetyl-p-benzoquinone imine. Normally it is neutralized by glutathione; at a large dose the glutathione stores are exhausted and the metabolite binds covalently to proteins, enhances oxidative stress and lipid peroxidation and damages the mitochondria — hepatocyte necrosis develops. The sharp rise in ALT and AST indicates damage to the hepatocyte membrane and release of enzymes from these cells into the blood — extensive necrosis; prolonged prothrombin time shows that the synthetic function of the liver is impaired. Acetylcysteine supplies cysteine for glutathione synthesis and restores glutathione stores, i.e. it restores the capacity to neutralize the toxic metabolite; it is most effective in the first 8–10 hours.

Case 2

A patient who had been in a closed room during a fire was brought in unconscious. The skin and mucous membranes are pinkish-red and breathing is rapid. Blood carboxyhemoglobin is 40 %.

Questions: What is the main mechanism of cell injury? Which tissues are damaged first, and why?

Answer: Carbon monoxide binds to hemoglobin about 200–250 times more strongly than oxygen, forming carboxyhemoglobin; the oxygen-carrying capacity of the blood decreases and the remaining oxyhemoglobin gives up oxygen less readily (the dissociation curve shifts to the left). In addition, CO partly blocks cytochrome oxidase. The result is tissue hypoxia → ATP deficiency → ion pumps stop, cell swelling, calcium accumulation and necrosis. The tissues with the highest oxygen demand and the greatest sensitivity to hypoxia — the brain and myocardium — are damaged first. The pinkish-red color is due to the color of carboxyhemoglobin and masks the severity of the hypoxia.

Case 3

A patient with cancer received a course of radiotherapy. After treatment the tumor decreased in size, but the patient developed leukopenia and diarrhea.

Questions: How does ionizing radiation kill tumor cells? Why are the complications related specifically to the bone marrow and intestine?

Answer: Ionizing radiation damages DNA directly and through free radicals formed by radiolysis of water (breaks in DNA strands). Damaged DNA triggers apoptosis via the p53 protein, or the cell dies

during division (mitotic death). Tumor cells divide rapidly and are therefore sensitive to radiation. According to the Bergonié–Tribondeau law, the rapidly renewing tissues of the body — the bone marrow and intestinal epithelium — are also sensitive to radiation: death of bone marrow cells leads to leukopenia, and damage to the epithelium of the intestinal crypts leads to impaired absorption and diarrhea.

Quick review

Table 13.8. *Key facts on Topic 13*

Question	Answer
Cell injury	A change in cell structure, metabolism and function that disrupts its vital activity
Most common cause	Hypoxia
ATP deficiency	Na^+/K^+ pump stops, cell swelling, lactic acidosis, Ca^{2+} rise
Free radicals	O_2^-, $OH^\bullet$, H_2O_2 — damage lipids, proteins and DNA
Antioxidants	Superoxide dismutase, catalase, glutathione peroxidase, vitamins E and C
Oxidative stress	Radical formation outweighs antioxidant defense
Calcium overload	Activation of phospholipases, proteases, endonucleases; mitochondrial damage
Irreversible injury	Calcium deposits in mitochondria, membrane rupture, nuclear changes
Necrosis	Forced death: the cell swells, the membrane ruptures, inflammation
Apoptosis	Programmed death: the cell shrinks, caspases, no inflammation
Pathways of apoptosis	Intrinsic (mitochondria, cytochrome C) and extrinsic (Fas, TNF receptor)
Cell markers	Troponin — myocardium, ALT — liver, lipase — pancreas
Forms of adaptation	Hypertrophy, hyperplasia, atrophy, metaplasia

TOPIC 14

Etiology and pathogenesis of tumor development in the body

Learning objectives

To explain to the student the nature of tumor growth, the differences between malignant and benign tumors, carcinogenic factors and their mechanisms of action, the stages and molecular basis of carcinogenesis (oncogenes, tumor suppressor genes), the types of tumor atypia, the stages of metastasis, the local and systemic changes caused by a tumor in the body, and the mechanisms of antiblastoma defense.

By the end of the session, the student should know:

- the concept of a tumor and the differences between malignant and benign tumors;
- the classification of carcinogenic factors (chemical, physical, biological) and their mechanism of action;
- the stages of carcinogenesis: initiation, promotion, progression;
- the role of proto-oncogenes, oncogenes and tumor suppressor genes;
- the types of tumor atypia (growth, metabolic, functional, morphological);
- the stages and routes of metastasis;
- the effects of the tumor on the body (cachexia, paraneoplastic syndromes) and antiblastoma defense.

The student should be able to:

- explain the methods of studying tumors experimentally;
- identify the carcinogenic factor and the likely mechanism of tumor growth in a clinical situation;
- distinguish a malignant from a benign tumor by its features.

1. The concept of a tumor

Tumor (neoplasia, blastoma) — a typical pathological process characterized by unlimited proliferation of cells that escapes the body's control and by impaired differentiation, resulting from changes in their genetic apparatus. These properties of tumor cells are passed on to subsequent cell generations.

The main features of a tumor:

- **autonomous (uncontrolled) growth** — tumor cells do not obey the signals that regulate growth;
- **atypia** — structure and function that differ from normal cells;
- **anaplasia** — loss of differentiation, resemblance to embryonic cells;
- **unlimited proliferation (“immortality”)** — due to telomerase activity;
- in malignant tumors, **invasive growth** and **metastasis**.

Table 14.1. *Differences between benign and malignant tumors*

Feature	Benign tumor	Malignant tumor
Differentiation	High, resembling the tissue of origin	Low, anaplasia
Growth rate	Slow	Fast

Feature	Benign tumor	Malignant tumor
Type of growth	Expansive (pushing the surrounding tissue aside), has a capsule	Infiltrative (invasive), no capsule
Metastasis	No	Present
Recurrence	Few	Many
Effect on the body	Mainly local (compression)	Local and systemic (cachexia, intoxication)
Nomenclature	Tissue name + “-oma” (lipoma, adenoma, fibroma)	Epithelial — carcinoma (cancer); connective tissue — sarcoma

CAUTION

A “benign” tumor can also be life-threatening depending on its location: a meningioma in the brain compresses the brain, and hormone-producing adenomas (pheochromocytoma, insulinoma) cause severe hormonal disturbances. Some benign tumors can become malignant (adenomatous polyps of the colon).

2. Carcinogenic factors

Carcinogens — factors that cause tumor development.

Table 14.2. *Classification of carcinogenic factors*

Group	Examples	Associated tumors
Chemical	Polycyclic aromatic hydrocarbons (benzo(a)pyrene — tobacco smoke, charred food), aromatic amines (aniline dyes), nitrosamines, aflatoxin B_1, asbestos, benzene, arsenic, vinyl chloride	Cancer of the lung, bladder, stomach, liver; mesothelioma; leukemias
Physical	Ionizing radiation, ultraviolet rays	Leukemias, thyroid cancer; skin cancer and melanoma
Biological — DNA viruses	Human papillomavirus (types 16, 18), Epstein–Barr virus, hepatitis B virus, Kaposi sarcoma herpesvirus (HHV-8)	Cervical cancer; Burkitt lymphoma, nasopharyngeal cancer; liver cancer; Kaposi sarcoma
Biological — RNA viruses	Hepatitis C virus, human T-lymphotropic virus (HTLV-1)	Liver cancer; T-cell leukemia
Bacteria	Helicobacter pylori	Gastric cancer, MALT lymphoma
Endogenous	Hormones (excess estrogens), chronic inflammation, hereditary mutations	Endometrial and breast cancer; colon cancer in ulcerative colitis

Chemical carcinogens by their mode of action:

- **direct-acting** — react with DNA themselves (alkylating agents, some cytostatics);
- **indirect-acting (procarcinogens)** — become carcinogenic after metabolic activation in the body (mainly in the liver by the cytochrome P450 system) (benzo(a)pyrene, aflatoxin, nitrosamines).

CLINICAL SIGNIFICANCE

Historical examples: in 1775 P. Pott linked the high frequency of scrotal skin cancer in London chimney sweeps to an occupational factor — soot. In 1915 K. Yamagiwa and K. Ichikawa first induced a tumor experimentally by applying coal tar to rabbits' ears for a long time. Today tobacco smoking is associated with 80–90 percent of lung cancer.

3. Stages and molecular mechanisms of carcinogenesis

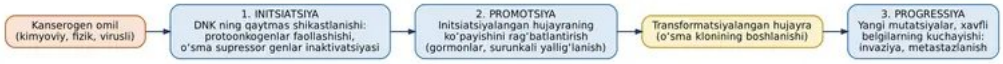


Figure 14.1. *Stages of carcinogenesis (based on Novitskiy V.V., 2009, figure 14.9)*

Oncogenes and tumor suppressor genes

Table 14.3. *Genes involved in carcinogenesis*

Gene group	Normal function	Change in tumors	Examples
Proto-oncogenes → oncogenes	Stimulate cell growth and division (growth factors, their receptors, signaling proteins, transcription factors)	Activated (dominant effect — one allele is enough): point mutation, amplification, translocation	RAS (many tumors), HER2/neu (breast cancer), MYC (Burkitt lymphoma), BCR-ABL (chronic myeloid leukemia)
Tumor suppressor genes	Stop cell division and trigger apoptosis when DNA is damaged	Inactivated (recessive — both alleles must be damaged, Knudson's "two-hit" hypothesis)	TP53 ("guardian of the genome" — in more than half of tumors), RB (retinoblastoma), APC (familial adenomatous polyposis), BRCA1/2 (breast and ovarian cancer)
DNA repair genes	Correct DNA errors	Inactivated → mutations accumulate rapidly	Lynch syndrome (colon cancer), xeroderma pigmentosum
Genes regulating apoptosis	Regulate cell death	Overexpression of antiapoptotic BCL2	Follicular lymphoma

REMEMBER

The main hallmarks of the tumor cell (D. Hanahan and R. Weinberg, "Hallmarks of cancer"): independence from growth signals, insensitivity to growth-inhibitory signals, evasion of apoptosis, unlimited replication (telomerase), sustained angiogenesis, invasion and metastasis; additional ones — reprogramming of energy metabolism, evasion of immune surveillance, genome instability and tumor-promoting inflammation.

Theories of carcinogenesis

- **Mutation theory** — a tumor is the result of the accumulation of several successive mutations (in colon cancer: APC → KRAS → TP53 mutations, the adenoma–carcinoma sequence).
- **Viral-genetic theory** (L.A. Zilber) — oncogenic viruses insert their genome into the cell DNA and transform it.
- **Epigenetic theory** — disturbed gene expression without changes in the DNA sequence (DNA methylation, histone modification).
- **Cancer stem cell theory** — a tumor develops from a small group of self-renewing cells, which are resistant to treatment.

4. Tumor atypia

Table 14.4. *Types of tumor atypia*

Type	Essence
Growth atypia	Uncontrolled proliferation, loss of contact inhibition, invasion and metastasis

Type	Essence
Metabolic (biochemical) atypia	Warburg effect: glycolysis predominates even when oxygen is sufficient (“aerobic glycolysis”) — high glucose consumption and lactate production; the tumor “draws” glucose, amino acids and lipids from the body (“nitrogen trap”, “glucose trap”)
Functional atypia	Loss or alteration of the function of the tissue of origin; ectopic hormone production
Morphological atypia	Cellular atypia (nuclear enlargement, pleomorphism, abnormal mitoses) and tissue atypia (disordered architecture)
Antigenic atypia	Tumor-specific antigens, reappearance of embryonic antigens (alpha-fetoprotein, carcinoembryonic antigen)

CLINICAL SIGNIFICANCE

Metabolic atypia is used in diagnosis: in **positron emission tomography** (PET), radiolabeled glucose (^{18}F-FDG) accumulates in the tumor and its metastases because they consume glucose rapidly. **Tumor markers** (alpha-fetoprotein — liver cancer, carcinoembryonic antigen — colon cancer, PSA — prostate cancer, CA-125 — ovarian cancer) are used to monitor treatment response and recurrence.

5. Metastasis

Metastasis — detachment of tumor cells from the primary focus, their migration to other organs and formation of new tumor foci (metastases) there. It is the main cause of death from malignant tumors.



Figure 14.2. *Stages of metastasis*

Table 14.5. *Routes of metastasis*

Path	Feature	Examples
Lymphogenic	First to the regional lymph nodes	Most carcinomas (gastric cancer — Virchow's metastasis to the left supraclavicular node)
Hematogenous	Via the blood, often to the lungs, liver, bone, brain	Sarcomas, kidney cancer, melanoma, breast cancer
Implantation	Along serous cavities	Gastric and ovarian cancer — peritoneal carcinomatosis, Krukenberg tumor
Canalicular	Along natural passages	Along the urinary tract, bronchi

MEMORY AID

Tumors that frequently metastasize to bone: “**P**rostate, **B**reast, **K**idney, **T**hyroid, **L**ung” — “**Pb KTL**” (“lead kettle”). Prostate cancer usually gives osteoblastic metastases (bone becomes denser), the others give osteolytic metastases.

6. Effects of the tumor on the body

Local effects: compression of surrounding tissues and organs, obstruction of vessels and ducts (jaundice, intestinal obstruction), tissue destruction (bleeding, perforation), pain.

Systemic effects:

- **Tumor cachexia** — loss of appetite and marked weight loss; mechanism: cytokines released by the tumor and immune cells (TNF-alpha — “cachectin”, IL-1, IL-6), proteolysis-inducing factor, the metabolic “trap” of the tumor, anorexia.

- **Paraneoplastic syndromes** — manifestations not related to the direct spread of the tumor, arising through substances it secretes or through immune reactions.
- **Anemia**, immunodeficiency, fever, tendency to thrombosis (Trousseau syndrome — migratory thrombophlebitis).

Table 14.6. *Examples of paraneoplastic syndromes*

Syndrome	Mechanism	Commonly associated tumor
Cushing syndrome	Ectopic ACTH	Small cell lung cancer
Syndrome of inappropriate ADH secretion	Ectopic ADH → hyponatremia	Small cell lung cancer
Hypercalcemia	Parathyroid hormone-related peptide	Squamous cell lung cancer, kidney cancer
Polycythemia	Ectopic erythropoietin	Kidney cancer, hepatocellular carcinoma
Lambert–Eaton myasthenic syndrome	Autoantibodies to calcium channels	Small cell lung cancer

7. Antiblastoma defense

Antiblastoma resistance — the system of the body's defense mechanisms against the emergence and development of tumors.

Table 14.7. *Mechanisms of antiblastoma defense*

Group	Mechanisms
Anticarcinogenic	Neutralization and excretion of carcinogens (liver enzymes), antioxidant systems
Antimutational	DNA repair systems, elimination of mutated cells by apoptosis (p53)
Anticellular (immune surveillance)	NK cells (natural killers), cytotoxic T lymphocytes, macrophages, antibodies, interferons, TNF

A tumor can escape immune surveillance: it reduces expression of HLA molecules, secretes immunosuppressive factors (TGF-beta, IL-10) and expresses immune “brake” molecules (PD-L1). Modern immunotherapy (PD-1 and CTLA-4 inhibitors) is based precisely on removing these “brakes”.

> **CAUTION**
> In immunodeficiency states (AIDS, immunosuppression after transplantation), tumors are common, especially virus-associated tumors (Kaposi sarcoma, lymphomas, cervical cancer) — this confirms the important role of immune surveillance in antiblastoma defense.

8. Practical part: methods of studying tumors experimentally

Table 14.8. *Methods of studying tumors experimentally*

Method	Essence	Examples
Induction	Inducing a tumor with a carcinogen	Applying coal tar to rabbits' ears (Yamagiwa and Ichikawa); applying benzo(a)pyrene to mouse skin; viral tumors (Rous sarcoma — in chickens)

Method	Essence	Examples
Transplantation	Transferring tumor cells from one animal to another	Ehrlich ascites carcinoma, Walker carcinosarcoma, Lewis carcinoma
Explantation	Growing tumor cells outside the body	Cell cultures (HeLa)
Genetic models	Transgenic and knockout animals	p53 knockout mice, mice expressing an oncogene

Modeling two-stage carcinogenesis (analysis using a scheme)

1. A small dose of a carcinogen is applied once to mouse skin (initiator) — no tumor develops.
2. Then a promoter (not itself carcinogenic, croton oil or phorbol esters) is applied to the same area for a long time — papillomas and then cancer develop.
3. If the promoter is applied **before** the initiator, or only the promoter is used, no tumor develops.

Analysis: initiation is an irreversible genetic change (DNA mutation); promotion is a process that stimulates the proliferation of initiated cells, requires prolonged and repeated exposure and is reversible up to a certain point. This experiment proves the multistage nature of carcinogenesis and explains why quitting smoking or treating chronic inflammation reduces the risk of tumors.

9. Practical skills

Skill: analyzing tumor risk factors

1. Identify exposure to occupational and household carcinogens (smoking, asbestos, dyes, radiation).
2. Identify chronic infections (hepatitis B and C, H. pylori, human papillomavirus).
3. Identify chronic inflammation and precancerous conditions (Barrett esophagus, ulcerative colitis, adenomatous polyps).
4. Check the family history (signs of BRCA, Lynch syndrome).
5. Take immunodeficiency states into account.
6. Write a conclusion and indicate the direction of screening or prevention.

Skill: distinguishing a malignant from a benign tumor

1. Assess the growth rate (history).
2. Determine the borders and whether there is a capsule (palpation, imaging).
3. Check for infiltration of surrounding tissues and mobility.
4. Palpate the regional lymph nodes.
5. Assess systemic signs: weight loss, weakness, anemia.
6. Indicate that histological examination is needed for the final diagnosis.

Review questions

1. What is called a tumor? What are its main features?
2. How do malignant and benign tumors differ?
3. How are carcinogenic factors classified?
4. Give examples of direct- and indirect-acting chemical carcinogens.

5. Which viruses cause tumors?
6. Explain the stages of carcinogenesis.
7. By what mechanisms do proto-oncogenes become oncogenes?
8. What is the role of tumor suppressor genes? Explain Knudson's “two-hit” hypothesis.
9. Why is the p53 protein called the “guardian of the genome”?
10. List the types of tumor atypia. What is the Warburg effect?
11. Explain the stages and routes of metastasis.
12. How does tumor cachexia develop?
13. Give examples of paraneoplastic syndromes.
14. List the mechanisms of antiblastoma defense.
15. What does the two-stage carcinogenesis experiment prove?

Test questions

1. What is the most important sign of a malignant tumor?
   A) Metastasis and invasive growth
   B) Being enclosed in a capsule
   C) Slow growth
   D) High differentiation
   E) Only local effects

2. What is the main carcinogen in tobacco smoke?
   A) Aflatoxin B_1
   B) Asbestos
   C) Vinyl chloride
   D) Benzo(a)pyrene
   E) Nitrosamines

3. What is the main etiological factor of cervical cancer?
   A) Hepatitis B virus
   B) Epstein–Barr virus
   C) Helicobacter pylori
   D) HTLV-1
   E) Human papillomavirus (types 16, 18)

4. Which gene is a tumor suppressor gene?
   A) RAS
   B) TP53
   C) MYC
   D) HER2
   E) BCR-ABL

5. How is the initiation stage of carcinogenesis characterized?
   A) Formation of metastases
   B) Rapid proliferation of cells
   C) Irreversible damage to DNA
   D) Angiogenesis
   E) Tumor cachexia

6. What does the Warburg effect mean?
   A) Susceptibility of tumor cells to apoptosis
   B) Decreased protein synthesis
   C) Absence of telomerase
   D) Increased contact inhibition
   E) Predominance of glycolysis even when oxygen is sufficient

7. Which oncogene is formed in chronic myeloid leukemia?
   A) TP53
   B) BCR-ABL (Philadelphia chromosome)
   C) RB
   D) APC
   E) BRCA1

8. To which genes does Knudson's “two-hit” hypothesis apply?
   A) Proto-oncogenes
   B) Ribosomal genes
   C) Tumor suppressor genes
   D) Immunoglobulin genes
   E) Mitochondrial genes

9. What is the cause of Cushing syndrome in small cell lung cancer?
   A) Adrenal metastasis
   B) Pituitary adenoma
   C) Glucocorticoid therapy
   D) Ectopic ACTH production
   E) Thyroid insufficiency

10. Which cytokine plays the main role in tumor cachexia?
    A) TNF-alpha
    B) IL-2
    C) Interferon gamma
    D) IL-4
    E) Erythropoietin

11. Which are the main cells of the anticellular mechanism of antiblastoma defense?
    A) Erythrocytes
    B) NK cells and cytotoxic T lymphocytes
    C) Platelets
    D) Fibroblasts
    E) Adipocytes

12. Which is an indirect-acting chemical carcinogen (procarcinogen)?
    A) Aflatoxin B_1 (activated in the liver)
    B) Alkylating agents
    C) Ionizing radiation
    D) Ultraviolet rays
    E) Human papillomavirus

13. What is an implantation metastasis of gastric cancer to the ovary called?
    A) Virchow's metastasis
    B) Schnitzler's metastasis
    C) Krukenberg tumor
    D) Kaposi sarcoma
    E) Burkitt lymphoma

14. Which example corresponds to studying tumors experimentally by the induction method?
    A) HeLa cell culture
    B) Transplantation of Ehrlich carcinoma
    C) Growing tumor cells in a test tube
    D) Adding a drug to a cell culture
    E) Applying coal tar to rabbits' ears

15. In the two-stage carcinogenesis experiment, what happens if the promoter is applied before the initiator?
    A) A tumor develops rapidly
    B) Metastases appear
    C) Only a malignant tumor develops
    D) No tumor develops
    E) Leukemia develops

Clinical cases

Case 1

A 62-year-old man has smoked a pack of cigarettes a day for 40 years. In recent months: cough, hemoptysis, 8 kg weight loss, weakness. Blood sodium 124 mmol/l, urine osmolarity high. Bronchoscopy revealed small cell lung cancer.

Questions: Which carcinogenic factor is leading in the development of the tumor? What is the mechanism of the hyponatremia? What is the weight loss due to?

Answer: The leading factor is the chemical carcinogens in tobacco smoke (benzo(a)pyrene and other polycyclic aromatic hydrocarbons, nitrosamines). After metabolic activation in the liver and lungs, they cause mutations in the DNA of the bronchial epithelium (TP53, KRAS and others) — initiation; many years of repeated exposure and chronic inflammation provide promotion and progression. The hyponatremia is the paraneoplastic syndrome of inappropriate ADH secretion: the tumor cells produce ectopic ADH, excess water is retained by the kidneys, the plasma is diluted and the urine is concentrated. The weight loss is tumor cachexia: anorexia and increased proteolysis and lipolysis under the action of TNF-alpha, IL-1 and IL-6, and the metabolic “trap” of the tumor.

Case 2

A 35-year-old woman's mother died of breast cancer at the age of 42, and her maternal aunt died of ovarian cancer. Genetic testing found a mutation in the BRCA1 gene, but so far there is no tumor.

Questions: To which group of genes does BRCA1 belong? Why is there no tumor yet despite the mutation?

Answer: BRCA1 is a tumor suppressor gene (involved in repairing DNA double-strand breaks). In a hereditary mutation, one allele is defective from birth in all cells (the “first hit”). For a tumor to develop, the second allele must also be damaged somatically in a particular cell (the “second hit”) —

Knudson's hypothesis. Therefore the tumor does not develop immediately, but it develops much earlier and with a much higher probability than in the general population. Such patients are advised to undergo regular screening (MRI, mammography) and preventive measures.

Case 3

A 55-year-old patient has had chronic hepatitis B for 20 years, with liver cirrhosis. The latest examination found a sharp rise in blood alpha-fetoprotein and a focus typical of a tumor in the liver.

Questions: By what mechanisms has the tumor developed? To which type of tumor atypia does the rise in alpha-fetoprotein belong?

Answer: Hepatocellular carcinoma. Hepatitis B virus DNA integrates into the hepatocyte genome and causes insertional mutagenesis; the viral protein HBx blocks p53 and stimulates cell growth (viral carcinogenesis). In addition, chronic inflammation, cycles of necrosis and regeneration, and cirrhosis create conditions for the accumulation of mutations through repeated cell division (promotion). Alpha-fetoprotein is an embryonic protein; its renewed synthesis by tumor cells is a manifestation of antigenic (biochemical) atypia and is used as a tumor marker.

Quick review

Table 14.9. *Key facts on Topic 14*

Question	Answer
Tumor	Uncontrolled proliferation and impaired differentiation of cells due to genetic changes
Signs of a malignant tumor	Anaplasia, invasive growth, metastasis, recurrence
Chemical carcinogens	Benzo(a)pyrene, aromatic amines, nitrosamines, aflatoxin, asbestos
Oncogenic viruses	Papillomavirus, Epstein–Barr, hepatitis B and C, HTLV-1, HHV-8
Stages of carcinogenesis	Initiation, promotion, progression
Oncogenes	Activated proto-oncogenes: RAS, MYC, HER2, BCR-ABL
Tumor suppressor genes	TP53, RB, APC, BRCA1/2 — “two hits”
Warburg effect	Aerobic glycolysis even when oxygen is present
Routes of metastasis	Lymphogenous, hematogenous, implantation, canalicular
Tumor cachexia	TNF-alpha, IL-1, IL-6, metabolic “trap”
Paraneoplastic syndromes	Ectopic ACTH, ADH, erythropoietin, hypercalcemia
Antiblastoma defense	Anticarcinogenic, antimutational, anticellular (NK, T lymphocytes)
Two-stage experiment	The initiator is irreversible; the promoter requires prolonged repeated exposure

TOPIC 15

Changes in the leukocyte differential in disease. Leukocytosis and leukopenia

Learning objectives

To explain to the student the functional and morphological features of the formed elements of the blood, hematopoiesis and leukopoiesis, the normal leukocyte count and differential, the types, causes and mechanisms of leukocytosis and leukopenia, the types of nuclear shift and the nuclear shift index; to teach the student to determine the leukocyte differential in blood smears and interpret it in various diseases.

By the end of the session, the student should know:

- the main stages of hematopoiesis and the factors that regulate leukopoiesis;
- the types of leukocytes, their functions and morphological features;
- the normal leukocyte count and the leukocyte differential;
- the types, causes and mechanisms of leukocytosis and leukopenia;
- agranulocytosis and its consequences;
- the left and right nuclear shifts and the nuclear shift index;
- the changes in the leukocyte differential in various diseases (pneumonia, viral infections, helminth infestations, atopy, trauma, myocardial infarction).

The student should be able to:

- distinguish the types of leukocytes in a blood smear;
- calculate the leukocyte differential and convert it to absolute counts;
- calculate and assess the nuclear shift index;
- interpret the leukogram in relation to the clinical situation.

1. Hematopoiesis and leukopoiesis

Hematopoiesis — the formation of blood cells. In adults it takes place mainly in the red bone marrow of the flat bones and the epiphyses of the long bones. All blood cells arise from a **pluripotent (stem) cell**, which gives rise to progenitor cells of the myeloid and lymphoid lineages.

The main factors regulating leukopoiesis are the **colony-stimulating factors**: granulocyte (G-CSF), granulocyte-macrophage (GM-CSF) and macrophage (M-CSF), as well as interleukins (IL-3 — multipotential, IL-5 — for eosinophils, IL-7 — for lymphocytes).

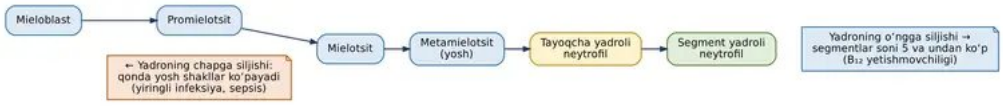


Figure 15.1. *Maturation of granulocytes and nuclear shift (based on Abdullayev N.H., 2008; Zayko N.N., 2002)*

REMEMBER

Neutrophils exist in three “pools” in the body: the **bone marrow pool** (maturing and reserve cells — the largest), the **circulating pool** (free in the bloodstream — measured in the blood test) and the **marginal (wall) pool** (close to the vessel wall, not measured in the blood test). Neutrophils circulate in the blood for only 6–10 hours and then move into the tissues.

2. Leukocytes and the leukocyte differential

In adults the normal leukocyte count in peripheral blood is **4.0–9.0 × 10^9/l**.

Leukocyte differential (leukogram) — the percentage ratio of the individual types of leukocytes in a blood smear.

Table 15.1. *Normal leukocyte differential in adults*

Cell type	Percentage	Absolute count, × 10^9/l	Main function
Neutrophils — band	1–6 %	0,04–0,3	Young form; released from the reserve
Neutrophils — segmented	47–72 %	2,0–5,5	Phagocytosis, defense against bacteria
Eosinophils	0,5–5 %	0,02–0,3	Defense against helminths, modulation of allergic reactions
Basophils	0–1 %	0–0,065	Release of histamine and heparin, type I allergic reactions
Lymphocytes	19–37 %	1,2–3,0	Specific immunity (B — humoral, T — cellular, NK — natural killers)
Monocytes	3–11 %	0,09–0,6	Become macrophages in the tissues, phagocytosis, antigen presentation

CAUTION

Assessing the leukocyte differential only in percentages can lead to a wrong conclusion. For example, in leukopenia the percentage of lymphocytes may be increased while their absolute count is normal — this is **relative lymphocytosis**. Always calculate the absolute count: absolute count = total leukocyte count × percentage / 100.

CLINICAL SIGNIFICANCE

In children the leukocyte differential changes with age: at birth neutrophils predominate; on day 4–6 of life neutrophils and lymphocytes become equal (**first crossover**); then lymphocytes predominate; at 4–6 years they become equal again (**second crossover**), after which the adult differential is established. Assessing a child's blood test against adult norms is a mistake.

3. Leukocytosis

Leukocytosis — a rise in the peripheral blood leukocyte count above 9.0 × 10^9/l.

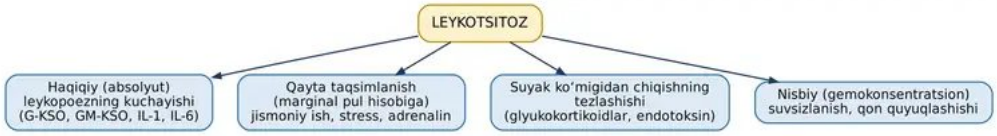


Figure 15.2. *Mechanisms of leukocytosis*

Table 15.2. *Types of leukocytosis*

Type	Essence	Examples
Physiological	In normal states	After meals (alimentary), physical work (myogenic), pregnancy and childbirth, newborns, emotional stress
Pathological (reactive)	In diseases	Infections, inflammation, necrosis, acute blood loss, tumors, intoxications
True (absolute)	Increased leukopoiesis	Bacterial infections, inflammation
Redistributive	Shift from the marginal pool to the circulating pool	Epinephrine, physical exertion, stress — develops quickly and passes quickly
Relative (hemoconcentration)	Hemoconcentration	Dehydration, burns, profuse vomiting

Table 15.3. *Main types of change in the leukocyte differential*

Change	Threshold	Causes
Neutrophilia	Neutrophils above 7.5×10^9/l	Bacterial (purulent) infections, inflammation, myocardial infarction, trauma, acute blood loss, glucocorticoids, diabetic ketoacidosis
Eosinophilia	Eosinophils above $0.4–0.5 \times 10^9$/l	Helminth infestations, allergic diseases (bronchial asthma, pollinosis), drug allergy, some skin diseases, eosinophilic leukemia, Hodgkin lymphoma
Basophilia	Basophils above $0.1–0.2 \times 10^9$/l	Chronic myeloid leukemia, allergic conditions, hypothyroidism
Lymphocytosis	Lymphocytes above 4.0×10^9/l	Viral infections (infectious mononucleosis, rubella, whooping cough), tuberculosis, chronic lymphocytic leukemia
Monocytosis	Monocytes above 0.8×10^9/l	Infectious mononucleosis, tuberculosis, septic endocarditis, malaria, monocytic leukemia, recovery phase of inflammation

MEMORY AID
Causes of eosinophilia — "**PAD**": **P**arasites (helminths), **A**llergy (asthma, pollinosis, drugs), **D**ermatoses.
Lymphocytosis — "**VTP**": **V**iruses, **T**uberculosis, **P**ertussis (whooping cough).

Nuclear shift

The nuclear shift is assessed according to the degree of maturity of the neutrophils.

Table 15.4. *Types of nuclear shift*

Type	Essence	Clinical significance
Regenerative left shift	Band forms and metamyelocytes increase, with leukocytosis	Active response of the bone marrow to infection; prognosis relatively good
Hyperregenerative left shift	Myelocytes and even promyelocytes appear	Severe purulent-septic process
Regenerative-degenerative	Many band forms, but	Reduced reactivity of the body;
(degenerative) left shift	no leukocytosis or leukopenia; degenerative changes in neutrophils (toxic granulation, vacuolization)	poor prognosis

Type	Essence	Clinical significance
Right shift	Hypersegmented neutrophils with 5 or more segments, fewer band forms	Vitamin B_{12} and folic acid deficiency, radiation sickness
Leukemic “gap” (hiatus leucaemicus)	Blast cells and mature neutrophils are present, intermediate forms are absent	Acute leukemia (see topic 16)

Nuclear shift index (NSI) = (myelocytes + metamyelocytes + band forms) / segmented neutrophils. Normally **0.05–0.08**. 0.3–1.0 — moderately severe, above 1.0 — severe process.

Toxic changes in neutrophils

In severe infections, neutrophils show **toxic granulation** (large dark granules), **vacuolization of the cytoplasm**, **Döhle bodies** and nuclear pyknosis — these indicate the severity of the intoxication.

4. Leukopenia

Leukopenia — a fall in the peripheral blood leukocyte count below 4.0×10^9/l. It usually develops through a decrease in neutrophils (**neutropenia**).

Table 15.5. *Mechanisms and causes of leukopenia*

Mechanism	Causes
Suppressed leukopoiesis	Ionizing radiation, cytostatics, benzene, aplastic anemia, infiltration of the bone marrow by tumor (leukemia, metastases), vitamin B_{12} and folic acid deficiency, hereditary neutropenias
Increased destruction of leukocytes	Immune (autoimmune, drug-induced — antibodies), hypersplenism (enlarged spleen), increased consumption in severe infections (sepsis)
Redistributive	Accumulation of leukocytes in the capillaries of the lungs and other organs in the early stage of anaphylactic and transfusion shock
Viral infections	Influenza, measles, viral hepatitis, HIV
Some bacterial infections	Typhoid fever, brucellosis

Agranulocytosis

Agranulocytosis — a sharp decrease or disappearance of granulocytes (neutrophils): granulocytes **below 0.75×10^9/l** or total leukocytes **below 1.0×10^9/l**.

- **Myelotoxic** — direct toxic action on the bone marrow (cytostatics, radiation); other blood cells also decrease.
- **Immune** — antibodies against granulocytes (hapten mechanism: metamizole sodium (analgin), thiamazole, sulfonamides, clozapine) or autoimmune.

Consequences: severe bacterial and fungal infections, necrotic tonsillitis, ulcers of the oral cavity, sepsis; fever is often the only sign.

> **CLINICAL SIGNIFICANCE**
> If fever develops in a patient receiving cytostatic chemotherapy (**febrile neutropenia**), this is an emergency: because neutrophils are low, signs of inflammation (pus, infiltrate) may be absent, and the infection quickly progresses to sepsis. Such patients are immediately given broad-spectrum antibiotics.

5. Changes in the leukogram in various diseases

Table 15.6. *Typical changes in the leukogram in some diseases*

Disease	Leukocytes	Differential
Lobar pneumonia, purulent infections	Leukocytosis	Neutrophilia, left nuclear shift, toxic granulation, eosinopenia
Influenza, most viral infections	Leukopenia or normal	Relative lymphocytosis
Infectious mononucleosis	Leukocytosis	Lymphomonocytosis, atypical mononuclear cells (more than 10 percent)
Whooping cough	High leukocytosis	Absolute lymphocytosis
Tuberculosis (active)	Moderate leukocytosis or normal	Lymphopenia and monocytosis (sign of poor prognosis); lymphocytosis during recovery
Helminth infestations (ascariasis, trichinosis)	Normal or leukocytosis	Eosinophilia (up to 50–60 percent in trichinosis)
Bronchial asthma, pollinosis	Normal	Eosinophilia
Myocardial infarction	Leukocytosis (from day 1, for 3–7 days)	Neutrophilia, left shift
Typhoid fever	Leukopenia	Neutropenia, aneosinophilia, relative lymphocytosis
Sepsis (severe, reduced reactivity)	Leukopenia or leukocytosis	Degenerative left shift

CAUTION

Disappearance of eosinophils (aneosinophilia) is seen in acute infection and stress (effect of glucocorticoids), and their reappearance is a sign of recovery (the “dawn of recovery”). Prolonged absence of eosinophils and lymphopenia in severe pneumonia or sepsis are signs of a poor prognosis.

6. Experimental part: studying a blood smear and determining the leukocyte differential

Aim: to distinguish the types of leukocytes in blood smears from a healthy person and from patients with various diseases, and to calculate and interpret the leukocyte differential.

Equipment: ready-stained blood smears (healthy person and patients with pneumonia, viral infection, helminth infestation, atopy, trauma, myocardial infarction), immersion microscope, immersion oil, 11-key counter (or a table).

Procedure

1. Place the smear under the immersion objective (stained by the Romanowsky–Giemsa or Pappenheim method).
2. At the edge of the smear (where leukocytes are evenly distributed), moving in a meander (zigzag) pattern, count at least **100** consecutive leukocytes, preferably 200.
3. Identify each cell by its morphological features (see the table) and record it on the counter.
4. Calculate the percentage of each type.
5. If the total leukocyte count is known, calculate the absolute counts.
6. Calculate the nuclear shift index.

7. Record toxic changes and atypical cells.
8. Compare the result with the normal values and write a conclusion.

Table 15.7. *Morphological features of leukocytes (Romanowsky–Giemsa stain)*

Cell	Volume	Nucleus	Cytoplasm
Segmented neutrophil	10–12 µm	3–5 segments joined by thin bridges	Pinkish-violet, fine granules
Band neutrophil	10–12 µm	Rod- or horseshoe-shaped, not divided into segments	As in the neutrophil
Eosinophil	12–14 µm	Usually 2 segments (“spectacle” shape)	Large, uniform, orange-red granules
Basophil	10–12 µm	Indistinct, covered by granules	Large, dark violet-black granules
Lymphocyte	7–10 µm (small), 10–12 µm (large)	Round, dense, occupies most of the cell	Narrow, sky-blue rim
Monocyte	14–20 µm (the largest)	Bean- or horseshoe-shaped, delicate chromatin	Abundant, gray-blue (“smoky”), fine dust-like granules

Table 15.8. *Example of recording a leukogram*

Parameter	Patient A	Patient B	Normal
Leukocytes, × 10^9/l			4,0–9,0
Myelocytes, %			0
Metamyelocytes, %			0
Band forms, %			1–6
Segmented forms, %			47–72
Eosinophils, %			0,5–5
Basophils, %			0–1
Lymphocytes, %			19–37
Monocytes, %			3–11
Nuclear shift index			0,05–0,08

7. Practical skills

Skill: calculating and interpreting the leukocyte differential

1. Count at least 100 leukocytes in the smear using the meander method.
2. Determine the percentage of each type.
3. Calculate the absolute count: (leukocyte count × percentage) / 100.
4. Compare the absolute values with the normal ranges: which type is truly increased or decreased.
5. Calculate the nuclear shift index and determine the type of shift.
6. Record toxic and atypical changes.
7. Relate the findings to the clinical situation and write a conclusion (e.g. “neutrophilic leukocytosis with a regenerative left shift — typical of acute bacterial inflammation”).

Skill: calculating the nuclear shift index

1. Add up the percentages of myelocytes, metamyelocytes and band neutrophils.
2. Divide the sum by the percentage of segmented neutrophils.
3. Compare the result with the normal value (0.05–0.08).
4. Assess it together with the leukocyte count: with leukocytosis — regenerative; with leukopenia — degenerative shift.
5. Write a conclusion about the severity of the process and its prognosis.

Review questions

1. Where does hematopoiesis take place and how is it regulated?
2. What “pools” of neutrophils are there?
3. What are the normal leukocyte count and leukocyte differential in adults?
4. Why should the leukocyte differential be assessed in absolute counts?
5. What are the “crossovers” of the leukocyte differential in children?
6. List the types and mechanisms of leukocytosis.
7. What are the causes of neutrophilia, eosinophilia, lymphocytosis and monocytosis?
8. Explain the types of nuclear shift.
9. How is the nuclear shift index calculated and assessed?
10. What do toxic changes in neutrophils indicate?
11. What are the mechanisms and causes of leukopenia?
12. What is agranulocytosis? State its types and consequences.
13. How do pneumonia and influenza differ in the blood test?
14. How does the leukogram change in helminth infestations and atopic diseases?
15. Describe the morphological features of leukocytes.

Test questions

1. What is the normal leukocyte count in adult peripheral blood?
   A) 1.0–3.0 × 10^9/l
   B) 10–15 × 10^9/l
   C) 4.0–9.0 × 10^9/l
   D) 15–20 × 10^9/l
   E) 0.5–1.0 × 10^9/l

2. What is the normal percentage of segmented neutrophils in the leukocyte differential?
   A) 47–72 %
   B) 19–37 %
   C) 3–11 %
   D) 80–95 %
   E) 0,5–5 %

3. What leukogram is seen in lobar pneumonia?
   A) Leukopenia and lymphocytosis
   B) Eosinophilia
   C) Right nuclear shift
   D) Neutrophilic leukocytosis and left nuclear shift
   E) Monocytosis and atypical mononuclear cells

4. Which change is typical of ascariasis and trichinosis?
   A) Basophilia
   B) Neutrophilia
   C) Lymphopenia
   D) Monocytosis
   E) Eosinophilia

5. Which leukogram is typical of whooping cough?
   A) Neutropenia
   B) High leukocytosis and absolute lymphocytosis
   C) Eosinophilia
   D) Agranulocytosis
   E) Monocytopenia

6. What is the normal nuclear shift index?
   A) 0,5–1,0
   B) 0,05–0,08
   C) 1,0–2,0
   D) 0,3–0,5
   E) above 2.0

7. Which type of nuclear shift is seen in vitamin B_{12} deficiency?
   A) Regenerative left shift
   B) Hyperregenerative left shift
   C) Degenerative left shift
   D) Right shift (hypersegmentation)
   E) Leukemic “gap”

8. If there are many band neutrophils but leukopenia, what kind of shift is this?
   A) Degenerative left shift
   B) Right shift
   C) Regenerative left shift
   D) Normal state
   E) Leukemic “gap”

9. Which sign is typical of infectious mononucleosis?
   A) Agranulocytosis
   B) Eosinophilia
   C) Lymphomonocytosis and atypical mononuclear cells
   D) Basophilia
   E) Neutrophilia and toxic granulation

10. What is the diagnostic criterion of agranulocytosis?
    A) Leukocytes below 4.0×10^9/l
    B) Lymphocytes below 1.0×10^9/l
    C) Absence of eosinophils
    D) Monocytes below 0.1×10^9/l
    E) Granulocytes below 0.75×10^9/l

11. What is the mechanism of the leukocytosis that develops rapidly under physical exertion and epinephrine?
    A) Increased leukopoiesis
    B) Hemoconcentration
    C) Bone marrow hyperplasia
    D) Redistribution from the marginal pool
    E) Leukemic transformation

12. What leukogram is seen in typhoid fever?
    A) Neutrophilic leukocytosis
    B) Monocytosis and atypical cells
    C) Eosinophilia
    D) Basophilia
    E) Leukopenia, aneosinophilia, relative lymphocytosis

13. Which is the largest leukocyte in a blood smear?
    A) Lymphocyte
    B) Monocyte
    C) Basophil
    D) Eosinophil
    E) Band neutrophil

14. Which is a sign of poor prognosis in active tuberculosis?
    A) Lymphopenia and monocytosis
    B) Lymphocytosis
    C) Eosinophilia
    D) Basophilia
    E) Normal leukogram

15. Leukopenia 3.0×10^9/l, lymphocytes 50 %. What is the absolute lymphocyte count and how is it assessed?
    A) 6.0×10^9/l — absolute lymphocytosis
    B) 0.5×10^9/l — lymphopenia
    C) 1.5×10^9/l — relative lymphocytosis (absolute count normal)
    D) 3.0×10^9/l — absolute lymphocytosis
    E) 0.15×10^9/l — severe lymphopenia

Clinical cases

Case 1

On day 5 after surgery for acute appendicitis, a 30-year-old patient has a fever of 39 °C and increasing abdominal pain. Blood test: leukocytes 18.5×10^9/l; myelocytes 2 %, metamyelocytes 6

%, band forms 18 %, segmented forms 60 %, eosinophils 0 %, lymphocytes 10 %, monocytes 4 %. Toxic granulation in neutrophils.

Questions: Interpret the leukogram. Calculate the nuclear shift index. What complication can be suspected?

Answer: Neutrophilic leukocytosis (absolute neutrophils ≈ 15.9 × 10^9/l), hyperregenerative left shift (myelocytes and metamyelocytes have appeared), aneosinophilia, relative lymphopenia, toxic granulation. NSI = (2 + 6 + 18) / 60 ≈ 0.43 — increased, indicating a severe purulent inflammatory process. A purulent complication (abdominal abscess or peritonitis) can be suspected. Cytokines and colony-stimulating factors released from the purulent focus have sharply increased neutrophil production and the release of young forms from the bone marrow; aneosinophilia reflects the effect of stress and glucocorticoids.

Case 2

An 8-year-old child has itchy skin, restless sleep and abdominal pain. Blood test: leukocytes 8.2 × 10^9/l; eosinophils 18 %, other values close to normal. Ascaris eggs were found in the stool.

Questions: Assess the change in the leukogram. What is the mechanism of the eosinophilia?

Answer: Absolute eosinophilia: 8.2 × 18 / 100 ≈ 1.48 × 10^9/l (normal up to 0.3 × 10^9/l). Helminth antigens induce a Th2-type immune response: Th2 lymphocytes secrete IL-5, which enhances the production and maturation of eosinophils in the bone marrow; IL-4 and IL-13 increase IgE synthesis. Eosinophils adhere to helminths coated with IgE and IgG and damage them by releasing major basic protein and other toxic substances. The itching is the allergic (IgE-mediated) component.

Case 3

A 60-year-old woman is taking thiamazole for an overactive thyroid. In week 6 of treatment she developed a fever of 39.5 °C, severe sore throat and necrotic ulcers on the tonsils. Blood test: leukocytes 1.1 × 10^9/l, granulocytes 0.2 × 10^9/l, erythrocytes and platelets normal.

Questions: What condition has developed? What are its mechanism and danger?

Answer: Drug-induced immune agranulocytosis (granulocytes far below 0.75 × 10^9/l; erythrocytes and platelets normal — not a myelotoxic but an immune mechanism). Thiamazole, acting as a hapten, binds to proteins on the neutrophil surface and causes the formation of antibodies against them and destruction of neutrophils in the blood and bone marrow. Because of the absence of phagocytes, the oral microflora has caused necrotic tonsillitis; the infection can quickly progress to sepsis. The drug is stopped immediately, the patient is placed in an isolated (aseptic) ward, and broad-spectrum antibiotics and granulocyte colony-stimulating factor are prescribed.

Quick review

Table 15.9. *Key facts on Topic 15*

Question	Answer
Normal leukocyte count	4.0–9.0 × 10^9/l
Leukocyte differential	Band 1–6 %, segmented 47–72 %, eosinophils 0.5–5 %, basophils 0–1 %, lymphocytes 19–37 %, monocytes 3–11 %
Absolute count	Leukocyte count × percentage / 100
Leukocytosis	Above 9.0 × 10^9/l; true, redistributive, relative

Question	Answer
Neutrophilia	Bacterial infection, inflammation, infarction, trauma
Eosinophilia	Helminths, allergy, dermatoses
Lymphocytosis	Viral infections, tuberculosis, whooping cough, lymphocytic leukemia
Monocytosis	Mononucleosis, tuberculosis, septic endocarditis, malaria
Left shift	Increase in young neutrophils: regenerative, hyperregenerative, degenerative
Right shift	Hypersegmentation: vitamin B_{12} and folic acid deficiency
NSI	(Myelocytes + metamyelocytes + band forms) / segmented forms; normal 0.05–0.08
Leukopenia	Below 4.0×10^9/l
Agranulocytosis	Granulocytes below 0.75×10^9/l; myelotoxic or immune
Toxic granulation	Large dark granules in neutrophils in severe infection

TOPIC 16

Leukemias: definition, etiology, pathogenesis and types. Leukemoid reactions

Learning objectives

To explain to the student the definition, etiology and pathogenesis of leukemias, their classification, the features of acute and chronic leukemias, the changes in peripheral blood and bone marrow in leukemia, the main hematological phenomena, the local and systemic changes in the body, and the difference between leukocytosis and leukemia (leukemoid reactions); to teach the student to study and analyze peripheral blood smears in acute and chronic myeloid leukemia, lymphocytic leukemia and undifferentiated leukemia.

By the end of the session, the student should know:

- the definition of leukemia and its neoplastic nature;
- the etiological factors and pathogenesis of leukemias (clonal theory);
- the classification of leukemias (acute and chronic, myeloid and lymphoid);
- the changes in blood and bone marrow in acute leukemias, the leukemic “gap”;
- the hematological signs of chronic myeloid leukemia and chronic lymphocytic leukemia;
- the main clinical syndromes of leukemias and their mechanisms;
- leukemoid reactions and how they differ from leukemia.

The student should be able to:

- identify blast cells and the leukemic “gap” by studying the blood smear in leukemias;
- distinguish smears of chronic myeloid leukemia and lymphocytic leukemia;
- distinguish leukemia from a leukemoid reaction by the blood test.

1. The concept of leukemia

Leukemia (hemoblastosis) — a neoplastic disease of the hematopoietic tissue, characterized by uncontrolled proliferation of tumor cells (a leukemic clone) derived from a single mutated hematopoietic cell in the bone marrow and other organs, and by displacement of normal hematopoiesis.

REMEMBER

The three main features of leukemia:

- **neoplastic nature** — leukemic cells have all the properties of tumor cells (uncontrolled proliferation, atypia, metastasis — leukemic infiltrates);
- **clonal origin** — all leukemic cells are descendants of a single mutated cell;
- **location** — the tumor starts in the bone marrow and spreads to the blood (leukemic cells do not have to be numerous in the blood).

The term “leukemia” (“white blood”) is historical: it refers to the whitish color the blood takes on when leukocytes are greatly increased (R. Virchow, 1845). However, only some leukemias have leukocytosis in the blood.

2. Etiology of leukemias

Table 16.1. *Etiological factors of leukemias*

Group	Examples
Ionizing radiation	Leukemias increased among the population of Hiroshima and Nagasaki, after the Chernobyl accident and after radiotherapy
Chemical factors	Benzene and its derivatives, cytostatics (alkylating agents), some pesticides, smoking
Viruses	HTLV-1 (adult T-cell leukemia), Epstein–Barr virus (Burkitt lymphoma)
Hereditary and genetic factors	Down syndrome (10–20 times higher risk of acute leukemia), Fanconi anemia, Bloom syndrome, ataxia-telangiectasia
Previous hematological diseases	Myelodysplastic syndrome, myeloproliferative diseases

3. Pathogenesis of leukemias

According to the modern **clonal theory**, leukemia begins with a mutation (often a chromosomal translocation) in a single hematopoietic stem or progenitor cell. As a result of the mutation:

- **proliferation** of the cell escapes control (oncogenes are activated);
- **differentiation** (maturation) is blocked — in acute leukemias the cells “get stuck” at the blast stage;
- **apoptosis** is impaired — the cells live long (especially in chronic lymphocytic leukemia).

As the leukemic clone develops, new mutations accumulate and the tumor becomes more malignant — **tumor progression**: transition from the monoclonal to the polyclonal stage, emergence of resistance to cytostatics, and development of a **blast crisis** in chronic leukemias.

> **CLINICAL SIGNIFICANCE**
> An example of a common chromosomal change in leukemias is the **Philadelphia chromosome** t(9;22): the ABL gene on chromosome 9 fuses with the BCR gene on chromosome 22, producing a permanently active tyrosine kinase. It is found in almost all cases of chronic myeloid leukemia. Imatinib, which selectively blocks this kinase, was the first successful example of targeted therapy in cancer.

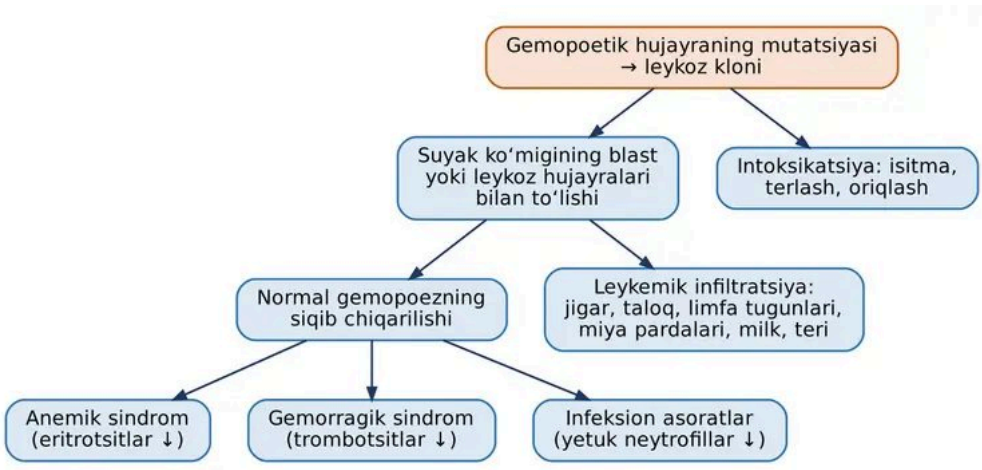


Figure 16.1. *Main clinical syndromes of leukemias and their mechanisms*

4. Classification of leukemias

Leukemias are classified by two main criteria: **the degree of cell maturity** (acute and chronic) and **the cell lineage** (myeloid and lymphoid).

Table 16.2. *Classification of leukemias*

Criterion	Groups
Cell differentiation	Acute — the tumor substrate consists of immature blast cells; chronic — the tumor substrate consists mainly of maturing and mature cells
Cell lineage	Myeloid (myeloblast, monoblast, erythroblast, megakaryoblast lineage), lymphoid (B- and T-lymphocyte lineage), undifferentiated
Peripheral blood leukocyte count	Leukemic (above 50–100 × 10^9/l), subleukemic (slightly increased), leukopenic (decreased), aleukemic (no leukemic cells in the blood)

CAUTION
The terms “acute” and “chronic” leukemia refer not to the duration of the disease but to the degree of maturity of the tumor cells. Acute leukemia never turns into chronic leukemia, whereas chronic myeloid leukemia may pass into an “acute leukemia-like” state with a blast crisis in its final stage.

5. Acute leukemias

In acute leukemias the leukemic cells have lost the ability to mature and have stopped at the blast stage. The bone marrow contains **more than 20 percent blast cells** (WHO criterion).

Table 16.3. *Main types of acute leukemia*

Type	Features
Acute lymphoblastic leukemia	The most common leukemia in children (2–5 years); lymph nodes, liver and spleen enlarge, the meninges are often involved; with modern treatment the cure rate in children exceeds 80–90 percent
Acute myeloblastic leukemia	More common in adults; Auer rods in blasts
Acute promyelocytic leukemia	t(15;17) translocation; accompanied by severe DIC; differentiation therapy with a vitamin A derivative (all-trans retinoic acid) is effective
Acute monoblastic leukemia	Gum and skin infiltration common
Undifferentiated leukemia	The blasts show no features of any lineage

Peripheral blood in acute leukemia

- the leukocyte count may vary — from leukopenia to high leukocytosis;
- **blast cells** appear;
- **leukemic “gap” (hiatus leucaemicus)** — the blood contains blast cells and mature (segmented) neutrophils, while the intermediate stages of maturation (promyelocyte, myelocyte, metamyelocyte) are absent or very few; this reflects the immaturity of the leukemic clone and the remnants of normal hematopoiesis;
- **normochromic anemia** and **thrombocytopenia**.

Table 16.4. *Morphological features of a blast cell*

Feature	Feature
Volume	Medium or large
Nucleus	Round, occupies most of the cell; chromatin delicate, reticular
Nucleoli	1–3, clearly visible
Cytoplasm	Narrow, sky-blue to blue (basophilic); Auer rods may be present in myeloblasts

6. Chronic leukemias

Chronic myeloid leukemia

Chronic myeloid leukemia is a myeloproliferative disease arising from a stem cell and characterized by the Philadelphia chromosome.

Peripheral blood: high leukocytosis (50–200 × 10^9/l or more); **all stages of granulocyte maturation** are present in the blood (myeloblasts, promyelocytes, myelocytes, metamyelocytes, band and segmented neutrophils) — a “marked left nuclear shift”, with **no** leukemic “gap”; **eosinophil–basophil association** (eosinophils and basophils increase together); thrombocytosis may be present at the onset; neutrophil alkaline phosphatase activity is reduced.

Clinical signs: marked enlargement of the spleen (splenomegaly), enlargement of the liver, weakness, sweating.

Table 16.5. *Phases of chronic myeloid leukemia*

Stage	Feature
Chronic (stable) phase	Few blasts in blood and bone marrow (below 10 percent); mature cells predominate; good response to treatment
Accelerated phase	Blasts and basophils increase, resistance to treatment appears
Blast crisis	Blasts above 20 percent; course resembling acute leukemia; poor prognosis

Chronic lymphocytic leukemia

Chronic lymphocytic leukemia is a tumor characterized by the accumulation of mature-looking but functionally defective B lymphocytes; it occurs mainly in the elderly (the most common leukemia in adults in Western countries).

Peripheral blood: leukocytosis; **absolute lymphocytosis** (lymphocytes up to 80–90 percent), mainly small mature lymphocytes; **Botkin–Gumprecht shadows** (smudge cells) — remnants of lymphocyte nuclei destroyed during smear preparation (because of the fragility of the lymphocytes).

Clinical signs: enlargement of lymph nodes in many regions (painless, soft-elastic), splenomegaly; recurrent infections due to hypogammaglobulinemia; autoimmune hemolytic anemia and thrombocytopenia.

MEMORY AID

Distinguishing chronic leukemias in a smear: **myeloid leukemia** — “all the steps of the granulocyte ladder + the eosinophil–basophil pair”; **lymphocytic leukemia** — “a sea of small lymphocytes + Botkin–Gumprecht shadows”.

7. Changes in the body in leukemias

Table 16.6. *Main syndromes of leukemias*

Syndrome	Mechanism	Manifestations
Anemic	Displacement of erythropoiesis, bleeding, hemolysis	Weakness, pallor, dyspnea, tachycardia
Hemorrhagic	Thrombocytopenia, DIC (in promyelocytic leukemia), infiltration of the vessel wall	Petechiae, ecchymoses, nose and gum bleeding, cerebral hemorrhage
Infectious	Decrease in mature neutrophils, functional defect of leukemic cells, immunodeficiency, cytostatic therapy	Pneumonia, sepsis, necrotic tonsillitis, fungal infections
Hyperplastic (infiltrative)	Accumulation of leukemic cells in organs (“leukemic metastases”)	Enlargement of lymph nodes, liver, spleen; bone pain; gum hyperplasia; neuroleukemia (meninges)
Intoxication (metabolic)	Tumor metabolism, massive cell breakdown	Fever, sweating, weight loss; hyperuricemia (gout, kidney stones); tumor lysis syndrome

CLINICAL SIGNIFICANCE

Tumor lysis syndrome develops when chemotherapy for leukemia is started, as a result of the simultaneous breakdown of many tumor cells: hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia and acute kidney failure. To prevent it, plenty of fluid and agents that reduce uric acid formation are given before treatment begins.

8. Leukemoid reactions

Leukemoid reactions — changes in the peripheral blood resembling leukemia (high leukocytosis, appearance of young cells), but they are not neoplastic in nature and disappear when the underlying disease is eliminated.

Table 16.7. *Types of leukemoid reactions*

Type	Causes	Blood picture
Myeloid (neutrophilic)	Severe bacterial infections (sepsis, pneumonia), tumor metastases, severe burns, acute hemolysis	High leukocytosis, left shift as far as myelocytes, toxic granulation
Eosinophil	Helminth infestations (trichinosis, echinococcosis), allergy	Many eosinophils
Lymphocytic	Infectious mononucleosis, whooping cough, cytomegalovirus infection	Lymphocytosis, atypical lymphocytes
Monocytic	Tuberculosis, septic endocarditis	Monocytosis

Table 16.8. *Differences between leukemia and a myeloid leukemoid reaction*

Feature	Chronic myeloid leukemia	Myeloid leukemoid reaction
Cause	Tumor (clonal)	Infection, tumor metastasis and others (reactive)
Philadelphia chromosome	Present	No
Neutrophil alkaline phosphatase	Decreased	Increased

Feature	Chronic myeloid leukemia	Myeloid leukemoid reaction
Eosinophil–basophil association	Present	No
Toxic granulation	Usually absent	Often present
Splenomegaly	Marked	Little or none
After treatment of the underlying disease	It does not change	Disappears

9. Experimental part: studying peripheral blood smears in leukemias

Aim: to identify the characteristic features of blood smears in acute and chronic leukemias.

Equipment: stained blood smears (acute myeloid leukemia, acute lymphoblastic leukemia, undifferentiated leukemia, chronic myeloid leukemia, chronic lymphocytic leukemia, myeloid leukemoid reaction), immersion microscope, hematology atlas.

Procedure

1. Examine each smear first at low magnification, then under immersion.
2. Roughly estimate the density of leukocytes (many or few).
3. Prepare a leukogram of at least 100 cells; count the blast cells separately.
4. Check whether the stages of granulocyte maturation are present (whether there is a leukemic “gap”).
5. Record eosinophils and basophils, Botkin–Gumprecht shadows and Auer rods.
6. Assess the state of the erythrocytes and platelets.
7. Write a conclusion for each smear and enter the main distinguishing features in a table.

Table 16.9. *Main features of the peripheral blood picture in leukemias*

Feature	Acute leukemia	Chronic myeloid leukemia	Chronic lymphocytic leukemia
Main cells	Blasts	All stages of granulocytes	Small mature lymphocytes
Leukemic “gap”	Present	No	No
Eosinophil–basophil association	No	Present	No
Botkin–Gumprecht shadows	No	No	Present
Anemia and thrombocytopenia	Early and marked	In the late phase	In the late phase or autoimmune

10. Practical skills

Skill: determining the type of leukemia from the smear

1. Determine the percentage of blast cells: if high — acute leukemia or blast crisis.
2. Check whether the intermediate stages of granulocytes are present (leukemic “gap”).
3. If granulocytes of all stages and an eosinophil–basophil association are present, suspect chronic myeloid leukemia.

4. If small lymphocytes predominate and Botkin–Gumprecht shadows are present, suspect chronic lymphocytic leukemia.
5. Assess the degree of anemia and thrombocytopenia.
6. Indicate that bone marrow examination, immunophenotyping and cytogenetic analysis are needed for the final diagnosis.

Skill: distinguishing leukemia from a leukemoid reaction

1. Assess the clinical situation: is there an obvious infection or other cause?
2. Check for toxic granulation and an eosinophil–basophil association.
3. Take into account the degree of splenomegaly.
4. Assess the results of neutrophil alkaline phosphatase and the Philadelphia chromosome.
5. Follow the change in the blood picture as the underlying disease is treated.
6. Write a conclusion.

Review questions

1. What is called leukemia? What are its main features?
2. List the etiological factors of leukemias.
3. Explain the clonal theory of leukemias.
4. How does tumor progression manifest itself in leukemias?
5. What is the Philadelphia chromosome and what is its significance?
6. How are leukemias classified?
7. How do acute and chronic leukemias differ?
8. What is the leukemic “gap” and why is it typical of acute leukemia?
9. What are the morphological features of a blast cell?
10. What is the blood picture in chronic myeloid leukemia?
11. What are the hematological and clinical signs of chronic lymphocytic leukemia?
12. Explain the main syndromes of leukemias and their mechanisms.
13. What is tumor lysis syndrome?
14. What are leukemoid reactions and what are their types?
15. How can chronic myeloid leukemia be distinguished from a myeloid leukemoid reaction?

Test questions

1. What is the essence of leukemia?
   A) Reactive increase in leukocytes
   B) A clonal neoplastic disease of the hematopoietic tissue
   C) Response of leukocytes to infection
   D) Increased destruction of erythrocytes
   E) Bone marrow aplasia

2. What is the most common leukemia in children?
   A) Chronic myeloid leukemia
   B) Chronic lymphocytic leukemia
   C) Acute promyelocytic leukemia
   D) Acute monoblastic leukemia
   E) Acute lymphoblastic leukemia

3. Which leukemia is the leukemic “gap” (hiatus leucaemicus) typical of?
   A) Chronic myeloid leukemia
   B) Chronic lymphocytic leukemia
   C) Acute leukemia
   D) Myeloid leukemoid reaction
   E) Infectious mononucleosis

4. Which chromosomal change is typical of chronic myeloid leukemia?
   A) Philadelphia chromosome t(9;22)
   B) Trisomy 21
   C) t(15;17)
   D) Monosomy X
   E) t(8;14)

5. What percentage of blast cells in the bone marrow is required to diagnose acute leukemia (WHO)?
   A) more than 5 %
   B) more than 1 %
   C) more than 10 %
   D) more than 20 %
   E) more than 50 %

6. Which leukemia are Botkin–Gumprecht shadows typical of?
   A) Chronic lymphocytic leukemia
   B) Chronic myeloid leukemia
   C) Acute myeloblastic leukemia
   D) Acute promyelocytic leukemia
   E) Erythroleukemia

7. Which blood picture is typical of chronic myeloid leukemia?
   A) Only blasts and mature neutrophils
   B) Absolute lymphocytosis
   C) All stages of granulocyte maturation and an eosinophil–basophil association
   D) Agranulocytosis
   E) Monocytosis and atypical mononuclear cells

8. Which severe complication is common in acute promyelocytic leukemia?
   A) Hypercalcemia
   B) Polyglobulia
   C) Hypoglycemia
   D) DIC
   E) Thrombocytosis

9. What is the main mechanism of anemia in leukemias?
   A) Iron deficiency only
   B) Vitamin B_{12} deficiency
   C) Increased destruction of erythrocytes in the spleen
   D) Excess erythropoietin
   E) Displacement of normal erythropoiesis by leukemic cells

10. What is the main cause of infectious complications in leukemias?
   A) Increased total leukocyte count
   B) Deficiency of functionally mature neutrophils and of immunity
   C) Increased erythrocytes
   D) Thrombocytosis
   E) Fever

11. Which sign distinguishes a myeloid leukemoid reaction from chronic myeloid leukemia?
   A) High leukocytosis
   B) Myelocytes in the blood
   C) Increased neutrophil alkaline phosphatase and absence of the Philadelphia chromosome
   D) Marked splenomegaly
   E) Eosinophil–basophil association

12. What is the cause of recurrent infections in chronic lymphocytic leukemia?
   A) Neutrophilia
   B) Hypogammaglobulinemia
   C) Erythrocytosis
   D) Thrombocytosis
   E) Eosinophilia

13. Which characteristic inclusions are found in myeloblasts?
   A) Auer rods
   B) Botkin–Gumprecht shadows
   C) Howell–Jolly bodies
   D) Cabot rings
   E) Toxic granulation

14. Which sign belongs to the hyperplastic (infiltrative) syndrome in leukemias?
   A) Pallor
   B) Petechiae
   C) Necrotic tonsillitis
   D) Gum hyperplasia and splenomegaly
   E) Fever

15. Which condition develops as a result of massive breakdown of tumor cells when chemotherapy is started?
   A) Blast crisis
   B) Leukemoid reaction
   C) Leukemic “gap”
   D) Neuroleukemia
   E) Tumor lysis syndrome

Clinical cases

Case 1

A 4-year-old child has had weakness, pallor, fever, leg pain and bruises on the skin for 3 weeks. The cervical lymph nodes, liver and spleen are enlarged. Blood test: hemoglobin 72 g/l, platelets 25 × 10^9/l, leukocytes 32 × 10^9/l, blast cells 68 %, segmented neutrophils 8 %, no intermediate stages, lymphocytes 24 %.

Questions: What disease can be suspected? Explain the changes in the blood test and the clinical signs.

Answer: Acute leukemia; given the child's age and enlarged lymph nodes, probably acute lymphoblastic leukemia (bone marrow examination and immunophenotyping are needed for confirmation). The blood contains many blast cells and mature neutrophils, with no intermediate stages — the leukemic “gap”: the leukemic clone does not mature, while the mature neutrophils come from the remnants of normal hematopoiesis. Anemia and thrombocytopenia — the bone marrow is filled with blasts and normal erythro- and thrombopoiesis are displaced; pallor and weakness — the anemic syndrome; bruises — the hemorrhagic syndrome. Fever — infectious complications and intoxication. Bone pain — bone marrow infiltration; enlargement of the lymph nodes, liver and spleen — the hyperplastic (infiltrative) syndrome.

Case 2

A 50-year-old man complains of heaviness under the left costal margin, sweating and weakness. The spleen is markedly enlarged. Blood test: leukocytes 150 × 10^9/l; myeloblasts 2 %, promyelocytes 4 %, myelocytes 15 %, metamyelocytes 12 %, band forms 16 %, segmented forms 38 %, eosinophils 6 %, basophils 5 %, lymphocytes 2 %. Platelets 520 × 10^9/l.

Questions: What is the disease? Which signs confirm it? By what test can the diagnosis be confirmed?

Answer: Chronic myeloid leukemia, chronic phase. Signs confirming it: very high leukocytosis; all stages of granulocyte maturation in the blood (from myeloblast to segmented form), no leukemic “gap”; eosinophil–basophil association (eosinophils 6 %, basophils 5 %); thrombocytosis; few blasts; marked splenomegaly (infiltration of the spleen by leukemic cells). The diagnosis is confirmed by detecting the Philadelphia chromosome t(9;22) on cytogenetic testing or the BCR-ABL gene on molecular testing; neutrophil alkaline phosphatase is reduced.

Case 3

A 45-year-old patient has severe right-sided pneumonia and sepsis. Blood test: leukocytes 48 × 10^9/l, myelocytes 6 %, metamyelocytes 10 %, band forms 22 %, segmented forms 54 %, lymphocytes 6 %, monocytes 2 %, no eosinophils or basophils. Toxic granulation in neutrophils. The spleen is not enlarged.

Questions: Is this leukemia or a leukemoid reaction? Justify your answer.

Answer: Myeloid (neutrophilic) leukemoid reaction. Reasons: an obvious cause — severe pneumonia and sepsis; toxic granulation in neutrophils (a sign of a reactive process); no eosinophil–basophil association (on the contrary, aneosinophilia — a sign of stress and infection); no splenomegaly. Neutrophil alkaline phosphatase is expected to be increased and the Philadelphia chromosome absent. After the infection is treated, the blood picture returns to normal.

Quick review

Table 16.10. *Key facts on Topic 16*

Question	Answer
Leukemia	A clonal neoplastic disease of the hematopoietic tissue
Etiology	Radiation, benzene, cytostatics, viruses (HTLV-1), Down syndrome
Clonal theory	All leukemic cells come from a single mutated cell
Acute leukemia	Blasts predominate (more than 20 percent in the bone marrow), differentiation block
Leukemic “gap”	Blasts and mature cells are present, intermediate stages absent — acute leukemia
Chronic myeloid leukemia	Philadelphia chromosome, all stages, eosinophil–basophil association, splenomegaly
Blast crisis	The final phase of chronic myeloid leukemia
Chronic lymphocytic leukemia	Absolute lymphocytosis, Botkin–Gumprecht shadows, the elderly
Main syndromes	Anemic, hemorrhagic, infectious, hyperplastic, intoxication
Tumor lysis syndrome	Hyperkalemia, hyperuricemia, kidney failure
Leukemoid reaction	Resembles leukemia but is reactive; disappears when the cause is removed
Distinguishing features	Alkaline phosphatase, Philadelphia chromosome, toxic granulation

TOPIC 17

Pathophysiology of erythrocytes. Classification of anemias. Posthemorrhagic and hemolytic anemias

Learning objectives

To explain to the student the stages of erythropoiesis and its regulation, the criteria for assessing cells of the erythroid series, the pathological forms of erythrocytes, the concept and classification of anemia, and the etiology, pathogenesis and blood picture of acute and chronic posthemorrhagic anemias and hemolytic anemias; to teach the student to identify pathological forms of erythrocytes in blood smears, to detect reticulocytes by supravital staining and to measure hemoglobin concentration by the colorimetric method.

By the end of the session, the student should know:

- the stages of erythropoiesis and the factors that regulate it (erythropoietin, iron, B_{12}, folic acid);
- the red blood cell indices and their normal values;
- the pathological forms of erythrocytes (anisocytosis, poikilocytosis, anisochromia, regenerative and degenerative forms);
- the definition and classification of anemia (by pathogenesis, color index, regeneration);
- the stages of compensation and the blood picture in acute posthemorrhagic anemia;
- the types of hemolytic anemia and the signs of intravascular and extravascular hemolysis.

The student should be able to:

- identify pathological forms of erythrocytes in a blood smear;
- count reticulocytes with supravital staining and assess the result;
- determine hemoglobin concentration and calculate the red blood cell indices;
- determine the type of anemia from the blood test.

1. Erythropoiesis and its regulation

Erythropoiesis — the formation of erythrocytes; in adults it takes place in the red bone marrow. Sequence: stem cell → erythroid progenitors (BFU-E, CFU-E) → proerythroblast → basophilic, polychromatophilic and oxyphilic (orthochromic) normoblasts (the nucleus is extruded) → **reticulocyte** (1–2 days in the bone marrow, 1 day in the blood) → **mature erythrocyte** (lifespan about 120 days).

Table 17.1. *Main factors regulating erythropoiesis*

Factor	Significance
Erythropoietin	Produced in the peritubular cells of the kidney in response to hypoxia; stimulates proliferation and differentiation of CFU-E
Iron	For hemoglobin (heme) synthesis
Vitamin B_{12} and folic acid	For DNA synthesis and cell division
Proteins, vitamin B_6, copper, vitamin C	Globin and heme synthesis, iron absorption
Androgens, thyroid hormones	Stimulate erythropoiesis

Table 17.2. *Normal red blood cell values (adults)*

Parameter	Men	Women
Hemoglobin (Hb)	130–160 g/l	120–140 g/l
Erythrocytes	4.0–5.0 × 10^{12}/l	3.7–4.7 × 10^{12}/l
Hematocrit	40–48 %	36–42 %
Color index	0,85–1,05	0,85–1,05
Mean corpuscular volume (MCV)	80–100 fl	80–100 fl
Mean corpuscular hemoglobin (MCH)	27–33 pg	27–33 pg
Reticulocytes	0,2–1,2 % (5–12 ‰)	0,2–1,2 %

2. Pathological forms of erythrocytes

Table 17.3. *Pathological changes in erythrocytes*

Group	Change	Significance
By size (anisocytosis)	Microcytes (smaller than 6.5 µm), macrocytes (larger than 8 µm), megalocytes (10–12 µm or more, oval)	Microcytosis — iron deficiency; megalocytosis — B_{12} and folate deficiency
By shape (poikilocytosis)	Spherocytes, elliptocytes, target cells, sickle cells, schistocytes (fragments), acanthocytes	Spherocytosis — Minkowski–Chauffard disease; sickle cells — hemoglobinopathy; schistocytes — microangiopathic hemolysis
By staining (anisochromia)	Hypochromia (widened central pallor), hyperchromia	Hypochromia — iron deficiency, thalassemia
Regenerative forms	Reticulocytes, polychromatophils, normoblasts (oxyphilic and polychromatophilic)	Active erythropoiesis — acute blood loss, hemolysis
Degenerative forms and inclusions	Howell–Jolly bodies (nuclear remnants), Cabot rings, basophilic stippling	Megaloblastic anemia, after splenectomy, lead poisoning
Megaloblastic forms	Megaloblasts and megalocytes	Vitamin B_{12} and folic acid deficiency

3. Erythrocytoses (polyglobulias)

Erythrocytosis — an increase in the erythrocyte count and hemoglobin in the blood.

- **Primary (polycythemia vera, Vaquez disease)** — a myeloproliferative neoplastic disease (JAK2 mutation), independent of erythropoietin; accompanied by leukocytosis and thrombocytosis.
- **Secondary absolute** — increased erythropoietin: hypoxia (high altitude, chronic lung diseases, cyanotic congenital heart defects, smoking), kidney diseases and tumors (ectopic erythropoietin).
- **Relative (hemoconcentration)** — reduced plasma volume (dehydration, burns); the total erythrocyte mass is unchanged.

4. The concept and classification of anemia

Anemia — a decrease in hemoglobin (and often erythrocytes) per unit volume of blood. WHO criterion: hemoglobin below 130 g/l in men, below 120 g/l in women, below 110 g/l in pregnant women.

Table 17.4. *Classification of anemias*

Basis of classification	Groups
By pathogenesis	1) Due to blood loss (posthemorrhagic — acute and chronic); 2) due to impaired erythropoiesis (iron, B_{12}, folate deficiency, aplastic, anemia of chronic disease); 3) due to increased destruction of erythrocytes (hemolytic)
By color index (or MCH)	Hypochromic (below 0.85), normochromic (0.85–1.05), hyperchromic (above 1.05)
By erythrocyte volume (MCV)	Microcytic (below 80 fl), normocytic (80–100 fl), macrocytic (above 100 fl)
By the regenerative capacity of the bone marrow (reticulocytes)	Regenerative (0.5–2 %), hyperregenerative (above 2 percent), hyporegenerative (below 0.5 percent), aregenerative (reticulocytes almost absent)
By severity	Mild (Hb above 90 g/l), moderate (70–90 g/l), severe (below 70 g/l)

Color index = (Hb, g/l × 3) / the first three digits of the erythrocyte count. For example, if Hb is 120 g/l and erythrocytes 4.0 × 10^{12}/l: (120 × 3) / 400 = 0.9.

REMEMBER

The general signs of anemia are related to hypoxia and compensatory reactions: weakness, fatigue, dizziness, tinnitus, “floaters” before the eyes, dyspnea, palpitations, tachycardia, pallor of the skin and mucous membranes, a functional systolic murmur at the cardiac apex (reduced blood viscosity and increased flow velocity).

5. Posthemorrhagic anemias

Acute posthemorrhagic anemia

Develops as a result of losing a large amount of blood in a short time. Loss of 10–15 percent of the circulating blood volume (CBV) is usually well tolerated, 25–30 percent is a severe condition, and more than 40–50 percent is life-threatening (hemorrhagic shock).

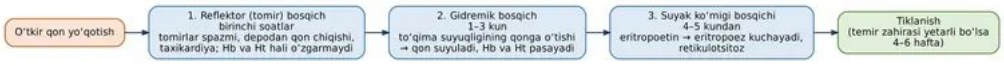


Figure 17.1. *Stages of compensation after acute blood loss (based on Abdullayev N.H., 2008)*

Table 17.5. *Dynamics of the blood picture in acute posthemorrhagic anemia*

Stage	Hemoglobin and erythrocytes	Other changes
Reflex (first hours)	Not yet changed (blood volume is reduced, but its composition is “proportional”)	Neutrophilic leukocytosis, thrombocytosis (release from depots)
Hydremic (days 1–3)	Decrease, normochromic anemia	Hematocrit and total protein decrease
Bone marrow (from day 4–5)	Gradual recovery	Reticulocytosis (up to 5–10 percent), polychromatophilia, normoblasts, slight decrease in the color index (hypochromia — iron consumption)

CAUTION

In the first hours of acute blood loss, hemoglobin and hematocrit may be normal, because plasma and erythrocytes are lost in the same proportion. Relying on these values and underestimating the volume of blood loss is dangerous: at this time pulse, arterial pressure, the shock index (pulse / systolic pressure) and clinical signs are more important.

Chronic posthemorrhagic anemia

Develops as a result of repeated loss of small amounts of blood over a long time (from the gastrointestinal tract — ulcer, tumor, hemorrhoids; heavy and prolonged menstruation; helminths — ancylostomiasis). The body's iron stores are exhausted and **iron deficiency anemia** (hypochromic, microcytic, hyporegenerative) develops (see topic 18).

6. Hemolytic anemias

Hemolytic anemias — anemias resulting from increased destruction of erythrocytes (shortened lifespan); even though the bone marrow increases erythropoiesis, it cannot keep up with the rate of destruction.

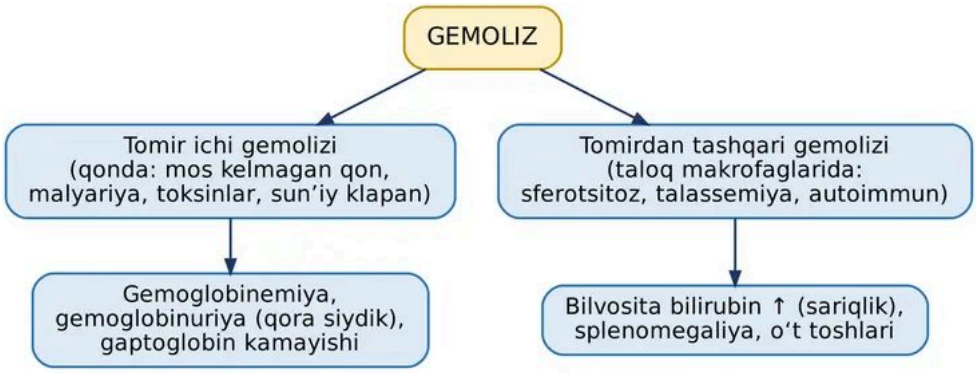


Figure 17.2. *Signs of intravascular and extravascular hemolysis*

Table 17.6. *Classification of hemolytic anemias*

Group	Type	Examples
Hereditary (intrinsic erythrocyte defect)	Membranopathies	Hereditary spherocytosis (Minkowski–Chauffard disease), hereditary elliptocytosis
	Enzymopathies	Glucose-6-phosphate dehydrogenase (G6PD) deficiency, pyruvate kinase deficiency
	Hemoglobinopathies	Sickle cell anemia (HbS), thalassemias (reduced synthesis of globin chains)
Acquired (mainly external factor)	Immune	Autoimmune hemolytic anemia, hemolytic disease of the newborn, incompatible blood transfusion
	Mechanical	Prosthetic heart valves, microangiopathic (DIC, hemolytic uremic syndrome), "march hemoglobinuria"
	Toxic and infectious	Hemolytic poisons (snake venom, arsine, lead), malaria, clostridial sepsis
	Acquired membrane defect	Paroxysmal nocturnal hemoglobinuria

General signs of hemolytic anemias

- **anemia** — normochromic, hyperregenerative (reticulocytosis);
- **jaundice** — increased indirect (unconjugated) bilirubin; stercobilin and urobilin increase in the stool and urine;
- **splenomegaly** — in extravascular hemolysis;
- **gallstone disease** — pigment stones;
- in intravascular hemolysis — **hemoglobinemia**, **hemoglobinuria** (black or red-brown urine), decreased haptoglobin, hemosiderinuria;
- erythroid hyperplasia in the bone marrow; in children with chronic hemolysis — skull deformity (“tower” skull).

Hereditary spherocytosis (Minkowski–Chauffard disease)

Inherited in an autosomal dominant manner. As a result of a defect in erythrocyte membrane proteins (spectrin, ankyrin), membrane is lost and the erythrocyte becomes spherical (a **spherocyte**). Spherocytes are small, lack central pallor and have low deformability; they cannot pass through the splenic sinuses and are destroyed by macrophages (extravascular hemolysis).

Signs: anemia, jaundice, splenomegaly, gallstones; microspherocytes and reticulocytosis in the blood; **decreased osmotic resistance** of erythrocytes (they hemolyze quickly in hypotonic solutions). Treatment — splenectomy (almost stops the hemolysis, but the membrane defect remains).

G6PD deficiency

Inherited in an X-linked manner. With the enzyme deficiency, little NADPH and reduced glutathione are formed in erythrocytes, and the erythrocytes become unable to withstand oxidative stress. **Hemolytic crises** develop after oxidizing drugs (primaquine, sulfonamides, nitrofurans), infections and **eating legumes (fava beans)** (favism). Heinz bodies (precipitated hemoglobin) are found in the blood.

Sickle cell anemia

A point mutation of the beta-globin gene (glutamic acid is replaced by valine at position 6) produces HbS. When oxygen falls, HbS polymerizes, the erythrocytes become sickle-shaped, block small vessels (vaso-occlusive crises, pain, infarcts) and undergo hemolysis. Heterozygous carriers are more resistant to falciparum malaria — which is why the mutation is common in areas where malaria is widespread.

> **CLINICAL SIGNIFICANCE**
> **Hemolytic disease of the newborn** — when the mother is Rh-negative and the fetus Rh-positive (or with ABO incompatibility), maternal IgG antibodies cross the placenta and destroy fetal erythrocytes (a type II allergic reaction). In severe cases indirect bilirubin accumulates in the brain nuclei — **bilirubin encephalopathy (kernicterus)**. Prevention — giving anti-D immunoglobulin to the Rh-negative woman within 72 hours after delivery.

7. Experimental part

Experiment 1: studying pathological forms of erythrocytes in a blood smear

1. Take stained smears from a healthy person and from patients with acute posthemorrhagic anemia, hereditary spherocytosis, sickle cell anemia and thalassemia.

2. Under the immersion objective, assess the size, shape and staining of the erythrocytes.
3. Find regenerative (polychromatophils, normoblasts) and degenerative (Howell–Jolly bodies, basophilic stippling) forms.
4. Draw each smear and write down the main changes.

Experiment 2: detecting reticulocytes by supravital staining

1. Put a few drops of brilliant cresyl blue (or new methylene blue) dye solution and an equal volume of blood into a test tube and mix.
2. Leave for 20–30 minutes at room temperature (or at 37 °C) — the living cells are stained (supravital staining).
3. Prepare a thin smear from the mixture and let it dry.
4. Under the immersion objective, count the reticulocytes among 1000 erythrocytes: their cytoplasm shows a blue network or granules (remnants of ribosomal RNA — the "granulofilamentous substance").
5. Express the result in per mille (‰) or percent.
6. Compare with the normal value (5–12 ‰) and assess the regenerative activity of the bone marrow.

REMEMBER

The reticulocyte count is the simplest indicator of the erythropoietic activity of the bone marrow. **Reticulocytosis** occurs after hemolysis and acute blood loss (from day 4–5); **reticulocytopenia** occurs in aplastic anemia and in iron, B_{12} and folate deficiency (before treatment). The "**reticulocyte crisis**" seen on day 5–7 of treatment of B_{12} deficiency is a sign of treatment effectiveness.

Experiment 3: determining hemoglobin concentration by the colorimetric (photometric) method

1. Put 5 ml of transforming solution (which converts hemoglobin into a stable compound — hemiglobincyanide) into a test tube.
2. Add 0.02 ml of blood with a micropipette, mix and leave for 10–20 minutes.
3. Measure the optical density in a photoelectric colorimeter (wavelength about 540 nm).
4. Calculate the hemoglobin concentration (g/l) using a calibration curve prepared with a standard solution, or a formula.
5. Determine hemoglobin and the color index in the blood of a patient with hemolytic anemia and assess the type of anemia.

CAUTION

The old Sahli method (converting hemoglobin to hematin with hydrochloric acid and comparing it visually with a standard color) is subjective and of low accuracy; today the hemiglobincyanide photometric method and automated analyzers are the standard. The reagents of the hemiglobincyanide method are toxic (cyanides) — observe safety precautions when working with them.

8. Practical skills

Skill: determining the type of anemia from the blood test

1. Determine from hemoglobin whether anemia is present and its severity.
2. Calculate the color index (or MCH): hypo-, normo- or hyperchromic.

3. Determine from MCV whether it is micro-, normo- or macrocytic.
4. Assess the regenerative activity from reticulocytes.
5. Examine the morphology of erythrocytes in the smear (spherocytes, target cells, megalocytes).
6. Assess bilirubin (indirect), haptoglobin and LDH as signs of hemolysis.
7. Determine the pathogenetic type and write a conclusion.

Skill: counting reticulocytes

1. Examine the supravitally stained smear under the immersion objective.
2. Moving from field to field in sequence, count 1000 erythrocytes.
3. Count the reticulocytes among them separately.
4. Record the result in per mille.
5. Compare with the normal value and write a conclusion about erythropoietic activity.

Review questions

1. List the stages of erythropoiesis and the factors that regulate it.
2. Where and under what conditions is erythropoietin produced?
3. What are the normal red blood cell values?
4. Describe the pathological forms of erythrocytes, dividing them into groups.
5. What are the types and causes of erythrocytosis?
6. What is called anemia? What are the WHO criteria?
7. How are anemias classified?
8. How is the color index calculated?
9. Explain the general signs of anemia and their mechanisms.
10. Explain the stages of compensation and the blood picture in acute posthemorrhagic anemia.
11. Why may hemoglobin be unchanged in the first hours of acute blood loss?
12. How are hemolytic anemias classified?
13. How do the signs of intravascular and extravascular hemolysis differ?
14. Explain the pathogenesis and blood picture of Minkowski–Chauffard disease.
15. What is the clinical significance of determining reticulocytes?

Test questions

1. Where is erythropoietin mainly produced?
   A) In the liver
   B) In the bone marrow
   C) In the spleen
   D) In the kidney
   E) In the pituitary

2. What is the average lifespan of erythrocytes?
   A) 120 days
   B) 30 days
   C) 10 days
   D) 1 year
   E) 7 days

3. Hb 90 g/l, erythrocytes 3.0×10^{12}/l. What is the color index?
   A) 0,6
   B) 0,9
   C) 1,2
   D) 0,75
   E) 1,5

4. Microspherocytosis and decreased osmotic resistance of erythrocytes are typical of which disease?
   A) Iron deficiency anemia
   B) B_{12} deficiency anemia
   C) Hereditary spherocytosis (Minkowski–Chauffard disease)
   D) Aplastic anemia
   E) Sickle cell anemia

5. In which stage after acute blood loss are dilution of the blood and a fall in hemoglobin observed?
   A) Reflex stage
   B) Bone marrow stage
   C) Recovery stage
   D) In the first hour
   E) Hydremic stage

6. Reticulocytosis is typical of which anemia?
   A) Aplastic anemia
   B) Untreated B_{12} deficiency anemia
   C) Hemolytic anemia
   D) Iron deficiency anemia (before treatment)
   E) Anemia of chronic kidney failure

7. Which sign is specific to intravascular hemolysis?
   A) Splenomegaly
   B) Microcytosis
   C) Hypochromia
   D) Gallstones
   E) Hemoglobinuria and decreased haptoglobin

8. What triggers a hemolytic crisis in G6PD deficiency?
   A) Iron preparations
   B) Oxidizing drugs and eating legumes (fava beans)
   C) Vitamin B_{12}
   D) Folic acid
   E) Glucose

9. What is the molecular cause of sickle cell anemia?
   A) A point mutation in the beta-globin gene (formation of HbS)
   B) Spectrin defect
   C) Deletion of alpha-globin genes
   D) Iron deficiency
   E) Autoimmune antibodies

10. Which bilirubin is responsible for jaundice in hemolytic anemias?
    A) Direct (conjugated)
    B) Urobilin only
    C) Stercobilin only
    D) Indirect (unconjugated)
    E) Biliverdin

11. What staining method is used to detect reticulocytes?
    A) Supravital staining (brilliant cresyl blue)
    B) Gram staining
    C) Ziehl–Neelsen staining
    D) Hematoxylin–eosin
    E) Romanowsky method after fixation

12. How is polycythemia vera (Vaquez disease) characterized?
    A) Only increased erythropoietin in response to hypoxia
    B) Hemoconcentration due to dehydration
    C) Only a kidney tumor
    D) An erythropoietin-independent myeloproliferative neoplasm (JAK2 mutation)
    E) Iron overload

13. Hemolytic disease of the newborn belongs to which type of allergic reaction?
    A) Type I
    B) Type III
    C) Type IV
    D) Type V
    E) Type II (cytotoxic)

14. What is the WHO criterion for anemia in men?
    A) Hemoglobin below 150 g/l
    B) Hemoglobin below 140 g/l
    C) Hemoglobin below 130 g/l
    D) Hemoglobin below 110 g/l
    E) Hemoglobin below 100 g/l

15. Megalocytes and Howell–Jolly bodies are typical of which anemia?
    A) Iron deficiency anemia
    B) Vitamin B_{12} and folic acid deficiency anemia
    C) Acute posthemorrhagic anemia
    D) Thalassemia
    E) Hereditary spherocytosis

Clinical cases

Case 1

A 35-year-old man lost blood from the femoral artery in a car accident. In hospital 1 hour after the accident: pulse 125 beats/min, arterial pressure 90/60 mm Hg, pale skin, cold sweat. Blood test: Hb 138 g/l, erythrocytes 4.4×10^{12}/l, hematocrit 42 %. The next day: Hb 92 g/l, hematocrit 28 %. On day 6, reticulocytes 6 %.

Questions: Why is the first test close to normal? Explain the subsequent changes stage by stage.

Answer: In the first hour, the reflex (vascular) stage: plasma and erythrocytes have been lost in equal proportion, so although blood volume has decreased, hemoglobin and hematocrit per unit volume have not yet changed; at the same time, the hemodynamic signs (tachycardia, hypotension, pallor, cold sweat) indicate severe blood loss. The next day, the hydremic stage: tissue fluid moves into the blood vessels and restores circulating blood volume, the blood is diluted — hemoglobin and hematocrit fall (normochromic anemia). From day 4–5, the bone marrow stage: in response to hypoxia, renal erythropoietin increases and erythropoiesis is enhanced — reticulocytosis (6 %), the anemia is hyperregenerative.

Case 2

A 12-year-old child has had episodes of jaundice and heaviness under the left costal margin since early childhood. His father has similar signs. The spleen is enlarged. Blood: Hb 95 g/l, color index 1.0, reticulocytes 8 %, microspherocytes in the smear, indirect bilirubin raised. Osmotic resistance of erythrocytes is decreased.

Questions: What is the disease? Explain its pathogenesis and the type of hemolysis. What treatment is effective?

Answer: Hereditary spherocytosis (Minkowski–Chauffard disease) — inherited in an autosomal dominant manner (the father is also affected). Because of a defect in erythrocyte membrane proteins (spectrin, ankyrin), membrane is lost, the erythrocyte becomes spherical and fragile, loses its deformability and, unable to pass through the splenic sinuses, is destroyed by macrophages — extravascular hemolysis. This is why splenomegaly and raised indirect bilirubin (jaundice) are seen; the bone marrow has increased erythropoiesis (reticulocytosis 8 %). Splenectomy almost stops the hemolysis, although the membrane defect persists.

Case 3

A 25-year-old man was prescribed a sulfonamide drug for pneumonia. 2 days later his urine turned black, and weakness and jaundice appeared. Blood: Hb 78 g/l, haptoglobin markedly decreased, Heinz bodies in the smear. A similar episode had occurred previously after eating fava beans.

Questions: What condition has developed? Explain its mechanism.

Answer: An acute hemolytic crisis due to G6PD deficiency. With the enzyme deficiency, little NADPH is formed in the pentose phosphate pathway, reduced glutathione is not regenerated and erythrocytes are not protected from oxidative stress. Under the action of an oxidizing drug (sulfonamide) or substances in fava beans, hemoglobin is oxidized and precipitates (Heinz bodies), the membrane is damaged and the erythrocytes are destroyed mainly within the vessels: hemoglobinemia, hemoglobinuria (black urine), decreased haptoglobin; raised indirect bilirubin leads to jaundice. The oxidizing drug must be stopped and such drugs avoided.

Quick review

Table 17.7. *Key facts on Topic 17*

Question	Answer
Erythropoietin	Produced in the kidney in response to hypoxia; stimulates erythropoiesis
Normal Hb	Men 130–160 g/l, women 120–140 g/l
Color index	(Hb × 3) / first three digits of the erythrocyte count; normal 0.85–1.05
Reticulocytes	0.2–1.2 % (5–12 ‰); indicator of erythropoietic activity

Question	Answer
Anisocytosis and poikilocytosis	Changes in size and shape
Anemia (WHO)	Hb below 130 g/l in men, below 120 g/l in women
Pathogenesis of anemias	Blood loss, impaired erythropoiesis, hemolysis
Acute posthemorrhagic anemia	Reflex, hydremic, bone marrow stages
Signs of hemolytic anemia	Anemia, jaundice (indirect bilirubin), reticulocytosis, splenomegaly
Intravascular hemolysis	Hemoglobinemia, hemoglobinuria, decreased haptoglobin
Minkowski–Chauffard disease	Membrane defect, microspherocytes, decreased osmotic resistance
G6PD deficiency	Hemolytic crisis after oxidizing drugs and fava beans, Heinz bodies
Sickle cell anemia	HbS, vaso-occlusive crises
Supravital staining	Detecting reticulocytes with brilliant cresyl blue

TOPIC 18

Etiology and pathogenesis of deficiency anemias. Iron, vitamin B12 and folic acid deficiency anemias

Learning objectives

To explain to the student the importance of iron, vitamin B_{12} and folic acid in erythropoiesis, the causes, pathogenesis, hematological signs and clinical manifestations of iron deficiency and megaloblastic (B_{12} and folate deficiency) anemias, as well as aplastic anemia and anemia of chronic disease; to teach the student to study blood smears of patients with iron and folate deficiency anemias, to determine hemoglobin concentration and to calculate the color index.

By the end of the session, the student should know:

- iron metabolism in the body and its regulation (hepcidin);
- the causes and stages of iron deficiency and the signs of iron deficiency anemia;
- the metabolism of vitamin B_{12} and folic acid and their role in DNA synthesis;
- the causes, pathogenesis and blood picture of megaloblastic anemias;
- the nature of pernicious (Addison–Biermer) anemia and the mechanism of funicular myelosis;
- the main features of aplastic anemia and anemia of chronic disease;
- how to distinguish deficiency anemias by laboratory values.

The student should be able to:

- distinguish smears of iron deficiency and megaloblastic anemia;
- determine hemoglobin concentration and calculate the color index;
- interpret iron metabolism parameters (serum iron, ferritin, transferrin saturation).

1. Iron metabolism

The adult body contains 3–5 g of iron: 65–70 percent in hemoglobin, 20–30 percent in stores (ferritin and hemosiderin), and the rest in myoglobin and enzymes (cytochromes, catalase).

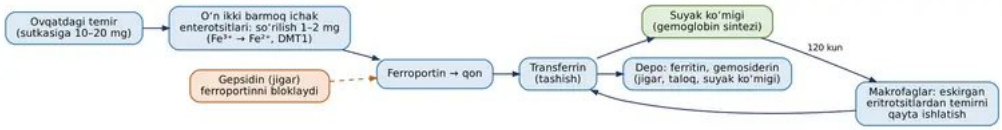


Figure 18.1. *Iron metabolism in the body (based on Novitskiy V.V., 2009; Kumar V. et al., 2015)*

REMEMBER

The body has **no** physiological mechanism for actively excreting iron — iron balance is maintained only by regulating intestinal absorption. Daily loss (epithelial shedding, sweating) is 1–2 mg, with an additional 0.5–1 mg in menstruating women. **Hepcidin** (produced in the liver) degrades ferroportin and stops iron absorption from the intestine and its release from macrophages; hepcidin increases with iron overload and inflammation (IL-6) and decreases with iron deficiency and hypoxia.

Heme iron (meat, liver) is well absorbed (20–30 %); non-heme iron (plant foods) is poorly absorbed. Vitamin C, gastric hydrochloric acid and animal proteins improve absorption; tea (tannins), phytates, calcium and antacids reduce it.

2. Iron deficiency anemia

Iron deficiency anemia (IDA) — a hypochromic microcytic anemia characterized by impaired hemoglobin synthesis due to iron deficiency in the body. It is the most common anemia in the world (about 80 percent of all anemias).

Table 18.1. *Causes of iron deficiency*

Group	Examples
Chronic blood loss (the main cause in adults)	Heavy and prolonged menstruation, occult gastrointestinal bleeding (ulcer, tumor, hemorrhoids, erosions, NSAID use), ancylostomiasis, frequent blood donation
Increased iron requirement	Pregnancy and breastfeeding, periods of rapid growth in children and adolescents
Reduced iron intake	Vegetarian diet, prolonged feeding of children with milk alone, starvation
Impaired absorption	Gastric resection, atrophic gastritis (achlorhydria), celiac disease, small intestinal diseases, H. pylori

Table 18.2. *Stages of iron deficiency*

Stage	Essence	Laboratory signs
Prelatent iron deficiency	Iron stores decrease	Ferritin decreased; serum iron and Hb normal
Latent iron deficiency (without anemia)	Stores are exhausted, the transport pool decreases	Ferritin, serum iron and transferrin saturation decreased; Hb normal
Iron deficiency anemia	Hemoglobin synthesis is impaired	Hb decreased, hypochromia, microcytosis

Signs of IDA

Anemic syndrome (general signs) and **sideropenic syndrome** (tissue iron deficiency — reduced activity of iron-containing enzymes):

- dry skin, hair loss and brittleness, thin, spoon-shaped concave nails (**koilonychia**);
- **disturbed taste and smell** — eating chalk, clay, rice, ice (**pica chlorotica**), liking the smell of gasoline or acetone;
- **angular stomatitis** (cracks at the corners of the mouth), glossitis, difficulty swallowing (sideropenic dysphagia);
- muscle weakness, urinary incontinence (weakened sphincters);
- delayed psychomotor development in children.

Blood picture: hypochromic (color index below 0.85, MCH decreased), microcytic (MCV below 80 fl) anemia; anisocytosis (microcytosis), poikilocytosis, ring-shaped erythrocytes (annulocytes — with a wide pale center); reticulocytes normal or decreased (hyporegenerative); serum iron, ferritin and transferrin saturation decreased, total iron-binding capacity of serum increased.

> **CLINICAL SIGNIFICANCE**
> When iron deficiency anemia is found in adults, especially men and postmenopausal women, its cause — above all occult gastrointestinal bleeding (colon cancer, gastric cancer, ulcer) — must always be sought. Treating with iron preparations without identifying the cause is a dangerous mistake.

3. Metabolism of vitamin B_{12} and folic acid

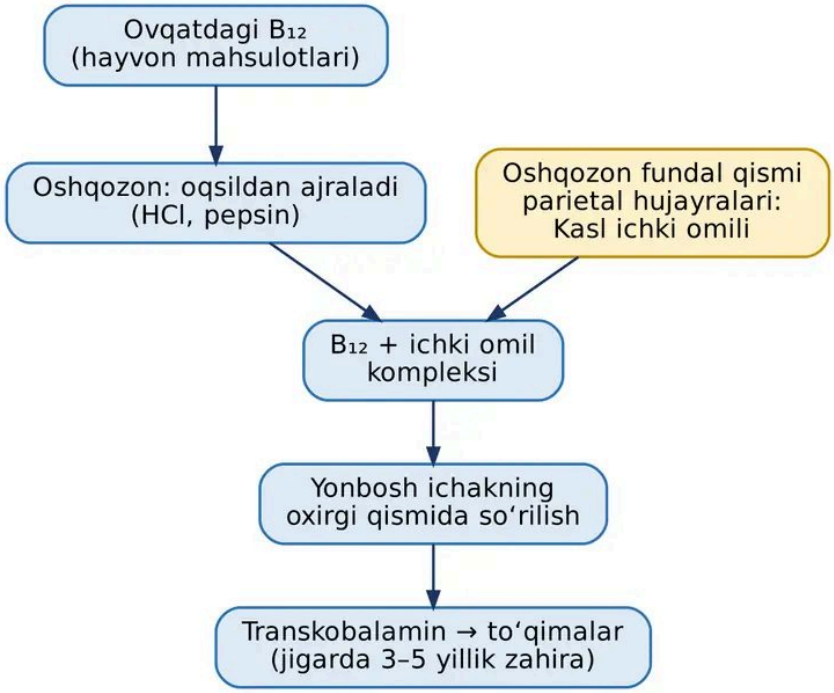


Figure 18.2. *Absorption of vitamin B_{12}*

Vitamin B_{12} (cobalamin) is found only in animal products (meat, liver, eggs, milk). The daily requirement is 2–3 µg; the store in the liver lasts 3–5 years — which is why deficiency develops very slowly.

Folic acid is found in green leaves, liver and legumes; it is quickly destroyed by cooking. The daily requirement is 200–400 µg; the store lasts only 3–4 months.

Role in DNA synthesis: folic acid (as tetrahydrofolate) carries one-carbon groups in the synthesis of thymidylate (the thymine nucleotide for DNA). B_{12}, as methylcobalamin, takes the methyl group from methyltetrahydrofolate and converts homocysteine to methionine, thereby "freeing" tetrahydrofolate. In B_{12} deficiency, folic acid is "trapped" in the methyl form (the **folate trap**) and DNA synthesis is impaired.

In addition, B_{12} (as adenosylcobalamin) takes part in the conversion of methylmalonyl-CoA to succinyl-CoA; in its deficiency methylmalonate accumulates and myelin synthesis is impaired — **neurological disturbances** are specific to B_{12} deficiency.

4. Megaloblastic anemias

Megaloblastic anemias — hyperchromic macrocytic anemias characterized by impaired DNA synthesis due to vitamin B_{12} or folic acid deficiency and a switch of erythropoiesis to the megaloblastic type.

Table 18.3. *Causes of B_{12} and folic acid deficiency*

Cause	B_{12} deficiency	Folic acid deficiency
Reduced intake	Strict vegetarianism (veganism)	Few fresh vegetables and fruits, alcoholism, diet of the elderly

Cause	B_{12} deficiency	Folic acid deficiency
Impaired absorption	Pernicious anemia (intrinsic factor deficiency), gastric resection (gastrectomy), resection or diseases of the ileum (Crohn), fish tapeworm (diphyllobothriasis), bacterial overgrowth in “blind loop” syndrome	Celiac disease, small intestinal diseases
Increased requirement	—	Pregnancy, hemolytic anemias, tumors, psoriasis
Drugs	Metformin, proton pump inhibitors (long-term), nitrous oxide (anesthesia)	Methotrexate, trimethoprim, phenytoin (act on dihydrofolate reductase or absorption)

Pernicious (Addison–Biermer) anemia

As a result of autoimmune atrophic gastritis, the gastric parietal cells die: autoantibodies against **parietal cells** and **Castle's intrinsic factor** are formed → intrinsic factor and hydrochloric acid are not produced (achlorhydria) → B_{12} is not absorbed. It occurs mainly after the age of 50–60; the risk of gastric cancer is increased.

Pathogenesis of megaloblastic anemia

1. DNA synthesis slows down while RNA and protein synthesis continue → **asynchrony of nuclear and cytoplasmic maturation** (the nucleus is “young”, the cytoplasm “mature”).
2. Large cells — **megaloblasts** — form in the bone marrow.
3. Most megaloblasts die in the bone marrow itself — **ineffective erythropoiesis** (intramedullary hemolysis); therefore indirect bilirubin and LDH rise.
4. The **megalocytes** released into the blood are large and saturated with hemoglobin, but fragile and short-lived.
5. Leukopoiesis and thrombopoiesis are also impaired — **pancytopenia**, hypersegmented neutrophils, large platelets.
6. All rapidly dividing cells (gastrointestinal epithelium) are affected — **Hunter's glossitis** (a smooth, “lacquered”, bright red tongue), dyspepsia.

Blood picture: hyperchromic (color index above 1.05, often 1.2–1.5) macrocytic (MCV above 100 fl) anemia; megalocytes (oval), megaloblasts; Howell–Jolly bodies, Cabot rings, basophilic stippling; aniso- and poikilocytosis; reticulocytes decreased (aregenerative); leukopenia, **hypersegmentation of neutrophils** (right nuclear shift), thrombocytopenia.

REMEMBER

Funicular myelosis (subacute combined degeneration) — a neurological complication specific to B_{12} deficiency: demyelination of the posterior and lateral columns of the spinal cord. Signs: paresthesias in the legs, loss of deep (vibration and joint-position) sensation, ataxia, spastic paresis, memory impairment. Folic acid deficiency does not cause such neurological signs.

CAUTION

Treating a patient with B_{12} deficiency with folic acid alone is dangerous: blood values may partly improve, but the neurological disturbances worsen and become irreversible. In megaloblastic anemia, B_{12} and folate levels should be determined first, and folic acid alone should not be given until B_{12} deficiency has been ruled out.

5. Other hyporegenerative anemias

Aplastic anemia — a sharp depression of the hematopoietic function of the bone marrow (hypoplasia or aplasia) with damage to stem cells. Causes: autoimmune (T lymphocytes damage the stem cells), drugs (chloramphenicol, cytostatics), benzene, radiation, viruses (hepatitis, parvovirus B19), hereditary (Fanconi anemia). Blood picture: pancytopenia (anemia, leukopenia, thrombocytopenia), normochromic anemia, reticulocytes markedly decreased; the bone marrow is replaced by adipose tissue.

Anemia of chronic disease (anemia of inflammation) — in chronic infections, autoimmune diseases (rheumatoid arthritis) and tumors. Mechanism: under the action of IL-6, hepcidin increases → iron is "locked" in macrophages and its intestinal absorption decreases (functional iron deficiency); erythropoietin production and sensitivity to it decrease; erythrocyte lifespan is shortened. Blood picture: normo- or hypochromic anemia; serum iron decreased, but **ferritin normal or increased** (iron is present in the stores).

Anemia in chronic kidney failure — mainly due to reduced erythropoietin production; treatment — erythropoiesis-stimulating agents.

Table 18.4. *Comparative characteristics of deficiency anemias*

Parameter	Iron deficiency anemia	B_{12} and folate deficiency anemia	Anemia of chronic disease	Aplastic anemia
Color index	below 0.85 (hypochromic)	above 1.05 (hyperchromic)	Normal or slightly low	Normal
MCV	Decreased	Increased	Normal or decreased	Normal or increased
Reticulocytes	Normal or low	Decreased	Decreased	Markedly decreased
Serum iron	Decreased	Normal or increased	Decreased	Increased
Ferritin	Decreased	Normal or increased	Normal or increased	Increased
Leukocytes and platelets	Normal	May be decreased	Normal	Markedly decreased
Characteristic signs	Koilonychia, pica, annulocytes	Megalocytes, hypersegmentation, glossitis, funicular myelosis (B_{12})	Signs of the underlying disease	Infections, bleeding

6. Experimental part

Experiment 1: studying blood smears of patients with iron and folate (B_{12}) deficiency anemias

1. Take stained smears from a healthy person and from patients with iron deficiency anemia and B_{12} (folate) deficiency anemia.
2. Under the immersion objective, assess the size (microcytes, megalocytes), staining (hypochromia, hyperchromia) and shape of the erythrocytes.
3. Find annulocytes, megaloblasts, Howell–Jolly bodies and Cabot rings.
4. Count the segments of the neutrophils (5 or more — hypersegmentation).
5. Draw the smears and write the distinguishing features in a table.

Experiment 2: determining hemoglobin concentration and calculating the color index

1. Determine the hemoglobin concentration in the patient's blood by the hemiglobincyanide method (see topic 17).
2. Determine the erythrocyte count (in a Goryaev chamber or on an automated analyzer).
3. Calculate the color index: (Hb, g/l × 3) / the first three digits of the erythrocyte count.
4. Calculate the mean corpuscular hemoglobin (pg) using the formula MCH = Hb (g/l) / erythrocytes (× 10^{12}/l).
5. Assess the anemia as hypo-, normo- or hyperchromic.

Table 18.5. *Example calculation*

Patient	Hb, g/l	Erythrocytes, × 10^{12}/l	Color index	MCH, pg	Conclusion
A	85	3,9	(85 × 3) / 390 ≈ 0,65	85 / 3,9 ≈ 22	Hypochromic anemia (iron deficiency)
B	80	1,8	(80 × 3) / 180 ≈ 1,33	80 / 1,8 ≈ 44	Hyperchromic anemia (B_{12} or folate deficiency)
C					

7. Practical skills

Skill: interpreting iron metabolism parameters

1. Assess serum iron (normal: 11.6–30.4 µmol/l in men, 8.9–30.4 µmol/l in women).
2. Assess ferritin (the main indicator of the body's iron stores; below 15–30 µg/l — stores exhausted).
3. Assess transferrin saturation with iron (normal 20–45 %; below 16 percent — iron deficiency).
4. Assess the total iron-binding capacity of serum (increased in iron deficiency).
5. Compare with markers of inflammation (C-reactive protein): ferritin, as an acute-phase protein, may rise in inflammation.
6. Distinguish true from functional iron deficiency and write a conclusion.

Skill: determining the cause of megaloblastic anemia

1. Confirm hyperchromic macrocytic anemia and hypersegmentation in the blood picture.
2. Determine serum B_{12} and folate levels.
3. Check for neurological signs (paresthesia, impaired deep sensation).
4. Clarify diet, alcohol, drugs and gastrointestinal surgery from the history.
5. In pernicious anemia, indicate the need to detect antibodies against parietal cells and intrinsic factor and atrophic gastritis on gastroscopy.
6. Write a conclusion.

Review questions

1. How is iron distributed in the body and how is its metabolism regulated?
2. What is the role of hepcidin?
3. List the main causes of iron deficiency.

4. Through what stages does iron deficiency pass?
5. What are the signs of the sideropenic syndrome and their mechanisms?
6. What is the blood picture in iron deficiency anemia?
7. How is vitamin B_{12} absorbed? What is the role of Castle's intrinsic factor?
8. Explain the role of B_{12} and folic acid in DNA synthesis. What is the “folate trap”?
9. List the causes of B_{12} and folic acid deficiency.
10. How does pernicious anemia develop?
11. Explain the pathogenesis of megaloblastic anemia. What is ineffective erythropoiesis?
12. What is the blood picture in megaloblastic anemia?
13. What is funicular myelosis and why is it specific to B_{12} deficiency?
14. What are the main signs of aplastic anemia and anemia of chronic disease?
15. How are the color index and MCH calculated?

Test questions

1. What is the most common cause of iron deficiency in adults?
   A) Vegetarian diet
   B) Gastric resection
   C) Pregnancy
   D) Celiac disease
   E) Chronic blood loss

2. Which hormone stops intestinal absorption of iron and its release from macrophages?
   A) Erythropoietin
   B) Transferrin
   C) Hepcidin
   D) Ferritin
   E) Castle's factor

3. What is the blood picture in iron deficiency anemia?
   A) Hypochromic microcytic anemia
   B) Hyperchromic macrocytic anemia
   C) Normochromic normocytic anemia, reticulocytosis
   D) Pancytopenia
   E) Megalocytosis

4. What is the most reliable indicator of the body's iron stores?
   A) Hemoglobin
   B) Erythrocyte count
   C) Reticulocytes
   D) Serum ferritin
   E) Color index

5. Which sign belongs to the sideropenic syndrome?
   A) Funicular myelosis
   B) Koilonychia and disturbed taste (pica)
   C) Hunter's glossitis
   D) Splenomegaly
   E) Jaundice

6. Where is Castle's intrinsic factor, which is needed for absorption of vitamin B_{12}, produced?
   A) In the pancreas
   B) In the liver
   C) In the small intestine
   D) In the gastric parietal cells
   E) In the kidney

7. What is the cause of pernicious anemia?
   A) An autoimmune process against parietal cells and intrinsic factor
   B) Iron deficiency
   C) Low intake of folic acid
   D) An erythrocyte membrane defect
   E) Bone marrow aplasia

8. What is the blood picture in megaloblastic anemia?
   A) Hypochromic microcytic, annulocytes
   B) Normochromic, reticulocytosis
   C) Microspherocytosis
   D) Sickle-shaped erythrocytes
   E) Hyperchromic macrocytic, megalocytes, hypersegmentation of neutrophils

9. Which sign is specific to B_{12} deficiency and absent in folate deficiency?
   A) Hyperchromic anemia
   B) Funicular myelosis
   C) Hunter's glossitis
   D) Hypersegmentation
   E) Ineffective erythropoiesis

10. What is the cause of the rise in indirect bilirubin and LDH in megaloblastic anemia?
    A) Hemolysis in the spleen
    B) Liver failure
    C) Ineffective erythropoiesis in the bone marrow (death of megaloblasts)
    D) Obstruction of the bile ducts
    E) Bleeding

11. Hb 80 g/l, erythrocytes 2.0×10^{12}/l. What are the color index and type of anemia?
    A) 0.6 — hypochromic
    B) 1.2 — hyperchromic
    C) 0.9 — normochromic
    D) 0.8 — hypochromic
    E) 1.0 — normochromic

12. What is the mechanism by which methotrexate causes megaloblastic anemia?
    A) It blocks intrinsic factor
    B) It breaks down B_{12}
    C) It reduces iron absorption
    D) It blocks erythropoietin
    E) It inhibits dihydrofolate reductase and disrupts folate metabolism

13. What is the main difference between anemia of chronic disease and iron deficiency anemia?
    A) Decreased serum iron
    B) Decreased hemoglobin
    C) Normal or increased ferritin
    D) Microcytosis
    E) Decreased reticulocytes

14. Which blood picture is typical of aplastic anemia?
    A) Leukocytosis and thrombocytosis
    B) Anemia only, reticulocytosis
    C) Erythrocytosis
    D) Pancytopenia and a marked decrease in reticulocytes
    E) Eosinophilia

15. Why is it dangerous to treat a patient with B_{12} deficiency with folic acid alone?
    A) Neurological disturbances may worsen and become irreversible
    B) Iron deficiency develops
    C) Hemolysis increases
    D) Leukemia develops
    E) It causes an allergic reaction

Clinical cases

Case 1

A 38-year-old woman has weakness, dizziness, hair loss and brittle nails; she craves chalk and likes the smell of acetone. Her periods last 8–9 days and are heavy. Blood test: Hb 78 g/l, erythrocytes 3.6 × 10^{12}/l, MCV 68 fl, reticulocytes 0.8 %, serum iron 4.2 µmol/l, ferritin 6 µg/l.

Questions: What anemia is this? Calculate the color index. Explain the mechanism of the clinical signs.

Answer: Iron deficiency anemia due to chronic blood loss (heavy and prolonged periods — menorrhagia). Color index = (78 × 3) / 360 ≈ 0.65 — hypochromic; MCV 68 fl — microcytic; ferritin markedly decreased (stores exhausted), serum iron low; reticulocytes normal — hyporegenerative. Weakness and dizziness — the anemic syndrome (hypoxia). Hair loss, brittle nails and disturbed taste and smell (eating chalk, liking the smell of acetone) — the sideropenic syndrome: reduced activity of iron-containing enzymes in the tissues and impaired trophism of the epithelium and its derivatives. Treatment — iron preparations, plus identifying and eliminating the cause of the menorrhagia.

Case 2

A 62-year-old man has weakness, dyspnea, numbness in the legs and an unsteady gait. His tongue is smooth, bright red and burning. The skin is slightly yellowish. Blood test: Hb 70 g/l, erythrocytes 1.7 × 10^{12}/l, MCV 118 fl, leukocytes 3.1 × 10^{9}/l, platelets 110 × 10^{9}/l, reticulocytes 0.2 %; megalocytes, Howell–Jolly bodies and hypersegmented neutrophils in the smear. Indirect bilirubin is raised. Antibodies to parietal cells are positive.

Questions: What is the disease? Explain the pathogenesis of the signs.

Answer: Pernicious (Addison–Biermer) anemia — absence of intrinsic factor and failure of B_{12} absorption due to autoimmune atrophic gastritis. Color index = (70 × 3) / 170 ≈ 1.24 — hyperchromic, macrocytic. In B_{12} deficiency DNA synthesis is impaired and erythropoiesis switches

to the megaloblastic type; megaloblasts die in the bone marrow (ineffective erythropoiesis) — indirect bilirubin rises and causes the yellowish skin; leuko- and thrombopoiesis are also impaired — pancytopenia, hypersegmentation of neutrophils. Hunter's glossitis — damage to the rapidly dividing epithelium of the tongue. Numbness in the legs and ataxia — funicular myelosis: accumulation of methylmalonyl-CoA and impaired myelin synthesis damage the posterior and lateral columns of the spinal cord. Treatment — lifelong parenteral (intramuscular) B_{12}.

Case 3

A 55-year-old woman has had rheumatoid arthritis for 10 years. Blood test: Hb 98 g/l, MCV 82 fl, serum iron 7 µmol/l, ferritin 240 µg/l, C-reactive protein raised.

Questions: What anemia is this? Why is serum iron low but ferritin high? Will iron preparations be effective?

Answer: Anemia of chronic disease (inflammation). In chronic inflammation, IL-6 increases hepcidin synthesis in the liver; hepcidin degrades ferroportin and stops the release of iron from macrophages and its absorption from the intestine. As a result, iron is “locked” in the stores (macrophages) — ferritin is high, but there is not enough iron in the blood for erythropoiesis — serum iron is low (functional iron deficiency). In addition, cytokines reduce erythropoietin production and sensitivity to it. Oral iron preparations are of little benefit because their absorption is blocked by hepcidin; the main treatment is to reduce the inflammatory activity (treat the underlying disease).

Quick review

Table 18.6. *Key facts on Topic 18*

Question	Answer
Iron balance	Regulated only by intestinal absorption; no active excretion
Hepcidin	Blocks ferroportin, reducing iron absorption and release from macrophages
Ferritin	Main indicator of iron stores
Causes of IDA	Chronic blood loss, increased requirement, low intake, impaired absorption
Blood picture of IDA	Hypochromic microcytic, annulocytes, low ferritin
Sideropenic syndrome	Koilonychia, pica, angular stomatitis, hair loss
Castle's intrinsic factor	In parietal cells; needed for B_{12} absorption
B_{12} store	3–5 years in the liver; folate — 3–4 months
Folate trap	In B_{12} deficiency, folate is trapped in the methyl form
Megaloblastic anemia	Hyperchromic macrocytic, megalocytes, hypersegmentation, pancytopenia
Ineffective erythropoiesis	Death of megaloblasts in the bone marrow
Pernicious anemia	Autoimmune atrophic gastritis, absence of intrinsic factor
Funicular myelosis	Only in B_{12} deficiency: demyelination of the spinal cord
Anemia of chronic disease	Increased hepcidin; low serum iron, high ferritin
Aplastic anemia	Pancytopenia, markedly low reticulocytes

TOPIC 19

Arrhythmias due to disturbances of cardiac automaticity and excitability

Learning objectives

To explain to the student the regulation of cardiovascular function, the properties and metabolism of the myocardium, the parameters of central hemodynamics, the main elements of the ECG and their origin, the classification of cardiac rhythm disturbances, and the causes, mechanisms and consequences of disturbances of automaticity and excitability (sinus tachycardia, bradycardia, arrhythmia, extrasystoles); to teach the student to record a normal ECG and to analyze the ECG in various disturbances of automaticity and excitability.

By the end of the session, the student should know:

- the main properties of the myocardium (automaticity, excitability, conductivity, contractility) and the structure of the conduction system;
- the parameters of central hemodynamics;
- the origin, amplitude and duration of the ECG waves, segments and intervals;
- the classification and mechanisms of arrhythmias (abnormal automaticity, triggered activity, re-entry);
- the causes and ECG signs of sinus tachycardia, bradycardia and arrhythmia;
- the types, causes, theories and ECG signs of extrasystoles.

The student should be able to:

- record an ECG and calculate its main parameters (heart rate, intervals);
- identify sinus rhythm;
- identify disturbances of automaticity and extrasystoles on the ECG.

1. Properties of the myocardium and regulation of cardiac function

Table 19.1. *Main properties of the myocardium*

Property	Essence	Disturbance
Automaticity	Ability to generate impulses without external stimulation (mainly in the sinus node)	Sinus tachycardia, bradycardia, arrhythmia, sick sinus syndrome, ectopic rhythms
Excitability	Ability to be excited in response to a stimulus	Extrasystoles, paroxysmal tachycardias
Conductivity	Conduction of the impulse through the heart	Blocks, pre-excitation syndromes
Contractility	Contraction in response to excitation	Heart failure
Refractoriness	Temporary inexcitability after excitation	Important in the re-entry mechanism

Table 19.2. *Centers of the conduction system*

Center	Location	Rate of automaticity
First order — sinus (sinoatrial) node	Right atrium, at the entry of the superior vena cava	60–80 beats/min
Second order — atrioventricular (AV) junction	Lower part of the interatrial septum	40–60 beats/min

Center	Location	Rate of automaticity
Third order — bundle branches of His and Purkinje fibers	Ventricles	20–40 beats/min

Cardiac activity is regulated by **sympathetic** nerves (norepinephrine, beta1-receptors: heart rate, force, conductivity and excitability increase) and **parasympathetic** nerves (vagus nerve, acetylcholine: heart rate and AV conduction decrease); humoral factors (catecholamines, thyroid hormones, K^+, Ca^{2+}) also have an effect.

Parameters of central hemodynamics: stroke volume (60–80 ml), cardiac output (4.5–5.5 l/min) = stroke volume × heart rate, cardiac index (2.5–4.0 l/min/m^2), total peripheral vascular resistance, central venous pressure (5–10 cm H_2O). The myocardium obtains its energy mainly from aerobic oxidation of fatty acids; oxygen extraction is high even at rest (70–75 %), so increased demand is met mainly by increased coronary blood flow.

2. Main elements of the ECG

Electrocardiography (ECG) — a method of recording from the body surface the electrical potentials generated during cardiac excitation. There are 12 standard leads: 3 standard limb leads (I, II, III), 3 augmented leads (aVR, aVL, aVF) and 6 chest leads (V_1–V_6).



Figure 19.1. *Normal sinus rhythm (schematic ECG)*

Table 19.3. *ECG elements and their normal values*

Element	What it reflects	Normal
P wave	Atrial depolarization	0.08–0.10 s, amplitude up to 2.5 mm; positive in lead II
PQ (PR) interval	Time for the impulse to pass from the atria to the ventricles	0.12–0.20 s
QRS complex	Ventricular depolarization	0.06–0.10 s
Q wave	Depolarization of the interventricular septum	Shorter than 0.03 s, smaller than 1/4 of R
ST segment	Fully excited state of the ventricles	On the isoelectric line (deviation of 0.5–1 mm is possible)
T wave	Ventricular repolarization	Asymmetric, positive in most leads
QT interval	Electrical systole of the ventricles	Depends on heart rate, usually 0.35–0.44 s
RR interval	Duration of the cardiac cycle	0.6–1.0 s (60–100 beats/min)

REMEMBER

At the standard paper speed of 25 mm/s: 1 mm (small square) = 0.04 s, 5 mm (large square) = 0.2 s. Heart rate = 60 / RR (in seconds) or 300 / the number of large squares between R waves. **Criteria of sinus rhythm:** a P wave before each QRS, P positive in lead II, constant PQ (0.12–0.20 s), equal RR intervals (difference below 10 percent), rate 60–90 beats/min.

3. Classification and mechanisms of arrhythmias

Arrhythmia — a disturbance of the rate, regularity or source of cardiac excitation or of the sequence of impulse conduction.

Table 19.4. *Classification of arrhythmias (by myocardial properties)*

Group	Types
Disturbance of automaticity (nomotopic)	Sinus tachycardia, sinus bradycardia, sinus arrhythmia, sick sinus syndrome
Disturbance of automaticity (heterotopic)	Escape (replacement) rhythms and complexes: atrial, AV junctional, idioventricular rhythms
Disturbance of excitability	Extrasystoles, paroxysmal tachycardias
Disturbance of conductivity	Blocks (sinoatrial, intra-atrial, AV, bundle branch), pre-excitation syndromes (WPW)
Mixed disturbances	Atrial and ventricular fibrillation and flutter (topic 20)

Table 19.5. *Electrophysiological mechanisms of arrhythmias*

Mechanism	Essence	Examples
Altered automaticity	Change in the rate of diastolic depolarization in the sinus node or in an ectopic focus	Sinus tachycardia, ectopic atrial tachycardia
Triggered activity	Early or delayed afterdepolarizations (an “extra wave” after the preceding potential)	Glycoside toxicity, “torsades de pointes” tachycardia in long QT syndrome
Re-entry of the excitation wave	The impulse is blocked in one pathway, returns through another pathway and re-excites a previously excited area; requires unidirectional block and slow conduction	Most extrasystoles and paroxysmal tachycardias, atrial flutter

4. Disturbances of automaticity

Sinus tachycardia

An increase in the rate of impulses from the sinus node at rest above **90 (100) beats/min**.

Causes: physiological — physical exertion, excitement; pathological — fever (about 8–10 beats/min for every 1 °C), anemia, hypoxia, hypovolemia, hypotension, thyrotoxicosis, heart failure, myocarditis, pheochromocytoma, drugs (atropine, adrenomimetics, caffeine).

Mechanism: increased sympathetic or decreased parasympathetic influence — diastolic depolarization in the sinus node cells accelerates.

ECG: sinus rhythm preserved, RR intervals shortened, rate above 90.

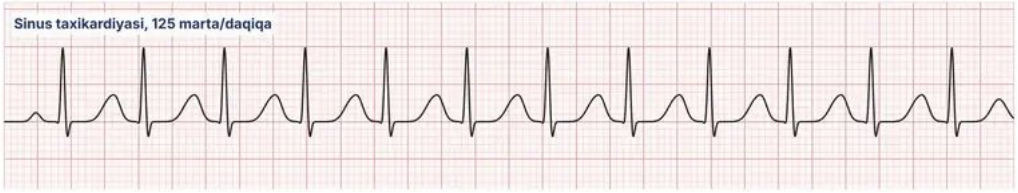


Figure 19.2. *Sinus tachycardia*

Consequences: diastole shortens → ventricular filling and coronary blood flow (which occurs mainly in diastole) decrease, myocardial oxygen demand increases → attacks of angina in patients

prone to ischemia.

Sinus bradycardia

A decrease in the rate of sinus node impulses below **60 beats/min**.

Causes: physiological — in athletes (vagotonia), during sleep; pathological — increased vagal tone (raised intracranial pressure, obstructive jaundice — effect of bile acids), hypothyroidism, hypothermia, sick sinus syndrome, myocardial infarction (inferior wall), drugs (beta-blockers, cardiac glycosides, verapamil).

ECG: sinus rhythm, RR intervals prolonged, rate below 60.

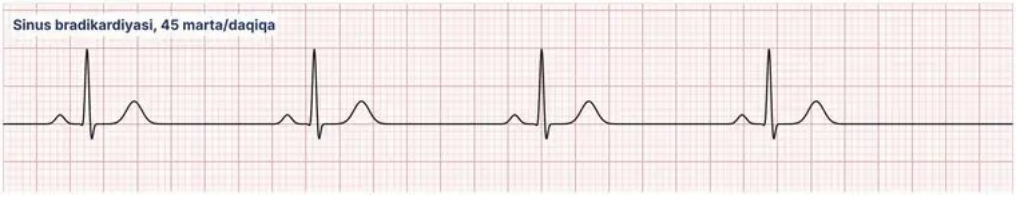


Figure 19.3. *Sinus bradycardia*

Sinus arrhythmia

Irregular discharge of impulses from the sinus node: the difference between RR intervals exceeds 0.1–0.15 s (or 10 percent).

- **Respiratory arrhythmia** — the rate increases on inspiration and decreases on expiration (variation of vagal tone with the phases of respiration); physiological in children and adolescents.
- **Non-respiratory arrhythmia** — organic myocardial changes, intoxications.



Figure 19.4. *Sinus arrhythmia*

> **CLINICAL SIGNIFICANCE**
> **Sick sinus syndrome** — reduced automaticity resulting from organic damage to the sinus node (ischemia, fibrosis, cardiomyopathies): persistent bradycardia, sinoatrial blocks, sinus arrest, alternation of bradycardia with tachyarrhythmias (“tachy-brady syndrome”). During long pauses there are attacks of fainting due to cerebral ischemia (**Morgagni–Adams–Stokes**); treatment — a permanent pacemaker.

Escape (replacement) rhythms

When sinus node activity is depressed or impulses fail to be conducted, lower-order centers begin to generate impulses: AV junctional rhythm (40–60 beats/min, P inverted or merged with QRS), idioventricular rhythm (20–40 beats/min, wide deformed QRS). This is a protective (compensatory) mechanism — it prevents asystole.

5. Disturbances of excitability. Extrasystole

Extrasystole — a premature (out-of-turn) contraction of the heart or part of it; the impulse arises from an ectopic focus (outside the sinus node).

Causes:

- **functional** — excitement, smoking, strong tea and coffee, alcohol, autonomic dysfunction, gastrointestinal diseases (reflex);
- **organic** — ischemic heart disease, myocardial infarction, myocarditis, cardiomyopathies, heart defects, arterial hypertension;
- **toxic and electrolyte** — cardiac glycoside toxicity, hypokalemia, hypomagnesemia, adrenomimetics, thyrotoxicosis.

Theories and mechanisms of extrasystole: re-entry of the excitation wave, increased automaticity of an ectopic focus, triggered activity (afterdepolarizations).

Compensatory pause — the prolonged interval after an extrasystole until the next normal contraction:

- **full compensatory pause** — the sum of the RR intervals before and after the extrasystole equals two normal RR intervals; typical of ventricular extrasystoles (the sinus node is not “discharged” and its rhythm is not disturbed; the sinus impulse following the extrasystole reaches refractory ventricles);
- **incomplete compensatory pause** — the sum is less than two RR intervals; typical of atrial and AV extrasystoles (the ectopic impulse also excites the sinus node and “resets” its cycle).

Table 19.6. *ECG signs of extrasystoles*

Feature	Atrial extrasystole	AV junctional extrasystole	Ventricular extrasystole
Premature	Yes	Yes	Yes
P wave	Present, altered in shape (or inverted)	Inverted P (before or after QRS) or absent	No
QRS complex	Narrow, unchanged	Narrow, unchanged	Wide (more than 0.12 s), deformed
T wave	Unchanged	Unchanged	Directed opposite to the main deflection of QRS
Compensatory pause	Incomplete	Usually incomplete	Full



Figure 19.5. *Atrial extrasystole*



Figure 19.6. *Ventricular extrasystole*

Types of extrasystole: **single**, **paired** (two in succession), **grouped** (three or more — “runs”, progressing to paroxysmal tachycardia); **allorhythmia** — regularly repeating: **bigeminy** (an extrasystole after every normal contraction), **trigeminy** (after every two); **monofocal** (from one focus) and **multifocal** (of different shapes, from different foci); early (“R on T” — falling on the T wave of the preceding contraction).

CAUTION

“Dangerous” signs of ventricular extrasystoles: multifocal, paired and grouped, early (“R on T”) extrasystoles, especially against the background of acute myocardial infarction — they indicate a risk of progression to ventricular fibrillation. The contraction that follows the prolonged pause after an extrasystole is stronger (Frank–Starling law) — the patient feels this as “the heart stopped and then beat hard”.

Hemodynamic consequences: during an extrasystole the ventricles do not fill adequately with blood — the stroke volume is small, and the pulse wave may not be felt at the periphery (**pulse deficit**); frequent extrasystoles reduce cardiac output and coronary blood flow.

6. Experimental part: recording and analyzing the ECG

Aim: to record a normal ECG and analyze the ECGs of patients with disturbances of automaticity and excitability.

Equipment: electrocardiograph, electrodes, contact gel, calipers (measuring tool), a set of ECG samples of various arrhythmias.

ECG recording procedure

1. Lay the subject on their back, with relaxed muscles and calm breathing.
2. Moisten the electrode sites with gel.
3. Apply the limb electrodes: red — right arm, yellow — left arm, green — left leg, black (ground) — right leg.
4. Apply the chest electrodes: V_1 — fourth intercostal space at the right sternal border; V_2 — fourth space at the left sternal border; V_4 — fifth space on the left midclavicular line; V_3 — midway between V_2 and V_4; V_5 — anterior axillary line at the level of V_4; V_6 — midaxillary line at the same level.
5. Check the calibration signal (1 mV = 10 mm) and the paper speed (25 or 50 mm/s).
6. Record all 12 leads; record lead II for longer (for rhythm analysis).

ECG analysis plan

1. Determine the rhythm: sinus or non-sinus, regular or irregular.
2. Calculate the heart rate.
3. Measure the duration of P, PQ, QRS and QT.
4. Determine the electrical axis of the heart.
5. Assess the shape of the P wave, QRS complex, ST segment and T wave.
6. Write a conclusion.

Table 19.7. *Example of recording the analysis results*

Parameter	ECG 1	ECG 2	ECG 3
Rhythm (sinus, regularity)			
Heart rate			
PQ, s			
QRS, s			
Premature complexes			
Compensatory pause			
Conclusion			

7. Practical skills

Skill: calculating heart rate from the ECG

1. Determine the paper speed (25 mm/s).
2. Measure the distance between two adjacent R waves in mm.
3. Multiply the distance by 0.04 to obtain RR in seconds.
4. Heart rate = 60 / RR.
5. If the rhythm is irregular, multiply the number of QRS complexes in a 6-second strip by 10.

Skill: identifying an extrasystole on the ECG and determining its type

1. Find the premature complex (shortened RR).
2. Check whether there is a P wave before it and what its shape is.
3. Measure the width of the QRS: below 0.12 s — supraventricular, above — ventricular.
4. Assess the direction of the T wave.
5. Measure the compensatory pause: compare the sum of the RR intervals before and after the extrasystole with two normal RR intervals.
6. Write a conclusion about the number, sequence (bigeminy, pairs) and shape (monofocal or multifocal) of the extrasystoles.

Review questions

1. List the main properties of the myocardium and their disturbances.
2. What are the centers of the conduction system and their rates of automaticity?
3. How do the sympathetic and parasympathetic nerves regulate cardiac activity?
4. List the parameters of central hemodynamics.
5. What do the P, QRS and T waves and the PQ and QT intervals of the ECG reflect?
6. What are the ECG criteria of sinus rhythm?
7. Explain the classification of arrhythmias.
8. Explain the electrophysiological mechanisms of arrhythmias (automaticity, triggered activity, re-entry).
9. What are the causes, mechanism and consequences of sinus tachycardia?
10. List the causes of sinus bradycardia.
11. What is sinus arrhythmia? What is the mechanism of respiratory arrhythmia?
12. How does sick sinus syndrome manifest itself?
13. What are the causes and mechanisms of extrasystole?
14. How do atrial and ventricular extrasystoles differ on the ECG?
15. Why do full and incomplete compensatory pauses occur?

Test questions

1. Which is the first-order center of automaticity?
   A) Sinus node
   B) AV junction
   C) Bundle of His
   D) Purkinje fibers
   E) Left ventricular myocardium

2. What does the P wave reflect on the ECG?
   A) Ventricular depolarization
   B) Atrial depolarization
   C) Ventricular repolarization
   D) Atrial repolarization
   E) Passage through the AV node

3. What is the normal PQ interval?
   A) 0.04–0.08 s
   B) 0.22–0.30 s
   C) 0.35–0.44 s
   D) 0.06–0.10 s
   E) 0.12–0.20 s

4. The RR interval on the ECG is 0.6 s. What is the heart rate?
   A) 60 beats/min
   B) 75 beats/min
   C) 120 beats/min
   D) 100 beats/min
   E) 50 beats/min

5. What is the mechanism of sinus bradycardia in athletes?
   A) Organic damage to the sinus node
   B) Hypothyroidism
   C) Increased vagal tone
   D) Glycoside toxicity
   E) Hypokalemia

6. What is the mechanism of sinus tachycardia in fever?
   A) Re-entry
   B) AV block
   C) Acceleration of diastolic depolarization in the sinus node
   D) Triggered activity
   E) Activity of an ectopic focus

7. How is respiratory arrhythmia characterized?
   A) The rate decreases on inspiration and increases on expiration
   B) The rate increases on inspiration and decreases on expiration
   C) It occurs only with ventricular extrasystoles
   D) It is always a sign of heart disease
   E) The P wave disappears

8. Which is an ECG sign of a ventricular extrasystole?
   A) Altered P wave and narrow QRS
   B) Inverted P and narrow QRS
   C) Incomplete compensatory pause
   D) No P wave, wide deformed QRS, T directed opposite
   E) Prolongation of the PQ interval

9. Why is the compensatory pause full after a ventricular extrasystole?
   A) The sinus node is “reset” by the ectopic impulse
   B) The atria are not excited
   C) The AV node is blocked
   D) The Purkinje fibers contract
   E) The sinus node rhythm is not disturbed; the next sinus impulse reaches refractory ventricles

10. What is it called when an extrasystole follows every normal contraction?
   A) Bigeminy
   B) Trigeminy
   C) Multifocal extrasystole
   D) Paroxysmal tachycardia
   E) Absence of allorhythmia

11. What condition is necessary for re-entry of the excitation wave?
   A) Sinus arrest
   B) Hypokalemia only
   C) Ventricular hypertrophy
   D) Unidirectional block and slow conduction of the impulse
   E) Decreased sympathetic tone

12. Which ventricular extrasystoles increase the risk of ventricular fibrillation?
   A) Multifocal, paired and early (“R on T”) extrasystoles
   B) Single monofocal late extrasystoles
   C) Atrial extrasystoles only
   D) Functional extrasystoles only
   E) Extrasystoles during sleep only

13. What is the main adverse consequence of sinus tachycardia for the heart?
   A) Ventricular fibrillation always develops
   B) Shortening of diastole and decreased coronary blood flow
   C) AV block
   D) Bradycardia
   E) Myocardial hypertrophy

14. What is the rate of the escape rhythm arising from the AV junction?
   A) 60–80 beats/min
   B) 20–40 beats/min
   C) 100–120 beats/min
   D) 10–20 beats/min
   E) 40–60 beats/min

15. What is the main mechanism of a Morgagni–Adams–Stokes attack?
    A) Arterial hypertension
    B) Hypoglycemia
    C) Cerebral ischemia due to prolonged asystole or severe bradycardia
    D) An epileptic focus
    E) Hyperthermia

Clinical cases

Case 1

A 24-year-old woman has frequent palpitations, trembling hands, sweating and weight loss. Pulse 118 beats/min, regular. ECG: a positive P before each QRS, PQ 0.14 s, QRS 0.08 s, RR 0.51 s. Blood thyroid hormones are raised.

Questions: Analyze the ECG. What are the type and mechanism of the arrhythmia? What is the danger for the heart?

Answer: Heart rate = 60 / 0.51 ≈ 118 beats/min; a P before each QRS, normal PQ, narrow QRS — sinus rhythm. This is sinus tachycardia (a nomotopic disturbance of automaticity). The cause is thyrotoxicosis: thyroid hormones increase the number of beta-adrenoceptors in the sinus node cells and their sensitivity to catecholamines, and act directly on ion channels to accelerate diastolic depolarization; in addition, metabolism increases. In tachycardia diastole shortens, ventricular filling and coronary blood flow decrease, and myocardial oxygen demand increases; if prolonged, atrial fibrillation and tachycardia-induced cardiomyopathy may develop.

Case 2

On day 2 of myocardial infarction, a 58-year-old man's cardiac monitor shows frequent premature wide QRS complexes: they have two different shapes, sometimes come in pairs, and some fall on the T wave of the preceding complex.

Questions: What arrhythmia is this? Why is it dangerous? What is the mechanism of its development?

Answer: Multifocal, paired and early (“R on T”) ventricular extrasystoles. In the ischemic myocardium of the infarct zone and its border, the membrane potential changes, conduction slows and refractoriness becomes uneven — conditions for re-entry are created; potassium leakage from damaged cells, increased catecholamines and reperfusion enhance triggered activity. Early extrasystoles falling into the relative refractory period of the ventricles (the “vulnerable phase”) can trigger ventricular fibrillation. Therefore the patient should be under monitor observation, electrolytes (potassium, magnesium) should be corrected and a defibrillator should be ready.

Case 3

A 72-year-old woman has had several attacks of sudden loss of consciousness, each lasting a few seconds and preceded by dizziness. 24-hour ECG monitoring revealed a heart rate of 38–42 beats/min, pauses of 4–5 seconds and episodes of atrial fibrillation between them.

Questions: What syndrome can be suspected? What is the mechanism of the fainting?

Answer: Sick sinus syndrome (the “tachy-brady” variant): as a result of organic damage to the sinus node (fibrosis in old age, ischemia), automaticity is reduced — persistent bradycardia and sinus arrest pauses, alternating with atrial tachyarrhythmia. During long pauses, if the escape centers do not start

in time, cardiac output stops and cerebral ischemia develops — Morgagni–Adams–Stokes attacks (fainting). Treatment — implantation of a permanent pacemaker.

Quick review

Table 19.8. *Key facts on Topic 19*

Question	Answer
Myocardial properties	Automaticity, excitability, conductivity, contractility
Centers of automaticity	Sinus node 60–80, AV junction 40–60, ventricles 20–40 beats/min
ECG normal values	P 0.08–0.10 s; PQ 0.12–0.20 s; QRS 0.06–0.10 s
Heart rate	60 / RR (s); at 25 mm/s, 1 mm = 0.04 s
Sinus rhythm	Positive P before each QRS, constant PQ, equal RR
Mechanisms of arrhythmia	Altered automaticity, triggered activity, re-entry
Sinus tachycardia	Above 90; fever, anemia, thyrotoxicosis, hypovolemia
Sinus bradycardia	Below 60; vagotonia, hypothyroidism, beta-blockers
Sinus arrhythmia	RR difference above 10 percent; respiratory arrhythmia is physiological
Extrasystole	A premature contraction from an ectopic focus
Ventricular extrasystole	No P, wide QRS, inverted T, full compensatory pause
Atrial extrasystole	Altered P, narrow QRS, incomplete pause
Bigeminy	An extrasystole after every normal contraction
Dangerous extrasystoles	Multifocal, paired, grouped, “R on T”
Morgagni–Adams–Stokes	Cerebral ischemia and fainting due to a long pause

TOPIC 20

Disturbances of cardiac conduction. Heart blocks. Atrial fibrillation

Learning objectives

To explain to the student the structure and function of the cardiac conduction system, the causes, mechanisms and ECG signs of conduction disturbances (sinoatrial, intra-atrial, atrioventricular and intraventricular blocks, pre-excitation syndromes), and the pathogenesis and consequences of atrial fibrillation (atrial fibrillation and flutter) and ventricular fibrillation; to teach the student to analyze the ECG in conduction disturbances and atrial fibrillation.

By the end of the session, the student should know:

- the parts of the conduction system and the sequence of impulse spread through the heart;
- the causes and mechanisms of conduction disturbances;
- the degrees and ECG signs of AV blocks;
- the ECG signs of bundle branch block;
- the nature of the pre-excitation (WPW) syndrome;
- the mechanisms and hemodynamic consequences of atrial fibrillation and flutter and ventricular fibrillation.

The student should be able to:

- determine the degree of AV block by measuring the PQ interval on the ECG;
- identify an intraventricular conduction disturbance by the width of the QRS;
- distinguish atrial fibrillation from atrial flutter on the ECG.

1. The conduction system and impulse spread

The impulse arises in the sinus node and reaches the **AV node** through the atrial myocardium and internodal pathways. In the AV node the impulse is physiologically **delayed** (about 0.1 s) — this is necessary so that atrial contraction is completed and the ventricles fill with blood. The impulse then spreads rapidly to the ventricular myocardium through the **bundle of His**, its right and left bundle branches (the left branch divides into anterior and posterior fascicles) and the **Purkinje fibers**.

Conduction disturbances are of two kinds:

- **slowing or arrest (block)** — the impulse is delayed in some part of the conduction system or does not pass at all;
- **acceleration** — the impulse reaches the ventricles prematurely via accessory (abnormal) pathways (pre-excitation syndromes).

Causes of conduction disturbances:

- organic — ischemic heart disease and myocardial infarction, myocarditis (rheumatic, diphtheritic), cardiosclerosis, cardiomyopathies, heart defects, idiopathic fibrosis of the conduction system (Lenègre and Lev diseases), surgical injuries;
- functional — a sharp increase in vagal tone;
- toxic and drug-induced — cardiac glycosides, beta-blockers, verapamil, antiarrhythmic drugs;
- electrolyte — hyperkalemia.

Mechanism: reduction of the resting potential of the cells (ischemia, hyperkalemia) leading to inactivation of sodium channels and reduced rate of rise and amplitude of the action potential; prolongation of the refractory period; replacement of conducting fibers by fibrosis.

2. Classification of heart blocks

Table 20.1. *Classification of heart blocks*

By location	By degree
Sinoatrial block	First degree — slowed conduction
Intra-atrial block	Second degree — some impulses are not conducted (incomplete block)
Atrioventricular (AV) block	Third degree — no impulses are conducted (complete block)
Intraventricular block (bundle branches and fascicles of His)	By duration: transient, intermittent, permanent

Sinoatrial block

The impulse generated in the sinus node does not pass to the atria. In second-degree sinoatrial block the ECG shows **dropping of the whole P–QRS–T complex**, and the resulting pause equals (or is close to) twice the preceding RR.

3. Atrioventricular blocks

First-degree AV block

Conduction of the impulse from the atria to the ventricles is slowed, but all impulses are conducted.

ECG: PQ interval **longer than 0.20 s**, a QRS follows every P. It has almost no effect on hemodynamics.

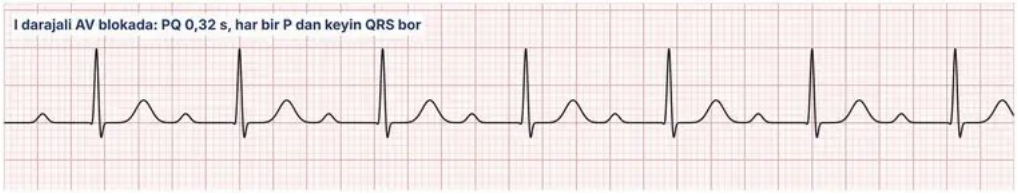


Figure 20.1. *First-degree AV block*

Second-degree AV block

Some atrial impulses are not conducted to the ventricles.

- **Mobitz type I (Samoilov–Wenckebach periods)** — the PQ interval gradually lengthens with each cycle, then one QRS is dropped (P present, no QRS); the cycle then starts again with a short PQ. Usually at the level of the AV node; prognosis relatively good (vagotonia, glycosides, inferior wall infarction).



Figure 20.2. *Second-degree AV block, Mobitz type I*

- **Mobitz type II** — the PQ interval is constant (normal or prolonged), but after some P waves the QRS complex is suddenly dropped; the ratio may be 2:1 or 3:1. Usually at or below the bundle of His — high risk of progression to third-degree block.



Figure 20.3. *Second-degree AV block, Mobitz type II*

Third-degree (complete) AV block

Not a single impulse is conducted from the atria to the ventricles. The atria are excited from the sinus node, and the ventricles from their own escape center — **atrioventricular dissociation**.

ECG: P waves and QRS complexes are unrelated to each other; the P waves occur at their own rate (60–80), the QRS complexes slowly (40–60 with a narrow QRS if from the AV junction; 20–40 with a wide deformed QRS if from the ventricles).



Figure 20.4. *Third-degree (complete) AV block*

CLINICAL SIGNIFICANCE

When complete AV block appears, the ventricular escape center starts a few seconds late (the "preautomatic pause"). During this time the brain is not supplied with blood — a **Morgagni–Adams–Stokes** attack: dizziness, loss of consciousness, pallor, then convulsions. Persistent bradycardia (30–40 beats/min) reduces cardiac output and leads to heart failure. Treatment — a pacemaker.

Table 20.2. *Distinguishing AV blocks*

Degree	PQ interval	P to QRS ratio
I	Constantly prolonged (more than 0.20 s)	1:1
II, Mobitz I	Gradually lengthens, then a QRS is dropped	One is dropped periodically
II, Mobitz II	Constant	Some QRS suddenly dropped (2:1, 3:1)
III	Not determined (P and QRS independent)	No relationship, fewer QRS than P

4. Intraventricular blocks and pre-excitation

Bundle branch block

If the impulse is not conducted along one bundle branch, the ventricle on that side receives the excitation slowly through the myocardium from the other ventricle — ventricular depolarization is prolonged and becomes asynchronous.

Table 20.3. *ECG signs of complete bundle branch block*

Feature	Right bundle branch block	Left bundle branch block
QRS width	0.12 s or more	0.12 s or more
Right chest leads (V_1–V_2)	rSR’ (“M” shape)	Wide deep S (QS)
Left chest leads (V_5–V_6)	Wide, deep S wave	Wide R with a notched peak, no Q
ST and T	Discordant (inverted) in V_1–V_2	Discordant in V_5–V_6
Significance	May also occur in healthy people; pulmonary embolism, right ventricular strain	Almost always organic disease (infarction, hypertension, aortic defect)

CAUTION

A new left bundle branch block in the setting of acute coronary syndrome is regarded as a sign as dangerous as myocardial infarction; left bundle branch block “masks” the signs of infarction on the ECG.

Pre-excitation syndrome (Wolff–Parkinson–White, WPW)

There is an accessory conducting pathway (the **Kent bundle**) between the atria and ventricles. The impulse passes through this pathway without the physiological delay of the AV node and excites part of the ventricles early.

ECG: shortened PQ (below 0.12 s), a **delta wave** (sloping upstroke) at the beginning of the QRS, widened QRS, discordant ST–T changes.

Significance: the accessory pathway and the AV node form a closed loop — attacks of paroxysmal tachycardia by **re-entry**; if atrial fibrillation develops, impulses can pass very rapidly to the ventricles via the accessory pathway and trigger ventricular fibrillation.

5. Atrial fibrillation

Atrial fibrillation (“ciliary arrhythmia”) — uncoordinated, very rapid (350–700 per minute) excitation of the atria; the atria do not contract effectively, and the ventricles contract irregularly. Two types are distinguished: **atrial fibrillation** and **atrial flutter**.

Causes: mitral defects (especially mitral stenosis — dilation of the left atrium), ischemic heart disease, arterial hypertension, thyrotoxicosis, cardiomyopathies, myocarditis, alcohol intoxication (“holiday heart”), chronic lung diseases, old age.

Mechanism:

- many small **re-entry** circuits in the atrial myocardium (the “multiple wavelet” theory) — excitation waves circulate chaotically in a dilated and fibrosed atrium;
- rapid impulses from **ectopic foci** at the mouths of the pulmonary veins (trigger);
- the AV node acts as a “filter” — it conducts only some of the impulses to the ventricles, which is why the ventricular rhythm is irregular.

ECG: no P waves; instead, small irregular **f waves** on the isoelectric line, completely irregular RR intervals, QRS usually narrow.

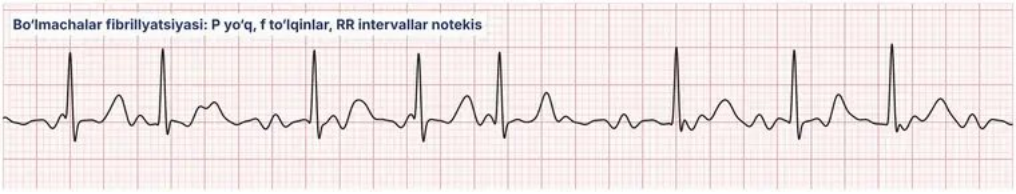


Figure 20.5. *Atrial fibrillation*

Atrial flutter — regular, rapid (250–350 per minute) excitation of the atria; mainly a single large re-entry circuit in the right atrium (macro re-entry). On the ECG there are regular "**sawtooth**" F waves, usually conducted to the ventricles in a 2:1, 3:1 or 4:1 ratio.



Figure 20.6. *Atrial flutter with 4:1 conduction*

Table 20.4. *Distinguishing atrial fibrillation from atrial flutter*

Feature	Fibrillation	Flutter
Atrial excitation	350–700, chaotic	250–350, regular
Waves on the ECG	Small irregular f waves	"Sawtooth" F waves
Ventricular rhythm	Completely irregular	Often regular (if the ratio is constant)
Main mechanism	Many small re-entry circuits, ectopic foci	One large re-entry circuit

Consequences of atrial fibrillation

1. **Loss of atrial systole** — ventricular filling decreases by about 20–30 percent; cardiac output falls significantly, especially in mitral stenosis and myocardial hypertrophy.
2. **Tachysystole** — diastole shortens, the ventricles do not fill adequately, coronary blood flow decreases and myocardial oxygen demand increases; if prolonged — heart failure.
3. **Pulse deficit** — contractions with a small stroke volume do not produce a pulse wave at the periphery: the number of contractions at the cardiac apex exceeds the radial pulse.
4. **Thromboembolism** — blood stagnates in the non-contracting atrium (especially the left atrial appendage) and a thrombus forms; a fragment of the thrombus enters the systemic circulation and causes **ischemic stroke** and embolism of the renal, intestinal and leg arteries.

> **MEMORY AID**
> Consequences of atrial fibrillation — "**STEP**": **S**ystole (atrial) loss, **T**achysystole, **E**mbolism (thrombus), **P**ulse deficit.

Ventricular fibrillation and flutter

Ventricular fibrillation — chaotic, uncoordinated excitation and contraction of individual fibers of the ventricular myocardium; the pumping function of the heart stops — this is the most common mechanism of **clinical death**. The ECG shows chaotic waves of varying shape and amplitude, with no distinguishable QRS or T. **Ventricular flutter** — regular, high-amplitude "sinusoidal" waves (200–300 per minute), which quickly turn into fibrillation.

Causes: acute myocardial infarction and ischemia, electric shock, hyper- and hypokalemia, glycoside toxicity, long QT syndrome, hypothermia, severe hypoxia.

Care: immediate cardiopulmonary resuscitation and **electrical defibrillation** — a strong electrical impulse depolarizes all myocardial cells simultaneously, after which the sinus node may restore the rhythm.

6. Experimental part: studying conduction disturbances

Stannius ligatures (on the frog heart)

Aim: to demonstrate the hierarchy of the centers of automaticity and the consequences of arrested impulse conduction.

1. Immobilize the frog, open the chest and leave the heart exposed; count the heart contractions.
2. **First ligature** — tie a thread at the border between the sinus venosus and the atria: the sinus venosus continues to contract at its previous rhythm, the atria and ventricle stop temporarily, then begin to contract at a slower rhythm (automaticity of the AV center).
3. **Second ligature** — tie a thread at the border between the atria and the ventricle (tightening it stimulates the AV region): the ventricle begins to contract at its own rhythm.
4. **Third ligature** — separate the lower third of the ventricle: the apical part does not contract (no center of automaticity), but it responds if stimulated.
5. At each stage, record the contraction rate of each part in a table and draw a conclusion.

Analyzing ECG samples

On the ECGs provided, measure the PQ interval, the P to QRS ratio, the QRS width and the regularity of RR, and determine the type of conduction disturbance.

> **REMEMBER**
> The exact procedure of the experiments, the animal species and the solutions should be checked against the department's methodological manual.

7. Practical skills

Skill: determining the degree of AV block from the ECG

1. Mark all the P waves and check their regularity.
2. Determine whether a QRS follows each P.
3. Measure successive PQ intervals.
4. If PQ is constantly prolonged and P:QRS = 1:1 — first degree.
5. If PQ gradually lengthens and a QRS is dropped — second degree Mobitz I; if PQ is constant and a QRS is suddenly dropped — second degree Mobitz II.
6. If P and QRS are unrelated — third degree; determine the location of the escape center from the width of the QRS.

Skill: identifying atrial fibrillation and calculating the pulse deficit

1. Find the absence of P waves and the f waves on the ECG.
2. Confirm the irregularity of the RR intervals with calipers.
3. Multiply the number of QRS complexes in a 6-second strip by 10 to find the average rate.

4. Simultaneously count the contractions at the cardiac apex (auscultation) and the radial pulse in the patient for 1 minute.
5. Pulse deficit = heart rate – pulse rate.

Review questions

1. What parts make up the cardiac conduction system? What is the significance of the delay in the AV node?
2. List the causes and mechanisms of conduction disturbances.
3. How are heart blocks classified by location and degree?
4. What is the ECG sign of sinoatrial block?
5. What are the ECG signs of first-degree AV block?
6. How do Mobitz type I and Mobitz type II AV blocks differ? Which is more dangerous?
7. What changes occur in complete AV block? How does a Morgagni–Adams–Stokes attack develop?
8. List the ECG signs of right and left bundle branch block.
9. What are the nature and ECG signs of WPW syndrome?
10. What are the causes and mechanism of atrial fibrillation?
11. How do atrial fibrillation and atrial flutter differ on the ECG?
12. Explain the hemodynamic consequences of atrial fibrillation.
13. Why does thromboembolism develop in atrial fibrillation?
14. What is the pulse deficit and how is it determined?
15. What are the causes, ECG signs and emergency measures of ventricular fibrillation?

Test questions

1. Why is the physiological slowing of impulse conduction in the AV node necessary?
   A) To accelerate the sinus node rhythm
   B) For ventricular repolarization
   C) So that atrial contraction is completed and the ventricles fill with blood
   D) To increase the automaticity of the Purkinje fibers
   E) To reduce coronary blood flow
2. Which is the ECG sign of first-degree AV block?
   A) Periodic dropping of the QRS complex
   B) Independent rhythms of P and QRS
   C) Delta wave
   D) Prolongation of the PQ interval beyond 0.20 s, a QRS after every P
   E) f waves
3. Gradual lengthening of the PQ interval followed by a dropped QRS is typical of which block?
   A) First-degree AV block
   B) Second-degree AV block, Mobitz type I
   C) Second-degree AV block, Mobitz type II
   D) Third-degree AV block
   E) Sinoatrial block

4. Which AV block has the highest risk of progressing to complete block?
   A) Second degree, Mobitz type II
   B) First degree
   C) Second degree, Mobitz type I
   D) Transient block in vagotonia
   E) First degree caused by glycosides

5. Which is the ECG sign of complete AV block?
   A) Shortened PQ, delta wave
   B) Completely irregular RR, no P
   C) “Sawtooth” waves
   D) Premature wide QRS
   E) P and QRS are unrelated, the QRS rate is lower than the P rate

6. What causes a Morgagni–Adams–Stokes attack in complete AV block?
   A) Atrial fibrillation
   B) A sharp rise in arterial pressure
   C) Hypoglycemia
   D) Tachycardia
   E) Cerebral ischemia during the pause before the escape center starts

7. A QRS wider than 0.12 s and rSR’ (“M” shape) in V_1 are typical of which disturbance?
   A) Complete right bundle branch block
   B) Complete left bundle branch block
   C) First-degree AV block
   D) Sinoatrial block
   E) Atrial flutter

8. Which is the ECG sign of WPW syndrome?
   A) Prolonged PQ, narrow QRS
   B) No P, f waves
   C) Shortened PQ, delta wave, widened QRS
   D) Periodic dropping of QRS
   E) Prolonged QT

9. Which is the ECG sign of atrial fibrillation?
   A) A positive P before each QRS
   B) “Sawtooth” F waves, regular RR
   C) Prolonged PQ
   D) No P, f waves, completely irregular RR
   E) Wide QRS, full compensatory pause

10. What is the main mechanism of atrial flutter?
    A) Increased automaticity of the sinus node
    B) A single large re-entry circuit in the right atrium
    C) AV block
    D) Triggered activity in the ventricles
    E) Bundle branch block

11. What is the main cause of ischemic stroke in atrial fibrillation?
    A) Embolism of a thrombus formed in the left atrial appendage
    B) Hypertensive crisis
    C) Hypoglycemia
    D) Venous thrombosis
    E) Ventricular hypertrophy

12. What is the pulse deficit?
    A) A fall in arterial pressure
    B) Acceleration of the pulse wave
    C) The difference between the heart rate and the radial pulse rate
    D) The difference between systolic and diastolic pressure
    E) A decrease of the pulse on inspiration

13. What is the main cause of the sharp fall in cardiac output when atrial fibrillation develops in a patient with mitral stenosis?
    A) Ventricular fibrillation
    B) Bundle branch block
    C) Arterial hypertension
    D) Loss of the contribution of atrial systole to ventricular filling
    E) Bradycardia

14. What is the first-line effective measure in ventricular fibrillation?
    A) Giving atropine
    B) Electrical defibrillation and cardiopulmonary resuscitation
    C) Cardiac glycosides
    D) Potassium preparations
    E) Observation

15. What is observed in the Stannius experiment when the first ligature is placed at the border between the sinus venosus and the atria?
    A) The whole heart continues to contract at the same rhythm
    B) The sinus venosus stops, the ventricle accelerates
    C) Only the apex of the ventricle contracts
    D) Atrial fibrillation
    E) The sinus venosus contracts at its previous rhythm, the atria and ventricle stop and then contract at a slower rhythm

Clinical cases

Case 1

A 66-year-old man has had several sudden losses of consciousness; when he fell his face turned pale, then convulsions occurred, and he came round after 20–30 seconds. Pulse 34 beats/min, regular. ECG: P waves 76 per minute, QRS 34 per minute, wide (0.14 s), no constant relationship between P and QRS.

Questions: What is the ECG conclusion? Explain the mechanism of the fainting attacks. What treatment is needed?

Answer: Third-degree (complete) AV block; the escape rhythm is ventricular (wide QRS, 34 beats/min — idioventricular rhythm). The atria are excited by the sinus node and the ventricles

independently by their own center (AV dissociation). When the block appears or deepens, the ventricular center starts a few seconds late; during this time cardiac output stops and cerebral ischemia develops — Morgagni–Adams–Stokes attacks (loss of consciousness, then hypoxic convulsions). Persistent bradycardia reduces cardiac output and leads to heart failure. Treatment — a permanent pacemaker.

Case 2

A 62-year-old woman has had mitral stenosis for many years. Over the last month irregular heartbeat and dyspnea have increased. Suddenly weakness of the left arm and leg and a speech disturbance appeared. Heart rate 118 beats/min, radial pulse 92 beats/min. ECG: no P, small irregular waves, RR intervals varying.

Questions: What arrhythmia is this? Calculate the pulse deficit and explain its mechanism. What are the neurological signs related to?

Answer: Atrial fibrillation (tachysystolic form). In mitral stenosis the left atrium dilates under pressure and becomes fibrotic — a substrate for many re-entry circuits is formed. Pulse deficit = 118 – 92 = 26: in contractions following a short diastole the ventricles do not fill adequately, the stroke volume is small and the pulse wave does not reach the wrist. Because the atrium does not contract effectively, blood stagnates in the left atrial appendage and a thrombus forms; a fragment of the thrombus has entered the right middle cerebral artery and caused an ischemic stroke (left-sided hemiparesis, speech disturbance) — a cardioembolic stroke. Such patients need anticoagulant prophylaxis and rhythm or rate control.

Case 3

A 20-year-old student has episodes of severe palpitations that start and end suddenly. ECG between attacks: PQ 0.10 s, a sloping upstroke at the beginning of the QRS, QRS 0.12 s.

Questions: What syndrome is this? What is the mechanism of the attacks? What dangerous complication may occur?

Answer: Wolff–Parkinson–White (WPW) syndrome: the impulse passes to the ventricles via an accessory pathway (the Kent bundle) without the delay of the AV node — shortened PQ, delta wave and wide QRS. The accessory pathway and the AV node form a closed loop — attacks of paroxysmal supraventricular tachycardia develop by the re-entry mechanism. If atrial fibrillation develops, very rapid impulses can pass to the ventricles via the accessory pathway (without the AV node “filter”) and lead to ventricular fibrillation and sudden death. Radical treatment — catheter ablation of the accessory pathway.

Quick review

Table 20.5. *Key facts on Topic 20*

Question	Answer
Heart block	Slowing or arrest of impulse conduction
Sinoatrial block	Dropping of the whole P–QRS–T complex
First-degree AV block	PQ longer than 0.20 s, P:QRS = 1:1
Mobitz I	PQ gradually lengthens, a QRS is dropped (Samoilov–Wenckebach period)
Mobitz II	PQ constant, QRS suddenly dropped; risk of progression to third degree

Question	Answer
Third-degree AV block	P and QRS independent; AV dissociation
Morgagni–Adams–Stokes	Cerebral ischemia during a pause — fainting
Bundle branch block	QRS wider than 0.12 s; right — rSR’ in V_1, left — wide R in V_6
WPW syndrome	Kent bundle: short PQ, delta wave, wide QRS
Atrial fibrillation	350–700 chaotic excitation; f waves, irregular RR
Atrial flutter	250–350 regular; “sawtooth” F waves
Pulse deficit	Heart rate minus radial pulse
Thromboembolism	Thrombus in the left atrial appendage — stroke
Ventricular fibrillation	Mechanism of clinical death; defibrillation
Stannius ligatures	Demonstrate the hierarchy of the centers of automaticity

TOPIC 21

Typical forms of cardiovascular dysfunction. Compensatory mechanisms (seminar)

Learning objectives

To explain to the student the typical forms of cardiovascular dysfunction — heart failure, heart defects, coronary insufficiency, arterial hypertension and hypotension, pericardial pathology — their causes, hemodynamic changes, cardiac and extracardiac compensatory mechanisms and the development of decompensation; to teach the student to apply this knowledge to clinical cases.

By the end of the session, the student should know:

- the definition and classification of circulatory failure;
- the causes (overload, myocardial damage, impaired diastolic filling) and pathogenesis of heart failure;
- the cardiac (rapid and long-term) and extracardiac compensatory mechanisms of the heart;
- the stages of myocardial hypertrophy and the features of the hypertrophied heart;
- the hemodynamic changes in heart defects;
- the pathogenesis of coronary insufficiency, myocardial infarction and arterial hypertension.

The student should be able to:

- explain the signs of heart failure in relation to their pathogenesis;
- depict the hemodynamic changes in heart defects as a diagram;
- distinguish signs of compensation and decompensation in clinical cases.

1. Circulatory failure and its forms

Circulatory failure — a state in which the cardiovascular system is unable to supply tissues and organs with blood in accordance with their needs. It develops for cardiac, vascular or mixed reasons.

Heart failure — a clinical syndrome in which, because of impaired pumping function of the heart, cardiac output cannot meet the needs of the tissues, or can do so only at the cost of increased ventricular filling pressure.

Table 21.1. *Classification of heart failure*

Criterion	Types
Course	Acute (hours, minutes — infarction, arrhythmia, tamponade), chronic (months, years)
Main part affected	Left ventricular, right ventricular, total (biventricular)
Phase of the cardiac cycle	Systolic (reduced contractility, low ejection fraction), diastolic (impaired relaxation and filling, preserved ejection fraction)
Origin	Myocardial (damage), from overload (volume or pressure overload), mixed
Direction of hemodynamics	Backward (congestion), forward (organ hypoperfusion)

Table 21.2. *Pathogenetic causes of heart failure*

Form	Mechanism	Examples
Volume overload (preload ↑)	Excess blood enters the ventricle in diastole	Valvular insufficiency, hypervolemia, left-to-right shunts
Pressure overload (afterload ↑)	Increased resistance to ejection	Aortic stenosis, arterial hypertension, pulmonary hypertension
Direct myocardial damage	Death of cardiomyocytes and reduced contractility	Myocardial infarction, myocarditis, cardiomyopathies, toxic effects (alcohol), vitamin B_1 deficiency
Impaired diastolic filling	The ventricles cannot relax and expand adequately	Pericardial tamponade, constrictive pericarditis, restrictive cardiomyopathy, hypertrophy

2. Compensatory mechanisms of the heart

Cardiac compensatory mechanisms

Rapid (emergency) mechanisms:

1. **Heterometric mechanism (Frank–Starling law)** — if myocardial fibers are stretched more in diastole (increased preload), the next contraction is stronger. It is effective up to a certain limit; with excessive stretching (myogenic dilation) the force of contraction decreases.
2. **Homeometric mechanism (Anrep phenomenon)** — without a change in fiber length, the force of contraction gradually increases when aortic resistance rises (through greater release of calcium ions).
3. **Tachycardia** — resulting from sympathetic influence and the Bainbridge reflex (stretching of the venae cavae and right atrium). If the rate rises excessively, diastole shortens, filling and coronary blood flow decrease, and the effect is lost.
4. **Inotropic effect** — catecholamines increase myocardial contractility.

Long-term mechanism — myocardial hypertrophy. With constant overload, cardiomyocyte size (hypertrophy) and heart mass increase; this normalizes the work and wall stress per unit mass.

- **Concentric hypertrophy** — with pressure overload (aortic stenosis, hypertension): the wall thickens, the cavity is unchanged or smaller (sarcomeres are added in parallel).
- **Eccentric hypertrophy** — with volume overload (valvular insufficiency): the cavity dilates and the wall thickens relatively (sarcomeres are added in series).

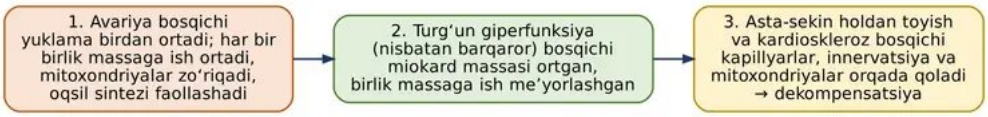


Figure 21.1. *Stages of compensatory myocardial hypertrophy (according to F. Z. Meerson)*

CAUTION

The hypertrophied myocardium is “imperfect” — **unbalanced growth**: the mass of cardiomyocytes increases, but the number of capillaries, nerve endings, mitochondria and the sarcoplasmic reticulum lag behind the growth of the myocardium. As a result the diffusion distance lengthens (relative coronary insufficiency), energy supply and calcium handling are impaired, the rates of contraction and relaxation decrease and fibrosis develops. This is why, in the long term, hypertrophy leads to decompensation and arrhythmias.

Cardiac reserve mechanisms of compensation — increases in heart rate, stroke volume and oxygen extraction; athletes show physiological hypertrophy (balanced growth).

Extracardiac compensatory mechanisms

- activation of the **sympathoadrenal system** — tachycardia, inotropic effect, constriction of arterioles (maintaining pressure), increased venous tone (increasing venous return), redistribution of blood in favor of vital organs;
- **renin–angiotensin–aldosterone system** — vasoconstriction, sodium and water retention, increased blood volume;
- **antidiuretic hormone** — water retention;
- **natriuretic peptides** (atrial and brain natriuretic peptides) — released from the stretched atria and ventricles, they enhance natriuresis and vasodilation (a counter-regulatory system; their blood level is used in the diagnosis of heart failure);
- **increased erythropoiesis** (due to hypoxia), increased oxygen extraction in the tissues, increased anaerobic glycolysis.

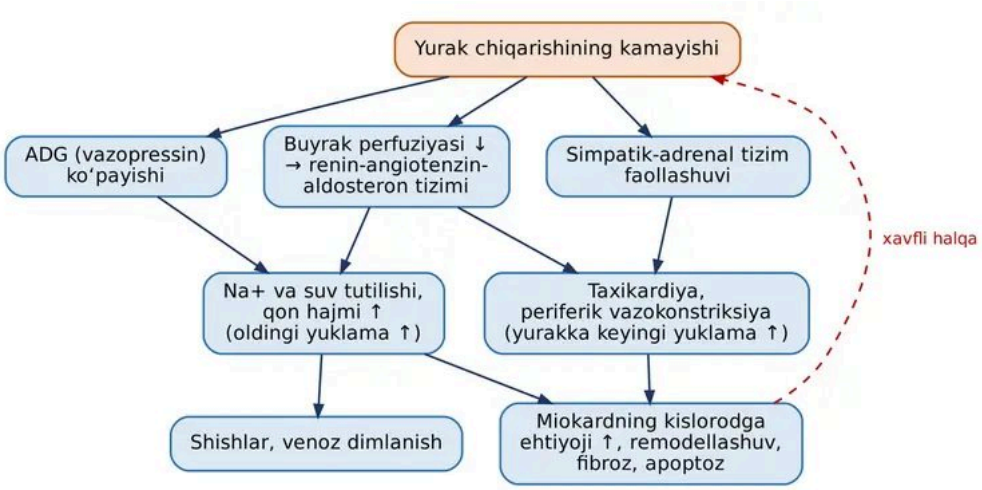


Figure 21.2. *Neurohumoral activation and the vicious circle in heart failure*

At first, neurohumoral mechanisms maintain blood pressure and perfusion of vital organs, but with prolonged activation they increase preload and afterload on the heart and have a direct harmful effect on the myocardium (remodeling, fibrosis, apoptosis) — a **vicious circle** forms. This is why the basis of treatment of chronic heart failure is drugs that block these systems (ACE inhibitors, angiotensin receptor blockers, beta-blockers, aldosterone antagonists).

3. Decompensation and its signs

Decompensation — exhaustion of the compensatory mechanisms, so that the heart can no longer meet the needs of the tissues.

Table 21.3. *Signs of left and right ventricular failure*

Type of sign	Left ventricular failure	Right ventricular failure
Site of congestion	Pulmonary circulation (lungs)	Systemic circulation (veins)
Main signs	Dyspnea, orthopnea, nocturnal attacks of suffocation (cardiac asthma), moist rales in the lungs, pulmonary edema	Distended neck veins, enlarged liver, leg edema, ascites, raised central venous pressure

Type of sign	Left ventricular failure	Right ventricular failure
Mechanism	Increased hydrostatic pressure in the pulmonary capillaries → interstitial and alveolar edema	Increased venous pressure → fluid moves into tissues, hepatic congestion
Forward failure	Weakness, fatigue, acrocyanosis, reduced renal perfusion (oliguria), cerebral hypoperfusion	Less blood reaches the pulmonary circulation — the preload on the left ventricle decreases

CLINICAL SIGNIFICANCE

Orthopnea — the patient's dyspnea increases when lying down and he is forced to sit up: in the lying position more blood returns to the heart from the legs and abdominal cavity, the weak left ventricle cannot pump it, and pulmonary congestion increases. **Acrocyanosis** — increased oxygen extraction at the periphery due to slowed blood flow, with more reduced hemoglobin.

Pathogenesis of edema in chronic heart failure: increased venous pressure (hydrostatic factor), reduced renal perfusion with RAAS activation and sodium and water retention, increased ADH, reduced albumin synthesis due to hepatic congestion (oncotic factor), impaired lymph flow. Edema first appears in the dependent parts (legs, ankles) and increases in the evening (topic 9).

4. Heart defects

Heart defect — a permanent change in the structure of the heart valves, septa or great vessels that leads to disturbed blood flow. Defects are **congenital** (atrial and ventricular septal defects, patent ductus arteriosus, coarctation of the aorta, tetralogy of Fallot) and **acquired** (mainly rheumatic fever, infective endocarditis, atherosclerosis, degenerative changes).

Two types of valvular defect:

- **stenosis (narrowing)** — the orifice narrows and resistance to blood flow increases → the preceding chamber is overloaded **by pressure** (concentric hypertrophy);
- **insufficiency** — the cusps do not close completely and part of the blood flows back (regurgitation) → the receiving chambers are overloaded **by volume** (eccentric hypertrophy and dilation).

Table 21.4. *Hemodynamics and compensation in the main acquired defects*

Defect	Hemodynamic disturbance	Compensating chamber	Consequence of decompensation
Mitral stenosis	Passage of blood from the left atrium to the left ventricle is impeded	Left atrium (hypertrophy, dilation), then the right ventricle	Pulmonary congestion, pulmonary hypertension, right ventricular failure; atrial fibrillation
Mitral insufficiency	In systole blood returns from the left ventricle to the left atrium	Left ventricle and left atrium (by volume, eccentric hypertrophy)	Left ventricular failure, pulmonary congestion
Aortic stenosis	Ejection from the left ventricle into the aorta is impeded	Left ventricle (concentric hypertrophy) — compensated for a long time	Angina, fainting (on exertion), left ventricular failure
Aortic insufficiency	In diastole blood returns from the aorta to the left ventricle	Left ventricle (by volume, eccentric hypertrophy — the largest heart, “cor bovinum”)	Diastolic pressure falls — coronary perfusion decreases; left ventricular failure

Defect	Hemodynamic disturbance	Compensating chamber	Consequence of decompensation
Tricuspid insufficiency	In systole blood returns from the right ventricle to the right atrium	Right ventricle and right atrium	Congestion in the systemic circulation

MEMORY AID

Stenosis is like "**a queue in front of a narrow door**": the load falls on the chamber in front of the door (pressure). Insufficiency is like "**guests who keep coming back**": both chambers receive an excess volume (volume).

Peripheral signs in **aortic insufficiency**: large pulse pressure (high systolic, low diastolic), a "quick and high" pulse, pulsation of the carotid arteries, head bobbing (de Musset sign).

5. Coronary insufficiency and myocardial infarction

Coronary insufficiency — a mismatch between the oxygen demand of the myocardium and the oxygen delivered by coronary blood flow.

Causes:

- reduced coronary blood flow — atherosclerosis of the coronary arteries (the main cause), spasm, thrombosis, embolism, fall in diastolic pressure (aortic insufficiency, hypotension), shortening of diastole in tachycardia;
- increased myocardial oxygen demand — physical and emotional stress (catecholamines), hypertrophy, tachycardia, thyrotoxicosis;
- reduced oxygen-carrying capacity of the blood — anemia, hypoxemia, carbon monoxide poisoning.

Consequences of myocardial ischemia: within seconds aerobic oxidation stops and is replaced by anaerobic glycolysis, lactate accumulates, ATP falls — contractility decreases; ion pumps fail, intracellular Na^+ and Ca^{2+} increase and K^+ leaves — electrical instability and arrhythmias; after 20–40 minutes irreversible injury — **necrosis (infarction)**. Pain arises from the action of ischemic metabolites (adenosine, lactate, bradykinin, K^+).

Table 21.5. *Main complications of myocardial infarction*

Complication	Mechanism
Arrhythmias, ventricular fibrillation	Re-entry and triggered activity at the border of the ischemic and necrotic zones
Acute left ventricular failure, pulmonary edema	Loss of contracting myocardial mass
Cardiogenic shock	Death of more than 40 percent of the left ventricular myocardium
Cardiac rupture, tamponade	Softening of the necrotic wall (days 3–7)
Cardiac aneurysm	Replacement of the necrotic zone by connective tissue and its bulging
Thromboembolism	Mural thrombus (damage to the endocardium)
Resorption-necrotic syndrome	Fever, leukocytosis, raised ESR, raised cardiac-specific markers (troponins, CK-MB)

6. Arterial hypertension and hypotension

Arterial hypertension — a persistent arterial pressure of 140/90 mm Hg or higher. Arterial pressure = cardiac output × total peripheral resistance.

- **Primary (essential) hypertension** — 90–95 percent of cases; hereditary predisposition, high salt intake, stress, obesity, insulin resistance, reduced renal sodium excretion, increased sympathetic and RAAS activity, endothelial dysfunction (decreased NO, increased endothelin).
- **Secondary (symptomatic) hypertension** — renal (glomerulonephritis, renal artery stenosis — renovascular), endocrine (primary hyperaldosteronism, Cushing syndrome, pheochromocytoma, thyrotoxicosis), hemodynamic (coarctation of the aorta), neurogenic, drug-induced.

Target organ damage: heart (left ventricular hypertrophy, ischemic heart disease, heart failure), brain (stroke, encephalopathy), kidney (nephrosclerosis), retina (retinopathy), vessels (atherosclerosis, aortic aneurysm).

Arterial hypotension — pressure below 100/60 (95/60 in women) mm Hg: physiological (athletes), acute (shock, collapse, blood loss), chronic (adrenal insufficiency, hypothyroidism, neurocirculatory dystonia), **orthostatic** (a sharp fall in pressure on standing — autonomic nervous system disorders, hypovolemia, drugs).

Collapse — acute vascular insufficiency resulting from an acute fall in vascular tone and a decrease in circulating blood volume: arterial and venous pressure fall, with hypoxia of the brain, heart and other organs.

7. Pericardial pathology

Pericardial tamponade — rapid accumulation of fluid (blood, exudate) in the pericardial cavity, compressing the heart. The ventricles cannot expand in diastole — filling and stroke volume decrease sharply and venous pressure rises. **Beck's triad:** arterial hypotension, distended neck veins, muffled heart sounds. **Pulsus paradoxus** — a fall in systolic pressure of more than 10 mm Hg on inspiration. Care — emergency pericardiocentesis.

8. Practical skills

Skill: drawing a diagram of the hemodynamic changes in a heart defect

1. Determine the type of defect: stenosis or insufficiency.
2. Mark where the blood flow is disturbed.
3. Determine which chamber is overloaded by pressure and which by volume.
4. Write down the compensating chamber and the type of hypertrophy (concentric or eccentric).
5. Show in which circulation blood stagnates in decompensation and what clinical signs appear.

Skill: grouping the signs of heart failure by pathogenesis

1. List the patient's signs (dyspnea, edema, enlarged liver, cyanosis, etc.).
2. Divide them into “backward” (congestion) and “forward” (hypoperfusion) signs.
3. Group the signs of congestion by the pulmonary and systemic circulations.
4. Write the mechanism of each sign in one sentence.
5. Conclude on the type of failure (left, right, total; systolic or diastolic).

Review questions

1. What is meant by circulatory failure and heart failure?
2. List the pathogenetic causes of heart failure.
3. How do systolic and diastolic heart failure differ?
4. What is the essence of the Frank–Starling and Anrep mechanisms?
5. When is tachycardia compensatory, and when is it harmful?
6. Explain the stages of myocardial hypertrophy.
7. Why is the hypertrophied myocardium considered “imperfect”?
8. In which cases do concentric and eccentric hypertrophy develop?
9. List the extracardiac compensatory mechanisms. What is the “vicious circle” of neurohumoral activation?
10. Explain the signs of left and right ventricular failure and their mechanisms.
11. What is the pathogenesis of edema in heart failure?
12. How do hemodynamics change in mitral stenosis and aortic insufficiency?
13. What are the causes of coronary insufficiency and the consequences of myocardial ischemia?
14. Explain the types and pathogenesis of arterial hypertension.
15. How are hemodynamics disturbed in pericardial tamponade?

Test questions

1. What type of overload does the left ventricle experience in aortic stenosis?
   A) Volume overload
   B) Pressure overload
   C) Impaired diastolic filling
   D) Direct myocardial damage
   E) No overload

2. What is the essence of the Frank–Starling mechanism?
   A) A gradual increase in the force of contraction when aortic resistance rises
   B) An increase in rate under sympathetic influence
   C) An increase in the force of contraction when the fibers are stretched more in diastole
   D) An increase in myocardial mass
   E) Release of natriuretic peptide

3. Which structure does NOT lag behind growth in the hypertrophied myocardium?
   A) Cardiomyocyte mass
   B) Number of capillaries
   C) Nerve endings
   D) Mitochondria
   E) Sarcoplasmic reticulum

4. What hypertrophy develops in aortic insufficiency?
   A) Concentric hypertrophy of the right ventricle only
   B) Left atrial hypertrophy
   C) No hypertrophy develops
   D) Right atrial hypertrophy
   E) Eccentric hypertrophy and dilation of the left ventricle

5. Which is a sign of left ventricular failure?
   A) Distended neck veins
   B) Enlarged liver
   C) Ascites
   D) Nocturnal attacks of suffocation and moist rales in the lungs
   E) Leg edema

6. What is the mechanism of orthopnea?
   A) Descent of the diaphragm
   B) Increased venous return when lying down and increased pulmonary congestion
   C) Bronchospasm
   D) Fluid in the pleural cavity
   E) Hypoglycemia

7. What is the adverse effect of prolonged RAAS activation in heart failure?
   A) Increased natriuresis
   B) Decreased blood volume
   C) Vasodilation
   D) Na^+ and water retention, vasoconstriction, myocardial fibrosis
   E) Normalization of cardiac output

8. Where are natriuretic peptides released from, and what are their effects?
   A) From the adrenal glands; they retain sodium
   B) From the pituitary; they retain water
   C) From the liver; they cause vasoconstriction
   D) From the kidney; they release renin
   E) From the stretched atria and ventricles; they cause natriuresis and vasodilation

9. Which chamber is overloaded first in mitral stenosis?
   A) Left atrium
   B) The left ventricle
   C) The aorta
   D) Right atrium
   E) Venae cavae

10. 3–7 days after myocardial infarction, the risk of which complication is especially high?
   A) Atherosclerosis
   B) Arterial hypertension
   C) Softening and rupture of the necrotic wall and tamponade
   D) Hypertrophy
   E) Anemia

11. What is the main mechanism of secondary (renovascular) hypertension?
   A) Decreased sympathetic tone
   B) Increased natriuretic peptide
   C) Decreased blood volume
   D) Hypothyroidism
   E) Activation of the renin–angiotensin system due to renal ischemia

12. What is included in Beck's triad of pericardial tamponade?
    A) Fever, leukocytosis, pain
    B) Hypotension, distended neck veins, muffled heart sounds
    C) Hypertension, bradycardia, respiratory disturbance
    D) Cyanosis, edema, ascites
    E) Jaundice, enlarged liver, ascites

13. What happens first in myocardial ischemia?
    A) Aerobic oxidation stops, with anaerobic glycolysis and a fall in ATP
    B) Immediate necrosis
    C) Hypertrophy
    D) Fibrosis
    E) Calcium leaves the cell

14. What is collapse?
    A) A sudden increase in cardiac contractility
    B) A persistent rise in arterial pressure
    C) Acute vascular insufficiency resulting from an acute fall in vascular tone and a decrease in circulating blood volume
    D) Pericardial tamponade
    E) Myocardial infarction

15. What is the mechanism of liver enlargement in right ventricular failure?
    A) Hypertrophy of liver cells
    B) Dilation of the hepatic artery
    C) Hepatitis
    D) Increased venous pressure in the systemic circulation and congestion of blood in the liver
    E) Fat accumulation in the liver

Clinical cases

Case 1

A 58-year-old man has had arterial hypertension for 15 years without regular treatment. Over the last year he has had dyspnea when walking, and at night he wakes up suffocating and has to sit up. On examination: moist rales in the lower parts of the lungs, BP 170/105 mm Hg. Echocardiography: left ventricular wall thickened, cavity dilated, ejection fraction 35 %.

Questions: What pathological condition has developed? By what mechanism was compensation achieved and why did decompensation develop? Explain the pathogenesis of the signs.

Answer: Chronic left ventricular (systolic) failure resulting from many years of arterial hypertension. The left ventricle was overloaded by pressure and was compensated for a long time by concentric hypertrophy. In the hypertrophied myocardium the capillaries, mitochondria and innervation lagged behind, and relative coronary insufficiency and fibrosis developed; in addition, prolonged activation of RAAS and the sympathetic system enhanced remodeling — the stage of exhaustion: the ventricle has dilated and its contractility has fallen. The weak left ventricle cannot pump the blood out — congestion in the pulmonary circulation: dyspnea, moist rales in the lungs; when lying down venous return increases and congestion worsens — orthopnea and nocturnal attacks of suffocation (cardiac asthma).

Case 2

A 34-year-old woman had rheumatic fever in childhood. She has dyspnea on exertion, cough and sometimes blood-streaked sputum. On her face there is a “mitral flush” (bluish-red cheeks), and her heart beats irregularly. Leg edema, liver enlarged by 4 cm.

Questions: Which defect can be suspected? Explain the sequence of hemodynamic changes.

Answer: Rheumatic mitral stenosis. The passage of blood from the left atrium to the left ventricle is impeded — the left atrium, overloaded by pressure, has hypertrophied and dilated (the dilated atrium is a substrate for atrial fibrillation, which is why the heart beats irregularly). Because the left atrium is weak, it cannot compensate for long — the pressure is transmitted to the pulmonary veins: pulmonary congestion (dyspnea, cough), rupture of capillaries (hemoptysis). Reflex spasm of the pulmonary arterioles (Kitaev reflex) increases pulmonary hypertension — the right ventricle, overloaded by pressure, hypertrophies and then decompensates: congestion in the systemic circulation (enlarged liver, leg edema). The left ventricle, for its part, does not receive enough blood — cardiac output is reduced.

Case 3

A 45-year-old man was brought in after a car accident with a chest injury. His condition is deteriorating: BP 75/50 mm Hg, pulse 130 beats/min and weak, neck veins distended, heart sounds very muffled, the pulse almost disappears on inspiration.

Questions: What condition has developed? Explain the mechanism of the hemodynamic disturbance. What care is needed?

Answer: Acute pericardial tamponade due to hemopericardium. Blood accumulates rapidly in the pericardial cavity and compresses the heart — the ventricles cannot expand in diastole (impaired diastolic filling), stroke volume and cardiac output fall sharply — hypotension, compensatory tachycardia. Venous blood cannot return to the heart — the neck veins are distended and central venous pressure rises; the fluid damps the sound — muffled heart sounds (Beck's triad). On inspiration the right ventricle fills more and pushes the interventricular septum to the left, further reducing left ventricular filling — pulsus paradoxus. Emergency care — pericardiocentesis (pericardial puncture) and surgical control of the source of bleeding.

Quick review

Table 21.6. *Key facts on Topic 21*

Question	Answer
Heart failure	Cardiac output unable to meet the needs of the tissues
Volume overload	Valvular insufficiency, hypervolemia — eccentric hypertrophy
Pressure overload	Stenosis, hypertension — concentric hypertrophy
Frank–Starling	Greater stretch — stronger contraction (heterometric)
Anrep phenomenon	Increased force of contraction when resistance rises (homeometric)
Stages of hypertrophy	Emergency, stable hyperfunction, exhaustion and cardiosclerosis
Imperfect hypertrophy	Capillaries, mitochondria, innervation lag behind
Extracardiac compensation	Sympathetic system, RAAS, ADH, erythropoiesis
Vicious circle	Neurohumoral activation increases the load and damages the myocardium

Question	Answer
Left ventricular failure	Pulmonary congestion: dyspnea, orthopnea, cardiac asthma
Right ventricular failure	Venous congestion: edema, enlarged liver, ascites
Mitral stenosis	Left atrium → pulmonary hypertension → right ventricle
Aortic insufficiency	“Cor bovinum”, large pulse pressure
Coronary insufficiency	Mismatch between oxygen demand and delivery
Beck's triad	Hypotension, distended neck veins, muffled heart sounds

TOPIC 22

Pathophysiology of the respiratory system. Hypoxia and its types

Learning objectives

To explain to the student the functions of the external respiration system, the definition, types and mechanisms of respiratory failure (disturbances of ventilation, diffusion, perfusion and ventilation–perfusion relationships), disturbances of respiratory regulation and pathological types of breathing, and the etiology, types, mechanisms of development and adaptive and compensatory reactions of hypoxia; to teach the student to assess external respiration parameters and to determine the type of hypoxia.

By the end of the session, the student should know:

- the main parameters of external respiration and their normal values;
- the definition, classification and stages of respiratory failure;
- the causes of and difference between obstructive and restrictive ventilatory disorders;
- the mechanisms of diffusion and perfusion disturbances and ventilation–perfusion mismatch;
- the pathological types of breathing and their causes;
- the classification of hypoxia, the changes in blood gases in each type, the adaptive mechanisms and the consequences.

The student should be able to:

- determine the type of ventilatory disorder from spirogram parameters;
- determine the type of hypoxia from blood gases;
- distinguish the pathological types of breathing.

1. External respiration and its parameters

External respiration — gas exchange between atmospheric air and the blood of the pulmonary capillaries. It consists of three processes: **ventilation** (movement of air into and out of the alveoli), **diffusion** (passage of gases across the alveolar–capillary membrane) and **perfusion** (blood flow through the pulmonary capillaries).

Table 22.1. *Main parameters of external respiration*

Parameter	Normal value
Respiratory rate	12–18 per minute
Tidal volume	400–600 ml
Minute ventilation	6–8 l
Vital capacity (VC)	3.5–5 l in men, 3–3.5 l in women (more than 80 percent of the predicted value)
Forced expiratory volume in the first second (FEV_1)	More than 80 percent of the predicted value
Tiffeneau index (FEV_1 / VC)	70–75 percent or more
Arterial pO_2	80–100 mm Hg
Arterial pCO_2	35–45 mm Hg
Arterial O_2 saturation (SaO_2)	95–98 percent

Parameter	Normal value
Venous pO_2	37–40 mm Hg

2. Respiratory failure

Respiratory failure — a state in which the external respiration system cannot maintain the normal blood gas composition, or maintains it only through the strain of compensatory mechanisms. The criterion is arterial pO_2 below 60 mm Hg and (or) pCO_2 above 45–50 mm Hg.

Table 22.2. *Classification of respiratory failure*

Criterion	Types
Course	Acute (minutes, hours), chronic (months, years)
Blood gases	Hypoxemic (type I — low pO_2, normal or low pCO_2), hypercapnic (type II — low pO_2, high pCO_2)
Compensation	Latent (only on exertion), compensated (dyspnea at rest, blood gases normal), decompensated (hypoxemia and hypercapnia even at rest)
Mechanism	Ventilatory, diffusional, perfusional, ventilation–perfusion mismatch

Dyspnea — a change in the rate, depth and rhythm of breathing accompanied by difficulty breathing and a sensation of air hunger. **Inspiratory** dyspnea (difficulty breathing in — narrowing of the upper airways, pulmonary edema), **expiratory** dyspnea (difficulty breathing out — bronchial asthma, emphysema), **mixed**.

3. Ventilatory disorders

Obstructive type

Impaired airway patency — resistance to air flow increases.

Causes: bronchospasm (bronchial asthma), swelling and inflammation of the mucosa, accumulation of sputum in the bronchi (chronic bronchitis), external compression of the bronchial wall, early closure of small bronchi on expiration in emphysema (reduced elastic recoil), obstruction of the upper airways by a foreign body, edema, tumor or laryngospasm.

Spirogram: FEV_1 and the **Tiffeneau index decreased** (below 70 percent), VC normal or slightly decreased, residual volume increased.

Restrictive type

The ability of the lungs to expand is limited.

Causes:

- extrapulmonary — fluid (hydrothorax) or air (pneumothorax) in the pleural cavity, pleural adhesions, chest deformities (kyphoscoliosis), paralysis of the respiratory muscles (poliomyelitis, myasthenia), obesity, increased intra-abdominal pressure;
- intrapulmonary — pneumonia, pulmonary fibrosis, pulmonary edema, atelectasis, surfactant deficiency, lung resection.

Spirogram: VC decreased, FEV_1 also decreased, but the Tiffeneau index normal or increased.

Table 22.3. *Distinguishing obstructive and restrictive disorders*

Parameter	Obstructive	Restrictive
VC	Normal or slightly decreased	Markedly decreased
FEV_1	Markedly decreased	Decreased (in proportion to VC)
Tiffeneau index	below 70 percent	Normal or increased
Residual volume	Increased (“air trapping”)	Decreased
Type of dyspnea	Expiratory	Inspiratory or mixed

Disturbances of respiratory regulation

When respiratory center function is impaired (brain trauma, stroke, tumors, encephalitis, poisoning with morphine and barbiturates, deep hypoxia, uremia, ketoacidosis), alveolar hypoventilation and pathological types of breathing are observed. When the excitability of the center increases (hysteria, pain, fever, metabolic acidosis) — hyperventilation and hypocapnia.

4. Pathological types of breathing

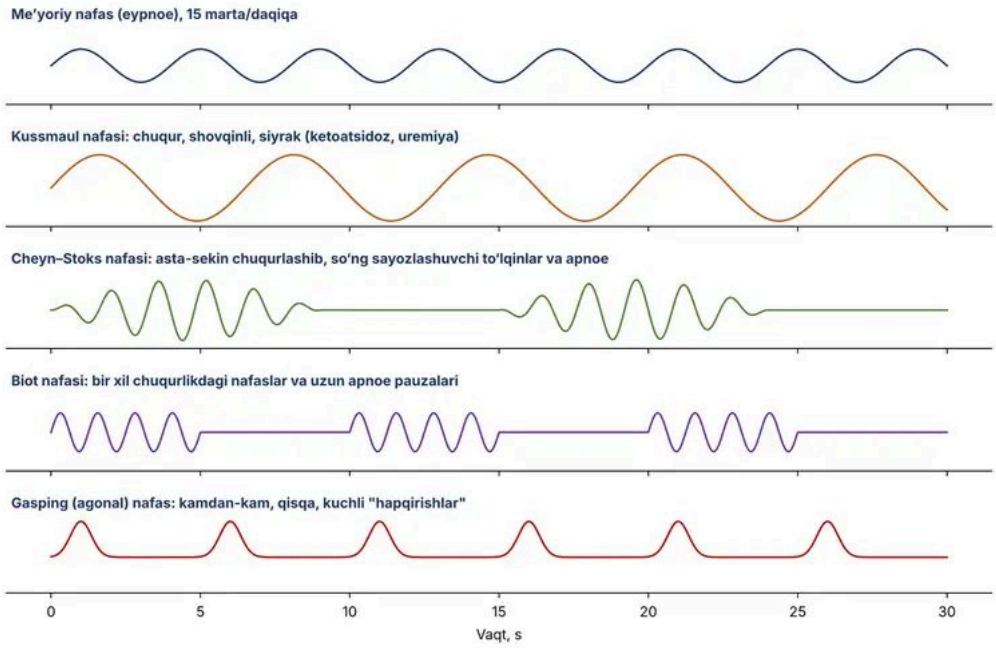


Figure 22.1. *Pathological types of breathing (schematic pneumograms)*

Table 22.4. *Pathological types of breathing*

Type	Description	Causes and mechanism
Tachypnea	Rapid, shallow breathing	Fever, pain, pneumonia, pulmonary edema (stimulation of stretch receptors)
Bradypnea	Slow breathing	Inhibition of the respiratory center (raised intracranial pressure, narcotic substances), stenosis of the upper airways
Kussmaul breathing	Deep, noisy, slow breathing	Metabolic acidosis (diabetic ketoacidosis, uremia): H^+ ions stimulate chemoreceptors, CO_2 elimination increases

Type	Description	Causes and mechanism
Cheyne–Stokes breathing	The depth of breathing gradually increases, then decreases, followed by apnea; this repeats	Reduced sensitivity of the respiratory center to CO_2 (impaired cerebral circulation, heart failure, intoxications, elderly people during sleep): during apnea CO_2 accumulates and stimulates the center; during hyperventilation CO_2 falls and the center is inhibited again
Biot breathing	Breaths of equal depth separated by long apneic pauses	Organic damage to the respiratory center (meningitis, encephalitis, brain trauma) — a more serious prognosis
Gasping (agonal) breathing	Rare, short, forceful inspirations	Terminal state: the last activity of the bulbar centers
Apneustic breathing	Prolonged inspiration and short expiration	Damage to the pons

MEMORY AID

Kussmaul — **K**etoacidosis ("blowing off" acid with deep breaths). **C**heyne–Stokes — **C**rescendo–decrescendo cycles (deepening and becoming shallow, then a pause). **B**iot — **B**ursts (equal breaths separated by long pauses).

5. Disturbances of diffusion and perfusion

Disturbance of diffusion

Passage of gases across the alveolar–capillary membrane is impeded. According to Fick's law, diffusion is directly proportional to the membrane area and the gas pressure difference and inversely proportional to the membrane thickness.

Causes: thickening of the membrane (interstitial pulmonary edema, pulmonary fibrosis, pneumoconioses, alveolitis), reduction of the diffusion area (emphysema, lung resection, atelectasis), too rapid passage of blood through the capillary (physical exertion).

Feature: CO_2 crosses the membrane about 20 times more easily than O_2 — which is why in diffusion disturbances **hypoxemia** develops first, while pCO_2 remains normal for a long time or falls because of hyperventilation.

Disturbance of perfusion

Reduced blood flow through the pulmonary capillaries: pulmonary thromboembolism, pulmonary hypertension, heart failure, shock, right-to-left shunts in congenital defects.

Disturbance of the ventilation–perfusion relationship

Normally the ratio of alveolar ventilation to blood flow (V/Q) is about **0.8–1.0**.

- **Decreased V/Q** (alveoli that are not ventilated but are perfused — atelectasis, pneumonia, bronchial obstruction) — the **shunt** effect: venous blood not saturated with oxygen is added to arterial blood; hypoxemia is poorly corrected even by breathing oxygen.
- **Increased V/Q** (ventilated but not perfused — pulmonary embolism) — **dead space** increases: ventilation is wasted.

CAUTION

Hypoxic vasoconstriction (**Euler–Liljestrand reflex**) — the arterioles constrict in a poorly ventilated part of the lung, and blood is directed to well-ventilated parts. This is a protective reaction that normalizes V/Q, but in chronic hypoxia of the whole lung (chronic obstructive pulmonary disease, high altitude) it leads to **pulmonary hypertension** and right ventricular hypertrophy — **cor pulmonale**.

6. Hypoxia: definition and classification

Hypoxia — a typical pathological process resulting from inadequate oxygen supply to tissues or impaired use of oxygen by them. **Hypoxemia** — a decrease in the oxygen content of the blood.

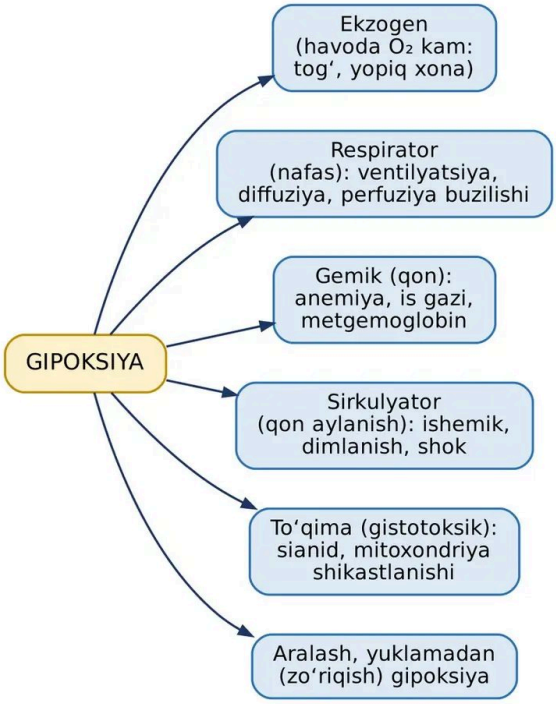


Figure 22.2. *Types of hypoxia (by etiology)*

Exogenous hypoxia

A decrease in the partial pressure of oxygen in the inspired air.

- **Hypobaric** — on ascent to altitude (mountain sickness, in pilots): total atmospheric pressure and pO_2 decrease.
- **Normobaric** — little oxygen in the air at normal pressure (mine, well, closed room, malfunction of a breathing apparatus).

Blood gases: arterial pO_2 and SaO_2 decreased; because of hyperventilation, **hypocapnia** and respiratory alkalosis (inhibition of the respiratory center, spasm of the cerebral vessels).

Respiratory hypoxia

Resulting from failure of external respiration: disturbances of ventilation, diffusion and perfusion and V/Q mismatch (sections 3 and 5). **Blood gases:** arterial pO_2 and SaO_2 decreased, often **hypercapnia** and respiratory acidosis (if ventilation is impaired).

Hemic (blood) hypoxia

The oxygen capacity of the blood decreases:

- **anemias** — decreased hemoglobin;
- **carbon monoxide (CO) poisoning** — carboxyhemoglobin is formed; CO binds to hemoglobin about 200–300 times more strongly than oxygen and also shifts the oxyhemoglobin dissociation curve to the left, impeding oxygen release; the skin is “cherry-red”;
- **methemoglobin-forming agents** (nitrites, nitrates, aniline, sulfonamides) — the iron passes into the Fe^{3+} state and does not carry oxygen; the blood is brown, the skin gray-bluish.

Blood gases: arterial pO_2 is **normal**, but the oxygen capacity and oxygen content of the blood are decreased; the arteriovenous difference is decreased.

Circulatory hypoxia

Reduced blood flow to the tissues:

- **ischemic** — reduced arterial blood flow (vascular spasm, thrombosis, blood loss, shock, heart failure);
- **congestive** — impaired venous outflow (heart failure, venous thrombosis).

Blood gases: arterial pO_2 and blood oxygen content normal (if the lungs are not damaged); because blood flows slowly in the capillaries, the tissue extracts more oxygen — the **arteriovenous difference increases**, venous pO_2 decreases; cyanosis.

Tissue (histotoxic) hypoxia

The tissues cannot use oxygen:

- inhibition of respiratory chain enzymes — **cyanides** (bind to the Fe^{3+} of cytochrome oxidase), sulfides, arsenic;
- uncoupling of oxidation and phosphorylation — excess thyroid hormones, 2,4-dinitrophenol, some toxins;
- impaired enzyme synthesis — deficiency of B vitamins (B_1, B_2, PP);
- damage to mitochondrial membranes — ionizing radiation, lipid peroxidation, alcohol.

Blood gases: arterial pO_2 normal, **venous pO_2 increased** — oxygen remains unused, **arteriovenous difference decreased**; the skin is pink.

Load (exertional) and mixed hypoxia

Exertional hypoxia — during heavy physical work the oxygen demand exceeds the delivery capacity (oxygen debt). **Mixed** — several types together (e.g. circulatory, respiratory and tissue hypoxia in shock; hemic and circulatory in blood loss).

Table 22.5. *Blood gases in the types of hypoxia*

Type	Arterial pO_2	O_2 capacity of the blood	Venous pO_2	Arteriovenous difference	pCO_2
Exogenous	↓	Normal	↓	↓ or normal	↓ (hyperventilation)
Respiratory	↓	Normal	↓	↓ or normal	↑ or normal
Hemic	Normal	↓	↓	↓	Normal
Circulatory	Normal	Normal	↓↓	↑	Normal or ↑
Tissue	Normal	Normal	↑	↓	Normal or ↓

7. Adaptive reactions and consequences of hypoxia

Emergency adaptive mechanisms

- **external respiration** — hyperventilation (stimulation of the chemoreceptors of the aortic and carotid bodies), opening of reserve alveoli;
- **circulation** — tachycardia, increased stroke volume, release of blood from depots, redistribution of blood in favor of vital organs (brain, heart) (**centralization of the circulation**);
- **blood** — release of erythrocytes from the spleen and other depots, enhanced oxyhemoglobin dissociation (acidosis, increased 2,3-diphosphoglycerate — the curve shifts to the right);
- **tissue** — increased oxygen extraction, increased anaerobic glycolysis.

Long-term adaptive mechanisms

- increased erythropoietin → **erythrocytosis** (in highlanders);
- hypertrophy of the lungs and respiratory muscles, increased number of alveoli;
- myocardial hypertrophy, increased capillary density (angiogenesis, the HIF-1 factor);
- increased number of mitochondria and activity of respiratory enzymes, increased tissue resistance to oxygen deficiency.

Consequences of hypoxia

Tissues differ in their sensitivity to hypoxia: the most sensitive is the **cerebral cortex** (irreversible changes after 4–6 minutes of complete ischemia), followed by the myocardium, kidney and liver; the most resistant are bone, cartilage and connective tissue.

At the cellular level: decreased ATP → impaired ion pumps → entry of Na^+ and water into the cell (cell swelling), Ca^{2+} accumulation → activation of phospholipases and proteases, lipid peroxidation → membrane damage; anaerobic glycolysis → lactic acidosis.

> **CLINICAL SIGNIFICANCE**
> **Cyanosis** (bluish discoloration) appears when reduced (deoxygenated) hemoglobin in capillary blood exceeds 50 g/l. Therefore in severe anemia (low total hemoglobin) there may be no cyanosis even with deep hypoxia, while in polycythemia cyanosis may appear even with mild hypoxemia. In carbon monoxide poisoning the skin is pink, and in methemoglobinemia gray-bluish — the degree of cyanosis does not accurately reflect the severity of hypoxia.

8. Experimental part

Studying exogenous hypobaric hypoxia

Aim: to compare the resistance to lowered atmospheric pressure of animals of different ages and of the body in different functional states.

1. Place mice (or a frog) in a Komovsky chamber (a glass bell jar with a pump).
2. Gradually pump out the air to lower the pressure; record the “altitude” from the manometer.
3. Observe the respiratory rate and pattern, motor activity, skin color, and the time of onset of convulsions and respiratory arrest.
4. Record the appearance of pathological types of breathing (Cheyne–Stokes, gasping).
5. Restore the pressure immediately and observe the recovery of the animal.
6. Record the results in a table and draw a conclusion.

Studying hemic hypoxia

Aim: to observe the formation of methemoglobin under the action of sodium nitrite.

1. Inject sodium nitrite solution intraperitoneally into a mouse.
2. Observe the respiratory rate, the color of the skin and mucous membranes, and motor activity.
3. Compare a drop of blood with a drop of control blood — in methemoglobinemia the blood turns brown.

> **REMEMBER**
> The doses and animal species used in the experiments follow the department's methodological guidelines; bioethics rules are observed when working with animals.

9. Practical skills

Skill: determining the type of ventilatory disorder from the spirogram

1. Compare VC with the predicted value (below 80 percent — decreased).
2. Compare FEV_1 with the predicted value.
3. Calculate the Tiffeneau index: FEV_1 / VC × 100 percent.
4. Tiffeneau below 70 percent — obstructive type.
5. VC markedly decreased, Tiffeneau normal or increased — restrictive type.
6. Both values impaired — mixed type.

Skill: determining the type of hypoxia from blood gases

1. Assess arterial pO_2: if decreased — exogenous or respiratory hypoxia.
2. Assess blood pCO_2: if decreased — exogenous (hyperventilation); if increased — respiratory (hypoventilation).
3. If arterial pO_2 is normal, assess the oxygen capacity of the blood: decreased — hemic.
4. If the oxygen capacity is normal, assess venous pO_2 and the arteriovenous difference: difference increased, venous pO_2 low — circulatory; difference decreased, venous pO_2 high — tissue hypoxia.
5. Compare the conclusion with the clinical data.

Review questions

1. What processes make up external respiration? List its main parameters.
2. What is respiratory failure? What are its classification and criteria?
3. What are the causes and spirographic signs of obstructive ventilatory disorder?
4. List the extrapulmonary and intrapulmonary causes of restrictive ventilatory disorder.
5. Why does hypoxemia develop first and hypercapnia later in diffusion disturbances?
6. What is ventilation–perfusion mismatch? How do the shunt and dead space effects differ?
7. What is the significance of the Euler–Liljestrand reflex and what is its adverse consequence in chronic hypoxia?
8. What are the mechanisms and causes of Kussmaul, Cheyne–Stokes and Biot breathing?
9. What is hypoxia? Into what types is it divided by etiology?
10. How do blood gases change in exogenous hypoxia?

11. What are the causes of hemic hypoxia and the mechanism of carbon monoxide poisoning?
12. How can circulatory and tissue hypoxia be distinguished by blood gases?
13. List the emergency and long-term adaptive mechanisms in hypoxia.
14. What are the mechanisms of cell injury in hypoxia?
15. When does cyanosis appear, and why does it not always reflect the severity of hypoxia?

Test questions

1. Which is a spirographic sign of obstructive ventilatory disorder?
   A) VC markedly decreased, Tiffeneau index increased
   B) All values normal
   C) Residual volume decreased
   D) Tiffeneau index below 70 percent
   E) Only the respiratory rate increased

2. Which condition leads to a restrictive ventilatory disorder?
   A) Bronchial asthma
   B) Chronic bronchitis
   C) A foreign body in a bronchus
   D) Laryngospasm
   E) Pneumothorax

3. Why does hypoxemia develop first in diffusion disturbances?
   A) O_2 crosses the membrane more easily than CO_2
   B) The respiratory center is inhibited
   C) CO_2 crosses the membrane about 20 times more easily than O_2
   D) Hemoglobin decreases
   E) Perfusion increases

4. How does the ventilation–perfusion ratio change in pulmonary thromboembolism?
   A) It decreases (shunt effect)
   B) It increases (dead space increases)
   C) It does not change
   D) It becomes zero
   E) Only ventilation decreases

5. What breathing is seen in diabetic ketoacidosis?
   A) Kussmaul breathing
   B) Biot breathing
   C) Cheyne–Stokes breathing
   D) Apneustic breathing
   E) Bradypnea

6. What is the mechanism of Cheyne–Stokes breathing?
   A) Reduced sensitivity of the respiratory center to CO_2
   B) Metabolic acidosis
   C) Bronchospasm
   D) Fluid in the pleura
   E) Damage to the pons

7. What is observed in the blood in exogenous hypobaric hypoxia?
   A) pO_2 normal, pCO_2 increased
   B) Oxygen capacity of the blood decreased
   C) pO_2 decreased, hypocapnia
   D) Venous pO_2 increased
   E) Methemoglobin increased

8. What type of hypoxia occurs in carbon monoxide poisoning?
   A) Exogenous
   B) Hemic
   C) Circulatory
   D) Respiratory
   E) Tissue

9. What change in the blood is observed in cyanide poisoning?
   A) Arterial pO_2 markedly decreased
   B) Oxygen capacity of the blood decreased
   C) Arteriovenous difference increased
   D) Venous pO_2 increased, arteriovenous difference decreased
   E) Carboxyhemoglobin increased

10. Which is a characteristic blood gas sign of circulatory hypoxia?
    A) Arterial pO_2 decreased
    B) Oxygen capacity of the blood decreased
    C) Venous pO_2 increased
    D) Arteriovenous difference decreased
    E) Arteriovenous difference increased, venous pO_2 decreased

11. Name a methemoglobin-forming substance.
    A) Cyanide
    B) Carbon monoxide
    C) Sodium nitrite
    D) 2,4-dinitrophenol
    E) Morphine

12. Which is a long-term mechanism of adaptation to hypoxia?
    A) Tachycardia
    B) Release of blood from depots
    C) Hyperventilation
    D) Increased erythropoietin and erythrocytosis
    E) Centralization of the circulation

13. Which tissue is most sensitive to hypoxia?
    A) Bone
    B) Cartilage
    C) Connective tissue
    D) Skin
    E) Cerebral cortex

14. What is the main mechanism of cor pulmonale development in chronic alveolar hypoxia?
    A) Pulmonary hypertension due to hypoxic vasoconstriction (Euler–Liljestrand reflex)
    B) Left ventricular hypertrophy
    C) Arterial hypotension
    D) Bradycardia
    E) Anemia

15. In which condition may there be no cyanosis even with deep hypoxia?
    A) Polycythemia
    B) Severe anemia
    C) Heart failure
    D) Pneumonia
    E) Mountain sickness

Clinical cases

Case 1

In winter, members of a family who had fallen asleep in a closed room with a coal stove burning were found in the morning with headache, nausea and severe weakness. One of them was unconscious, with pinkish-red skin and mucous membranes and shallow, rapid breathing. Arterial pO_2 92 mm Hg, carboxyhemoglobin 45 percent.

Questions: What type of hypoxia has developed? Explain its mechanism. Why is there no cyanosis and why is pO_2 normal?

Answer: Carbon monoxide (CO) poisoning — hemic hypoxia. CO binds to hemoglobin about 200–300 times more strongly than oxygen, forming carboxyhemoglobin — the oxygen capacity of the blood falls sharply; the dissociation curve of the remaining oxyhemoglobin shifts to the left, impeding oxygen delivery to the tissues; CO also partly inhibits cytochrome oxidase (a tissue component). Arterial pO_2 is determined by the oxygen dissolved in plasma and remains normal because lung function is not impaired — which is why pulse oximetry and pO_2 can be misleading. Because carboxyhemoglobin is bright red, the skin is pinkish-red and there is no cyanosis. Treatment — removal to fresh air, 100 percent oxygen (hyperbaric oxygenation if needed), which accelerates the dissociation of carboxyhemoglobin.

Case 2

A 64-year-old man has smoked for 40 years and has had a productive cough and expiratory dyspnea for many years. His lips are bluish, the neck veins distended, and there is leg edema. Spirogram: VC 78 percent of predicted, FEV_1 42 percent, Tiffeneau index 48 percent. Arterial pO_2 55, pCO_2 54 mm Hg. Hemoglobin 178 g/l.

Questions: What is the type of ventilatory disorder? Determine the type of respiratory failure. How did the right ventricular failure and erythrocytosis develop?

Answer: Chronic obstructive pulmonary disease: Tiffeneau index below 70 percent, FEV_1 markedly decreased — obstructive type. Hypoxemia and hypercapnia — type II (hypercapnic) chronic decompensated respiratory failure (respiratory hypoxia, respiratory acidosis). Chronic alveolar hypoxia causes persistent spasm of the pulmonary arterioles (Euler–Liljestrand reflex) and then their remodeling — pulmonary hypertension; the right ventricle, overloaded by pressure, hypertrophies and decompensates — cor pulmonale: distended neck veins, edema. Hypoxia increases erythropoietin

production in the kidney — secondary erythrocytosis (an adaptive reaction, but it increases blood viscosity); cyanosis is due to the large amount of reduced hemoglobin.

Case 3

A 28-year-old mountaineer developed headache, insomnia, dyspnea and palpitations after a rapid ascent to 4500 m. At night his breathing is periodic: several deep breaths followed by a pause. Pulse 112 beats/min, respiratory rate 26 per minute, SaO_2 80 percent.

Questions: What type of hypoxia is this? How do blood gases change? Explain the adaptive reactions and the cause of the periodic breathing at night.

Answer: Exogenous hypobaric hypoxia (acute mountain sickness). Atmospheric pressure and the pO_2 of the air are decreased — arterial pO_2 and SaO_2 are decreased. The chemoreceptors are stimulated and hyperventilation develops — a lot of CO_2 is exhaled: hypocapnia and respiratory alkalosis; hypocapnia leads to constriction of the cerebral vessels and inhibition of the respiratory center. Emergency adaptation: hyperventilation, tachycardia, release of blood from depots, centralization of the circulation. Periodic (Cheyne–Stokes-type) breathing at night: during hyperventilation CO_2 falls and the stimulus to the respiratory center is lost — apnea; during apnea CO_2 accumulates and O_2 falls, and the center is stimulated again. Long-term adaptation (over several weeks): erythropoietin and erythrocytosis, increased 2,3-diphosphoglycerate, increased capillary density and number of mitochondria. Gradual ascent (acclimatization) prevents mountain sickness.

Quick review

Table 22.6. *Key facts on Topic 22*

Question	Answer
Respiratory failure	pO_2 below 60 and (or) pCO_2 above 45–50
Obstructive type	Airway narrowing; Tiffeneau below 70 percent
Restrictive type	Lung expansion limited; VC decreased, Tiffeneau normal
Disturbance of diffusion	Membrane thickening; hypoxemia first
Decreased V/Q	Shunt effect (atelectasis, pneumonia)
Increased V/Q	Dead space (pulmonary embolism)
Euler–Liljestrand reflex	Hypoxic vasoconstriction; chronic — cor pulmonale
Kussmaul breathing	Deep and noisy; metabolic acidosis
Cheyne–Stokes breathing	Wave-like deepening and apnea; reduced sensitivity to CO_2
Biot breathing	Equal breaths and long pauses; organic brain damage
Exogenous hypoxia	Little O_2 in the air; low pO_2, hypocapnia
Hemic hypoxia	Oxygen capacity decreased; anemia, CO, methemoglobin
Circulatory hypoxia	Blood flow decreased; arteriovenous difference increased
Tissue hypoxia	O_2 not used; venous pO_2 increased (cyanide)
Cyanosis	Reduced Hb in capillary blood above 50 g/l

TOPIC 23

Pathophysiology of the digestive system

Learning objectives

To explain to the student the general etiology and pathogenesis of digestive disorders, the disturbances of secretory, motor-evacuatory and absorptive function of the oral cavity, esophagus, stomach, intestine and pancreas, disturbances of luminal and membrane (parietal) digestion, peptic ulcer disease, the maldigestion and malabsorption syndromes, intestinal obstruction and the features of digestive disorders in children; to teach the student to interpret the results of gastric juice analysis.

By the end of the session, the student should know:

- the causes and main syndromes of digestive disorders;
- the types of gastric secretion disorders and their consequences;
- disorders of the motor-evacuatory function of the stomach (vomiting, belching, heartburn);
- the pathogenesis of peptic ulcer disease (balance of aggressive and protective factors);
- disorders of luminal and membrane digestion and of absorption (maldigestion, malabsorption);
- disorders of intestinal motility (diarrhea, constipation, intestinal obstruction) and pancreatic insufficiency;
- the features of digestive disorders in children.

The student should be able to:

- assess the acidity of gastric juice and the type of secretion;
- identify digestive disorder syndromes by clinical signs.

1. General etiology of digestive disorders

Digestive insufficiency — a state in which digestion and absorption of nutrients do not meet the needs of the body.

Causes:

- **dietary factors** — irregular eating habits, poor-quality, very hot or spicy food, alcohol, dry food;
- **infections and parasites** — Helicobacter pylori, intestinal infections (Shigella, Salmonella, rotaviruses), Giardia, helminths;
- **drugs and toxins** — nonsteroidal anti-inflammatory drugs (NSAIDs), glucocorticoids, antibiotics (dysbiosis), cytostatics, radiation;
- **disturbances of nervous and humoral regulation** — stress, autonomic dysfunction, endocrine diseases (diabetes, thyrotoxicosis, hypothyroidism);
- **hereditary factors** — enzyme deficiencies (lactase, disaccharidases), celiac disease (gluten enteropathy), cystic fibrosis;
- **diseases of other organs** — liver, kidney (uremic gastroenteritis), heart failure (venous congestion), autoimmune processes;
- **tumors, surgical interventions** (gastric resection, intestinal resection).

2. Digestive disorders in the oral cavity and esophagus

Disorders of salivation:

- **hypersalivation (ptyalism)** — stomatitis, helminths, toxemia of pregnancy, mercury poisoning, parkinsonism; large amounts of swallowed saliva reduce gastric acidity;
- **hyposalivation (xerostomia)** — fever, dehydration, diabetes mellitus, Sjögren syndrome, atropine; dry mouth, difficulty chewing and swallowing, development of caries and stomatitis (loss of the bactericidal properties of saliva).

Disorders of the esophagus:

- **dysphagia** — difficulty swallowing: mechanical (stenosis, tumor, post-burn scar, diverticulum) or functional (achalasia — failure of the lower sphincter to relax, esophageal spasm);
- **gastroesophageal reflux** — incompetence of the lower esophageal sphincter, increased intragastric pressure, hiatal hernia; acidic contents flow back into the esophagus — **heartburn**, reflux esophagitis, and if prolonged, Barrett esophagus (metaplasia, a precancerous condition).

3. Disorders of gastric secretory function

Gastric juice secretion is regulated in three phases: **complex reflex (cephalic)** — via the vagus nerve; **gastric (neurohumoral)** — gastrin, histamine (H_2 receptors), acetylcholine; **intestinal** — secretin, cholecystokinin and somatostatin inhibit gastric secretion.

Table 23.1. *Normal values of gastric juice*

Parameter	Normal value
Basal secretion volume (fasting, per hour)	50–100 ml
Stimulated secretion volume (per hour)	100–150 ml
Total acidity (after stimulation)	40–60 titration units
Free hydrochloric acid	20–40 titration units
Fasting pH	1,5–2,0

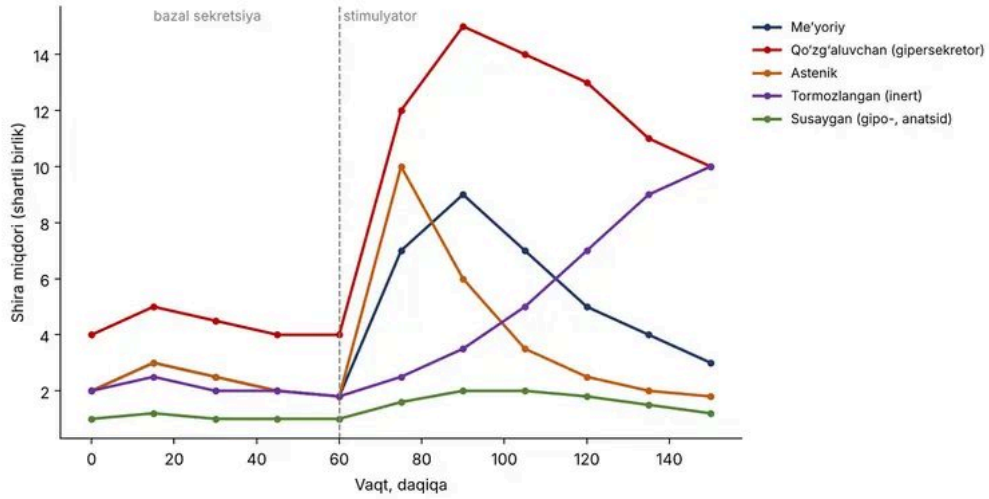


Figure 23.1. *Types of gastric secretion (schematic curves, based on textbooks)*

Types of secretion (by the form of the response to a stimulant): **normal**; **excitable** — secretion rises quickly and strongly and lasts long; **asthenic** — rises quickly and declines quickly (the glands tire quickly);

inhibited (inert) — rises slowly and late and lasts long; **depressed** — secretion is low and does not rise adequately.

Secretion disorders are assessed quantitatively (increased or decreased juice volume) and qualitatively (changes in acidity and enzyme activity); they often occur together.

Table 23.2. *Disorders of gastric secretion*

Type	Causes	Consequences
Hyperacidity, hypersecretion	Increased vagal tone, increased parietal cell mass, increased gastrin (Zollinger–Ellison syndrome — gastrinoma), Helicobacter pylori (antral gastritis), hypercalcemia	Heartburn, sour belching, pyloric spasm and delayed evacuation, constipation, peptic ulcer disease (duodenal)
Hypoacidity, anacidity (achylia — absence of acid and pepsin)	Atrophic gastritis (autoimmune — antibodies to parietal cells), gastric cancer, resection, chronic intoxications, hypothyroidism, cachexia	Protein digestion is impaired, the bactericidal effect is lost (fermentation, putrefaction, dysbiosis), the pylorus stays open — rapid evacuation and diarrhea, reduced iron absorption (iron deficiency); if Castle's intrinsic factor is lost — B_{12} deficiency anemia

4. Disorders of gastric motor-evacuatory function

- **Hypertonus and hyperkinesia** — rapid emptying of food from the stomach, strong contractions and pain; **pylorospasm** — delayed evacuation.
- **Hypotonus and hypokinesia** — dilation of the stomach, stagnation of food, fermentation and putrefaction.
- **Disturbed evacuation**: accelerated (achylia, after resection) and delayed (pyloric stenosis — ulcer scar, tumor; pyloric spasm in hyperacidity; diabetic gastroparesis).

Belching (eructation) — expulsion of air or food from the stomach through the mouth; in cardiac incompetence and increased gastric pressure. Belching smelling of “rotten eggs” — putrefaction of proteins in pyloric stenosis (hydrogen sulfide).

Nausea and vomiting — a protective reflex resulting from stimulation of the **vomiting center** in the medulla oblongata. The center is stimulated via the **trigger (chemoreceptor) zone** in the floor of the fourth ventricle by blood-borne toxins, drugs (cytostatics, morphine, glycosides), uremia and ketoacidosis, as well as by impulses from the gastrointestinal tract, vestibular apparatus and cerebral cortex, and by raised intracranial pressure.

> **CLINICAL SIGNIFICANCE**
>
> **Consequences of repeated vomiting:** dehydration (hypovolemia), **hypochloremic metabolic alkalosis** due to loss of hydrochloric acid, hypokalemia (renal potassium loss), possible alkalotic tetany; tearing of the mucosa at the esophagogastric junction (**Mallory–Weiss syndrome**), aspiration pneumonia.

5. Pathogenesis of peptic ulcer disease

Gastric and duodenal peptic ulcer disease — a disease characterized by the formation of a chronic recurrent ulcer in the mucosa. Its pathogenesis is explained by an **imbalance between aggressive and protective factors**.

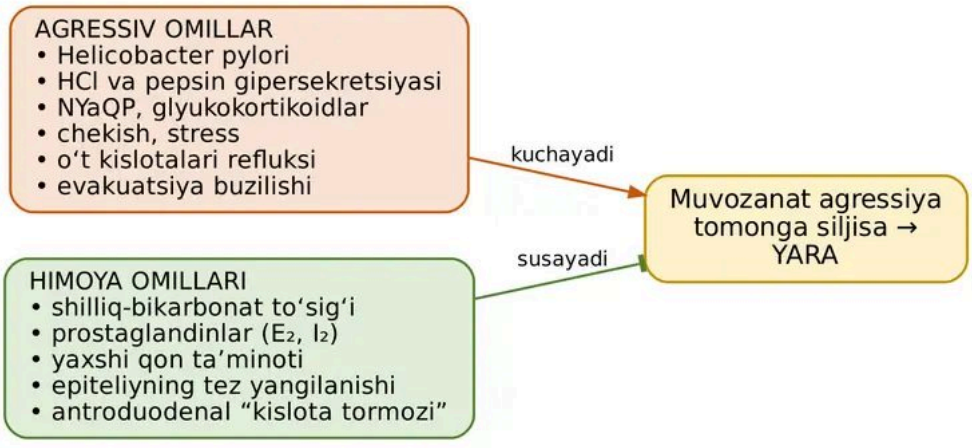


Figure 23.2. *Pathogenesis of peptic ulcer disease: balance of aggressive and protective factors*

Role of Helicobacter pylori: using the enzyme **urease**, the bacterium breaks down urea, forms a “cloud” of ammonia around itself and survives in the acidic environment; with its flagella it penetrates the mucus layer and produces cytotoxins (CagA, VacA) — inflammation (gastritis); somatostatin-releasing D cells in the antrum decrease → gastrin increases → hyperacidity → gastric metaplasia in the duodenum and ulcer.

NSAIDs inhibit cyclooxygenase and reduce the synthesis of protective prostaglandins — mucus, bicarbonate and blood supply decrease.

Stress ulcers (severe burns — Curling, brain trauma — Cushing, shock) — mucosal ischemia and the effects of catecholamines and glucocorticoids.

Complications: bleeding, perforation (peritonitis), penetration (into a neighboring organ), pyloric stenosis, malignant transformation (in gastric ulcer).

6. Disorders of intestinal digestion and absorption

Luminal and membrane digestion

- **Luminal digestion** — breakdown of large molecules in the intestinal lumen by the enzymes of pancreatic and intestinal juice and by bile.
- **Membrane (parietal) digestion** — the stage associated with final breakdown and absorption by enzymes located on the surface of the microvilli of the intestinal epithelium, in the glycocalyx (disaccharidases, peptidases) (A. M. Ugolev).

Maldigestion syndrome — impaired digestion: pancreatic insufficiency, failure of bile acids to reach the intestine (obstructive jaundice), disaccharidase deficiency (lactase — milk intolerance), change in intestinal pH.

Malabsorption syndrome — impaired absorption: damage to the intestinal mucosa (celiac disease — villous atrophy, Crohn disease, enteritis), reduced absorptive surface (intestinal resection — “short bowel” syndrome), impaired lymph flow, impaired blood circulation in the intestinal wall.

Consequences: steatorrhea (fat in the stool), creatorrhea (undigested muscle fibers), amylorrhea (starch), weight loss, hypoproteinemic edema, anemia (iron, B_{12}, folate), deficiency of fat-soluble vitamins (A, D, E, K) — hypocalcemia, osteomalacia, impaired blood clotting.

CAUTION

Lactase deficiency: undigested lactose enters the colon and is broken down by bacteria into lactic acid and gases — osmotic diarrhea, flatulence, abdominal pain. The osmotic pressure of the colonic contents rises and draws water into the intestinal lumen.

Disorders of intestinal motility

Diarrhea — frequent passage of liquid stools. Mechanisms:

- **secretory** — increased secretion of water and electrolytes into the intestinal lumen (cholera toxin activates adenylate cyclase and raises cAMP — several liters of fluid are lost per day);
- **osmotic** — non-absorbable osmotically active substances in the intestine (lactase deficiency, magnesium sulfate);
- **exudative (inflammatory)** — inflammation and ulceration of the mucosa, with discharge of blood, mucus and protein (dysentery, ulcerative colitis);
- **motor (hyperkinetic)** — accelerated peristalsis (thyrotoxicosis, irritable bowel syndrome).

Consequences: dehydration, hypokalemia, **metabolic acidosis** due to loss of bicarbonate, hypovolemic shock (especially in children).

Constipation — retention of stool for more than 3 days: alimentary (little fiber and fluid), hypokinetic (physical inactivity, hypothyroidism, old age), spastic (autonomic dysfunction, lead poisoning), mechanical obstruction (tumor, scar), proctogenic (anal fissure, hemorrhoids — reflex inhibition due to pain). Consequences: intoxication (absorption of putrefaction products — indole, skatole, phenol), hemorrhoids.

Intestinal obstruction

Intestinal obstruction (ileus) — arrest of the movement of intestinal contents.

- **Mechanical** — obturation (tumor, gallstone, a ball of worms, foreign body), strangulation (volvulus, knotting, incarcerated hernia — the mesenteric vessels are also compressed, rapid necrosis), intussusception (more common in children).
- **Dynamic** — paralytic (peritonitis, after abdominal surgery, hypokalemia) and spastic (lead poisoning, hysteria).

Pathogenesis: above the obstruction the intestine is distended and the circulation in its wall is impaired; fluid and electrolytes accumulate in the intestinal lumen and are lost with vomiting — **dehydration and hypovolemia**; bacteria multiply, and toxins and bacteria pass through the wall into the blood (**endotoxemia**), peritonitis; pain impulses — pain shock. The result is water–electrolyte disturbances, intoxication and shock.

7. Disorders of pancreatic function

Exocrine pancreatic insufficiency — chronic pancreatitis, cystic fibrosis, resection and tumors of the gland. Lipase deficiency manifests first: **steatorrhea** (fatty, shiny, bulky stools), creatorrhea, amylorrhea, weight loss.

Acute pancreatitis — activation of the gland's enzymes within the gland and digestion of its own tissue (autolysis).

- **Causes:** gallstones (obstruction of the common bile–pancreatic duct and bile reflux), alcohol, large fatty meals, trauma, hypertriglyceridemia, hypercalcemia, infections, drugs.

- **Pathogenesis:** trypsinogen is converted to trypsin within the gland → it activates other enzymes: **phospholipase A_2** and **lipase** — fat necrosis (steatonecrosis, formation of soaps with calcium — hypocalcemia), **elastase** — damage to vessel walls and bleeding, activation of the **kallikrein–kinin system** — vasodilation, increased vascular permeability, pain, shock. The enzymes pass into the blood (**enzymemia** — amylase and lipase rise) and damage distant organs (lungs, kidneys).

8. Features of digestive disorders in children

- The acidity of digestive juices and enzyme activity are low and the permeability of the intestinal mucosa is high — infections and allergens pass through easily.
- The lower esophageal sphincter is immature and the stomach lies horizontally — **regurgitation** is frequent.
- **Pyloric stenosis** — congenital hypertrophy of the pyloric muscle: projectile ("fountain") vomiting from 2–4 weeks of age, hypochloremic alkalosis, weight loss.
- **In diarrhea** water and electrolyte balance is disturbed quickly — exsiccosis and toxicosis; the risk of hypovolemic shock is much higher than in adults.
- Hereditary enzymopathies (lactase deficiency, celiac disease, cystic fibrosis) often manifest when switching to formula feeding or introducing complementary foods.
- Intussusception is the most common cause of intestinal obstruction in children.

9. Experimental part

Determining the acidity of gastric juice (titration method)

Aim: to determine the total acidity and free hydrochloric acid of gastric juice.

1. Put 5 ml of filtered gastric juice into a flask.
2. Add the indicators: dimethylaminoazobenzene (for free hydrochloric acid) and phenolphthalein (for total acidity).
3. Titrate with 0.1 N sodium hydroxide solution: when the red color turns yellowish-orange — free hydrochloric acid; when a pink color appears — total acidity.
4. Multiply the milliliters of alkali used by 20 (calculating per 100 ml of juice) and express the result in titration units.
5. Compare the result with the normal values and write a conclusion (hyperacid, normacid, hypoacid, anacid).

Studying the activity of digestive enzymes

In test tubes, observe the breakdown of starch by salivary amylase (iodine test) and the digestion of protein (fibrin, egg white) by gastric juice, and the effect of various factors (temperature, pH, bile) on them.

> **REMEMBER**
> The indicators, reagents and calculation procedure should be checked against the department's methodological manual; in modern practice gastric secretion is assessed mainly by (intragastric) pH-metry.

10. Practical skills

Skill: interpreting gastric juice analysis

1. Compare the volumes of basal and stimulated secretion with the normal values.
2. Assess total acidity and the amount of free hydrochloric acid.
3. No free hydrochloric acid and very low total acidity — anacid state; if pepsin is also absent — achylia.
4. Determine the type of secretion from the shape of the secretion curve.
5. Relate the result to the clinical signs (ulcer, atrophic gastritis, anemia) and write a conclusion.

Skill: determining the mechanism of diarrhea

1. Assess the volume and character of the stools (watery, mixed with blood and mucus, fatty).
2. Does the diarrhea stop with fasting? If it stops — osmotic mechanism; if not — secretory.
3. If blood, mucus and leukocytes are present — exudative (inflammatory).
4. Fatty, shiny, bulky stools — maldigestion or malabsorption (steatorrhea).
5. Assess the degree of dehydration and electrolyte disturbance.

Review questions

1. List the main causes of digestive disorders.
2. What are the causes and consequences of hyper- and hyposalivation?
3. What are the mechanisms of dysphagia and gastroesophageal reflux?
4. For what reasons do hyperacid and hypoacid states of gastric secretion develop, and what are their consequences?
5. What disturbances are observed in achylia?
6. Explain the disorders of gastric motor-evacuatory function.
7. By what mechanism does vomiting occur? What are the consequences of repeated vomiting?
8. Explain the pathogenesis of peptic ulcer disease in terms of aggressive and protective factors.
9. What is the role of Helicobacter pylori in ulcer formation?
10. What are luminal and membrane digestion? How do the maldigestion and malabsorption syndromes differ?
11. List the pathogenetic types of diarrhea.
12. What are the causes and consequences of constipation?
13. Explain the types and pathogenesis of intestinal obstruction.
14. What is the pathogenesis of acute pancreatitis?
15. What are the features of digestive disorders in children?

Test questions

1. What is the cause of increased gastric secretion in Zollinger–Ellison syndrome?
   A) Decreased vagal tone
   B) Increased somatostatin
   C) Atrophic gastritis
   D) Increased secretin
   E) A gastrin-producing tumor (gastrinoma)

2. Which is a consequence of achylia?
   A) Pyloric spasm and constipation
   B) Loss of the bactericidal effect, rapid evacuation and diarrhea
   C) Duodenal ulcer
   D) Heartburn
   E) Hypochloremic alkalosis

3. What does the loss of Castle's intrinsic factor lead to?
   A) Iron deficiency anemia
   B) Steatorrhea
   C) Lactase deficiency
   D) B_{12} deficiency anemia
   E) Hypocalcemia

4. What acid–base disturbance develops with repeated vomiting?
   A) Hypochloremic metabolic alkalosis
   B) Metabolic acidosis
   C) Respiratory acidosis
   D) Respiratory alkalosis
   E) No change

5. What enables Helicobacter pylori to survive in an acidic environment?
   A) Production of lipase
   B) Its capsule
   C) Formation of ammonia by urease
   D) Formation of spores
   E) Anaerobic respiration

6. By what mechanism do NSAIDs cause gastric ulcer?
   A) They increase gastrin
   B) They activate Helicobacter pylori
   C) They activate pepsin
   D) They inhibit cyclooxygenase and reduce protective prostaglandins
   E) They narrow the pylorus

7. What is the mechanism of diarrhea in lactase deficiency?
   A) Secretory
   B) Exudative
   C) Hyperkinetic
   D) Malabsorptive
   E) Osmotic

8. What is the mechanism of diarrhea in cholera?
   A) The toxin activates adenylate cyclase and increases secretion of water and electrolytes
   B) Osmotically active substances accumulate
   C) Ulceration of the mucosa
   D) Slowing of peristalsis
   E) Enzyme deficiency

9. How does the acid–base state change in profuse watery diarrhea?
   A) Metabolic alkalosis
   B) Respiratory alkalosis
   C) Metabolic acidosis due to loss of bicarbonate
   D) Respiratory acidosis
   E) It does not change

10. What is characteristic of strangulation intestinal obstruction?
   A) Only the intestinal lumen is blocked
   B) The mesenteric vessels are compressed together with the intestine, and necrosis develops rapidly
   C) Intestinal peristalsis increases
   D) It occurs only in hypokalemia
   E) There is never necrosis

11. Which sign manifests first in exocrine pancreatic insufficiency?
   A) Diabetes mellitus
   B) Steatorrhea
   C) Jaundice
   D) Bleeding
   E) Hypoglycemia

12. What is the mechanism of hypocalcemia in acute pancreatitis?
   A) Binding of calcium as soaps in foci of fat necrosis
   B) Decreased calcitonin
   C) Renal loss of calcium
   D) Increased vitamin D
   E) Increased parathyroid hormone

13. Where does membrane (parietal) digestion take place?
   A) In the gastric lumen
   B) In the oral cavity
   C) On the surface of the microvilli of the intestinal epithelium
   D) In the liver
   E) In the colon with the help of bacteria

14. What disturbance develops in a newborn with congenital pyloric stenosis?
   A) Metabolic acidosis
   B) Hyperkalemia
   C) Hypernatremia
   D) Diarrhea
   E) Projectile (“fountain”) vomiting and hypochloremic alkalosis

15. What is the main cause of intoxication in constipation?
   A) Hypoglycemia
   B) Increased bile acids
   C) Hypokalemia
   D) Absorption of protein putrefaction products (indole, skatole, phenol)
   E) Loss of ammonia

Clinical cases

Case 1

A 32-year-old man has had epigastric pain on an empty stomach and at night for 3 years; the pain decreases after eating; heartburn, sour belching, constipation. He smokes heavily. Endoscopy: an ulcer in the duodenal bulb, positive urease test. Gastric juice: acidity increased.

Questions: Which aggressive and protective factors were involved in the development of the ulcer? Explain the constipation and the reduction of pain after eating.

Answer: Duodenal ulcer. Aggressive factors: Helicobacter pylori (antral gastritis, loss of D cells → decreased somatostatin and increased gastrin → hyperacidity), hypersecretion of hydrochloric acid and pepsin, smoking (reduces prostaglandin and bicarbonate secretion and worsens blood supply). The protective factors (mucus–bicarbonate barrier, blood supply) are weakened. On an empty stomach, highly acidic contents enter the duodenum and cause pain; food neutralizes the acid — the pain decreases. Hyperacidity causes pyloric spasm and reflex slowing of intestinal motility — constipation. Treatment — eradication of Helicobacter pylori and reduction of acid secretion.

Case 2

A 4-year-old child at a kindergarten has contracted cholera: watery stools resembling “rice water” 15–20 times a day and vomiting. The skin is dry, the skin fold returns slowly, the eyeballs are sunken, urine is scanty, BP is decreased, the pulse rapid and weak. Blood: hematocrit raised, K^+ decreased, pH 7.26, bicarbonate decreased.

Questions: Explain the mechanism of the diarrhea. Interpret the water–electrolyte and acid–base disturbances. Why is this condition especially dangerous in children?

Answer: Cholera toxin permanently activates adenylate cyclase in the intestinal epithelium and raises cAMP — chloride and bicarbonate ions, and water with them, are secreted into the intestinal lumen in large amounts and sodium absorption decreases — secretory diarrhea (the mucosa is not damaged, which is why there is no blood in the stool). The result is isotonic dehydration (hypovolemia — raised hematocrit, oliguria, hypotension, hypovolemic shock), potassium loss (hypokalemia), and metabolic acidosis due to loss of bicarbonate and lactate accumulation with tissue hypoperfusion. In children the proportion of body water and of extracellular fluid is higher, water turnover is faster and the concentrating ability of the kidneys is lower — dehydration and shock develop very quickly. The main treatment is oral or intravenous administration of water–electrolyte solutions (rehydration).

Case 3

After a festive meal (much fatty food, alcohol), a 48-year-old man developed severe girdle-like epigastric pain and repeated vomiting. BP 90/60 mm Hg, pulse 120 beats/min, bluish patches on the abdominal skin. Blood amylase and lipase are sharply raised, calcium decreased, leukocytosis.

Questions: What is the disease? Explain its pathogenesis. By what mechanism did the hypotension and hypocalcemia develop?

Answer: Acute pancreatitis. Alcohol and fatty food stimulate the gland's secretion, cause spasm of the sphincter of Oddi and increased duct pressure, and premature activation of trypsinogen in the acinar cells — trypsin activates the other enzymes and self-digestion of the gland (autolysis) begins. Lipase and phospholipase A_2 cause fat necrosis; in the necrotic foci fatty acids form soaps with calcium — hypocalcemia (a sign of a severe course). Elastase damages vessel walls and causes hemorrhages (the patches on the abdominal skin). Activation of the kallikrein–kinin system leads to

vasodilation and increased vascular permeability, accumulation of fluid in the abdominal cavity and retroperitoneal tissue, and water loss with vomiting — hypovolemia, which together with pain and enzymemia causes shock (hypotension, tachycardia). The enzymes pass into the blood (raised amylase and lipase) and may damage the lungs and kidneys.

Quick review

Table 23.3. *Key facts on Topic 23*

Question	Answer
Hyperacid state	Increased HCl; heartburn, pylorospasm, constipation, ulcer
Achylia	No HCl or pepsin; rapid evacuation, diarrhea, anemia
Zollinger–Ellison	Gastrinoma — marked hypersecretion, multiple ulcers
Consequences of vomiting	Dehydration, hypochloremic alkalosis, hypokalemia
Pathogenesis of ulcer	Imbalance between aggressive and protective factors
Helicobacter pylori	Urease, ammonia; gastritis, increased gastrin
NSAIDs	Reduced prostaglandins — weakened protection
Membrane digestion	Enzymes on the surface of the microvilli (Ugolev)
Maldigestion	Impaired digestion (enzyme or bile deficiency)
Malabsorption	Impaired absorption (celiac disease, resection)
Secretory diarrhea	Cholera: increased cAMP, large losses of water and electrolytes
Osmotic diarrhea	Lactase deficiency, magnesium sulfate
Intestinal obstruction	Mechanical and dynamic; dehydration, endotoxemia, shock
Acute pancreatitis	Activation of enzymes within the gland, autolysis
In children	Regurgitation, pyloric stenosis, rapid exsiccosis

TOPIC 24

Pathophysiology of the hepatobiliary system

Learning objectives

To explain to the student the main functions of the liver, the causes, types and pathogenesis of liver failure, the changes in metabolism, blood clotting and detoxification when liver function is impaired, the mechanisms of hepatic coma, portal hypertension and liver cirrhosis, the types of jaundice and their laboratory differentiation, disorders of bile formation and excretion, and the features in children; to teach the student to detect bile pigments in urine and to determine the type of jaundice from laboratory parameters.

By the end of the session, the student should know:

- the main functions of the liver and the signs of their impairment;
- the causes, types and pathogenesis of liver failure;
- the types and pathogenesis of hepatic coma;
- the syndromes that develop in portal hypertension and liver cirrhosis;
- bilirubin metabolism and the types of jaundice, and their laboratory differences;
- the cholestasis and acholia syndromes;
- the features of neonatal jaundice.

The student should be able to:

- detect bilirubin and urobilin in urine;
- determine the type of jaundice from blood, urine and stool parameters.

1. Main functions of the liver and their impairment

Table 24.1. *Liver functions and changes in liver failure*

Function	Normal	In failure
Carbohydrate metabolism	Glycogen synthesis and breakdown, gluconeogenesis	Fasting hypoglycemia, postprandial hyperglycemia
Protein metabolism	Synthesis of albumin, fibrinogen, prothrombin and transport proteins; deamination of amino acids; urea formation	Hypoalbuminemia (edema, ascites), increased blood ammonia, decreased urea, aminoacidemia
Lipid metabolism	Synthesis of lipoproteins, cholesterol, bile acids	Fatty liver degeneration, reduced cholesterol esterification
Blood clotting	Synthesis of clotting factors (I, II, V, VII, IX, X), with the participation of vitamin K	Hypocoagulation, bleeding (prolonged prothrombin time)
Detoxification	Inactivation of ammonia, phenol, indole, skatole, drugs and hormones (estrogens, aldosterone, insulin)	Intoxication, hyperestrogenemia (spider angiomas, palmar erythema, gynecomastia), secondary hyperaldosteronism
Bile formation	Excretion of bile acids and bilirubin	Jaundice, cholestasis, impaired fat digestion
Vitamins and trace elements	Storage of A, D, B_{12}, iron and copper; 25-hydroxylation of vitamin D	Hypovitaminoses, osteomalacia
Immune and barrier	Kupffer cells (macrophages) — capture of bacteria and endotoxin coming from the intestine	Endotoxemia, susceptibility to infections

2. Liver failure

Liver failure — a condition with impairment of one or more functions of the liver.

Causes:

- **infectious** — viral hepatitis (B, C, D, A, E), leptospirosis, echinococcosis, alveococcosis, opisthorchiasis;
- **toxic** — alcohol, hepatotropic poisons (carbon tetrachloride, phosphorus, poisonous mushrooms — death cap), drugs (a large dose of paracetamol, antituberculosis and cytostatic drugs);
- **circulatory disturbance** — chronic heart failure ("nutmeg liver"), shock, hepatic vein thrombosis (Budd–Chiari syndrome);
- **impaired bile outflow** — gallstones, tumors (secondary biliary cirrhosis);
- **hereditary metabolic disorders** — hemochromatosis (iron), Wilson–Konovalov disease (copper), alpha1-antitrypsin deficiency;
- **autoimmune** — autoimmune hepatitis, primary biliary cholangitis;
- **tumors** — hepatocellular carcinoma, metastases;
- **nutritional** — protein deficiency, obesity (fatty hepatosis).

Table 24.2. *Types of liver failure*

Criterion	Types
Course	Acute (fulminant — days, weeks), chronic
By function	Partial (one function), total (all functions)
By pathogenesis	Hepatocellular (death of liver cells), portosystemic shunting (blood bypasses the liver), mixed

Syndromes of hepatocellular failure

- **Cytolysis syndrome** — the hepatocyte membrane is damaged and enzymes are released into the blood: **ALT and AST** rise (ALT is more specific to the liver).
- **Mesenchymal-inflammatory syndrome** — increased gamma globulins, positive thymol test.
- **Hepatodepressive (synthetic failure)** syndrome — decreased albumin, prothrombin, cholinesterase, cholesterol.
- **Cholestasis syndrome** — increased alkaline phosphatase, GGT (gamma-glutamyl transpeptidase), direct bilirubin, bile acids and cholesterol.

3. Hepatic coma (hepatic encephalopathy)

Hepatic encephalopathy — impaired function of the central nervous system under the action of toxic substances that are not neutralized in liver failure; its severe degree is **hepatic coma**.

Types:

- **endogenous (hepatocellular)** — extensive necrosis of the liver parenchyma (fulminant hepatitis, poisoning);
- **exogenous (portosystemic, bypass)** — in portal hypertension, blood from the intestine bypasses the liver through portocaval anastomoses and enters the systemic circulation directly;
- **mixed**.

Pathogenesis:

1. **Ammonia** — formed mainly from the breakdown of proteins by bacteria in the intestine; it must be converted to urea in the liver. In the brain, excess ammonia binds alpha-ketoglutarate to form glutamate and then glutamine — **alpha-ketoglutarate in the Krebs cycle decreases and ATP formation falls**; glutamine accumulates in astrocytes and causes their swelling (cerebral edema); GABA-ergic inhibition increases.
2. **False neurotransmitters** — aromatic amino acids (phenylalanine, tyrosine, tryptophan) increase and branched-chain amino acids (valine, leucine, isoleucine) decrease; **octopamine** and **phenylethylamine** are formed in the brain and replace the true transmitters (dopamine, norepinephrine).
3. Other toxins — mercaptans ("fetor hepaticus" from the mouth), short-chain fatty acids, phenols, endogenous benzodiazepines.
4. Additional factors — hypoglycemia, hypokalemia, alkalosis (ammonia passes into the brain easily), hypoxia.

CLINICAL SIGNIFICANCE

Factors that precipitate hepatic encephalopathy: gastrointestinal bleeding (blood in the intestine is a source of protein — ammonia increases), a high-protein diet, constipation, infections, diuretics (hypokalemia and alkalosis), sedatives, alcohol. Clinical signs: disturbed sleep rhythm, behavioral changes, "flapping" tremor (asterixis), stupor, coma.

4. Portal hypertension and liver cirrhosis

Portal hypertension — a rise in pressure in the portal vein system above 10–12 mm Hg.

Table 24.3. *Types of portal hypertension*

Type	Causes
Prehepatic	Portal vein thrombosis, external compression
Intrahepatic	Liver cirrhosis (the most common), schistosomiasis, tumors
Posthepatic	Hepatic vein thrombosis (Budd–Chiari), constrictive pericarditis, right ventricular failure

Consequences:

- **dilation of portocaval anastomoses** — varicose veins of the esophagus and gastric cardia (dangerous bleeding), rectal veins (hemorrhoids), periumbilical veins ("caput medusae");
- **splenomegaly and hypersplenism** — anemia, leukopenia, thrombocytopenia;
- **ascites**;
- portosystemic encephalopathy.

Pathogenesis of ascites

1. Portal hypertension — hydrostatic pressure in the hepatic sinusoids and intestinal capillaries rises and fluid passes into the abdominal cavity; hepatic lymph increases.
2. Hypoalbuminemia — oncotic pressure decreases.
3. Dilation of the visceral arteries (increased NO) — the effective blood volume decreases → **activation of RAAS, the sympathetic system and ADH** → sodium and water are retained by the kidneys; aldosterone inactivation in the liver decreases (secondary hyperaldosteronism).

Liver cirrhosis — a chronic diffuse disease of the liver characterized by death of hepatocytes, fibrosis and replacement of the normal architecture by **regenerative nodules**. The main causes are chronic viral hepatitis (B, C), alcohol, metabolic dysfunction-associated fatty liver disease, and autoimmune and biliary diseases.

Fibrosis develops as a result of activation of hepatic stellate cells (Ito cells) and their production of collagen. Cirrhosis combines the syndromes of liver failure and portal hypertension.

5. Bilirubin metabolism and jaundice

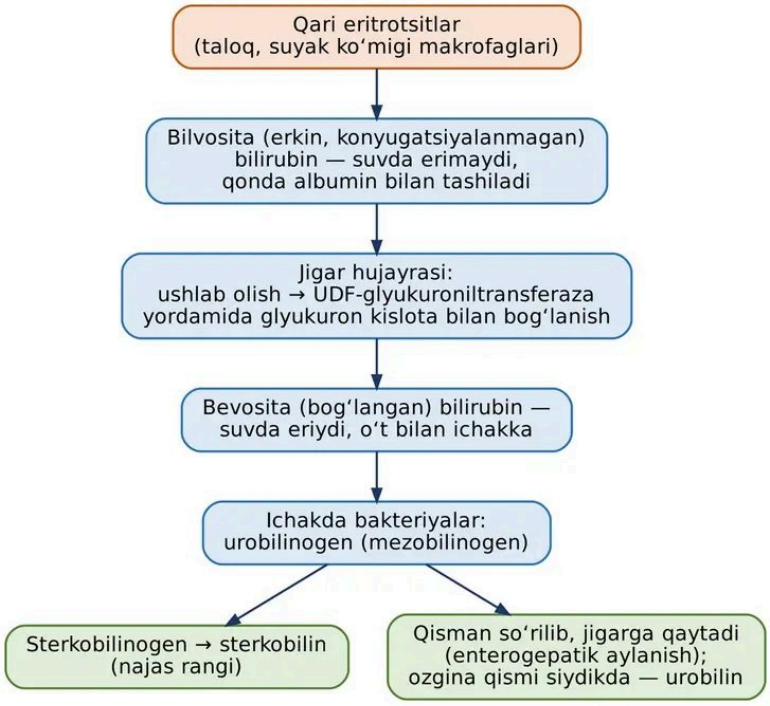


Figure 24.1. *Bilirubin metabolism (diagram)*

Aged erythrocytes are broken down in macrophages: the heme of hemoglobin is converted via **biliverdin** into **indirect (free) bilirubin**. It is insoluble in water, is carried in the blood by albumin and does not pass the renal filter (it is absent from urine). In the liver it is conjugated with glucuronic acid to form **direct bilirubin**, is excreted with bile into the intestine and is converted by bacteria to **urobilinogen** and then to **stercobilinogen** (which colors the stool). Part of the urobilinogen is absorbed and returns to the liver; a healthy liver retains it almost completely.

Normally total blood bilirubin is 8.5–20.5 µmol/l, mainly the indirect fraction (about 75 percent).

Jaundice (icterus) — yellow coloring of the skin, mucous membranes and sclerae due to increased blood bilirubin (hyperbilirubinemia). Yellowing of the sclerae becomes visible when bilirubin exceeds about 34 µmol/l.



Figure 24.2. *Types of jaundice and their laboratory signs*

Prehepatic (hemolytic) jaundice

Cause: increased breakdown of erythrocytes (hemolytic anemias, incompatible blood transfusion, hemolytic poisons, malaria, hemolytic disease of the newborn) — formation of indirect bilirubin exceeds the conjugating capacity of the liver.

Signs: indirect bilirubin raised in the blood; no bilirubin in the urine (indirect bilirubin is not filtered); a lot of bilirubin enters the intestine — **urobilin** in the urine and **stercobilin** in the stool are increased (dark stool); the skin is lemon-yellow, no itching; anemia, reticulocytosis.

Hepatic (parenchymal) jaundice

Cause: damage to hepatocytes — viral and toxic hepatitis, cirrhosis, leptospirosis.

Mechanism: uptake, conjugation and excretion of bilirubin into the bile capillaries are impaired in the liver cells; the cells die and conjugated bilirubin returns to the blood (the barrier between the bile capillaries and blood vessels is broken); the liver cannot retain urobilinogen coming from the intestine.

Signs: both fractions raised in the blood; direct bilirubin present in the urine (urine “the color of beer”) and urobilin increased; stool decolorized (less stercobilin); ALT and AST sharply raised, synthetic function of the liver impaired; the skin is saffron-yellow.

By stage of hepatitis: urobilinuria in the initial (prodromal) period; at the peak, if cholestasis develops, urobilin may temporarily disappear; it reappears during recovery.

Posthepatic (obstructive, mechanical) jaundice

Cause: obstruction of bile outflow — bile duct stones, cancer of the head of the pancreas, tumor of the papilla of Vater, cicatricial narrowing of the bile ducts, helminths, external compression.

Mechanism: pressure in the bile ducts rises, the bile capillaries rupture and conjugated bilirubin, bile acids and cholesterol pass into the blood (**cholemia**); bile does not enter the intestine (**acholia**).

Signs: direct bilirubin sharply raised in the blood; much bilirubin in the urine (dark urine, “the color of beer”), **no urobilin** (since bile does not enter the intestine, no urobilinogen is formed); stool **white-gray (acholic)**, fatty (steatorrhea); alkaline phosphatase, GGT and cholesterol raised; the skin is greenish-yellow, with severe itching.

Table 24.4. *Laboratory differences between the types of jaundice*

Parameter	Prehepatic	Liver	Posthepatic
Indirect bilirubin in the blood	↑↑	↑	Normal or slightly ↑
Direct bilirubin in the blood	Normal	↑	↑↑
Bilirubin in the urine	No	Present	Many
Urobilin in the urine	↑↑	↑	No
Stercobilin in the stool	↑ (dark stool)	↓	Absent (acholic stool)
ALT, AST	Normal	↑↑	Slightly ↑
Alkaline phosphatase, GGT	Normal	Slightly ↑	↑↑
Skin itching	No	May be present	Severe

6. Cholestasis, cholemia and acholia syndromes

Cholemia — entry of bile components (mainly bile acids) into the blood:

- skin itching (action of bile acids and other pruritogens on nerve endings);
- bradycardia and arterial hypotension (action of bile acids on the sinus node and the vagal nuclei, reduced vascular tone);
- depression of the CNS — weakness, depression, drowsiness;
- hemolysis of erythrocytes, reduced leukocyte activity;
- hypercholesterolemia, xanthomas on the skin.

Acholia — bile does not enter the intestine:

- emulsification and digestion of fats are impaired — **steatorrhea**;
- fat-soluble vitamins (A, D, E, K) are not absorbed — **vitamin K deficiency** (reduced synthesis of factors II, VII, IX, X, bleeding), osteomalacia, night blindness;
- intestinal peristalsis and the bacteriostatic action of bile decrease — putrefaction and fermentation in the intestine, flatulence, dysbiosis;
- the stool is decolorized.

> **MEMORY AID**
> To remember obstructive jaundice: “urine the color of **beer**, stool the color of **clay**, the skin **itches**, the blood **does not clot**” (bilirubinuria, acholia, cholemia, vitamin K deficiency).

7. Features of hepatobiliary disorders in children

Physiological jaundice of the newborn — appears on day 2–3 and disappears by day 7–10 (by 2–3 weeks in premature infants). Causes:

- the short life of erythrocytes containing fetal hemoglobin (HbF) and their massive breakdown after birth;
- low activity of hepatic **UDP-glucuronyltransferase** (immaturity of the enzyme);
- absence of bacteria in the intestine and intense enterohepatic circulation.

Pathological jaundice:

- **Hemolytic disease of the newborn** — maternal antibodies against the fetus's Rh-positive (or AB0) antigens destroy fetal erythrocytes; it appears on the first day and increases rapidly.
- **Bilirubin encephalopathy (kernicterus)** — in newborns the permeability of the blood–brain barrier is high and albumin is low; indirect bilirubin (which is fat-soluble) accumulates in the subcortical nuclei of the brain and damages them. The risk is high when bilirubin exceeds about 340 µmol/l (even lower in premature infants). Treatment — phototherapy, exchange transfusion.
- **Hereditary pigmentary hepatoses: Gilbert syndrome** (mild impairment of bilirubin uptake and conjugation — indirect bilirubin rises with fasting and stress, harmless), **Crigler–Najjar syndrome** (glucuronyltransferase absent or markedly reduced — severe indirect hyperbilirubinemia, risk of kernicterus), **Dubin–Johnson syndrome** (impaired excretion of direct bilirubin).
- **Biliary atresia** — congenital obstructive jaundice; early surgery is required.

8. Experimental part: detecting bile pigments in urine

Detecting bilirubin in urine (Rosin test)

1. Pour 4–5 ml of urine into a test tube.

2. Carefully layer 1 percent alcoholic iodine solution along the wall of the test tube.
3. If a **green ring** appears at the boundary of the two liquids, bilirubin is present in the urine (positive test).

Detecting urobilin in urine

A reagent for urobilinogen (e.g. Ehrlich's reagent) is added to the urine: the appearance of a red color indicates increased urobilinogen (urobilin); normally the color is very faint or appears only on heating.

Interpreting the results

Detect bilirubin and urobilin in the urine samples of the “patients” provided, compare the results with the blood and stool parameters and determine the type of jaundice.

> **REMEMBER**
> The concentrations of reagents and the procedure of the tests should be checked against the department's methodological manual; in a modern laboratory bilirubin and urobilinogen in urine are detected with test strips.

9. Practical skills

Skill: determining the type of jaundice from laboratory parameters

1. Assess total blood bilirubin and its fractions: which fraction predominates?
2. Determine whether bilirubin is present in the urine: if present, direct bilirubin is raised (hepatic or posthepatic).
3. Assess urobilin in the urine: much — prehepatic or hepatic; absent — posthepatic.
4. Assess the stool color: dark — prehepatic; acholic — posthepatic.
5. Assess the enzymes: ALT and AST predominate — hepatic; alkaline phosphatase and GGT predominate — posthepatic.
6. Write a conclusion and suggest the underlying disease.

Skill: identifying the syndromes of liver failure

1. Assess the signs of cytolysis (ALT, AST).
2. Assess synthetic function (albumin, prothrombin index or INR, cholesterol).
3. Assess the signs of cholestasis (alkaline phosphatase, GGT, direct bilirubin).
4. Identify the signs of portal hypertension (ascites, splenomegaly, varicose veins).
5. Assess the signs of encephalopathy (consciousness, sleep, asterixis).

Review questions

1. What are the main functions of the liver and what changes are observed when they are impaired?
2. List the causes of liver failure.
3. What are the types of liver failure?
4. What are the laboratory signs of the cytolysis, cholestasis and hepatodepressive syndromes?
5. Explain the types and pathogenesis of hepatic coma.
6. What is the mechanism of the toxic effect of ammonia on the brain?

7. List the types and consequences of portal hypertension.
8. How does ascites develop in liver cirrhosis?
9. Explain the stages of bilirubin metabolism.
10. What are the causes and laboratory signs of prehepatic jaundice?
11. What are the mechanism and laboratory signs of hepatic jaundice?
12. What are the causes and laboratory signs of posthepatic jaundice?
13. What are the signs and mechanisms of the cholemia and acholia syndromes?
14. Why does physiological jaundice of the newborn develop? How does kernicterus develop?
15. How are bilirubin and urobilin detected in urine, and what is their diagnostic significance?

Test questions

1. What is the cause of impaired blood clotting in liver failure?
   A) Decreased synthesis of clotting factors (prothrombin, fibrinogen and others)
   B) Increased platelets
   C) Weakened fibrinolysis
   D) Increased synthesis of anticoagulants
   E) Hemolysis

2. What is the cause of gynecomastia and spider angiomas in men with liver cirrhosis?
   A) Increased androgens
   B) Hypoglycemia
   C) Reduced inactivation of estrogens
   D) Hypoalbuminemia
   E) Increased ammonia

3. What is the mechanism of the toxic effect of ammonia on the brain in hepatic coma?
   A) Increased glucose synthesis
   B) Increased dopamine
   C) Dilation of cerebral vessels
   D) Binding to albumin
   E) Binding of alpha-ketoglutarate and slowing of the Krebs cycle, swelling of astrocytes

4. What are the “false neurotransmitters” in hepatic encephalopathy?
   A) Dopamine and norepinephrine
   B) Acetylcholine
   C) Valine and leucine
   D) Octopamine and phenylethylamine
   E) Glutamine

5. How does gastrointestinal bleeding precipitate hepatic encephalopathy?
   A) Through hyperglycemia
   B) Through the formation of large amounts of ammonia from blood proteins in the intestine
   C) Through hypertension
   D) Through increased albumin
   E) Through decreased bilirubin

6. Which factor is NOT involved in the pathogenesis of ascites in liver cirrhosis?
   A) Portal hypertension
   B) Hypoalbuminemia
   C) Secondary hyperaldosteronism
   D) ADH activation
   E) Increased oncotic pressure of the blood

7. What is found in the urine in hemolytic jaundice?
   A) No bilirubin, urobilin increased
   B) Much bilirubin, no urobilin
   C) Bilirubin present, urobilin increased
   D) Neither
   E) Bile acids only

8. What is the stool color in obstructive jaundice?
   A) Dark brown
   B) White-gray (acholic)
   C) Black
   D) Green
   E) It does not change

9. What is the main laboratory sign of hepatic (parenchymal) jaundice?
   A) Only indirect bilirubin raised, ALT normal
   B) No urobilin in the urine
   C) Both fractions raised in the blood and ALT and AST sharply raised
   D) Increased stercobilin in the stool
   E) Only alkaline phosphatase raised

10. What is the cause of bradycardia in cholemia?
    A) The effect of bilirubin on the heart
    B) Hypokalemia
    C) Anemia
    D) The effect of bile acids on the sinus node and the vagal nuclei
    E) Hyperthyroidism

11. What is the cause of the bleeding tendency in acholia?
    A) Breakdown of platelets
    B) Increased fibrinogen
    C) Inflammation of the vessel wall
    D) Failure to absorb vitamin K and decreased synthesis of factors II, VII, IX, X
    E) Hemophilia

12. Bilirubin appears in the urine when which fraction rises?
    A) Indirect (free) bilirubin
    B) Albumin-bound bilirubin
    C) Biliverdin
    D) Stercobilin
    E) Direct (conjugated) bilirubin

13. What is the main cause of physiological jaundice of the newborn?
    A) Biliary atresia
    B) Low UDP-glucuronyltransferase activity and massive breakdown of HbF-containing erythrocytes
    C) Rh conflict
    D) Hepatitis
    E) Obstructive jaundice

14. Which bilirubin causes kernicterus?
    A) Indirect (free) bilirubin
    B) Direct bilirubin
    C) Stercobilin
    D) Urobilin
    E) Biliverdin

15. What indicates the presence of bilirubin in urine in the Rosin test?
    A) A red color
    B) A blue precipitate
    C) A green ring at the boundary of the two liquids
    D) Foam formation
    E) Turbidity

Clinical cases

Case 1

A 62-year-old man has had gradually increasing jaundice for 2 months, severe skin itching, loss of appetite and weight loss. The skin is greenish-yellow with scratch marks; pulse 56 beats/min. Blood: total bilirubin 280 µmol/l (direct 230), alkaline phosphatase and GGT sharply raised, ALT slightly raised, prothrombin index decreased. Urine dark, bilirubin positive, urobilin negative. Stool white-gray and fatty. CT: a tumor in the head of the pancreas.

Questions: Determine the type of jaundice and explain the laboratory signs. What is the mechanism of the itching, bradycardia and decreased prothrombin index?

Answer: Posthepatic (obstructive) jaundice — a tumor of the head of the pancreas has compressed the common bile duct. Pressure in the bile ducts rises and conjugated bilirubin and bile acids pass into the blood (cholemia): direct bilirubin predominates and the cholestatic enzymes (alkaline phosphatase, GGT) are sharply raised; direct bilirubin is water-soluble and is excreted by the kidneys — dark urine, bilirubin positive. Bile does not enter the intestine (acholia) — no urobilinogen is formed (no urobilin in the urine), and the stool is acholic and fatty (steatorrhea). Itching and bradycardia — the action of bile acids on the nerve endings of the skin, the sinus node and the vagal centers. Vitamin K is not absorbed without bile acids — synthesis of factors II, VII, IX and X decreases and the prothrombin index falls (vitamin K is given parenterally before surgery).

Case 2

A 25-year-old student developed jaundice a week after fever and weakness. Urine dark, stool pale. The liver is enlarged and tender. Blood: bilirubin 140 µmol/l (direct 80, indirect 60), ALT 1800 U/l (normal up to 40), AST 1200 U/l. Urine: bilirubin and urobilin positive. HBsAg positive.

Questions: Determine the type of jaundice. Why are both fractions raised in the blood? Explain the mechanism of urobilinuria.

Answer: Hepatic (parenchymal) jaundice — acute viral hepatitis B. Hepatocytes are damaged (cytolysis — ALT and AST sharply raised): uptake and conjugation of indirect bilirubin are impaired — the indirect fraction rises; excretion of conjugated bilirubin into the bile capillaries is impaired and direct bilirubin returns to the blood through the dead cells — the direct fraction also rises and is found in the urine. Part of the bile still enters the intestine, urobilinogen is formed and absorbed, but the damaged liver cannot capture and reprocess it — urobilinogen passes into the systemic circulation and is excreted in the urine (urobilinuria). Because less bile enters the intestine, the stool is pale.

Case 3

A 55-year-old man has drunk alcohol for 20 years and has been diagnosed with liver cirrhosis. In recent days he has had black (tarry) stools, followed by drowsiness, confusion, a “flapping” tremor when he stretches out his hands and a characteristic sweetish-musty odor from the mouth. The abdomen is enlarged (ascites), with dilated veins around the navel.

Questions: What complication has developed? Explain the precipitating factor and the pathogenesis. How did the ascites and the periumbilical veins arise?

Answer: Hepatic encephalopathy (precoma) — mixed (hepatocellular failure and portosystemic shunting). The precipitating factor is bleeding from esophageal varices (black stools): bacteria form large amounts of ammonia from blood proteins in the intestine; the cirrhotic liver cannot convert it into urea, and in addition blood bypasses the liver through portocaval anastomoses. In the brain, ammonia binds alpha-ketoglutarate and reduces energy production, and glutamine accumulates in astrocytes and causes swelling; false neurotransmitters (octopamine) and mercaptans (“fetor hepaticus”) also accumulate — impaired consciousness, asterixis. Ascites is the result of portal hypertension (increased hydrostatic pressure), hypoalbuminemia (decreased oncotic pressure), secondary hyperaldosteronism and sodium and water retention. The periumbilical veins (“caput medusae”) are dilated portocaval anastomoses.

Quick review

Table 24.5. *Key facts on Topic 24*

Question	Answer
Cytolysis syndrome	Raised ALT, AST — damage to hepatocytes
Cholestasis syndrome	Raised alkaline phosphatase, GGT, direct bilirubin
Hepatodepressive syndrome	Decreased albumin, prothrombin, cholinesterase
Hepatic coma	Accumulation of ammonia, false neurotransmitters, mercaptans
Portal hypertension	Portal vein pressure above 10–12 mm
Ascites	Portal hypertension + hypoalbuminemia + sodium retention
Liver cirrhosis	Fibrosis and regenerative nodules; failure + portal hypertension
Indirect bilirubin	Free, water-insoluble, absent from urine
Direct bilirubin	Conjugated with glucuronide, excreted in urine
Hemolytic jaundice	Indirect ↑, urobilin ↑, no bilirubin in urine
Parenchymal jaundice	Both fractions ↑, ALT ↑↑, urobilin and bilirubin present
Obstructive jaundice	Direct ↑↑, no urobilin, acholic stool, itching
Cholemia	Itching, bradycardia, hypotension

Question	Answer
Acholia	Steatorrhea, vitamin K deficiency, bleeding
Kernicterus	Accumulation of indirect bilirubin in the brain nuclei

TOPIC 25

Pathophysiology of the kidneys

Learning objectives

To explain to the student the causes and mechanisms of renal dysfunction — disturbances of glomerular ultrafiltration and of tubular reabsorption and secretion, changes in diuresis and urine composition, extrarenal signs (edema, hypertension, anemia), the nephrotic and nephritic syndromes, the pathogenesis of acute and chronic kidney failure and uremia, and the features in children; to teach the student the concept of clearance and to interpret renal function tests, especially the Zimnitsky test.

By the end of the session, the student should know:

- the functions of the nephron and the methods of examining renal function (clearance, functional tests);
- the causes and mechanisms of impaired glomerular filtration;
- disturbances of tubular reabsorption and secretion (tubulopathies);
- changes in diuresis and urine composition (polyuria, oliguria, proteinuria, hematuria, etc.);
- the pathogenesis of the nephrotic and nephritic syndromes;
- the causes and stages of acute and chronic kidney failure and the pathogenesis of uremia.

The student should be able to:

- interpret the results of the Zimnitsky test;
- calculate the glomerular filtration rate from clearance;
- interpret pathological changes in urinalysis.

1. Renal functions and their assessment

Main functions of the kidneys: excretory (excretion of nitrogenous waste, drugs and toxins), homeostatic (water–electrolyte balance, osmotic pressure, acid–base balance), endocrine (renin, erythropoietin, calcitriol — the active form of vitamin D, prostaglandins, kinins), metabolic (gluconeogenesis, breakdown of protein hormones).

Urine formation consists of three processes:

1. **Glomerular ultrafiltration** — primary urine is filtered from the capillaries into Bowman's capsule (about 180 l per day). Effective filtration pressure = capillary hydrostatic pressure – (plasma oncotic pressure + intracapsular pressure) ≈ 70 – (30 + 20) = 20 mm Hg.
2. **Tubular reabsorption** — 99 percent of the water and solutes are returned to the blood.
3. **Tubular secretion** — certain substances (H^+, K^+, ammonia, drugs) are transferred from the blood into the urine.

Clearance — the volume of plasma cleared of a given substance by the kidneys in 1 minute: C = (U × V) / P, where U is the urine concentration of the substance, V is the minute diuresis and P is the plasma concentration.

The clearance of a substance that is only filtered and neither reabsorbed nor secreted (inulin; in practice **endogenous creatinine** — the Rehberg–Tareev test) equals the **glomerular filtration rate (GFR)**.

Table 25.1. *Main parameters of renal function*

Parameter	Normal value
Glomerular filtration rate	80–120 ml/min (per 1.73 m^2)
Tubular reabsorption	97–99 percent
Daily diuresis	1–2 l (70–80 percent of fluid intake)
Urine specific gravity	1012–1025 (may vary from 1001 to 1040 over a day)
Blood creatinine	53–106 µmol/l (approximately)
Blood urea	2.5–8.3 mmol/l
Urine protein	up to 0.033 g/l (up to 150 mg per day)

2. Disturbances of glomerular filtration

Table 25.2. *Causes of changes in glomerular filtration*

Factor	Filtration decreases	Filtration increases
Capillary hydrostatic pressure	Arterial hypotension (systolic pressure below 80 mm — shock, collapse, blood loss), spasm of the afferent arteriole (sympathetic influence, catecholamines), renal artery stenosis	Constriction of the efferent arteriole, early stage of arterial hypertension
Plasma oncotic pressure	Hemoconcentration (dehydration)	Hypoproteinemia
Intracapsular pressure	Obstruction of urine outflow (stones, tumor, prostatic adenoma)	—
Filtering surface and membrane	Reduced number of functioning nephrons (chronic glomerulonephritis, nephrosclerosis), thickening of the membrane (diabetic nephropathy)	Increased membrane permeability (protein passes through)

Consequences of reduced filtration: oliguria, accumulation of nitrogenous substances (creatinine, urea, uric acid) in the blood — **azotemia**, sodium and water retention.

Increased permeability of the filtering membrane (immune damage, podocyte injury, loss of the negative charge of the membrane) — protein (**proteinuria**) and erythrocytes (**hematuria**) pass into the urine.

3. Disturbances of tubular function

Causes: ischemia, nephrotoxins (mercury, lead, aminoglycosides, radiocontrast agents), interstitial inflammation (pyelonephritis), hereditary enzymopathies and transporter protein defects, disturbed hormonal regulation (deficiency of ADH or aldosterone, or insensitivity to them).

Table 25.3. *Disturbances of tubular reabsorption (tubulopathies)*

Disorder	Mechanism	Signs
Renal glucosuria	Defect of glucose transporters	Glucosuria with normal blood glucose
Phosphate diabetes (hypophosphatemic rickets)	Reduced phosphate reabsorption	Hypophosphatemia, vitamin D-resistant rickets
Aminoacidurias (cystinuria)	Defect of amino acid transporters	Amino acids in urine, cystine stones

Disorder	Mechanism	Signs
De Toni–Debré–Fanconi syndrome	Impaired reabsorption of many substances in the proximal tubule	Glucosuria, aminoaciduria, phosphaturia, bicarbonaturia, rickets, acidosis
Renal tubular acidosis	Impaired H^+ secretion (distal) or bicarbonate reabsorption (proximal)	Hyperchloremic metabolic acidosis, high urine pH
Nephrogenic diabetes insipidus	Insensitivity of the collecting ducts to ADH	Polyuria, hyposthenuria, thirst
Pseudohypoaldosteronism	Insensitivity to aldosterone	Sodium loss, hyperkalemia

Disturbance of the urine-concentrating function:

- **hyposthenuria** — urine specific gravity constantly low (below 1012), reduced concentrating ability;
- **isosthenuria** — urine specific gravity fixed and equal to that of primary urine (plasma filtrate) (about 1010); neither concentration nor dilution — a sign of chronic kidney failure;
- **hypersthenuria** — high specific gravity (glucosuria, protein, dehydration, oliguria).

4. Changes in diuresis and urine composition

Table 25.4. *Changes in diuresis*

Term	Essence	Causes
Polyuria	Daily diuresis above 2 l	Diabetes mellitus (osmotic diuresis), diabetes insipidus (ADH deficiency), polyuric stage of chronic kidney failure, resorption of edema, diuretics
Oliguria	Daily diuresis below 500 ml	Shock, dehydration, acute kidney failure, acute glomerulonephritis, heart failure
Anuria	Daily diuresis below 50–100 ml	Acute kidney failure, complete obstruction of the urinary tract
Nocturia	Night diuresis exceeds day diuresis	Chronic kidney failure, heart failure, prostatic adenoma
Pollakiuria	Frequent urination	Cystitis, prostate diseases

Pathological changes in urine composition:

- **Proteinuria** — protein in the urine: **glomerular** (increased membrane permeability; selective — mainly albumin, nonselective — larger proteins as well), **tubular** (impaired reabsorption of low-molecular-weight proteins), **overflow** (increased small proteins in the blood — Bence Jones protein in myeloma), **postrenal** (inflammation of the urinary tract). Functional proteinuria — heavy physical exertion, fever, orthostatic.
- **Hematuria** — erythrocytes in the urine: glomerular (altered, dysmorphic erythrocytes — glomerulonephritis) and non-glomerular (unaltered erythrocytes — stones, tumors, trauma).
- **Leukocyturia** — increased leukocytes in the urine (pyelonephritis, cystitis); **pyuria** — a large amount of pus.
- **Cylindruria** — “casts” formed in the tubules from protein and cells: hyaline, granular, waxy, erythrocyte (glomerulonephritis), leukocyte (pyelonephritis).
- **Glucosuria** — when blood glucose exceeds the renal threshold (about 8.9–10 mmol/l) or in a tubular defect.

5. Extrarenal signs

Renal edema

- **Nephrotic edema** — massive proteinuria → hypoalbuminemia → oncotic pressure falls → fluid moves into the tissues → circulating blood volume decreases → RAAS and ADH are activated → sodium and water are retained. The edema is large and soft, starting on the face and progressing to anasarca.
- **Nephritic edema** — reduced filtration and primary retention of sodium and water (hypervolemia), increased capillary permeability. The edema is on the face and eyelids, more in the morning.

Renal hypertension

- **Renovascular** — renal ischemia → increased renin → angiotensin II (vasoconstriction) and aldosterone (sodium and water retention).
- **Renoparenchymal** — sodium and water retention (hypervolemia), RAAS activation, reduction of the renal depressor systems (prostaglandins, kinins).

Renal anemia

Reduced **erythropoietin** production as renal parenchyma decreases; additional factors — the effect of uremic toxins on the bone marrow, shortened erythrocyte lifespan, bleeding.

Renal osteodystrophy

Reduced calcitriol synthesis (hypocalcemia) and phosphate retention (hyperphosphatemia) → increased parathyroid hormone (**secondary hyperparathyroidism**) → release of calcium from bones, osteomalacia, osteoporosis.

6. Nephritic and nephrotic syndromes

Table 25.5. *Distinguishing the nephritic and nephrotic syndromes*

Feature	Nephritic syndrome	Nephrotic syndrome
Main mechanism	Immune inflammation of the glomeruli, damage to the capillaries	Damage to the filtering membrane (podocytes), sharp increase in protein permeability
Proteinuria	Moderate (below 3 g/day)	Massive (above 3.5 g/day)
Hematuria	Marked (“meat-washings”-colored urine)	Absent or few
Edema	On the face and eyelids, moderate	Large, anasarca
Arterial pressure	Raised	Usually normal
Oliguria, azotemia	Often	Usually absent
Blood	—	Hypoalbuminemia, hyperlipidemia, hypercoagulation
Example	Acute poststreptococcal glomerulonephritis	Minimal change disease (in children), membranous nephropathy, diabetic nephropathy, amyloidosis

Pathogenesis of acute poststreptococcal glomerulonephritis: 1–3 weeks after infection with group A beta-hemolytic streptococcus (tonsillitis, pyoderma), **immune complexes** formed from

streptococcal antigens and antibodies are deposited on the glomerular basement membrane (a type III allergic reaction) → complement is activated (C3 decreases) and neutrophils accumulate → the capillaries are damaged and inflamed → filtration falls (oliguria, azotemia), erythrocytes and protein pass into the urine, and sodium and water are retained (edema, hypertension).

Hyperlipidemia in nephrotic syndrome — increased hepatic synthesis of lipoproteins in response to hypoalbuminemia and reduced lipoprotein lipase activity; **hypercoagulation** — urinary loss of antithrombin III and increased hepatic synthesis of clotting factors (risk of thrombosis).

7. Kidney failure

Acute kidney failure (acute kidney injury)

Acute kidney failure — a sudden (within hours or days) sharp decline in renal function, characterized by a rise in blood creatinine and (or) a decrease in diuresis.



Figure 25.1. *Types of acute kidney failure*

Stages:

1. **Initial** — action of the cause (shock, poisoning), hours or days.
2. **Oligoanuric** (1–3 weeks) — oliguria or anuria; **azotemia**, **hyperkalemia** (risk of arrhythmia and cardiac arrest — the most dangerous complication), metabolic acidosis, hyperhydration (pulmonary and cerebral edema), hyperphosphatemia, hypocalcemia, uremic intoxication.
3. **Polyuric (recovery of diuresis)** — filtration recovers, but the tubular epithelium is still immature, reabsorption and concentration are low — polyuria, hyposthenuria; risk of dehydration, **hypokalemia** and hyponatremia.
4. **Recovery** — from several months to a year.

> **CAUTION**
> **Mechanisms of oliguria in acute tubular necrosis:** spasm of the afferent arterioles (RAAS, tubuloglomerular feedback — the macula densa senses an increased sodium load and reduces filtration), obstruction of the tubules by dead cells and casts (intratubular pressure rises and filtration falls), back-leak of filtrate through the damaged tubular wall into the blood, interstitial edema.

Chronic kidney failure (chronic kidney disease)

Chronic kidney failure — a progressive decline in renal function resulting from a gradual, irreversible reduction in the number of functioning nephrons (GFR below 60 ml/min or signs of kidney damage for more than 3 months).

Causes: diabetes mellitus (diabetic nephropathy), arterial hypertension (nephrosclerosis), chronic glomerulonephritis, chronic pyelonephritis, polycystic disease, urinary tract obstruction, amyloidosis.

Pathogenesis: as nephrons are lost, the remaining nephrons work in a hyperfiltration mode (**adaptive hyperfiltration**) — glomerular pressure rises and they gradually become sclerotic (**progressive nephrosclerosis**) — a “vicious circle”. Proteinuria and angiotensin II enhance interstitial fibrosis.

Stages (by GFR): latent (GFR 90–60) → compensated → polyuria, nocturia, isosthenuria (GFR 60–30; osmotic diuresis in the surviving nephrons) → increasing azotemia (GFR 30–15) → **terminal (uremic)** stage (GFR below 15) — oliguria; hemodialysis or kidney transplantation is required.

Pathogenesis of uremia

Uremia — poisoning of the body by metabolic products in end-stage kidney failure:

- **uremic toxins** — urea, creatinine, guanidines, phenols, indole, “middle molecules”, parathyroid hormone (one of the most important “uremic toxins”);
- **nervous system** — uremic encephalopathy (weakness, drowsiness, coma), polyneuropathy;
- **cardiovascular** — hypertension, uremic pericarditis (pericardial friction rub), heart failure;
- **respiratory** — Kussmaul breathing (acidosis), uremic pneumonitis, ammonia odor from the mouth;
- **digestive** — uremic gastroenteritis (urea is broken down to ammonia in the intestine and irritates the mucosa), nausea, vomiting;
- **blood** — anemia, platelet dysfunction (bleeding), immunodeficiency;
- **skin** — dryness, itching, “uremic frost” (urea crystals excreted through the skin);
- **bone** — renal osteodystrophy;
- **water–electrolyte** — hyperkalemia, metabolic acidosis, hyperphosphatemia, hypocalcemia.

8. Features of renal disorders in children

- In infants GFR and concentrating ability are low (they mature by 1–2 years) — they are sensitive to dehydration and to excessive fluid administration.
- Tubular bicarbonate reabsorption and H^+ secretion are low — a tendency to metabolic acidosis.
- Congenital defects (urinary tract anomalies, vesicoureteral reflux) are the main cause of pyelonephritis.
- The main cause of nephrotic syndrome in children is **minimal change disease** (responds well to glucocorticoids).
- Hereditary tubulopathies (Fanconi syndrome, phosphate diabetes, renal tubular acidosis) often present in childhood with rickets-like changes and growth retardation.
- A frequent cause of acute kidney failure in children is **hemolytic uremic syndrome** (after diarrhea associated with Shiga toxin-producing Escherichia coli).

9. Renal function tests. The Zimnitsky test

Zimnitsky test — a method of assessing the **concentrating and diluting** ability of the kidneys under an ordinary drinking regimen.

Procedure:

1. The patient follows the usual diet and drinking regimen (diuretics are stopped); the fluid intake is recorded.
2. At 6 a.m. the bladder is emptied (this urine is discarded).

3. Then urine is collected every 3 hours into separate containers: at 9, 12, 15, 18, 21, 24, 3 and 6 o'clock — 8 portions in total.
4. The volume and specific gravity (with a urometer) of each portion are determined.
5. Day (9–18) and night (21–6) diuresis and daily diuresis are calculated.

Normally:

- daily diuresis is 70–80 percent of the fluid intake;
- day diuresis is about 2/3 of the daily total, night diuresis — 1/3;
- portion volumes vary within 40–350 ml;
- the difference between the highest and lowest specific gravity is 8–10 or more (e.g. 1008–1025), and the maximum specific gravity is above 1020.

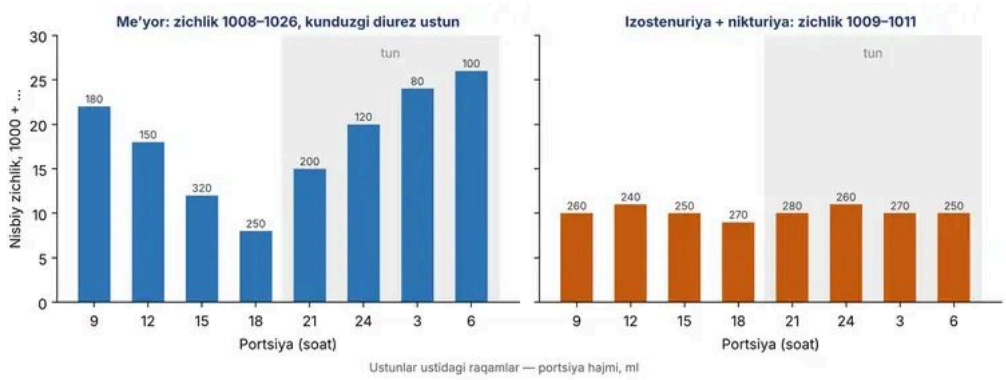


Figure 25.2. *Zimnitsky test: normal and isosthenuria (sample data)*

Table 25.6. *Interpreting the results of the Zimnitsky test*

Result	Meaning
The maximum specific gravity in all portions does not exceed 1012–1015	Hyposthenuria — reduced concentrating ability
Specific gravity almost the same (1009–1011), variation of 1–2 units	Isosthenuria — concentrating and diluting ability lost (chronic kidney failure)
Night diuresis exceeds day diuresis	Nocturia
Daily diuresis above 2 l	Polyuria
Specific gravity constantly high, diuresis low	Hypersthenuria with oliguria (dehydration, acute glomerulonephritis, heart failure)
Specific gravity high, diuresis high	Glucosuria (diabetes mellitus)
Specific gravity very low (1001–1005), diuresis very high	Diabetes insipidus

REMEMBER

Other functional tests: the **Rehberg–Tareev test** (determining GFR and tubular reabsorption from endogenous creatinine clearance), the **Nechiporenko test** (number of leukocytes, erythrocytes and casts in 1 ml of urine), the concentration test (dry diet) and the dilution test (water load). In modern practice GFR is usually calculated from blood creatinine using formulas (CKD-EPI).

10. Practical skills

Skill: interpreting the results of the Zimnitsky test

1. Calculate the daily diuresis and compare it with the fluid intake.
2. Calculate day and night diuresis and determine their ratio.
3. Find the highest and lowest specific gravity among the portions.
4. If the maximum specific gravity is below 1020 — concentrating ability is reduced (hyposthenuria).
5. If the difference in specific gravity is 1–2 units and the values are around 1010 — isosthenuria.
6. Write a conclusion: polyuria or oliguria, nocturia, hypo- or isosthenuria.

Skill: calculating creatinine clearance

1. Determine the minute diuresis: V = daily urine (ml) / 1440.
2. Take the urine (U) and blood (P) creatinine concentrations in the same units.
3. Calculate GFR = (U × V) / P.
4. Compare the result with the normal value (80–120 ml/min) and determine the stage of kidney failure.

Review questions

1. List the main functions of the kidneys.
2. What is clearance? How is the glomerular filtration rate determined?
3. Explain the factors that decrease and increase glomerular filtration.
4. Give examples of disturbances of tubular reabsorption.
5. What are hyposthenuria, isosthenuria and hypersthenuria?
6. What are the causes of polyuria, oliguria, anuria and nocturia?
7. Explain the types and mechanisms of proteinuria.
8. How does the pathogenesis of nephrotic and nephritic edema differ?
9. What are the mechanisms of renal hypertension and renal anemia?
10. Compare the nephritic and nephrotic syndromes.
11. Explain the pathogenesis of acute poststreptococcal glomerulonephritis.
12. What are the types and stages of acute kidney failure?
13. What are the pathogenesis and stages of chronic kidney failure?
14. What are the mechanisms of damage to organs and systems in uremia?
15. How is the Zimnitsky test performed and interpreted?

Test questions

1. What does endogenous creatinine clearance show?
   A) Tubular secretion
   B) Renal blood flow
   C) Glomerular filtration rate
   D) Urine-concentrating ability
   E) Erythropoietin production

2. Which condition reduces glomerular filtration?
   A) Hypoproteinemia
   B) Constriction of the efferent arteriole
   C) Increased membrane permeability
   D) Early stage of arterial hypertension
   E) A fall in arterial pressure below 80 mm Hg

3. What is isosthenuria?
   A) Constantly high urine specific gravity
   B) Night diuresis exceeding day diuresis
   C) Daily diuresis above 2 l
   D) A fixed urine specific gravity equal to that of primary urine (about 1010)
   E) Protein in the urine

4. What is the main sign of nephrotic syndrome?
   A) Marked hematuria and hypertension
   B) Massive proteinuria (above 3.5 g/day), hypoalbuminemia, large edema
   C) Leukocyturia only
   D) Polyuria
   E) Glucosuria

5. What is the initial mechanism of nephrotic edema?
   A) A fall in plasma oncotic pressure due to hypoalbuminemia
   B) Increased capillary hydrostatic pressure
   C) Impaired lymph flow
   D) Venous congestion
   E) Histamine release

6. To which type of allergic reaction does acute poststreptococcal glomerulonephritis belong?
   A) Type I (anaphylactic)
   B) Type III (immune complex)
   C) Type II (cytotoxic)
   D) Type IV (cellular)
   E) It is not an allergic reaction

7. What is the most dangerous disturbance in the oligoanuric stage of acute kidney failure?
   A) Hypokalemia
   B) Hypernatremia
   C) Hyperkalemia
   D) Hypoglycemia
   E) Hypocalcemia

8. What danger is there in the polyuric stage of acute kidney failure?
   A) Hyperkalemia
   B) Hyperhydration
   C) Oliguria
   D) Dehydration and hypokalemia
   E) Hypertension

9. What is the main mechanism of renal anemia?
   A) Reduced erythropoietin production
   B) Iron loss
   C) B_{12} deficiency
   D) Hereditary impairment of hemoglobin synthesis
   E) Splenic enlargement

10. What is the cause of secondary hyperparathyroidism in chronic kidney failure?
   A) Hypercalcemia
   B) Increased calcitonin
   C) Hypophosphatemia
   D) Increased vitamin D
   E) Hypocalcemia due to decreased calcitriol and hyperphosphatemia

11. In the Zimnitsky test, what proportion of the daily diuresis does day diuresis normally make up?
   A) About 2/3
   B) About 1/3
   C) Half
   D) 1/10
   E) Almost all of it

12. In the Zimnitsky test, if the specific gravity is 1009–1011 in all portions, what does this indicate?
   A) Normal
   B) Diabetes mellitus
   C) Isosthenuria — loss of concentrating and diluting ability
   D) Hypersthenuria
   E) Dehydration

13. What is the significance of “adaptive hyperfiltration” in the progression of chronic kidney failure?
   A) It completely restores renal function
   B) It reduces proteinuria
   C) It lowers hypertension
   D) It increases erythropoietin
   E) Pressure rises in the remaining nephrons, leading to their sclerosis

14. What is the cause of Kussmaul breathing in uremia?
   A) Hypokalemia
   B) Anemia
   C) Respiratory alkalosis
   D) Metabolic acidosis
   E) Hypoglycemia

15. Which is a sign of glomerular hematuria?
   A) Unaltered erythrocytes and blood clots
   B) Altered (dysmorphic) erythrocytes and erythrocyte casts
   C) Leukocytes only
   D) Bacteria
   E) Glucose

Clinical cases

Case 1

2 weeks after tonsillitis, a 9-year-old child's face and eyelids are swollen, urine output has decreased and the urine is “meat-washings”-colored. BP 145/95 mm Hg. Urine: protein 1.5 g/l, altered erythrocytes covering the whole field of view, erythrocyte casts. Blood: creatinine slightly raised, high ASO titer, complement C3 decreased.

Questions: What syndrome and disease is this? Explain the pathogenesis. What is the mechanism of the edema and hypertension?

Answer: Nephritic syndrome — acute poststreptococcal glomerulonephritis. Group A streptococcal antigens and antibodies against them form immune complexes that are deposited on the glomerular basement membrane (a type III allergic reaction); complement is activated (C3 is consumed and decreases) and neutrophils accumulate and damage the capillaries. Membrane permeability increases and erythrocytes (altered, with casts — glomerular hematuria) and protein pass into the urine. Filtration falls in the inflamed glomeruli — oliguria, azotemia; because tubular sodium reabsorption is preserved, sodium and water are retained — hypervolemia: edema (on the face and eyelids — loose tissue) and hypertension (increased circulating blood volume, RAAS activation).

Case 2

A 56-year-old woman has had diabetes mellitus for 20 years. Weakness, loss of appetite, nausea, skin itching, frequent urination at night. The skin is pale-yellowish and dry. BP 170/100. Blood: Hb 82 g/l, creatinine 520 µmol/l, urea 28 mmol/l, K^+ 5.8 mmol/l, pH 7.28, phosphates raised, calcium decreased, parathyroid hormone raised. Zimnitsky test: specific gravity 1008–1011, night diuresis exceeds day diuresis.

Questions: What condition has developed? Interpret the laboratory values. What is the mechanism of the anemia and bone disorders?

Answer: Chronic kidney failure resulting from diabetic nephropathy, uremic stage. The number of functioning nephrons is sharply reduced — azotemia (raised creatinine and urea), hyperkalemia, metabolic acidosis (reduced excretion of H^+ and ammonium), hyperphosphatemia. Isosthenuria and nocturia in the Zimnitsky test — the remaining nephrons work in an osmotic diuresis mode and cannot concentrate the urine. Anemia — reduced erythropoietin production, the effect of uremic toxins on erythropoiesis and shortened erythrocyte lifespan. Calcitriol synthesis decreases and calcium absorption falls, phosphates are retained — hypocalcemia stimulates parathyroid hormone (secondary hyperparathyroidism) — renal osteodystrophy. Nausea and itching — the effect of uremic toxins. Hypertension — sodium and water retention, RAAS activation. The patient needs hemodialysis or kidney transplantation.

Case 3

A 35-year-old man's legs were trapped for a long time in a car accident. After rescue his BP fell, his urine became brown-red, and over the following days daily diuresis was 150 ml. Blood: creatinine 480 µmol/l, K^+ 6.9 mmol/l, CK sharply raised, myoglobinuria. ECG: tall peaked T waves.

Questions: What condition has developed? Explain the mechanism of the acute kidney failure. Which disturbance is the most life-threatening?

Answer: Acute kidney failure (renal, acute tubular necrosis) due to crush syndrome, oligoanuric stage. Myoglobin, potassium, phosphates and toxins pass from the damaged muscles into the blood

(rhabdomyolysis — raised CK). Myoglobin precipitates in the tubules in an acidic environment, obstructs them and has a direct toxic effect on the epithelium; hypovolemia and shock aggravate renal ischemia (spasm of the afferent arterioles). As a result filtration falls sharply — anuria, azotemia. The most dangerous is hyperkalemia (release of potassium from muscles and failure of renal excretion): myocardial excitability and conduction are disturbed, there are tall peaked T waves on the ECG and the risk of cardiac arrest is high. Emergency measures: calcium preparations, glucose–insulin, hemodialysis.

Quick review

Table 25.7. *Key facts on Topic 25*

Question	Answer
Clearance	The volume of plasma cleared of a substance in 1 minute; C = U × V / P
Normal GFR	80–120 ml/min
Reduced filtration	Hypotension, afferent arteriole spasm, obstruction, loss of nephrons
Tubulopathies	Glucosuria, phosphate diabetes, Fanconi syndrome, tubular acidosis
Isosthenuria	Specific gravity fixed at about 1010; chronic kidney failure
Nocturia	Night diuresis exceeds day diuresis
Nephritic syndrome	Hematuria, hypertension, oliguria, moderate proteinuria
Nephrotic syndrome	Massive proteinuria, hypoalbuminemia, edema, hyperlipidemia
Renal anemia	Decreased erythropoietin
Renal osteodystrophy	Decreased calcitriol, secondary hyperparathyroidism
Acute kidney failure	Prerenal, renal, postrenal; hyperkalemia is the most dangerous
Polyuric stage	Risk of dehydration and hypokalemia
Chronic kidney failure	Loss of nephrons, hyperfiltration, nephrosclerosis
Uremia	Toxins: encephalopathy, pericarditis, gastroenteritis, itching
Zimnitsky test	8 portions; concentrating and diluting ability

TOPIC 26

Pathophysiology of the endocrine system

Learning objectives

To explain to the student the structure and regulatory hierarchy of the endocrine system, the mechanisms of hormone action, the general etiology and pathogenesis of endocrine disorders (central, glandular and extraglandular mechanisms), the main forms of dysfunction of the hypothalamic–pituitary system, the adrenal glands, the thyroid and parathyroid glands and the gonads, and the methods of examining the endocrine system, including the Galli Mainini test.

By the end of the session, the student should know:

- the hierarchy of endocrine regulation and the feedback principle;
- the mechanisms of hormone action and the types of receptors;
- the central, glandular and peripheral mechanisms of endocrine disorders;
- the main forms of hypothalamic–pituitary disorders (gigantism, acromegaly, pituitary dwarfism, panhypopituitarism, diabetes insipidus, Cushing disease);
- hyper- and hypofunction of the adrenal, thyroid and parathyroid glands;
- the methods of examining the endocrine system and the essence of the Galli Mainini test.

The student should be able to:

- determine the level of the disorder (primary or secondary) from hormone levels and functional tests;
- relate the clinical signs of endocrinopathies to their pathogenesis.

1. Hierarchy of endocrine regulation

The endocrine system is organized **hierarchically**: cerebral cortex and limbic system → **hypothalamus** (liberins and statins) → **pituitary** (tropic hormones) → **peripheral glands** (adrenal cortex, thyroid, gonads) → target tissues. Glands that are independent of the pituitary (pancreatic islets, parathyroid glands, adrenal medulla) are regulated mainly by the controlled variable itself (glucose, calcium) or by the nervous system.

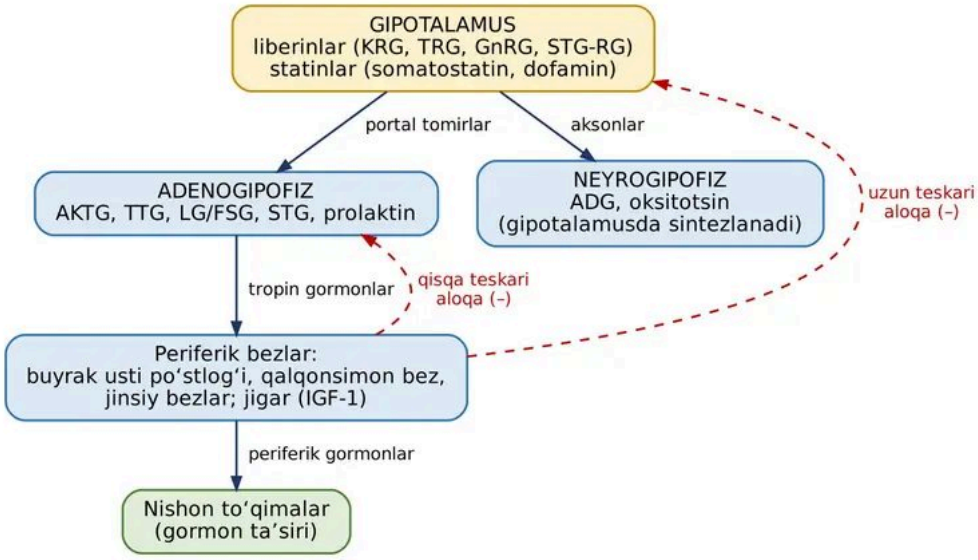


Figure 26.1. *The hypothalamic–pituitary system and feedback*

Feedback principle: when the level of a peripheral hormone rises, it inhibits the hypothalamus and pituitary (**negative feedback**) — the hormone level is kept normal. In some cases positive feedback operates (in the menstrual cycle the estrogen peak triggers a surge of LH release).

> **CAUTION**
> The feedback principle is the key to determining the level of a disorder. In **primary** (peripheral gland) insufficiency the peripheral hormone is low and the tropic hormone is **high** (the inhibition is lost). In **secondary** (pituitary) insufficiency both are **low**. For example: in primary hypothyroidism T_4 ↓, TSH ↑; in secondary hypothyroidism T_4 ↓, TSH ↓.

2. Mechanisms of hormone action

Table 26.1. *Hormone receptors and mechanisms of action*

Receptor location	Hormones	Mechanism
Cell membrane (G protein-coupled)	Epinephrine (beta), glucagon, ACTH, TSH, LH, FSH, PTH, ADH (V_2)	Adenylate cyclase → cAMP → protein kinase A
Cell membrane (Gq protein)	Epinephrine (alpha1), angiotensin II, ADH (V_1), oxytocin	Phospholipase C → IP_3 and DAG → Ca^{2+}, protein kinase C
Cell membrane (tyrosine kinase)	Insulin, IGF-1	Receptor autophosphorylation, intracellular signaling cascades
Cell membrane (JAK–STAT)	Growth hormone, prolactin, erythropoietin	Activation of Janus kinases
Intracellular (cytoplasm and nucleus)	Steroid hormones (cortisol, aldosterone, sex hormones), thyroid hormones, calcitriol	The hormone–receptor complex binds to DNA and alters gene transcription

3. General pathogenesis of endocrine disorders

Endocrine disorders manifest as **hyperfunction** (excess hormone action), **hypofunction** (deficiency) and **dysfunction** (imbalance between different hormones).



Figure 26.2. *Main mechanisms of endocrine disorders*

Table 26.2. *Mechanisms of endocrine disorders*

Degree	Mechanisms	Examples
Central (hypothalamus, pituitary)	Altered secretion of liberins and tropic hormones: tumors, trauma, hemorrhage, infections, autoimmune processes, nervous system disorders (stress)	Cushing disease, acromegaly, pituitary dwarfism, Sheehan syndrome
In the gland itself (primary)	Destruction of glandular tissue (autoimmune, infection, hemorrhage), tumor (adenoma — hyperfunction), hereditary defect of hormone-synthesizing enzymes, lack of raw material (iodine)	Addison disease, Graves (Basedow) disease, endemic goiter, Conn syndrome, adrenogenital syndrome

Degree	Mechanisms	Examples
Extraglandular (peripheral)	Altered binding of the hormone to transport proteins; impaired inactivation of the hormone in the liver and kidney; impaired conversion to the active form ($T_4 \rightarrow T_3$); antibodies against the hormone; receptor defects or resistance; ectopic hormone production (tumors)	Type 2 diabetes mellitus (insulin resistance), nephrogenic diabetes insipidus, androgen insensitivity syndrome, hyperestrogenemia in liver cirrhosis, ectopic ACTH in lung cancer

4. Disorders of the hypothalamic–pituitary system

Hyperfunction of the adenohypophysis

- **Excess somatotropin (growth hormone)** (usually a somatotroph adenoma) before closure of the bone growth plates (childhood and adolescence) — **gigantism** (height above 200 cm in men, above 190 cm in women).
- **Excess growth hormone after closure of the growth plates** — **acromegaly**: the hands, feet, nose, lower jaw, ears and tongue enlarge, the internal organs enlarge (splanchnomegaly), and the facial features coarsen; counter-insulin effect — **hyperglycemia** (secondary diabetes), arterial hypertension, cardiomyopathy. The effects are mediated mainly by **IGF-1**, produced in the liver.
- **Excess ACTH — Cushing disease** (pituitary adenoma): secondary hypercortisolism.
- **Excess prolactin** (prolactinoma) — in women galactorrhea, amenorrhea, infertility; in men impotence, gynecomastia (prolactin inhibits GnRH).

Hypofunction of the adenohypophysis

- **Pituitary dwarfism** — growth hormone deficiency in childhood: marked growth retardation, but body proportions are preserved and mental development is normal (unlike hypothyroid dwarfism); sexual development is also often delayed.
- **Panhypopituitarism** — deficiency of all hormones of the adenohypophysis: tumors, trauma, surgery, radiation, postpartum pituitary necrosis (**Sheehan syndrome** — ischemia of the pituitary, enlarged during pregnancy, due to heavy blood loss at delivery), **Simmonds disease** (pituitary cachexia). Signs: secondary hypothyroidism, secondary adrenal insufficiency, hypogonadism (amenorrhea, loss of libido, loss of body hair), failure of lactation, weakness, hypoglycemia, hypotension.

Disorders of the neurohypophysis

- **Diabetes insipidus** — ADH deficiency (**central**: tumors of the hypothalamus and pituitary, trauma, infections, surgery) or insensitivity of the kidney to ADH (**nephrogenic**). Water reabsorption in the collecting ducts decreases — **polyuria** (5–20 l per day), very low urine specific gravity (1001–1005), intense thirst (polydipsia), risk of dehydration and hypernatremia.
- **Syndrome of inappropriate ADH secretion** (Parhon syndrome) — tumors (small cell lung cancer — ectopic ADH), CNS diseases, lung diseases, drugs: water is retained — **water intoxication**, dilutional **hyponatremia**, cerebral edema; the urine specific gravity is high, and there is usually no edema.

5. Disorders of the adrenal glands

Hyperfunction of the adrenal cortex

Hypercortisolism (Cushing syndrome) — excess glucocorticoids.

Table 26.3. *Types of hypercortisolism*

Type	Cause	ACTH	Cortisol
Cushing disease (central)	Pituitary corticotroph adenoma	↑	↑
Cushing syndrome (primary)	Adrenocortical tumor (corticosteroma)	↓	↑
Ectopic ACTH syndrome	Lung cancer and other tumors	↑↑	↑↑
Iatrogenic	Long-term glucocorticoid therapy	↓	Exogenous ↑

Signs and mechanisms:

- **centripetal obesity** (“moon face”, fat on the back of the neck — “buffalo hump”, abdominal obesity, thin limbs) — different sensitivity of adipose tissue in different regions to cortisol and insulin;
- **protein catabolism** — muscle atrophy and weakness, thinning of the skin, purple-red striae (stretch marks), osteoporosis (pathological fractures), slow wound healing;
- **steroid diabetes** — increased gluconeogenesis and insulin resistance;
- **arterial hypertension** — mineralocorticoid effect, increased angiotensinogen, increased vascular sensitivity to catecholamines;
- **immunosuppression** — lymphopenia, eosinopenia, susceptibility to infections;
- mental changes; in women, hirsutism and menstrual disorders (increased androgens).

Primary hyperaldosteronism (Conn syndrome) — an aldosterone-producing adenoma: sodium and water retention → **hypertension**; potassium loss → **hypokalemia** (muscle weakness, arrhythmias, polyuria), metabolic alkalosis; renin is **decreased** (the difference from secondary hyperaldosteronism, in which renin is increased).

Adrenogenital syndrome — a hereditary defect of cortisol-synthesizing enzymes (most often 21-hydroxylase): cortisol decreases → ACTH increases → hyperplasia of the adrenal cortex and excess androgen production → virilization of the external genitalia in girls, precocious puberty in boys; in the severe form aldosterone is also low — salt-wasting syndrome.

Hypofunction of the adrenal cortex

Primary chronic insufficiency (Addison disease) — destruction of more than 90 percent of the adrenal cortex: autoimmune adrenalitis (the main cause), tuberculosis, metastases, hemorrhage.

- glucocorticoid deficiency — weakness, weight loss, **hypoglycemia**, hypotension, intolerance of stress;
- mineralocorticoid deficiency — loss of sodium and water (hypovolemia, hypotension), **hyperkalemia**, craving for salty food;
- **hyperpigmentation of the skin and mucous membranes** (“bronze disease”) — the fall in cortisol, through feedback, leads to an increase in **ACTH and melanocyte-stimulating hormone** (whose common precursor is proopiomelanocortin). In secondary (pituitary) insufficiency there is no hyperpigmentation.

Acute adrenal insufficiency (Addisonian crisis) — Waterhouse–Friderichsen syndrome (adrenal hemorrhage in meningococcal sepsis), sudden withdrawal of glucocorticoids, stress in Addison disease: collapse, hypoglycemia, hyperkalemia, dehydration — life-threatening.

Hyperfunction of the adrenal medulla

Pheochromocytoma — a tumor of the chromaffin cells with excess catecholamines: paroxysmal (crisis) or persistent **arterial hypertension**, tachycardia, headache, sweating, pallor, hyperglycemia, weight loss.

6. Disorders of the thyroid gland

Table 26.4. *Comparison of hyperthyroidism and hypothyroidism*

Feature	Hyperthyroidism (thyrotoxicosis)	Hypothyroidism
Main causes	Diffuse toxic goiter (Graves–Basedow disease — autoantibodies that stimulate TSH receptors), toxic adenoma, early phase of thyroiditis	Autoimmune thyroiditis (Hashimoto), iodine deficiency, thyroidectomy, radioiodine therapy; secondary — pituitary
Basal metabolism	↑ (uncoupling of oxidation and phosphorylation, increased Na^+/K^+-ATPase activity)	↓
Body weight	Decreases (even with a good appetite)	Increases
Temperature	Heat intolerance, sweating	Cold intolerance, dry and cold skin
Heart	Tachycardia, atrial fibrillation, systolic hypertension (increased sensitivity of beta-receptors)	Bradycardia, pericardial fluid
Nervous system	Excitability, hand tremor, insomnia	Sluggishness, drowsiness, memory impairment
Digestion	Diarrhea	Constipation
Lipids	Cholesterol decreased	Hypercholesterolemia, atherosclerosis
Characteristic signs	Exophthalmos (autoimmune ophthalmopathy in Graves disease)	Mucous edema (myxedema) — accumulation of glycosaminoglycans under the skin
Laboratory (primary form)	T_3, T_4 ↑, TSH ↓	T_4 ↓, TSH ↑

Congenital hypothyroidism in children — cretinism: thyroid hormones are essential for brain development, so if it is not detected early — severe mental retardation, dwarfism (with disproportionate body), delayed teething, a large tongue and prolonged jaundice. This is why **neonatal screening** (TSH) is performed in all newborns.

Endemic goiter — iodine deficiency (some regions of Uzbekistan are also iodine-deficient areas): thyroid hormone synthesis decreases → TSH increases → the glandular tissue undergoes compensatory hyperplasia (goiter). Prevention — iodized salt.

7. Disorders of the parathyroid glands

- **Hyperparathyroidism** — excess parathyroid hormone: primary (adenoma), secondary (chronic kidney failure, vitamin D deficiency — in response to hypocalcemia). In the primary form: **hypercalcemia**, hypophosphatemia, release of calcium from bones (osteoporosis, osteitis fibrosa, fractures), kidney stones, peptic ulcer, muscle weakness, polyuria. To remember: "**bones, stones, abdominal groans and psychic moans**".

- **Hypoparathyroidism** — most often damage to the parathyroid glands during thyroid surgery: **hypocalcemia** → increased neuromuscular excitability → **tetany** (cramps, “obstetrician's hand”, laryngospasm), positive Chvostek and Trousseau signs, QT prolongation.

8. Disorders of the gonads

- **Hypogonadism** — primary (damage to the gonads: Klinefelter syndrome 47,XXY; Shereshevsky–Turner syndrome 45,X0; orchitis, trauma, radiation — LH and FSH increased) and secondary (damage to the hypothalamus–pituitary — LH and FSH decreased).
- Androgen deficiency before puberty — **eunuchoidism** (long limbs, undeveloped secondary sexual characteristics).
- **Precocious puberty** — true (hypothalamic tumors, early activation of GnRH) and pseudo- (adrenal and gonadal tumors, adrenogenital syndrome).

9. Methods of examining the endocrine system

- **Determining hormone levels** — radioimmune, enzyme immunoassay and chemiluminescent methods (in blood, urine, saliva).
- **Stimulation tests** — when hypofunction is suspected (ACTH test — adrenal reserve; insulin hypoglycemia test — growth hormone and ACTH reserve).
- **Suppression tests** — when hyperfunction is suspected (dexamethasone test — distinguishing Cushing disease from Cushing syndrome; suppression of growth hormone by a glucose load — acromegaly).
- **Glucose tolerance test** (topic 10).
- **Imaging** — US, CT, MRI (pituitary adenoma), scintigraphy (thyroid gland).
- **Detecting autoantibodies** (to thyroid peroxidase, TSH receptors, the adrenal cortex).
- **Biological tests** — assessing hormone activity in animals (historical methods; e.g. the Galli Mainini test).
- **Experimental methods** — removal of the gland (extirpation), administration of a gland extract or hormone, transplantation of glandular tissue.

The Galli Mainini test

The Galli Mainini test (1947) — a biological method for the early detection of pregnancy. The urine of a pregnant woman contains **chorionic gonadotropin** (produced by the placenta); it acts on the germ cells of a male frog (or toad) and stimulates the **release of spermatozoa**.

Procedure:

1. Examine the fluid from the cloaca of a male frog (toad) under the microscope and confirm that it contains no spermatozoa.
2. Inject a few milliliters of the morning urine of the woman being tested into the dorsal lymph sac of the frog.
3. After 1–2 hours, take fluid from the cloaca with a pipette and examine it under the microscope.
4. **If spermatozoa are found, the test is positive** (chorionic gonadotropin is present in the urine — pregnancy is likely); if not — negative.

Diagnostic significance: chorionic gonadotropin rises sharply in pregnancy, as well as in **trophoblastic tumors** (hydatidiform mole, choriocarcinoma) and some testicular tumors. The test is of historical importance; today it has been replaced by **immunological detection of β-chorionic**

gonadotropin in blood and urine (test strips, enzyme immunoassay) — these are more sensitive and faster.

> **REMEMBER**
> The exact procedure of the Galli Mainini test (animal species, urine volume, observation time) should be checked against the department's methodological manual. Other experiments demonstrated in class: comparing basal metabolism and resistance to an oxygen-free environment in animals given thyroid preparations; observing hypoglycemic convulsions after insulin administration (topic 10).

10. Practical skills

Skill: determining the level of an endocrine disorder by the feedback principle

1. Assess the level of the peripheral hormone (increased or decreased).
2. Assess the level of the corresponding tropic hormone.
3. Peripheral hormone low, tropic hormone high — primary (glandular) insufficiency.
4. Peripheral hormone and tropic hormone both low — secondary (pituitary or hypothalamic) insufficiency.
5. Peripheral hormone high, tropic hormone low — primary hyperfunction (gland tumor or autoimmune stimulation).
6. Both high — central (pituitary) hyperfunction or ectopic secretion.

Skill: interpreting the result of the Galli Mainini test

1. Before injecting the urine, confirm that there are no spermatozoa in the cloacal fluid.
2. After the specified time, examine the cloacal fluid under the microscope.
3. Spermatozoa present — positive result; absent — negative.
4. Compare the result with the clinical data (pregnancy, suspected trophoblastic tumor).
5. Write down the limitations of the method (possibility of false-positive and false-negative results) and modern alternatives.

Review questions

1. Explain the hierarchy of endocrine regulation and the feedback principle.
2. What hormone receptors and mechanisms of action exist?
3. Explain the central, glandular and peripheral mechanisms of endocrine disorders with examples.
4. How do primary and secondary endocrine insufficiency differ in laboratory terms?
5. How do gigantism and acromegaly develop? What is the cause of hyperglycemia in acromegaly?
6. How do pituitary dwarfism and hypothyroid dwarfism differ?
7. What are the causes and signs of panhypopituitarism? What is Sheehan syndrome?
8. What are the mechanisms of diabetes insipidus and the syndrome of inappropriate ADH secretion?
9. Explain the types of hypercortisolism and the pathogenesis of its signs.
10. What is the pathogenesis of Conn syndrome and adrenogenital syndrome?
11. How do hyperpigmentation and hyperkalemia develop in Addison disease?
12. Compare the causes and signs of hyperthyroidism and hypothyroidism.
13. What disturbances are observed in hyper- and hypoparathyroidism?

14. List the main methods of examining the endocrine system.
15. What are the essence and diagnostic significance of the Galli Mainini test?

Test questions

1. What changes in the blood are observed in primary hypothyroidism?
   A) T_4 ↑, TSH ↓
   B) T_4 ↓, TSH ↑
   C) T_4 ↓, TSH ↓
   D) T_4 ↑, TSH ↑
   E) No change

2. Where are the receptors for steroid hormones located?
   A) Inside the cell (cytoplasm and nucleus)
   B) Only on the cell membrane
   C) In the mitochondria
   D) In the extracellular matrix
   E) In the lysosomes

3. When does acromegaly develop?
   A) With growth hormone deficiency in childhood
   B) With growth hormone deficiency in adults
   C) With excess growth hormone after closure of the bone growth plates
   D) With excess thyroid hormones
   E) With excess cortisol

4. What is the mechanism of nephrogenic diabetes insipidus?
   A) Decreased ADH synthesis
   B) Insulin deficiency
   C) Increased aldosterone
   D) Pituitary tumor
   E) Insensitivity of the collecting ducts to ADH

5. What disturbance develops in the syndrome of inappropriate ADH secretion?
   A) Hypernatremia and dehydration
   B) Polyuria and urine of low specific gravity
   C) Hyperkalemia
   D) Water intoxication and dilutional hyponatremia
   E) Hyperglycemia

6. What change in the blood occurs in Cushing syndrome caused by an adrenocortical tumor?
   A) ACTH ↑, cortisol ↑
   B) ACTH ↓, cortisol ↓
   C) ACTH ↓, cortisol ↑
   D) ACTH ↑, cortisol ↓
   E) No change

7. What is the cause of striae and muscle atrophy in hypercortisolism?
    A) Increased lipogenesis
    B) Hyperkalemia
    C) Increased thyroid hormones
    D) Increased protein catabolism
    E) Sodium retention

8. Which are the signs of Conn syndrome?
    A) Hypotension, hyperkalemia, increased renin
    B) Hypoglycemia, hyperpigmentation
    C) Tetany, hypocalcemia
    D) Polyuria, hypernatremia, hypotension
    E) Hypertension, hypokalemia, decreased renin

9. What is the cause of hyperpigmentation in Addison disease?
    A) A hereditary defect of melanin synthesis
    B) Increased ACTH and melanocyte-stimulating hormone due to decreased cortisol
    C) Increased aldosterone
    D) Ultraviolet radiation
    E) Liver failure

10. What is the cause of increased androgens in adrenogenital syndrome (21-hydroxylase deficiency)?
    A) Decreased cortisol with increased ACTH and hyperplasia of the adrenal cortex
    B) A gonadal tumor
    C) Increased LH
    D) Decreased estrogens
    E) Increased prolactin

11. What is the mechanism of Graves–Basedow disease?
    A) Iodine deficiency
    B) Pituitary adenoma
    C) Thyroid peroxidase defect
    D) Resistance to thyroid hormones
    E) Autoantibodies that stimulate TSH receptors

12. What is the main sign of hypoparathyroidism?
    A) Hypercalcemia and kidney stones
    B) Hypocalcemia and tetany
    C) Osteoporosis and fractures
    D) Polyuria
    E) Hypertension

13. Why is timely detection of congenital hypothyroidism especially important?
    A) To prevent obesity
    B) To prevent goiter
    C) To prevent heart failure
    D) Thyroid hormones are essential for brain development — to prevent cretinism
    E) To prevent exophthalmos

14. What is the Galli Mainini test based on detecting?
    A) The release of spermatozoa in a male frog under the action of chorionic gonadotropin in the urine
    B) Glucose in the urine
    C) Estrogens in the blood
    D) Protein in the urine
    E) Progesterone in the blood

15. What laboratory changes are observed in secondary (pituitary) hypogonadism?
    A) Sex hormones ↓, LH and FSH ↑
    B) Sex hormones ↑, LH and FSH ↓
    C) Sex hormones ↓, LH and FSH ↓
    D) Sex hormones ↑, LH and FSH ↑
    E) Prolactin ↓ only

Clinical cases

Case 1

Over the last 2 years a 38-year-old woman has gained weight, mainly on the face and abdomen, while her arms and legs have become thinner. Broad purple-red striae on the abdominal skin, a round, flushed face, increased hair growth, menstrual disturbance. BP 165/100 mm Hg, blood glucose 8.2 mmol/l, potassium slightly decreased, lymphopenia. Cortisol high, ACTH high. A low dose of dexamethasone does not suppress cortisol, but a high dose does. MRI: a pituitary microadenoma.

Questions: What is the disease? Determine the level of the disorder. Explain the pathogenesis of the main signs.

Answer: Cushing disease — central (pituitary) hypercortisolism: a pituitary corticotroph adenoma produces excess ACTH, which stimulates both adrenal cortices (cortisol and ACTH both raised; the adenoma is partly sensitive to feedback — a high dose of dexamethasone suppresses it). Centripetal obesity — different sensitivity of adipose tissue in different regions to cortisol and insulin; striae, thin skin, thin limbs — protein catabolism; steroid diabetes — increased gluconeogenesis and insulin resistance; hypertension and hypokalemia — the mineralocorticoid effect of cortisol and increased vascular sensitivity; lymphopenia — immunosuppression; hirsutism and menstrual disturbance — increased adrenal androgens under the action of ACTH.

Case 2

A 42-year-old man has weakness, fatigue, weight loss, lack of appetite, dizziness and a craving for salty food. The skin, especially the palmar creases and elbows, and the oral mucosa are dark brown. BP 90/55 mm Hg. Blood: Na^+ 128, K^+ 5.9 mmol/l, glucose 3.2 mmol/l, cortisol low, ACTH sharply raised, antibodies to the adrenal cortex positive.

Questions: What is the disease? What is the level of the disorder? How did the hyperpigmentation, hypotension, electrolyte disturbances and hypoglycemia develop?

Answer: Primary chronic adrenocortical insufficiency (Addison disease), autoimmune adrenalitis. The gland itself is destroyed — cortisol is low and, through feedback, ACTH is sharply raised (primary level). ACTH and melanocyte-stimulating hormone are formed from a common precursor — proopiomelanocortin; their increase stimulates melanocytes — hyperpigmentation. Aldosterone deficiency leads to loss of sodium and water (hyponatremia, hypovolemia, hypotension) and potassium retention (hyperkalemia); craving for salt — sodium deficiency. In cortisol deficiency

gluconeogenesis decreases — hypoglycemia; vascular sensitivity to catecholamines falls — hypotension worsens. With stress, infection or surgery an Addisonian crisis (collapse) may develop — the patient needs lifelong hormone replacement therapy.

Case 3

A 29-year-old woman has had palpitations, sweating, hand tremor, nervousness and insomnia for several months, and has lost 8 kg despite a good appetite; diarrhea. Her eyes are protruding and the thyroid gland is diffusely enlarged. Pulse 120 beats/min, BP 150/70 mm Hg. Blood: free T_4 and T_3 sharply raised, TSH very low, antibodies to TSH receptors positive.

Questions: What is the disease? Explain its pathogenesis. What is the mechanism of the weight loss, tachycardia and large pulse pressure?

Answer: Diffuse toxic goiter (Graves–Basedow disease). Autoantibodies that stimulate the TSH receptors continuously stimulate the gland in place of TSH — the gland enlarges diffusely and produces excess hormones; excess T_3 and T_4 suppress TSH through feedback (primary hyperfunction). Exophthalmos — autoimmune inflammation of the orbital tissues. Thyroid hormones increase basal metabolism (Na^+/K^+-ATPase activity, partial uncoupling of oxidation and phosphorylation, heat production) and enhance protein and fat catabolism — weight loss despite a good appetite, sweating. The number and sensitivity of beta-adrenoceptors increase — tachycardia, tremor, nervousness; stroke volume increases (systolic pressure rises) and peripheral vessels dilate (diastolic pressure falls) — large pulse pressure. Intestinal peristalsis accelerates — diarrhea.

Quick review

Table 26.5. *Key facts on Topic 26*

Question	Answer
Hierarchy	Hypothalamus → pituitary → peripheral gland → target tissue
Negative feedback	The peripheral hormone inhibits the hypothalamus and pituitary
Primary insufficiency	Peripheral hormone ↓, tropic hormone ↑
Secondary insufficiency	Peripheral hormone ↓, tropic hormone ↓
Peripheral mechanism	Receptor resistance, disturbed transport or inactivation
Gigantism and acromegaly	Excess growth hormone (before and after closure of the growth plates)
Sheehan syndrome	Postpartum pituitary necrosis — panhypopituitarism
Diabetes insipidus	ADH deficiency or resistance; polyuria, low specific gravity
Cushing	Excess cortisol: centripetal obesity, striae, diabetes
Conn syndrome	Aldosterone ↑, renin ↓: hypertension, hypokalemia
Addison disease	Cortisol and aldosterone ↓, ACTH ↑: hyperpigmentation
Graves disease	Antibodies that stimulate TSH receptors
Cretinism	Congenital hypothyroidism; mental retardation and dwarfism
Hypoparathyroidism	Hypocalcemia — tetany
The Galli Mainini test	Chorionic gonadotropin — release of spermatozoa in the male frog

TOPIC 27

Pathophysiology of extreme states (seminar)

Learning objectives

To explain to the student the concept of extreme and terminal states, the general pathogenesis of conditions associated with exhaustion of the body's adaptive mechanisms, the causes, stages and mechanisms of shock, collapse, coma, crush syndrome and terminal states (preagony, agony, clinical and biological death), and the pathophysiological basis of resuscitation and post-resuscitation disease; to teach the student to apply this knowledge to clinical cases.

By the end of the session, the student should know:

- the definition of extreme and terminal states and their common features;
- the types, stages and general pathogenesis of shock;
- the mechanisms of impaired microcirculation and multiple organ failure in shock;
- the difference between collapse and shock;
- the types, general pathogenesis and degrees of depth of coma;
- the pathogenesis of crush syndrome;
- the stages of terminal states, the factors affecting the duration of clinical death, and post-resuscitation disease.

The student should be able to:

- calculate the shock index and assess the severity of shock;
- assess the level of consciousness using the Glasgow Coma Scale;
- determine the type of shock and coma in clinical cases.

1. Extreme and terminal states

Extreme state — a state resulting from extraordinarily strong influences on the body (injury, blood loss, infection, poisoning, etc.) with excessive strain and exhaustion of the adaptive mechanisms and a risk of profound disturbance of vital functions. These include **shock, collapse, coma**, crush syndrome and acute organ failure.

Terminal state — states on the border between life and death that do not resolve spontaneously and lead to death without special help: preagony, terminal pause, agony and clinical death.

Common features of extreme states:

- disturbance of vital functions (circulation, respiration, CNS activity);
- initial hyperactivity of the adaptive mechanisms followed by their exhaustion (stress reaction → exhaustion);
- **hypoxia** — the common pathogenetic link of all extreme states;
- metabolic acidosis, endogenous intoxication;
- formation of **vicious circles** and multiple organ failure.

2. Shock

Shock — an acute, severe generalized pathological process developing as a result of extremely strong influences, characterized by a sharp decrease in the blood supply (perfusion) of organs and tissues,

cellular hypoxia and metabolic disturbance, followed by multiple organ failure.

Table 27.1. *Types of shock (by pathogenesis)*

Type	Main mechanism	Examples
Hypovolemic	Decreased circulating blood volume	Blood loss (hemorrhagic), plasma loss (burns), dehydration (vomiting, diarrhea)
Cardiogenic	Sharp decline in the pumping function of the heart	Extensive myocardial infarction, severe arrhythmias, acute valvular insufficiency
Obstructive	Mechanical obstruction to blood flow	Pericardial tamponade, massive pulmonary embolism, tension pneumothorax
Distributive (vasogenic)	Sharp fall in vascular tone, redistribution of blood to the periphery	Septic (endotoxin), anaphylactic, neurogenic (spinal cord injury)
Traumatic and burn shock	Mixed: pain, blood and plasma loss, toxemia	Multiple bone fractures, extensive burns

General pathogenesis of shock

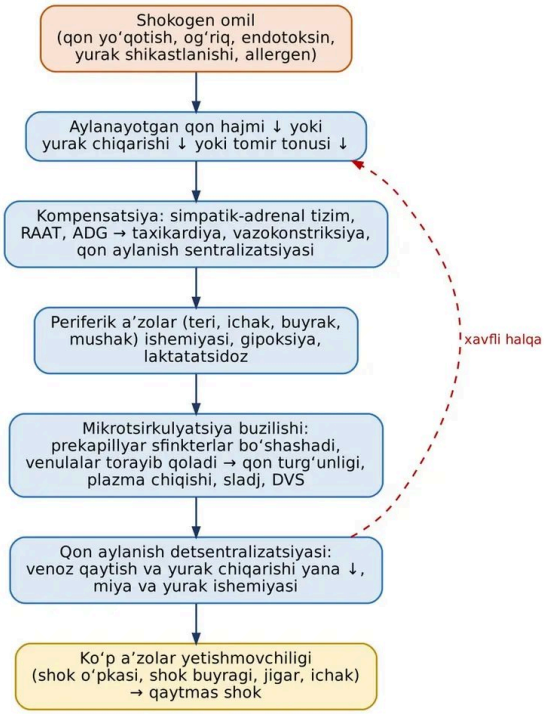


Figure 27.1. *Pathogenesis of shock and the vicious circle*

Stages of shock (according to N. I. Pirogov):

1. **Erectile (excitation) stage** — short (minutes), especially marked in traumatic and burn shock: strong pain impulses, excitation of the CNS and the sympathoadrenal system — the patient is restless, agitated and talkative, BP normal or raised, tachycardia, pale skin.
2. **Torpid (inhibition) stage** — inhibition of the CNS: the patient is apathetic, immobile and shows no interest in the surroundings, BP decreased, pulse rapid and weak (thready), skin pale-gray, cold sweat, oliguria.

Stages by circulatory changes:

- **Compensated shock** — under the action of catecholamines, angiotensin II and vasopressin the vessels of the skin, muscles and internal organs (intestine, kidney) constrict and blood is directed

mainly to the brain and heart — **centralization of the circulation**; BP is maintained.

- **Decompensated (progressive) shock** — with prolonged ischemia, lactate, H^+, adenosine and histamine accumulate in the tissues; the **precapillary sphincters**, being sensitive to them, relax, while the **postcapillary venules** remain constricted — blood enters the capillaries but cannot leave (**stasis**, "sequestration"); hydrostatic pressure rises and plasma leaks into the tissue, the blood thickens, erythrocytes aggregate (**sludge**) and DIC develops. Venous return decreases and BP falls — **decentralization**: the brain and heart also become ischemic.
- **Irreversible shock** — cells die en masse, and lysosomal enzymes, myocardial depressant factor (formed in pancreatic ischemia) and endotoxins (which enter the blood when the intestinal barrier fails) accumulate; treatment is ineffective.

> **CAUTION**
> **Multiple organ failure syndrome** — the main cause of death in shock: **shock lung** (acute respiratory distress syndrome — damage to the alveolar–capillary membrane, pulmonary edema, hypoxemia), **shock kidney** (acute kidney failure — prerenal, then acute tubular necrosis), shock liver (cytolysis, jaundice), intestine (mucosal ischemia, translocation of bacteria and endotoxins), heart (reduced coronary perfusion, myocardial depression), CNS (encephalopathy), blood (DIC).

Shock index (Allgöwer index) = pulse / systolic BP. Normally about 0.5–0.7; around 1.0 — moderately severe shock (about 20–30 percent of blood volume lost); 1.5 or higher — severe shock.

Some types of shock

- **Hemorrhagic shock** — when more than 20–25 percent of the circulating blood volume is lost rapidly (topic 17).
- **Burn shock** — with deep burns of more than 15–20 percent of the body surface (more than 10 percent in children): severe pain, **plasma loss** due to a sharp increase in capillary permeability — hemoconcentration (hematocrit and hemoglobin rise), hypovolemia; toxins from the burned tissues (burn toxemia).
- **Septic shock** — bacterial endotoxin (lipopolysaccharide) activates macrophages → cytokines (TNF-alpha, IL-1, IL-6) → activation of **inducible NO synthase** → excess NO formation → marked vasodilation and hypotension; endothelial damage, increased capillary permeability, DIC, mitochondrial dysfunction. There is an initial "warm" phase (hyperdynamic — warm skin, increased cardiac output) and a later "cold" phase (hypodynamic).
- **Anaphylactic shock** — a type I allergic reaction: histamine, leukotrienes and platelet-activating factor are released from mast cells and basophils → marked vasodilation, increased vascular permeability (plasma leakage), bronchospasm, laryngeal edema (topic 8). Care — immediate epinephrine.
- **Cardiogenic shock** — when more than 40 percent of the left ventricular myocardium is damaged (topic 21): cardiac output falls, with pulmonary congestion (pulmonary edema) together with peripheral hypoperfusion.

3. Collapse

Collapse — acute vascular insufficiency resulting from an acute fall in vascular tone and (or) a decrease in circulating blood volume: arterial and venous pressure fall sharply, with hypoxia of the brain, heart and other organs.

Causes: infections and intoxications (critical fall of fever, typhoid fever, pneumonia), blood loss, orthostatic position (standing up quickly, antihypertensive drugs), adrenal insufficiency, severe hypoxia, poisoning.

Table 27.2. *Distinguishing shock from collapse*

Feature	Shock	Collapse
Onset	Staged (erectile, torpid), with vicious circles	Sudden
Initial mechanism	Strain of compensatory reactions, followed by impaired microcirculation	Primary fall in vascular tone and (or) decrease in blood volume
Consciousness	Excitation in the erectile stage, inhibition in the torpid stage	Often clouded, or fainting
Microcirculation and multiple organ failure	Main pathogenetic link	Usually does not develop quickly
Prognosis	Serious; urgent care is needed	Usually recovers quickly if the cause is removed

CAUTION

Collapse is not shock, but prolonged collapse can progress to shock through tissue hypoxia and impaired microcirculation. **Fainting (syncope)** — a brief (seconds, minutes) loss of consciousness with spontaneous recovery, resulting from short-term cerebral hypoperfusion (vasovagal, orthostatic, cardiogenic — arrhythmias).

4. Coma

Coma — an extremely severe state resulting from profound inhibition of the CNS, with complete loss of consciousness, weakening or loss of responses to external stimuli, and disturbed reflexes and regulation of vital functions (respiration, circulation).

Table 27.3. *Types of coma*

Group	Types
Primary cerebral comas	Traumatic, stroke (hemorrhagic, ischemic), tumors, meningitis and encephalitis, epileptic
Endocrine comas	Diabetic ketoacidotic, hyperosmolar, lactic acidotic, hypoglycemic (topic 10), hypothyroid (myxedema), adrenal (Addisonian crisis), thyrotoxic crisis
Toxic comas	Endogenous: uremic (topic 25), hepatic (topic 24); exogenous: poisoning with alcohol, opioids, barbiturates, carbon monoxide, organophosphorus compounds
Hypoxic comas	Respiratory, hemic (anemic), circulatory (cardiogenic)
Comas due to water–electrolyte and thermal disturbances	Hypo- and hypernatremic, heat stroke (topic 2), hypothermic

General pathogenesis of coma:

- **disturbed energy supply of neurons** — hypoxia, hypoglycemia (the brain depends mainly on glucose for energy), mitochondrial damage;
- **direct action of toxins** — ammonia, uremic toxins, ketone bodies, alcohol, drugs;

- **water–electrolyte and acid–base disturbances** — osmotic imbalance between the intracellular and extracellular compartments (cerebral edema or dehydration of brain cells), acidosis;
- **cerebral edema and raised intracranial pressure** — reduced cerebral blood flow, compression of the brainstem (herniation);
- **disturbed neurotransmitter metabolism** (false neurotransmitters, increased GABA-ergic inhibition);
- as a result, the activity of cortical and subcortical structures is inhibited and the activating influence of the reticular formation is lost — consciousness is lost; if the process deepens, the respiratory and vasomotor centers in the medulla oblongata are damaged.

Degrees of impaired consciousness: clear consciousness → obtundation (somnolence, pre-sopor) → **sopor** (protective movements only in response to strong painful stimuli) → **coma** (I — moderate, II — deep, III — terminal, transcendental coma).

Table 27.4. *Glasgow Coma Scale (abbreviated)*

Criterion	Response and score
Eye opening	Spontaneous — 4; to speech — 3; to pain — 2; none — 1
Verbal response	Oriented — 5; confused — 4; inappropriate words — 3; incomprehensible sounds — 2; none — 1
Motor response	Obeys commands — 6; localizes pain — 5; withdraws from pain — 4; abnormal flexion — 3; abnormal extension — 2; none — 1
Interpretation	15 — clear consciousness; 13–14 — obtundation; 9–12 — sopor; 8 or less — coma; 3 — terminal coma

5. Crush syndrome (prolonged compression syndrome)

Crush syndrome — a severe condition that develops after soft tissues (mainly limb muscles) have been compressed for a long time (more than 4–6 hours) and then released (earthquakes, car accidents, building collapses).

Main links of the pathogenesis:

1. **Pain factor** — prolonged pain impulses during compression, elements of traumatic shock.
2. **Plasma loss** — after release, blood flows into the damaged muscles and, because capillary permeability is increased, plasma leaks into the tissue — the limb swells sharply, with **hypovolemia** and hemoconcentration.
3. **Toxemia** — **myoglobin**, **potassium**, phosphates, creatine, lactate and other toxins are released from the ischemic muscles into the blood (**rhabdomyolysis**); **hyperkalemia** can cause cardiac arrest.
4. **Acute kidney failure** — myoglobin precipitates in the tubules in an acidic environment, obstructs them and damages the epithelium; hypovolemia aggravates renal ischemia (topic 25).

Periods: early (days 1–3, shock and hemodynamic disturbances predominate), intermediate (days 3–12, acute kidney failure), late (recovery, local complications — infection, muscle atrophy, contractures).

CLINICAL SIGNIFICANCE

Care: start intravenous fluids **before** releasing the compressed limb; at the moment of release, bandage the limb and cool it (to reduce the rapid absorption of toxins), give analgesia and alkaline solutions (alkalinizing the urine reduces myoglobin precipitation), monitor for hyperkalemia, and use hemodialysis if needed.

6. Terminal states and death



Figure 27.2. *Sequence of terminal states*

Table 27.5. *Stages of terminal states*

Stage	Signs	Pathophysiological essence
Preagonal state	Consciousness clouded or absent, BP sharply low (systolic below 60–70 mm or undetectable), thready pulse, breathing rapid and shallow or slow, skin pale-bluish	Increasing hypoxia, inhibition of the higher parts of the CNS
Terminal pause	Breathing stops for a few seconds up to 2–4 minutes, bradycardia, the pupils dilate	Extinction of cortical activity, predominance of vagal influence
Agony	Gasping (agonal) breathing, a brief slight rise in BP, temporary increase in cardiac activity, convulsions, relaxation of the sphincters	The last activity of the bulbar centers, released from the control of the higher parts; anaerobic glycolysis predominates
Clinical death	Breathing and cardiac activity have stopped, no reflexes, the pupils dilated and unresponsive to light	Metabolism continues at a minimal level; the changes are still reversible
Biological death	Irreversible changes; then signs of death (livor mortis, rigor mortis, algor mortis, the “cat's eye” sign)	Death of the neurons of the cerebral cortex first of all

Clinical death

Duration — 4–6 minutes at normal temperature: during this time the cerebral cortex tolerates complete ischemia.

Factors that increase its duration: hypothermia (metabolism and oxygen demand decrease — in drowning in cold water clinical death may last 30 minutes or more), early childhood, sudden death (electric current, arrhythmia — the preagony is short and energy reserves are preserved).

Factors that shorten its duration: prolonged preagony and agony (severe chronic diseases, prolonged blood loss — energy reserves exhausted), hyperthermia, old age.

Resuscitation and post-resuscitation disease

Resuscitation — restoration of vital functions: opening the airway, artificial ventilation, closed-chest cardiac massage, defibrillation (in ventricular fibrillation), drugs (epinephrine).

Post-resuscitation disease (V. A. Negovsky) — a specific pathological state after successful recovery from clinical death:

- **reperfusion injury** — when blood flow is restored, large amounts of reactive oxygen species are formed and damage membranes through lipid peroxidation; massive entry of Ca^{2+} into the cells;
- cerebral edema and posthypoxic encephalopathy, delayed death of some neurons (apoptosis);
- cardiac dysfunction, arrhythmias;
- multiple organ failure (lungs, kidneys, liver);
- translocation of endotoxins from the intestine, DIC, immunodepression.

> **MEMORY AID**
> The duration of clinical death: "**cold prolongs it, heat shortens it, a long agony steals it**".

7. Practical skills

Skill: calculating the shock index and assessing the severity of shock

1. Count the patient's pulse rate for 1 minute.
2. Measure the systolic arterial pressure.
3. Shock index = pulse / systolic BP.
4. Assess the result: 0.5–0.7 — normal; about 1.0 — moderately severe; 1.5 or more — severe shock.
5. Assess additional signs: skin color and temperature, level of consciousness, hourly diuresis (below 0.5 ml/kg/h — renal hypoperfusion), capillary refill time (more than 2 seconds).

Skill: assessing the level of consciousness using the Glasgow Coma Scale

1. Assess the eye-opening response (1–4 points).
2. Assess the verbal response (1–5 points).
3. Assess the motor response (1–6 points).
4. Add up the points (3–15).
5. Determine the degree of impaired consciousness: 15 — clear, 13–14 — obtundation, 9–12 — sopor, 8 or less — coma.

6. To determine the cause of the coma, assess the history, glucose, breath odor, pupils and other signs.

Review questions

1. What is meant by extreme and terminal states? What are their common features?
2. What is shock? List its types by pathogenesis.
3. How do the erectile and torpid stages of shock differ?
4. What is centralization of the circulation and what is its significance?
5. How does the disturbance of microcirculation develop in shock?
6. Explain the components of the multiple organ failure syndrome.
7. What is the role of NO in the pathogenesis of septic shock?
8. What are the features of burn shock?
9. What is collapse and how does it differ from shock?
10. List the types of coma.
11. Explain the general pathogenesis of coma.
12. What is the pathogenesis of crush syndrome?
13. What are the stages of terminal states and their pathophysiological essence?
14. What factors affect the duration of clinical death?
15. What are the mechanisms of post-resuscitation disease?

Test questions

1. What is the common pathogenetic link of all extreme states?
   A) Hyperthermia
   B) Hyperglycemia
   C) Arterial hypertension
   D) Hypoxia
   E) Alkalosis
2. Which sign is typical of the erectile stage of shock?
   A) Apathy and immobility of the patient
   B) Excitation, restlessness, BP normal or raised
   C) Thready pulse and marked hypotension
   D) Anuria
   E) Unconsciousness
3. What is the essence of centralization of the circulation?
   A) Redistribution of blood in favor of the brain and heart
   B) Dilation of all vessels
   C) Accumulation of blood at the periphery
   D) Increased cardiac output
   E) Decreased venous return

4. What is the mechanism of the microcirculatory disturbance in decompensated shock?
   A) Constriction of all vessels
   B) Dilation of the venules
   C) Relaxation of the precapillary sphincters while the venules remain constricted
   D) Increased arteriolar spasm
   E) Accelerated blood flow

5. Which condition causes cardiogenic shock?
   A) Extensive burns
   B) Anaphylaxis
   C) Blood loss
   D) Sepsis
   E) Extensive myocardial infarction

6. What is the main mediator of the marked vasodilation in septic shock?
   A) NO produced by inducible NO synthase
   B) Epinephrine (adrenaline)
   C) Angiotensin II
   D) Vasopressin
   E) Endothelin

7. What change in the blood is observed in burn shock?
   A) Hemodilution
   B) Hemoconcentration due to plasma loss
   C) Decreased hemoglobin
   D) Hypervolemia
   E) No hypoproteinemia

8. What does a shock index (Allgöwer) of 1.5 indicate?
   A) Normal
   B) Mild shock
   C) Severe shock
   D) Arterial hypertension
   E) Bradycardia

9. What is the main difference between collapse and shock?
   A) It always begins with pain
   B) Its erectile stage is pronounced
   C) Multiple organ failure develops rapidly
   D) It occurs only in trauma
   E) It develops suddenly, and its initial mechanism is a primary fall in vascular tone

10. What does a score of 8 or less on the Glasgow Coma Scale mean?
    A) Clear consciousness
    B) Obtundation
    C) Sopor
    D) Coma
    E) Normal

11. What is the main cause of acute kidney failure in crush syndrome?
    A) Hypertension
    B) Hypoglycemia
    C) Obstruction of the tubules by myoglobin and hypovolemia
    D) Hemolysis
    E) Glomerulonephritis

12. What is the main cause of sudden death in crush syndrome?
    A) Hypoglycemia
    B) Hypokalemia
    C) Alkalosis
    D) Hyperkalemia
    E) Hyperthermia

13. What is the duration of clinical death at normal temperature?
    A) 4–6 minutes
    B) 30–40 minutes
    C) 1–2 hours
    D) 15–20 seconds
    E) 12 hours

14. Which factor increases the duration of clinical death?
    A) Hyperthermia
    B) Hypothermia
    C) Prolonged agony
    D) Old age
    E) Prolonged blood loss

15. What is the pathophysiological essence of agony?
    A) Increased activity of the cerebral cortex
    B) Onset of irreversible changes
    C) Complete cessation of metabolism
    D) Cessation of cardiac activity only
    E) The last activity of the bulbar centers released from the control of the higher parts

Clinical cases

Case 1

A 30-year-old man suffered second- and third-degree burns over 40 percent of his body surface at work. In the first minutes he was agitated and shouting, BP 140/90. 2 hours later he is apathetic and answers questions slowly, BP 85/50, pulse 128 beats/min, pale skin, cold sweat, hourly diuresis 15 ml. Blood: hematocrit 0.58, hemoglobin 190 g/l, K^+ raised.

Questions: What condition has developed? Determine its stages. Explain the mechanism of the hemoconcentration, oliguria and hyperkalemia. Calculate the shock index.

Answer: Burn shock: first the erectile stage (pain impulses, sympathoadrenal excitation — restlessness, raised BP), then the torpid stage (CNS inhibition, hypotension, thready pulse). In the burned area capillary permeability increases sharply — plasma leaks into the tissue and out through the burn surface — hypovolemia and hemoconcentration (raised hematocrit and hemoglobin). Under the action of hypovolemia and catecholamines the renal vessels constrict and filtration decreases —

oliguria (risk of shock kidney). Hyperkalemia — release of potassium from the breakdown of burned tissues and erythrocytes and reduced renal excretion. Shock index = 128 / 85 ≈ 1.5 — severe shock. Treatment — rapid infusion therapy of adequate volume, analgesia.

Case 2

A 68-year-old woman has had a fever for 3 days (urinary tract infection). Her condition suddenly worsened: temperature 39.5 °C, BP 80/40, pulse 125 beats/min, respiratory rate 28, the skin at first warm and flushed, later cold and mottled. Scanty urine. Blood: leukocytes 22 × 10^9/l, lactate raised, platelets decreased, creatinine raised. BP did not rise even after infusion.

Questions: What condition is this? Explain the pathogenesis. What are the “warm” and “cold” phases related to? What is the cause of the thrombocytopenia and the rise in lactate?

Answer: Septic shock (urosepsis). Endotoxin of gram-negative bacteria activates macrophages and endothelium — inflammatory cytokines (TNF-alpha, IL-1, IL-6) are released; they activate inducible NO synthase — excess NO causes marked vasodilation (distributive shock, no response to infusion). The initial “warm” (hyperdynamic) phase — the peripheral vessels are dilated and cardiac output is compensatorily increased; then, because of myocardial depression and plasma leakage from the capillaries, it passes into the “cold” (hypodynamic) phase — peripheral vasoconstriction, mottled skin. Endothelial damage and activation of the clotting system — DIC (platelets are consumed). Tissue hypoperfusion and mitochondrial dysfunction — anaerobic glycolysis, lactic acidosis. Oliguria and raised creatinine — shock kidney; tachypnea — a response to acidosis and early shock lung: multiple organ failure is developing.

Case 3

In winter a 7-year-old child fell into a frozen lake and was pulled out of the water after 20 minutes. No breathing or cardiac activity, pupils dilated, body temperature 28 °C. Resuscitation restored cardiac activity; over the following days cerebral edema and impaired lung and kidney function were observed, with gradual recovery.

Questions: Why was resuscitation successful even after 20 minutes? What condition are the post-recovery disturbances related to, and what is their mechanism?

Answer: At normal temperature clinical death lasts 4–6 minutes, but hypothermia (28 °C) sharply reduces the metabolism and oxygen demand of the tissues, especially the brain neurons; in children tissue resistance to hypoxia is higher, and the death was sudden (there was no prolonged agony, and energy reserves were preserved) — this is why clinical death was prolonged and resuscitation was successful. The later disturbances are post-resuscitation disease: reperfusion injury when blood flow is restored (reactive oxygen species, lipid peroxidation, entry of Ca^{2+} into cells), posthypoxic cerebral edema, lung injury due to aspiration and hypoxia (respiratory distress), renal ischemia — multiple organ failure. Such patients need prolonged observation in the intensive care unit.

Quick review

Table 27.6. *Key facts on Topic 27*

Question	Answer
Extreme state	A severe state with exhaustion of the adaptive mechanisms
Shock	Sharp decrease in perfusion, hypoxia, multiple organ failure
Types of shock	Hypovolemic, cardiogenic, obstructive, distributive

Question	Answer
Erectile stage	Excitation, BP normal or high
Torpid stage	Inhibition, hypotension, thready pulse
Centralization	Blood is redistributed in favor of the brain and heart
Microcirculatory disturbance	Precapillaries relax, venules remain constricted — sludge, DIC
Septic shock	Endotoxin → cytokines → NO → vasodilation
Shock index	Pulse / systolic BP; above 1.5 — severe shock
Collapse	Acute vascular insufficiency, sudden onset
Coma	Complete loss of consciousness; Glasgow 8 or less
Crush syndrome	Plasma loss, myoglobinuria, hyperkalemia, kidney failure
Agony	The last activity of the bulbar centers, gasping breathing
Clinical death	4–6 minutes, a reversible state; prolonged in hypothermia
Post-resuscitation disease	Reperfusion injury, cerebral edema, multiple organ failure

Answers to the test questions

Topic 1: 1-C; 2-A; 3-E; 4-B; 5-D; 6-A; 7-C; 8-E; 9-B; 10-D; 11-B; 12-E; 13-A; 14-C; 15-D.

Topic 2: 1-B; 2-D; 3-A; 4-E; 5-C; 6-B; 7-E; 8-A; 9-D; 10-C; 11-A; 12-C; 13-E; 14-B; 15-D.

Topic 3: 1-D; 2-A; 3-C; 4-E; 5-B; 6-D; 7-A; 8-B; 9-E; 10-C; 11-D; 12-A; 13-C; 14-E; 15-B.

Topic 4: 1-A; 2-C; 3-B; 4-E; 5-D; 6-B; 7-A; 8-D; 9-C; 10-E; 11-A; 12-E; 13-B; 14-D; 15-C.

Topic 5: 1-D; 2-B; 3-E; 4-A; 5-C; 6-E; 7-C; 8-A; 9-B; 10-D; 11-B; 12-D; 13-A; 14-C; 15-E.

Topic 6: 1-C; 2-E; 3-A; 4-D; 5-B; 6-A; 7-D; 8-B; 9-E; 10-C; 11-E; 12-B; 13-C; 14-A; 15-D.

Topic 7: 1-B; 2-A; 3-D; 4-C; 5-E; 6-D; 7-B; 8-E; 9-A; 10-C; 11-C; 12-A; 13-E; 14-B; 15-D.

Topic 8: 1-E; 2-D; 3-C; 4-B; 5-A; 6-C; 7-E; 8-A; 9-D; 10-B; 11-D; 12-C; 13-B; 14-E; 15-A.

Topic 9: 1-A; 2-E; 3-D; 4-C; 5-B; 6-B; 7-C; 8-E; 9-A; 10-D; 11-A; 12-B; 13-D; 14-C; 15-E.

Topic 10: 1-C; 2-B; 3-A; 4-E; 5-D; 6-A; 7-E; 8-C; 9-D; 10-B; 11-B; 12-D; 13-E; 14-A; 15-C.

Topic 11: 1-D; 2-C; 3-E; 4-B; 5-A; 6-E; 7-A; 8-D; 9-B; 10-C; 11-C; 12-E; 13-A; 14-D; 15-B.

Topic 12: 1-B; 2-D; 3-C; 4-A; 5-E; 6-C; 7-B; 8-A; 9-E; 10-D; 11-D; 12-A; 13-C; 14-B; 15-E.

Topic 13: 1-E; 2-A; 3-B; 4-D; 5-C; 6-D; 7-C; 8-B; 9-A; 10-E; 11-E; 12-C; 13-D; 14-A; 15-B.

Topic 14: 1-A; 2-D; 3-E; 4-B; 5-C; 6-E; 7-B; 8-C; 9-D; 10-A; 11-B; 12-A; 13-C; 14-E; 15-D.

Topic 15: 1-C; 2-A; 3-D; 4-E; 5-B; 6-B; 7-D; 8-A; 9-C; 10-E; 11-D; 12-E; 13-B; 14-A; 15-C.

Topic 16: 1-B; 2-E; 3-C; 4-A; 5-D; 6-A; 7-C; 8-D; 9-E; 10-B; 11-C; 12-B; 13-A; 14-D; 15-E.

Topic 17: 1-D; 2-A; 3-B; 4-C; 5-E; 6-C; 7-E; 8-B; 9-A; 10-D; 11-A; 12-D; 13-E; 14-C; 15-B.

Topic 18: 1-E; 2-C; 3-A; 4-D; 5-B; 6-D; 7-A; 8-E; 9-B; 10-C; 11-B; 12-E; 13-C; 14-D; 15-A.

Topic 19: 1-A; 2-B; 3-E; 4-D; 5-C; 6-C; 7-B; 8-D; 9-E; 10-A; 11-D; 12-A; 13-B; 14-E; 15-C.

Topic 20: 1-C; 2-D; 3-B; 4-A; 5-E; 6-E; 7-A; 8-C; 9-D; 10-B; 11-A; 12-C; 13-D; 14-B; 15-E.

Topic 21: 1-B; 2-C; 3-A; 4-E; 5-D; 6-B; 7-D; 8-E; 9-A; 10-C; 11-E; 12-B; 13-A; 14-C; 15-D.

Topic 22: 1-D; 2-E; 3-C; 4-B; 5-A; 6-A; 7-C; 8-B; 9-D; 10-E; 11-C; 12-D; 13-E; 14-A; 15-B.

Topic 23: 1-E; 2-B; 3-D; 4-A; 5-C; 6-D; 7-E; 8-A; 9-C; 10-B; 11-B; 12-A; 13-C; 14-E; 15-D.

Topic 24: 1-A; 2-C; 3-E; 4-D; 5-B; 6-E; 7-A; 8-B; 9-C; 10-D; 11-D; 12-E; 13-B; 14-A; 15-C.

Topic 25: 1-C; 2-E; 3-D; 4-B; 5-A; 6-B; 7-C; 8-D; 9-A; 10-E; 11-A; 12-C; 13-E; 14-D; 15-B.

Topic 26: 1-B; 2-A; 3-C; 4-E; 5-D; 6-C; 7-D; 8-E; 9-B; 10-A; 11-E; 12-B; 13-D; 14-A; 15-C.

Topic 27: 1-D; 2-B; 3-A; 4-C; 5-E; 6-A; 7-B; 8-C; 9-E; 10-D; 11-C; 12-D; 13-A; 14-B; 15-E.

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